HODGE CLASSES BEYOND THE WEIL LINES

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Abstract. We prove that the Hodge conjecture is equivalent to two statements about algebraic loci, one on abelian varieties and one on the classes reached from them by algebraic correspondences, each the existence of an interior point in a dense locus that is either everything or meagre; we prove the second on every variety whose cohomology is reached from abelian varieties, on every hypersurface of simplex type and on every complete intersection of diagonal hypersurfaces of Vandermonde type; we prove the Hodge conjecture itself for the very general such intersection of quadrics in every dimension, of cubics and quartics up to dimension seven and of sextics up to dimension five; and we study the first instance of the statement on abelian varieties, the exceptional classes of a Mumford fourfold.

Contents

1. Introduction 2
2. The reach of the obstructions 5
3. Hodge classes beyond the Weil lines 83
4. The main theorem in every family 137
5. The closure theorem 147
Acknowledgements 152
Declarations 152
References 153

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This paper and its companions [I, II] together replace the earlier paper *Algebraic loci of Weil classes on abelian varieties* by the authors and Priyabrata Mandal (archive [BMB26]); every result of that paper appears in one of the three with its proof. The preprint *Relative secant cycles and Hodge classes* [BB26a] contains errors and incomplete proofs and is replaced by them.

Every proof is given in the text. Programs that repeat much of the finite arithmetic, and a Lean 4 certificate of part of it, are archived as described in the statement on data availability at the end of the paper; no statement rests on them.

1. Introduction

Let X be a smooth complex projective variety. The Hodge conjecture asserts that the cycle class map

$$\mathrm{cl}^p_X\colon \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^p(X) \coloneqq H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$$

is surjective for every p. Beyond degree two the best understood Hodge classes are those of abelian varieties, and the first of them that divisors do not produce are the Weil classes, which Weil exhibited on abelian varieties with an action of an imaginary quadratic field whose two eigenspaces on the tangent space have the same dimension [Wei77]; the construction works for every CM field F. Markman proved that the Weil classes are algebraic on the abelian sixfolds of split Weil type for every imaginary quadratic field, and, through a product argument of Schoen [Sch98, Proposition 10], on every abelian fourfold of Weil type [Mar25a, Mar25b]. His sheaves are built from secant sheaves on an abelian variety of half the dimension, and they extend along the family because they satisfy an equivariant form of semiregularity.

The companion paper [I] studies the set of members of a family on which a flat Hodge class is algebraic, its *algebraic locus*, and determines it for the Weil classes of every CM field and every dimension at once: for a CM field F, an integer $n \geq 2$ and a discriminant δ, the abelian varieties of (F, n, δ)-Weil type form one family over a connected period domain $\mathcal{D}_F$, and the locus Σ_F of members on which the Weil classes are algebraic contains an explicit point of complex multiplication, is dense, is a countable union of closed algebraic strata of the Shimura variety of the family, and is either all of $\mathcal{D}_F$ or meagre; so the Weil classes are algebraic on every member exactly when Σ_F is closed ([I, Theorem 1.1], which we call the main theorem throughout). The companion paper [II] determines what closing Σ_F requires of a single object at a single point: a perfect complex at a base point whose Chern character is a nonzero multiple of a Weil class plus a polynomial in the polarisation and whose semiregularity map is injective makes Σ_F everything ([II, Theorem 2.3]), the criterion is never met when the Chern character is exactly a Weil class ([II, Theorem 3.5], [II, Theorem 5.4]), and with powers of the polarisation added it becomes one number ([II, Theorem 3.9]). The notation of those papers is kept here.

This paper places the Weil classes inside the whole conjecture. It determines the reach of the obstructions of [I], shows that the Hodge conjecture is two statements about algebraic loci, proves each on the families where it can be proved, proves the conjecture itself on a family of varieties of every dimension, and studies the first instance of the statement on abelian varieties, the exceptional classes of a Mumford fourfold.

1.1. Diagonal complete intersections.

The paper also proves the Hodge conjecture itself on a family of varieties of every dimension. Let $X \subset \mathbb{P}^N$ be a complete intersection of c diagonal hypersurfaces $\sum_i a_{ki} x_i^d = 0$ of degree d. It is *of Vandermonde type* if $a_{ki} = \mu_i \sum_{l=1}^{c} g_{kl} \lambda_i^{l-1}$ for distinct numbers $\lambda_0, \ldots, \lambda_N$, nonzero numbers μ_i and an invertible matrix (g_{kl}); for $c \geq 2$ this says that the columns of (a_{ki}) are distinct points of a rational normal curve, and every smooth complete intersection of two diagonal hypersurfaces of the same degree is of this type. Such an X is isomorphic to a variety $X_{d,N-c}(\lambda)$ that depends only on λ (Theorem 2.55), and *very general* refers to λ. For $d = 2$ we call X a *Vandermonde quadric intersection.*

Theorem 1.1 (Diagonal complete intersections)**.** *Every rational Hodge class is algebraic on the very general Vandermonde quadric intersection, in every dimension and in every projective space. The same holds for the very general complete intersection of Vandermonde*

type of diagonal cubics or quartics of dimension at most seven, or of diagonal sextics of dimension at most five, and for every smooth complete intersection of two diagonal cubics, of two diagonal quartics or of two diagonal sextics in $\mathbb{P}^6$.

The first two statements are Theorem 2.66(iii) and the third is Corollary 2.56(ii). The variety X is a quotient of a power of a generalised Fermat curve, so its Hodge classes come from abelian subvarieties of the Jacobian of that curve. For quadrics the monodromy of the curve along a chain of vanishing cycles is the full symplectic group on every eigenspace, so every Hodge class comes from powers of polarisations and no theorem on abelian varieties of higher dimension is needed. For cubics, quartics and sextics we compute the monodromy by an induction on the number of branch points (Proposition 2.64), and the classes it leaves are divisor classes, Weil classes of abelian fourfolds of Weil type, and Weil classes of abelian sixfolds of split Weil type, all algebraic by the theorems of Lefschetz and Markman [Mar25a, Mar25b]. In odd dimension every Hodge class is a multiple of a power of the hyperplane class. So the bound seven for cubics and quartics is the point where abelian varieties of dimension eight appear, and the bound five for sextics comes from abelian sixfolds of non-split Weil type, which occur in dimension six.

1.2. **Weil classes and the Hodge conjecture.** Sections 2 to 4 place the Weil classes inside the conjecture. Two statements carry all of it. The first, (F2), asks that every Hodge class on an abelian variety outside the subring generated by divisor classes and Weil classes of quotients be algebraic; multiplying by an exceptional class on the square of a Mumford fourfold moves every Hodge class of every abelian variety outside that subring, so (F2) is equivalent to the Hodge conjecture for all abelian varieties (Proposition 3.49). The second, (F3′), asks that every Hodge class on every smooth projective variety be algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences (Definition 2.34). The Hodge conjecture is equivalent to (F2) and (F3′) together (Proposition 2.35, Theorem 5.1).

We prove (F3′) for every variety whose cohomology is reached from abelian varieties by correspondences, a class closed under products, surjective images, blow-ups, projective bundles and Hilbert schemes of points on surfaces, which contains every Fermat hypersurface and every K3 surface whose Kuga–Satake correspondence is algebraic (Proposition 2.35, Proposition 2.42); for every variety of dimension at most three, and on a variety of dimension k in the degrees 0, 2, $2k-2$ and $2k$; for every fourfold and fivefold whose zero-cycles are supported in dimension at most three, among them every uniruled fourfold and every rationally connected fivefold (Proposition 2.43, Proposition 2.44); for every fourfold and fivefold dominated by a variety of that class (Proposition 2.46); for every smooth ample divisor of odd dimension in a variety of that class, and in even dimension outside the middle degree and for a very general member (Proposition 2.47); and, by a toric compactification whose boundary is again of the same kind, for every hypersurface of simplex type, so for every smooth Delsarte hypersurface of every dimension and every cyclic cover branched along one, whose Hodge conjecture then follows from that of Fermat varieties of one degree (Theorem 2.52, Corollary 2.53). A complete intersection of diagonal hypersurfaces whose coefficients are of Vandermonde type, which holds for every smooth complete intersection of two of them, is a quotient of a power of a generalised Fermat curve, so it satisfies (F3′), and its Hodge conjecture reduces to abelian varieties attached to the characters of the curve; for two diagonal hypersurfaces of degree 3, 4 or 6 in $\mathbb{P}^6$ these have dimension at most five, and the Hodge conjecture holds, the exceptional classes being Weil classes of abelian fourfolds of Weil type (Theorem 2.55, Corollary 2.56). At

a very general configuration the monodromy of the curve leaves only the Hodge classes that come from polarisations and from Weil classes; we compute that monodromy for cyclic covers of degree three, four and six by an induction on the number of branch points (Proposition 2.64), so the Hodge conjecture holds for the very general such intersection of quadrics in every dimension, of cubics and quartics up to dimension seven, and of sextics up to dimension five, in every projective space (Theorem 2.66). On the powers, Hilbert schemes and moduli spaces of a K3 surface whose transcendental field is a CM field, and on the fourfolds of $\mathrm{K3}^{[2]}$ type with field $\mathbb{Q}$ or a CM field, the Hodge conjecture itself holds (Proposition 2.39, Proposition 2.40).

The first instance of (F2) is a pair of classes on the square of a Mumford fourfold, and Section 3 studies it in detail. The two classes are the real multiplication by a totally real cubic field on the primitive part of H^2, and they are algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen is algebraic (Proposition 3.2, Theorem 3.4); they are algebraic at every CM point (Theorem 2.29); they are rigid, their Hodge locus being the compact Mumford curve (Theorem 3.21); and they are represented by algebraic cycles at every closed point of every reduction of the curve modulo a prime, so that their algebraicity on every member is a bound on the Hilbert data of those cycles (Proposition 3.42, Theorem 3.43). For an abelian scheme over a curve whose invariant classes are algebraic the Lefschetz standard conjecture holds on the total space (Theorem 2.21), and over a compact Shimura curve the total space of a Mumford family satisfies the Hodge conjecture (Proposition 2.26).

The structure that [I, Theorem 1.1] finds in one family is present in every family that carries (F2) or (F3′) (Section 4). Every Hodge class of every abelian variety lies in an abelian scheme with a member of CM type, and (F2) is equivalent to the statement that in every such scheme the algebraic locus of a flat Hodge class that is algebraic at a CM member has an interior point (Proposition 4.4, Theorem 4.5); the base points of the Weil families are the seeds, through André's theorem on the Hodge classes of abelian varieties of CM type. (F3′) is equivalent to the absence of Hodge classes off the part of the cohomology reached from abelian varieties by algebraic correspondences (Proposition 4.7). On the squares of K3 surfaces with real multiplication, where (F3′) is first open, the algebraic locus of the real multiplication contains every CM point, is dense, is stable under rational Hodge isometries, and is either the whole family or meagre (Theorem 4.9). In each case the property still missing is the interior point (Theorem 4.10).

Finally, the implications of the three papers and of the literature form a finite rule set, and its consequences can be determined by hand. They give the minimal sets of statements from which the conjecture follows (Theorem 2.70, Remark 3.50): (F3′) together with the Lefschetz standard conjecture, with the variational Hodge conjecture for algebraic classes, or with (F2), besides two sets that are each equivalent to the conjecture.[1] Section 5 collects the results in one theorem.

Every proof is written to be checked by hand. Where an argument rests on a finite calculation, the calculation is given where it is used, as an explicit identity, a determinant or a case analysis. The archive [BMB26] holds programs that repeat much of this arithmetic and a Lean 4 certificate of part of it; no statement of the paper rests on them.

[1]The Lefschetz standard conjecture asks that the inverse of the hard Lefschetz isomorphism $L^{\dim X-k}\colon H^k(X,\mathbb{Q}) \to H^{2\dim X-k}(X,\mathbb{Q})$ be induced by an algebraic correspondence. The variational Hodge conjecture asks that a flat section of Hodge classes over the base of a smooth projective family that is algebraic at one fibre be algebraic at every fibre.

2. The reach of the obstructions

An obstruction theorem is useful only if its boundary is drawn exactly. This section draws the boundary of each of the main results, then states the statements through which the conjecture is reached and proves the implications of the literature that join them.

2.1. **Limits of the divisor obstruction.**

Remark 2.1 (Markman's construction). The objects through which Markman proves the Weil classes of the sixfolds of trivial discriminant algebraic [Mar25a, Mar25b] are not obtained by restriction from an ambient projective space, and they are not K-equivariant. [I, Section 13] analyses exactly which hypotheses they escape and why, and [I, Proposition 13.5] lists the three. No obstruction of this paper bears on those objects, and none needs to: their conclusion is a theorem. The candidates proposed for cases that remain open, in [Mar25b, Section 12] and [Mar25c, Example 11.2.7], are another matter, and [II, Theorem 4.61] and [II, Corollary 5.30] decide them.

Remark 2.2 (Twisted objects and gerbes). The hypothesis of [I, Theorem 7.3] is that $\mathcal{F}$ is an object of $D^b(\mathbb{P}^{N-1})$, so a twisted sheaf on a gerbe over $\mathbb{P}^{N-1}$, or an object of a derived category of twisted sheaves, is not covered. What is covered, and what the proof uses, is that the Chern character of the restricted object lies in the subring of $H^*(A^\vee, \mathbb{Q})$ generated by divisor classes. Any construction whose input has that property is subject to the conclusion, whatever category it is phrased in.

Remark 2.3 (Other ambient varieties). Replacing $\mathbb{P}^{N-1}$ by another ambient variety P changes the conclusion exactly when $H^*(P, \mathbb{Q})$ is not generated by divisor classes, since the proof only uses $\mathrm{ch}(\mathcal{F}) \in D^*(P)$ and the fact that i^* carries $D^*(P)$ into $D^*(A^\vee)$. For P a Grassmannian, a flag variety, or any variety whose cohomology is generated in degree two, nothing changes. For P with exceptional Hodge classes of its own the argument does not apply, but then the problem has been moved rather than solved.

2.2. **Limits of the character obstruction.**

Remark 2.4 (Constructions that are not equivariant). [I, Theorem 5.6] constrains eigenclasses. A construction whose output is not a $K^\times$-eigenclass is not constrained, and by [I, Remark 5.8] such a construction must break the naturality of the K-action in an essential way: it must depend on a choice that is not K-invariant, and its output must have components in both extreme summands of the grading at once. These obstructions therefore point to constructions of that kind.

Remark 2.5 (Abelian varieties that are not very general). At special points of the moduli of Weil type the endomorphism algebra grows, the Néron–Severi group grows with it, and the Weil line can meet the divisor subring. This is the split case, where Weil classes are known to be algebraic by elementary means, and it is consistent with everything here: our statements about $W(A) \cap D^{2n}(A)$ assume $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$, which fails there. [I, Theorem 5.6] itself makes no such assumption and holds at every point, but at special points a class can be a sum of eigenclasses for several characters and the conclusion applies to each summand separately.

2.3. **Limits of the semiregularity statement.**

Remark 2.6 (Objects of positive rank). [I, Theorem 9.7] and [I, Corollary 9.18] both require the Chern character to lie on the Weil line, so that $z \mapsto z \cdot \mathrm{ch}(\mathcal{E})$ has no degree-two term. An object of nonzero rank has $\mathrm{rk}(\mathcal{E})\, z$ in that degree, the map is injective, and nothing

here applies. Markman's sheaves are of this kind, and [I, Remark 9.9] records why the distinction is not incidental to his argument.

Remark 2.7 (The weaker semiregularity hypothesis). What [I, Corollary 9.18] obstructs is the injectivity of $\sigma_{\mathcal{E}}$ on the whole of $\mathrm{Ext}^2(\mathcal{E},\mathcal{E})$. The variant proposed in [Mar25b, Question 11.4], injectivity on the image of an evaluation map, asks less. What we prove about it is confined to three groups of results. In [I, Theorem 9.19], [II, Proposition 4.18], [I, Theorem 9.20] and [II, Corollary 4.21] it fails. In [II, Theorem 4.56], for a secant object of positive rank, it is shown to be equivalent to two conditions on the factor. In [II, Theorem 4.61] and [II, Corollary 4.63] it fails for Markman's candidate in dimension eight and for every secant object whose support has two transversal branches of codimension two through a point. We do not decide the variant for any other object of positive rank, nor whether an affirmative answer to the question would follow from it.

Remark 2.8 (Deformation of the class itself). A kernel is not an obstruction class. [I, Corollary 9.18] says that a sufficient criterion for the Weil class to stay algebraic is unavailable at the objects in question; it does not say the class fails to stay algebraic, and no statement in this paper does. [I, Question 15.13] is asked in both directions.

2.4. **Limits of the hyperplane obstruction.**

Remark 2.9 (Families that are not linear systems on X). [I, Theorem 10.3] is about smooth members of an ample linear system on X, and about the maps j^* and j_* they carry. It does not obstruct an argument that deforms X itself, that passes to a different variety by a correspondence, or that uses a family in which X appears as a fibre and the other fibres have the same dimension. In the last case the inductive parameter does not decrease, which is a different difficulty and the one recorded in [I, Remark 10.10].

Remark 2.10 (Coniveau). A Hodge class of positive coniveau is supported on a proper closed subvariety, and the reduction to that subvariety is legitimate and lowers the dimension. Nothing here obstructs it. The obstruction concerns classes of coniveau zero, which is where the primitive middle cohomology of a general variety sits, and the generalised Hodge conjecture predicts that these are precisely the ones a coniveau argument cannot reach.

Remark 2.11 (The Kuga–Satake route). For a weight-two Hodge structure of K3 type the Kuga–Satake construction produces an abelian variety $\mathrm{KS}(T)$ and an injection of T, twisted once, into the endomorphisms of its first cohomology, carrying the Hodge class in question to a class of type $(0,0)$, that is to an endomorphism, whose graph is a cycle. The conclusion is transported back by a correspondence, not by a restriction, so [I, Theorem 10.3] does not apply; this is [I, Proposition 11.9]. The route is limited by the requirement $h^{2,0}=1$, and [I, Proposition 11.11] locates that limit exactly for abelian varieties of Weil type: no rational sub-Hodge structure of $H^2(A,\mathbb{Q})$ is of K3 type unless $n=1$. [I, Section 11] treats the construction in full, including the algebraicity of the correspondence itself.

2.5. **The Weil statements and the closure map.**

The results above close several routes and narrow one. The statements through which the conjecture is reached form a list, and we record each entry with its exact statement and with what this paper proves about it. Proposition 2.13 records the one way in which the entries on Weil classes are linked.

Fix an imaginary quadratic field $K=\mathbb{Q}(\sqrt{-d})$, an integer $n\geq 2$, and a class $\delta\in\mathbb{Q}^{\times}_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^{\times})$, the discriminant of the hermitian form. Write

> $\mathbf{W}(K,n,\delta)$: every abelian $2n$-fold of $(K,-1,n)$-Weil type with discriminant δ has algebraic Weil classes.

Proposition 2.12 (The Weil statements, and how they are linked)**.** *For each triple (K, n, δ) the following hold.*

(i) $\mathbf{W}(K, n, \delta)$ *is equivalent to the algebraicity of the single class ω_s at every point s of the period domain $\mathcal{D}_{n,n}$, and that domain is connected; this is* [I, Corollary 15.35].

(ii) $\mathbf{W}(K, 3, \delta_0)$ *holds for the trivial discriminant δ_0* [Mar25a, Mar25b]*, and with it, by Proposition 2.13, $\mathbf{W}(K, 2, \delta)$ for every K and every δ, which is how Markman obtains the fourfolds.*

(iii) *The period domains attached to distinct triples are disjoint, and* [I, Theorem 15.14] *and* [I, Corollary 15.35] *are statements about one triple at a time. The triples are nevertheless not independent: by Proposition 2.13, $\mathbf{W}(K, N, \delta'')$ implies $\mathbf{W}(K, n, \delta)$ for every $n < N$ and every δ, so the families of trivial discriminant in infinitely many dimensions give every imaginary quadratic family, and the families of a CM field of degree at least four give those of the imaginary quadratic fields it contains.*

Proof. (i) is [I, Corollary 15.35] together with [I, Proposition B.8]. (ii) is the cited work together with Proposition 2.13(i); [I, Remark 15.20] records that the construction of [I, Section 15.1] reaches only δ_0. For (iii), the discriminant is a discrete invariant of the hermitian form and is constant on $\mathcal{D}_{n,n}$ by [I, Proposition 15.19(ii)], while n and K are fixed by the endomorphism algebra, so the domains are disjoint; the links are Proposition 2.13, which passes between families through a product with a Weil surface or through a scalar extension, and not through a deformation. The product argument of Schoen that Markman uses to pass from sixfolds to fourfolds is an instance of such a link. ∎

Proposition 2.13 (Descent and scalar extension)**.** *Let F be a CM field of degree $2m$ and write $\mathbf{W}(F, n, \delta)$ for the statement that every abelian variety of (F, n, δ)-Weil type has algebraic Weil classes, so that $\mathbf{W}(K, n, \delta)$ is the statement above when $m = 1$.*

(i) Descent. *Let B be of (F, n, δ)-Weil type and Y of $(F, 1, \delta')$-Weil type. Then $X = B \times Y$, with the diagonal action of F and the product polarisation, is of $(F, n+1, \delta\delta')$-Weil type, and the correspondence*

$$P(x) = pr_{B*}(x \cdot pr_Y^*(\eta_Y^{2m-2}\, y')),$$

with η_Y the polarisation of Y and y' a nonzero Weil class of Y, maps the Weil classes of X isomorphically onto those of B. Since y' lies in $H^2(Y)$ it is algebraic, so P is algebraic, and every discriminant, a totally positive class in $F_0^\times/\mathrm{N}_{F/F_0}(F^\times)$, is that of some Y. Hence $\mathbf{W}(F, N, \delta'')$ for one N and one δ'' implies $\mathbf{W}(F, n, \delta)$ for every $n < N$ and every δ.

(ii) Scalar extension. *Let $K \subset F$ be CM fields and B of (K, n, δ)-Weil type. Then $B_F = B \otimes_{\mathcal{O}_K} \mathcal{O}_F$ is of $(F, n, \iota(\delta))$-Weil type, and for $j\colon B \to B_F$, $b \mapsto b \otimes 1$, one has $j^* \operatorname{Tr}_{F/\mathbb{Q}}(a \det_F) = \operatorname{Tr}_{K/\mathbb{Q}}(\operatorname{Tr}_{F/K}(a) \det_K)$, so j^* maps the Weil classes of B_F onto those of B. Hence $\mathbf{W}(F, n, \iota(\delta))$ implies $\mathbf{W}(K, n, \delta)$; every imaginary quadratic K lies in the quartic CM field $K(\sqrt{2})$.*

Proof. (i) $H^1(X) = H^1(B) \oplus H^1(Y)$ as F-spaces and the hermitian form of X is the orthogonal sum, of signature $(n+1, n+1)$ at every real place; its normalised discriminant is the product, and $H_Y = \operatorname{diag}(1, -t)$ with t totally positive realises every class. The Weil classes of X are $\bigoplus_\sigma \bigwedge^{2n} V_{B,\sigma} \otimes \bigwedge^2 V_{Y,\sigma}$, because the top exterior power of a direct sum is the tensor product of the top powers. On the σ-component, P is multiplication by $\int_Y w_Y^\sigma\, \eta_Y^{2m-2}\, y'^{\bar\sigma}$, which is nonzero since $w_Y^\sigma \wedge y'^{\bar\sigma}$ fills the $(\sigma, \bar\sigma)$ block and η_Y is nondegenerate on the others; the remaining pairings vanish for weight reasons under the norm-one torus

of F. So P is rational, diagonal over $\mathbb{C}$ with nonzero entries, and algebraic, and $\omega(B)$ is algebraic whenever $\omega(X)$ is; [I, Lemma 16.3] passes from one Weil class to all of them.

(ii) $H^1(B_F) = H^1(B) \otimes_K F$, each eigenspace of F is an eigenspace of K of signature (n, n), the hermitian form is $H_B \otimes F$ with determinant $\det H_B$, and a norm from K is a norm from F. The identity for j^* is the multiplicativity of the determinant under restriction of scalars, and $\mathrm{Tr}_{F/K}$ is onto. ∎

The case K imaginary quadratic, $n = 2$ and $N = 3$ of (i) is Proposition 10 of Schoen's addendum [Sch98], the argument through which Markman obtains every Weil fourfold, at every discriminant, from the sixfolds of trivial discriminant [Mar25a]. Schoen multiplies a Weil fourfold A by a Weil surface A' of the complementary discriminant, identifies $A \times A'$ up to isogeny with a member of a family of Weil sixfolds of fixed discriminant, and pushes forward the cycles $z \cdot (A \times D)$, for D a divisor on A' whose class spans its Weil classes; this is the correspondence P of (i) with $m = 1$, and it involves no degeneration. What (i) adds is that the same argument works in every dimension and for every CM field. As a consequence the triples (K, n, δ) are not separate problems. The trivial discriminant families in infinitely many dimensions give every imaginary quadratic family, and by (ii) the split families of the CM fields of degree at least four give the imaginary quadratic fields as well. In particular the base points of nontrivial discriminant supplied by [I, Theorem 15.27] are not needed on any route to the conjecture, since the split families already carry base points [I, (34)]; what [I, Theorem 15.27] shows is that propagation can be started in every family, not that it has to be. Figure 1 draws the web of families that Proposition 2.13 links.

The list is as follows.

(C1) *One secant object on one fourfold.* There should exist a principally polarised abelian fourfold (X, Θ) and a perfect complex $\mathcal{F}$ on it with $\mathrm{ch}(\mathcal{F}) = u_\Theta + 3v_\Theta$ satisfying [II, (19)]. By [II, Theorem 4.56] and [II, Corollary 4.57] the transform $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ would then satisfy the weakened criterion, and by [II, Proposition 4.52] the three conditions of Markman's strategy that concern only the Chern character hold at the point $(1, 3)$ for any such $\mathcal{F}$, since they depend on $\mathrm{ch}(\mathcal{F})$ alone. Turning this into algebraicity needs one further input. The weakened criterion enters in place of the hypothesis Markman verifies, and the survival of the semiregularity theorem under the substitution is exactly [Mar25b, Question 11.4]. So (C1) is a demand on a construction, and an affirmative answer to that question is a separate prerequisite standing beside it; Theorem 2.70 places both. What this paper contributes is the exact form of the demand. The local part of the criterion is a condition on depth and nothing else: if the support is Cohen–Macaulay there is no local obstruction at any point ([II, Theorem 4.75]); if it is not, the local Ext^2 is nonzero at the bad points, and where two smooth branches of codimension two meet transversally the criterion fails outright ([II, Theorem 4.61], [II, Corollary 4.63], [II, Corollary 4.77]). The candidate of [Mar25b, Section 12] is of the second kind, and it cannot be moved to the first: separating the branches is what would restore the depth, and [II, Remark 4.78] shows that doing so moves the Chern character off the secant plane. What is needed is therefore a support without such points. For a Cohen–Macaulay support the identity [II, (19)] is the single vanishing of the image of $A_\theta A_{\theta'}$ in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$, and the sheaf must in addition extend to first order along the polarised deformations ([II, Theorem 4.56(iii)]). Such a support has one of two shapes ([II, Corollary 4.94]): a smooth one, where the invariants are forced

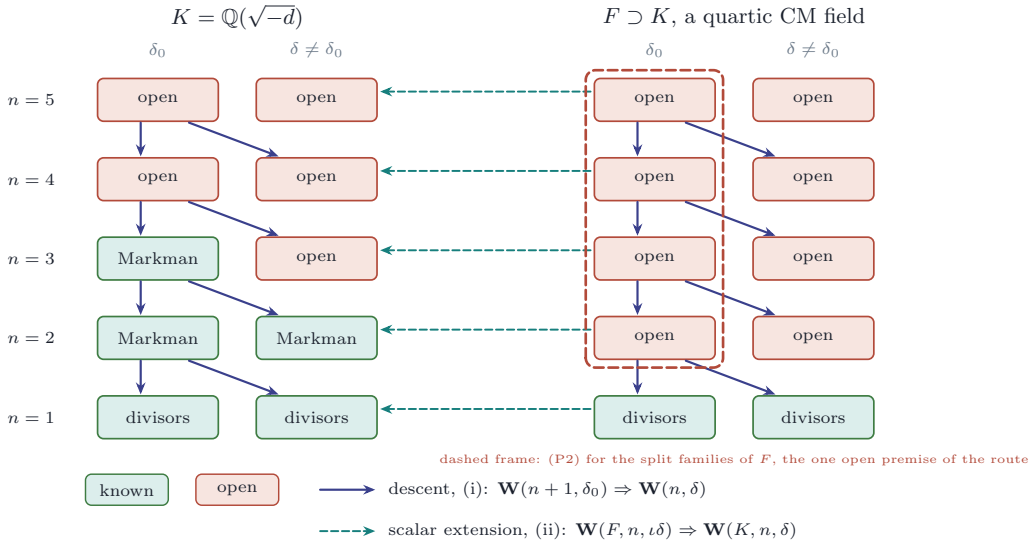


FIGURE 1. The Weil families of an imaginary quadratic field K and of a quartic CM field $F \supset K$, one cell for each n and each kind of discriminant, grass where the Weil classes are proved algebraic (divisor classes at $n = 1$; Markman [Mar25a, Mar25b] at $n = 2$ and at $n = 3$, $\delta = \delta_0$) and clay where they rest on the open propagation statement. The indigo arrows are the descent of Proposition 2.13(i), from the trivial discriminant in dimension $n + 1$ to every discriminant in dimension n; the dashed teal arrows are the scalar extension of Proposition 2.13(ii). The dashed frame is the propagation statement for the split families of the field of higher degree, which by Theorem 2.70(ii) is the only form of (P2) that any minimal route to the conjecture uses: every clay cell of the left panel lies below it.

outright and $d \leq b^2$ for the twist $b\Theta$, so $d \leq 9$ for the twist 3Θ of [II, Setup 4.51] ([II, Theorem 4.67], [II, Corollary 4.68]), and a singular one. In neither may the Hilbert–Burch resolution split into powers of Θ or into simple semi-homogeneous bundles ([II, Theorem 4.89], [II, Theorem 4.92]), and no complete intersection occurs ([II, Theorem 4.81]).

(C2) *The same for every n and every discriminant.* A positive answer to (C1) reaches one triple, and the discriminant of that triple is the trivial one, because the object lives on $X \times \widehat{X}$ and every split member has trivial discriminant by [I, Proposition 15.19(ii)]. The other triples are reached through Proposition 2.13: a positive answer in infinitely many dimensions gives $\mathbf{W}(K, n, \delta)$ for every n and every δ, so only uniformity in n is asked for, and (C1) at $n = 4$, with Question 11.4, already gives every sixfold. No answer of this kind reaches the CM fields of degree at least four. What this paper contributes: three of the four ingredients of [I, Question 13.7] are supplied for every n and every K ([I, Theorem 14.15]), including the existence of the secant sheaves themselves ([I, Theorem 14.13]); the fourth is isolated exactly ([I, Remark 14.16]).

(C3) *CM fields of degree larger than two.* For a CM field F of degree $2m > 2$ acting on an abelian variety, the Weil classes span a space of dimension $2m$, and the secant framework of [I, Section 14], which is built on a plane meeting the spinor variety in

a conjugate pair, does not apply to them as it stands. Markman has extended it to these fields [Mar25c]: for an abelian variety X with real multiplication by the totally real subfield F_0 of F, a polarisation $\Theta \in \bigwedge^2_{F_0} H^1(X,\mathbb{Q})$ and $q \in F_0$ with $F = F_0(\sqrt{-q})$, he makes F act on $X \times \widehat{X}$ with split Weil type, replaces the plane by a rational subspace B of dimension $2^{[F_0:\mathbb{Q}]}$ of the half-spin representation that is secant to the spinor variety, and shows that for sheaves F_1, F_2 on X with characters in B the class $\kappa(E) = \mathrm{ch}(E)\exp(-c_1(E)/\operatorname{rk} E)$ of $E = \Phi(F_1 \boxtimes F_2^\vee)$ stays of Hodge type on every member of the family, and that where it stays algebraic all Weil classes are algebraic [Mar25c, Theorem 1.1.2, Proposition 10.2.1]. He exhibits such a pair meeting his non-degeneracy criterion when F is biquadratic and X is the Jacobian of a curve of genus four with real multiplication by F_0 and an automorphism acting through a unit of norm one [Mar25c, Example 11.2.7, Lemma 11.2.8]. That $\kappa(E)$ stays algebraic would follow from a semiregularity property of E, which is not addressed there [Mar25c, Question 11.2.2]; for the pair of that example the weaker property asked in the question fails at the points of the glued curves ([II, Corollary 5.30]). No result of this paper depends on a theorem of [Mar25c]; [II, Section 5.2] and [II, Section 5.3] use its construction as their setting. What this paper contributes: everything except the propagation. By [I, Section 17] each such family has a connected period domain, a flat space of Weil classes, a dense set of rational orbits, a base point at a CM member where the classes are written as polynomials in divisor classes ([I, Theorem 17.5]), and the two propagation criteria ([I, Theorem 17.7]); so the higher CM fields are the same propagation problem as the quadratic ones, and not a separate entry. For a quartic field at $n = 2$, the smallest case of higher degree, [II, Section 5.2] determines how far divisor classes, scalar extension and correspondences reach, and computes the number that the polarised criterion asks of one complex. For a sextic field, [II, Section 5.3] shows that the dimension count, which excludes a minimal Orlov product over a quartic field, does not exclude one over a sextic field, and gives the profile such an object would have. [I, Appendix C] places the imaginary quadratic case inside Albert's classification.

(C4) *The non-split sixfolds.* $\mathbf{W}(K,3,\delta)$ for $\delta \neq \delta_0$; the case $\delta = \delta_0$ is Markman's [Mar25a]. By Proposition 2.13(i) it would follow from $\mathbf{W}(K,4,\delta_0)$, and by Proposition 2.13(ii) from $\mathbf{W}(K(\sqrt{2}),3,\iota(\delta))$. What this paper contributes: a base point in every family, at a member with quaternionic multiplication ([I, Theorem 15.27]); the locus of such members is a Lagrangian Grassmannian of codimension $n(n-1)/2$ ([I, Proposition 15.32]), and the rest of the family can be reached from it only by propagation.

(C5) *Exceptional classes that are not of Weil type.* On an abelian variety the Hodge ring can contain exceptional classes that the Weil construction does not produce. Through dimension five it does not [MZ99], and in dimension eight it does: the self-product of an abelian fourfold of Mumford type carries two Hodge classes that are neither products of divisor classes nor Weil classes of any quotient (Proposition 2.74). So the Hodge conjecture for abelian varieties does not reduce to Weil classes, and the statement needed is the algebraicity of these classes themselves. What this paper contributes: the example, made exact, and the forms of their algebraicity studied in Section 3.

(C6) *Varieties that are not abelian.* A reduction of the Hodge conjecture for arbitrary smooth projective varieties to abelian varieties cannot be of Hodge-theoretic kind: the H^3 of a very general quintic threefold lies outside the tensor category generated by the cohomology of abelian varieties (Proposition 2.75), so its Hodge structure cannot be realised inside that of an abelian variety by any correspondence. [I, Theorem 10.3] and [I, Theorem 10.8] close the two routes through hyperplane sections that would be the natural candidates. What this paper contributes: those three obstructions and, on the positive side, special cases only (Proposition 2.26, Remark 2.68). Nor is this entry the complement of the others. The varieties that are not abelian include $A \times \mathbb{P}^1$ for every abelian variety A, and the conjecture for $A \times \mathbb{P}^1$ gives it for A, so the conjecture for the varieties that are not abelian is the whole conjecture (Proposition 2.33).

2.6. **Implications from the literature.** The theorems of this paper are not the only implications that end in the conjecture. The literature supplies others, and a frontier drawn without them overstates what is needed. This subsection proves those implications, or quotes them where they are theorems of the literature, so that Section 2.9 runs over both. Nothing in it is new except the assembly, but the assembly changes the answer.

Definition 2.14 (Three statements about all varieties)**.** Here a variety is smooth, projective and defined over $\mathbb{C}$, except the base S in (V).

(L) $B(X)$ holds for every variety X ([I, Definition 10.6]).

(M) Every Hodge class on every variety X is *motivated* in the sense of André [And96b, §2.1]: it equals $\mathrm{pr}_*(\alpha \cup *\beta)$ for some variety Y, algebraic classes α, β on $X \times Y$, the projection $\mathrm{pr}\colon X \times Y \to X$, and the Lefschetz involution $*$ of $H^*(X \times Y, \mathbb{Q})$ with respect to some ample class.

(V) For every smooth connected complex variety S, every smooth projective morphism $f\colon \mathcal{X} \to S$ and every global section ξ of $R^{2p}f_*\mathbb{Q}$ over S^{an}: if ξ_s is algebraic at one point $s \in S$, then ξ_t is algebraic at every point $t \in S$.

The Lefschetz involution of a variety Z of dimension e with ample class L sends $L^r x$, for x primitive of degree j and $0 \le r \le e - j$, to $\pm L^{e-j-r}x$, with a sign that depends only on j [And96b]; the sign plays no role below.

The statement (V) is the variational Hodge conjecture for algebraic classes; it asks nothing about Hodge classes, only that algebraicity, once present, survives deformation.

Lemma 2.15 (Kleiman)**.** *Let Z be a variety of dimension e with ample class L, and suppose that $B(Z)$ holds. Then the Künneth projectors π_k of $H^*(Z, \mathbb{Q})$, the projectors onto the summands $L^r P^j$ of the Lefschetz decomposition, and the Lefschetz involution $*$ are all induced by algebraic cycles on $Z \times Z$.*

Proof. Operators induced by algebraic cycles on $Z \times Z$ are closed under sums and rational multiples, and under composition, because the composite of two algebraic correspondences is again algebraic [Kle68]. The operator L is induced by the image of a divisor of class L under the diagonal, and Λ by $B(Z)$. With $[L, \Lambda] = H$ and H acting on H^k by $k - e$ (the proof of [I, Lemma 10.7]), $H = L\Lambda - \Lambda L$ is induced by an algebraic cycle, and since the numbers $k - e$ for $0 \le k \le 2e$ are distinct,

$$\pi_k = \prod_{i \neq k} \frac{H - (i - e)}{k - i}$$

is as well. Put $\Omega = H^2 + 2H + 4\Lambda L$. For x primitive of degree j put $m = e - j \geq 0$. Then $\Lambda x = 0$ and $Hx = -mx$, and the relation

$$\Lambda L^s x = s(m - s + 1)\, L^{s-1} x \qquad (s \geq 1) \tag{1}$$

follows by induction from $\Lambda L = L\Lambda - H$: the case $s = 1$ is [I, Lemma 10.7], and $\Lambda L^s x = L\Lambda L^{s-1}x - HL^{s-1}x = \big((s-1)(m-s+2) - (2s-2-m)\big)L^{s-1}x = s(m-s+1)L^{s-1}x$. So $\Omega x = (m^2 - 2m + 4m)x = m(m+2)x$, and Ω commutes with L, since $[L, H] = -2L$ and $[L, \Lambda L] = HL$ give $[L, \Omega] = -2(LH + HL) - 4L + 4HL = 0$; hence Ω acts on $L^r P^j$ by $m(m+2)$. Inside H^k the summands are $L^r P^{k-2r}$, with $m = e - k + 2r$, and $m \mapsto m(m+2)$ is injective on $m \geq 0$, so Lagrange interpolation in Ω composed with π_k gives the projector $p_{k,r}$ onto $L^r P^{k-2r}$ as an algebraic operator. Finally let $x \in P^j$ and $0 \leq r \leq m$. If $2r \leq m$, then $L^{m-2r}(L^r x) = L^{m-r}x$. If $2r > m$, put $t = 2r - m$, so $1 \leq t \leq r$; by (1),

$$\Lambda^t L^r x = \prod_{i=0}^{t-1} (r - i)(m - r + i + 1)\; L^{m-r} x,$$

and every factor is at least 1, because $r - i \geq r - t + 1 \geq 1$ and $m - r + i + 1 \geq m - r + 1 \geq 1$. So on $L^r P^j$ the involution is $\pm L^{m-2r}$ or a nonzero rational multiple of Λ^{2r-m}, and $* = \sum_{k,r} c_{k,r} T_{k,r}\, p_{k,r}$ with rational $c_{k,r}$ and $T_{k,r}$ a power of L or of Λ. Every term is algebraic. ∎

Proposition 2.16 (The conjecture is (L) and (M))**.** *The Hodge conjecture for every variety is equivalent to* (L) *and* (M) *together.*

Proof. Assume the conjecture. Then (L) holds by [I, Theorem 12.2]. An algebraic class ξ on X of dimension d is motivated: take Y a point, $\alpha = \xi$, and $\beta = \varepsilon h^d$ with h the ample class and $\varepsilon = \pm 1$ the sign for which $*(h^d) = \varepsilon$, which exists because the unit $1 \in H^0$ is primitive of degree 0; then $\alpha \cup *\beta = \xi$. Every Hodge class is algebraic, so (M) holds.

Conversely assume (L) and (M), and let ξ be a Hodge class on X. By (M), $\xi = \mathrm{pr}_*(\alpha \cup *\beta)$ with α, β algebraic on $X \times Y$. By (L), $B(X \times Y)$ holds, so $*$ is induced by an algebraic cycle (Lemma 2.15); hence $*\beta$, the product $\alpha \cup *\beta$ and its push-forward ξ are algebraic. This is André's observation that motivated classes are algebraic as soon as the Lefschetz involutions are [And96b, §2.1]. ∎

Proposition 2.17 (Two sources of (V))**.** *The Hodge conjecture implies* (V)*, and so does* (L)*.*

Proof. Let $f\colon \mathcal{X} \to S$, ξ and s be as in (V), with ξ_s algebraic. As S is smooth and connected it is irreducible, so two affine open subsets meet, and it suffices to treat S affine, joining s to an arbitrary t through a point of the intersection of an affine neighbourhood of each. Then $\mathcal{X}$ is quasi-projective and, by resolution of singularities, has a smooth projective compactification $\overline{\mathcal{X}}$.

Assume the conjecture. Write $n = 2p$ and $\rho\colon H^n(\overline{\mathcal{X}}, \mathbb{Q}) \to H^0(S^{\mathrm{an}}, R^n f_* \mathbb{Q})$ for restriction; it is surjective by the theorem of the fixed part [Del71, 4.1.1(ii)]. For $t \in S$ the restriction $r_t\colon H^n(\overline{\mathcal{X}}, \mathbb{Q}) \to H^n(\mathcal{X}_t, \mathbb{Q})$ is ρ followed by evaluation at t, and evaluation is injective because S is connected; so $K = \ker r_t = \ker \rho$ does not depend on t. It is a sub-Hodge structure, and the Hodge structure $H^n(\overline{\mathcal{X}}, \mathbb{Q})$ is polarisable, hence semisimple, so $H^n(\overline{\mathcal{X}}, \mathbb{Q}) = K \oplus K'$ with K' a sub-Hodge structure. Choose $\eta \in K'$ with $\rho(\eta) = \xi$. Then $r_s|_{K'}$ is an injective morphism of Hodge structures and $r_s(\eta) = \xi_s$ is of type (p, p), so every component of η of type other than (p, p) maps to zero and is zero: η is a Hodge class. So $\xi_t = r_t(\eta)$ is a Hodge class on $\mathcal{X}_t$ for every t, and algebraic by the conjecture.

Assume (L). The algebraic class ξ_s is motivated (proof of Proposition 2.16), so ξ_t is motivated for every t by André's deformation theorem [And96b, Théorème 0.5], and a motivated class is algebraic under (L) (proof of Proposition 2.16). ∎

Theorem 2.18 (Deligne, André; Abdulali, André)**.** (V) *implies the Hodge conjecture for every abelian variety, and so does* (L)*.*

Proof. Milne calls a class *accessible* if it lies in the least family of rational classes that contains the classes of algebraic cycles, is stable under pull-back by morphisms of varieties, and contains t_t for every t whenever $(t_s)_{s\in S}$ is a global section of $R^r f_*\mathbb{Q}$ for a smooth projective family over a smooth connected base and contains t_s for one s; and he proves, following Deligne and André, that every Hodge class on an abelian variety is accessible [Mil20, §5.1, Theorem 3]. The algebraic classes contain the classes of algebraic cycles and are stable under pull-back, and under (V) they have the third property. So they contain every accessible class, hence every Hodge class on an abelian variety; this is [Mil20, Remark 2]. The statement for (L) follows by Proposition 2.17, and is the theorem of Abdulali and André in the form of [Mil20, Theorem 4]. ∎

Remark 2.19 (The three in the literature)*.* $B(X)$ is known for curves, where it is immediate, and for abelian varieties [Kle68, §2], and Corollary 2.22 below adds the total space of every Mumford family. (M) holds for the Hodge classes on abelian varieties [And96b, Théorème 0.6.2]. (V) holds in degree 2, where by the argument of Proposition 2.17 the class ξ_t is a Hodge class and the Lefschetz theorem on $(1,1)$ classes applies; the one general mechanism for it in higher degree is the semiregularity theorem [Blo72, BF03, Pri24], which is the mechanism of [II, Theorem 2.3]. Theorem 2.18 says what is at stake for this paper: the single statement (V), which the conjecture itself implies, gives the Weil classes of every CM field and the classes beyond the Weil lines at once, by a route that uses none of the theorems proved here.

2.7. **The Lefschetz standard conjecture for abelian schemes over curves.** The statement $B(X)$ is known for abelian varieties X. This subsection proves it for the total space of an abelian scheme over a curve whose invariant cycles are algebraic, and so for the total space of every Mumford family. The proof assembles the operator Λ from a relative part, which is a Pontryagin product and is algebraic on every abelian scheme, and a part along the base, which is algebraic exactly when the invariant cycles are.

Proposition 2.20 (Λ as a Pontryagin product)**.** *Let A be an abelian variety of dimension g, $\ell \in H^2(A,\mathbb{Q})$ the class of an ample line bundle, $D = \int_A \ell^g/g!$ and $\gamma = \ell^{g-1}/(g-1)!$. Let $\mu\colon A\times A\to A$ be the addition and $x\star y = \mu_*(p_1^*x \cup p_2^*y)$ the Pontryagin product. Then the operator Λ of* [I, Definition 10.6] *for the class ℓ is*

$$\Lambda(x) = D^{-1}\, x\star\gamma.$$

In particular Λ is induced by the algebraic cycle $D^{-1}(p_1,\mu)_*(p_2^*Z)$ on $A\times A$, for any cycle Z of class γ.

Proof. Put $V = H_1(A,\mathbb{Q})$, so that $H^*(A,\mathbb{Q}) = \bigwedge^* V^\vee$ under cup product and $H_*(A,\mathbb{Q}) = \bigwedge^* V$ under the Pontryagin product of homology, and let $\mathrm{PD}(x) = x\cap[A]$. Cap product makes H_* a module over H^*, and for a torus it is contraction; so $\mathrm{PD}(\ell\cup x) = \ell\cap\mathrm{PD}(x)$ is the contraction ι_ℓ applied to $\mathrm{PD}(x)$. Poincaré duality commutes with μ_* and carries $p_1^*x\cup p_2^*y$ to the cross product of $\mathrm{PD}(x)$ and $\mathrm{PD}(y)$, without sign when y has even degree, so $\mathrm{PD}(x\star\gamma) = \mathrm{PD}(x)\cdot\beta$ with $\beta = \mathrm{PD}(\gamma)\in\bigwedge^2 V$.

Choose a basis $x_1, y_1, \dots, x_g, y_g$ of $V^\vee$ with $\ell = \sum_i d_i\, x_i \wedge y_i$, $d_i > 0$, oriented so that $\int_A x_1 \wedge y_1 \wedge \cdots \wedge x_g \wedge y_g = 1$; then $\int_A \ell^g = g!\, d_1 \cdots d_g$, so $D = d_1 \cdots d_g$. Let e_i, f_i be the dual basis of V. Then $\gamma = \sum_j (\prod_{i \neq j} d_i) \prod_{i \neq j} x_i \wedge y_i$ and $\beta = \varepsilon D \sum_j d_j^{-1} e_j \wedge f_j$ with $\varepsilon = \pm 1$ independent of j, since the permutations of the pairs preserve the orientation and exchange the terms. The algebra $\bigwedge^* V$ is the graded tensor product of the algebras $\bigwedge^* \langle e_i, f_i \rangle$, and ι_ℓ and multiplication by $D^{-1}\beta$ are sums over i of even operators on the i-th factor, namely $d_i\, \iota_{x_i \wedge y_i}$ and $\varepsilon d_i^{-1} e_i \wedge f_i$. Operators on different factors commute, and on the i-th factor, with basis $1, e_i, f_i, e_i \wedge f_i$ and $\iota_{x_i \wedge y_i}(e_i \wedge f_i) = \sigma$ for a sign σ fixed by the convention for contraction, the commutator $[\iota_\ell, D^{-1}\beta]$ acts by $\varepsilon\sigma$ on 1, by 0 on e_i and f_i, and by $-\varepsilon\sigma$ on $e_i \wedge f_i$, that is by $\varepsilon\sigma(1 - k_i)$ in degree k_i. Summing over i, $[\iota_\ell, D^{-1}\beta] = \varepsilon\sigma(g - k)$ on $\bigwedge^k V$. As PD carries H^q to $\bigwedge^{2g-q} V$, the operator $\Lambda'(x) = D^{-1} x \star \gamma$ satisfies $[L, \Lambda'] = \varepsilon\sigma(q - g) = \varepsilon\sigma H$ on $H^q(A)$.

The sign is fixed by one value. For x and y of complementary degrees, $x \star y \in H^0(A)$ is $\int_{A \times A} p_1^* x \cup p_2^* y \cup \mu^*[\mathrm{pt}]$, and $\mu^*[\mathrm{pt}]$ is the class of the antidiagonal $\{(a, -a)\}$, so $x \star y = \int_A x \cup [-1]^* y$; for $y = \gamma$, of even degree, this is $\int_A x \cup \gamma$. Hence $\Lambda'(\ell) = D^{-1} \int_A \ell \cup \gamma = g$, while $\Lambda(\ell) = \Lambda(L \cdot 1) = g$ by [I, Lemma 10.7]. If $\varepsilon\sigma = -1$, then $-\Lambda'$ would be the operator of [I, Definition 10.6] and $\Lambda'(\ell) = -g$. So $\varepsilon\sigma = 1$, and $\Lambda' = \Lambda$ by the uniqueness in [I, Definition 10.6]. Finally, the correspondence $(p_1, \mu)_*(p_2^* Z)$ acts by $x \mapsto p_{2*}(p_1^* x \cup (p_1, \mu)_* p_2^* [Z]) = \mu_*(p_1^* x \cup p_2^* \gamma)$, by the projection formula. ∎

Theorem 2.21 (The Lefschetz operator of an abelian scheme over a curve; cf. Tankeev [Tan03])**.** *Let C be a smooth projective connected curve and $\pi\colon W \to C$ an abelian scheme of relative dimension g with W projective. Suppose that every class in $H^*(W_c, \mathbb{Q})$ invariant under the monodromy of $R^*\pi_*\mathbb{Q}$ is the restriction of an algebraic class on W. Then there is an ample class L_W on W whose operator Λ is induced by an algebraic cycle on $W \times W$: $B(W)$ holds.*

Proof. Step 1: the decomposition. Let $[n]\colon W \to W$ be multiplication by n over C. It acts on $R^q\pi_*\mathbb{Q} = \bigwedge^q R^1\pi_*\mathbb{Q}$ by n^q. The Leray spectral sequence degenerates [Del71, 4.1.1(i)], and $[2]^*$ preserves the Leray filtration and acts on its graded pieces $H^p(C, R^q\pi_*\mathbb{Q})$ by 2^q. So $H^k(W, \mathbb{Q}) = \bigoplus_q H^k_{(q)}$, where $H^k_{(q)}$ is the 2^q-eigenspace of $[2]^*$ and maps isomorphically onto $H^{k-q}(C, R^q\pi_*\mathbb{Q})$; only $k - q \in \{0, 1, 2\}$ occurs. The projectors $P_q = \prod_{q' \neq q} ([2]^* - 2^{q'})/(2^q - 2^{q'})$ are induced by algebraic cycles, polynomials in the graph of $[2]$. As $[2]^*$ is a ring homomorphism, $H_{(q)} \cup H_{(q')} \subset H_{(q+q')}$; as $\int_W [2]^* z = 2^{2g} \int_W z$, the integral vanishes on $H_{(q)}$ for $q \neq 2g$, so Poincaré duality pairs $H_{(q)}$ with $H_{(2g-q)}$ only; and $[2]_* = 2^{2g}([2]^*)^{-1}$ acts on $H_{(q)}$ by 2^{2g-q}.

Step 2: the ample class. Let $\mathcal{L}$ be a relatively ample line bundle, replaced by $\mathcal{L} \otimes [-1]^*\mathcal{L}$ so that it is symmetric. Then $[-1]^*$ fixes $c_1(\mathcal{L})$ and acts on $H_{(q)}$ by $(-1)^q$, so $c_1(\mathcal{L})$ has no component in $H^2_{(1)}$, while $H^2_{(0)} = H^2(C, \mathbb{Q})$ is spanned by the class f of a fibre. So $c_1(\mathcal{L}) = \ell + af$ with $\ell \in H^2_{(2)}$ algebraic and $a \in \mathbb{Q}$. Put $L_W = \ell + mf = c_1(\mathcal{L}) + (m - a)f$ with $m \in \mathbb{Q}$ so large that L_W is a positive rational multiple of the class of an ample line bundle, which is possible because $\mathcal{L}$ is relatively ample. Write L_{rel} and L_C for cup product with ℓ and with mf, so that $L_W = L_{\mathrm{rel}} + L_C$; on $H^p(C, R^q\pi_*\mathbb{Q})$ the first raises q by two and the second raises p by two.

Step 3: the relative operator. Put $\gamma = \ell^{g-1}/(g-1)!$, which is algebraic and lies in $H_{(2g-2)}$, and $D = \int_{W_c} \ell^g/g!$. Let $\mu, \mathrm{pr}_1, \mathrm{pr}_2\colon W \times_C W \to W$ be the addition and the projections, and $T(x) = \mu_*(\mathrm{pr}_1^* x \cup \mathrm{pr}_2^* \gamma)$. As in Proposition 2.20, T is induced by the algebraic cycle $(\mathrm{pr}_1, \mu)_*(\mathrm{pr}_2^* Z)$ on $W \times W$. Since $\pi \circ \mathrm{pr}_1 = \pi \circ \mu$, the projection formula

gives $T(x\cup\pi^*b)=T(x)\cup\pi^*b$, so T commutes with L_C. Since $[2]\circ\mu=\mu\circ([2]\times_C[2])$, $[2]_*T(x)=\mu_*(\mathrm{pr}_1^*[2]_*x\cup\mathrm{pr}_2^*[2]_*\gamma)$, and for $x\in H_{(q)}$ this is $2^{2g-q}\,2^2\,T(x)$; so T maps $H_{(q)}$ into $H_{(q-2)}$. A relative correspondence acts on the Leray spectral sequence through the maps it induces on the local systems, so on $H^p(C,R^q\pi_*\mathbb{Q})$ the operator T is $H^p(C,T_\bullet)$, with T_c the Pontryagin product with $\gamma|_{W_c}$ on the fibre. By Proposition 2.20, $D^{-1}T_c$ is the operator Λ of the fibre, so $\Lambda_{\mathrm{rel}}=D^{-1}T$ satisfies $[L_{\mathrm{rel}},\Lambda_{\mathrm{rel}}]=H_{\mathrm{rel}}$, where H_{rel} acts on $H_{(q)}$ by $q-g$.

Step 4: the invariant classes. Restriction maps $H^q(W,\mathbb{Q})$ onto the monodromy invariants $I_q\subset H^q(W_c,\mathbb{Q})$ [Del71, 4.1.1(ii)], and it kills $H^q_{(q')}$ for $q'<q$, which lies in positive Leray degree; so it maps $H^q_{(q)}$ isomorphically onto I_q. Every element of $H^q_{(q)}$ is algebraic: if $v\in I_q$ is the restriction of an algebraic class z, then P_qz is algebraic, lies in $H^q_{(q)}$ and restricts to v. The subspace I_q is a sub-Hodge structure, as the image of a morphism of Hodge structures, and it is stable under cup product with ℓ_c and under the operator Λ of the fibre, which are the restrictions of L_{rel} and Λ_{rel}. The polarisation of $H^q(W_c,\mathbb{Q})$ attached to ℓ_c has the form $(v,w)\mapsto\int_{W_c}v\cup S(w)$, with S a polynomial in cup product with ℓ_c, the operator Λ and the projectors of the fibre, and its restriction to the sub-Hodge structure I_q is again a polarisation [Voi02]. Since $S(I_q)\subset I_{2g-q}$, the pairing $I_q\times I_{2g-q}\to\mathbb{Q}$ is nondegenerate.

Step 5: the operator along the base. For $y\in H^q_{(q)}$ and $w\in H^{2g-q}_{(2g-q)}$ one has $\int_W(y\cup f)\cup w=\int_{W_c}y|_{W_c}\cup w|_{W_c}$. By Step 4 and Step 1 this shows that $L_C\colon H^q_{(q)}\to H^{q+2}_{(q)}$ is injective, and it is onto because $\dim H^{q+2}_{(q)}=\dim H^{2g-q}_{(2g-q)}=\dim I_{2g-q}=\dim I_q$, by Poincaré duality on W and on the invariants. Choose bases (a_j) of $H^q_{(q)}$ and (b_k) of $H^{2g-q}_{(2g-q)}$, all algebraic by Step 4, let $M_{jk}=\int_W a_j\cup mf\cup b_k$, which is invertible, and put

$$\Lambda_C(x)=\sum_{j,k}(M^{-1})_{kj}\Big(\int_W x\cup b_k\Big)a_j\qquad(x\in H^{q+2}(W,\mathbb{Q})),$$

for every q. It is induced by the algebraic cycle $\sum_{j,k}(M^{-1})_{kj}\,b_k\times a_j$. By Step 1, $\int_W x\cup b_k$ vanishes for $x\in H^{q+2}_{(q')}$ with $q'\neq q$, so Λ_C is zero on $H^p(C,R^q\pi_*\mathbb{Q})$ for $p\neq2$, and $\Lambda_C(L_Ca_i)=a_i$. Hence $[L_C,\Lambda_C]$ is -1 on $H^q_{(q)}$, 0 on $H^{q+1}_{(q)}$ and $+1$ on $H^{q+2}_{(q)}$: it is the operator H_C that acts on $H^p(C,R^q\pi_*\mathbb{Q})$ by $p-1$.

Step 6: the cross relations and the conclusion. $[L_C,\Lambda_{\mathrm{rel}}]=0$ by Step 3. Both $L_{\mathrm{rel}}\Lambda_C$ and $\Lambda_CL_{\mathrm{rel}}$ vanish off the summands with $p=2$, and on L_Cy with $y\in H^q_{(q)}$ both give $\ell\cup y$, because $\ell\cup y\in H^{q+2}_{(q+2)}$ and L_C commutes with L_{rel}; so $[L_{\mathrm{rel}},\Lambda_C]=0$. Put $\Lambda=\Lambda_{\mathrm{rel}}+\Lambda_C$, of degree -2. Then

$$[L_W,\Lambda]=[L_{\mathrm{rel}},\Lambda_{\mathrm{rel}}]+[L_C,\Lambda_C]=H_{\mathrm{rel}}+H_C,$$

which acts on $H^p(C,R^q\pi_*\mathbb{Q})\subset H^{p+q}(W)$ by $(q-g)+(p-1)=(p+q)-\dim W$. So Λ is the operator of [I, Definition 10.6] for L_W, by its uniqueness, and it is induced by an algebraic cycle by Steps 3 and 5. ∎

Figure 2 draws the summands of the proof in relative dimension four.

Corollary 2.22 (Mumford families)**.** *Let $\pi\colon W\to C$ be a non-isotrivial abelian scheme of relative dimension four over a smooth projective connected curve, W projective, whose very general fibre X is a fourfold as in Proposition 2.74, with Hodge group the group G of Mumford's construction* [Mum69]*, the restriction of scalars from a totally real cubic field F of the norm one group of a quaternion algebra over F. Then $B(W)$ holds. Such families exist, over the compact Shimura curves of* [Mum69].

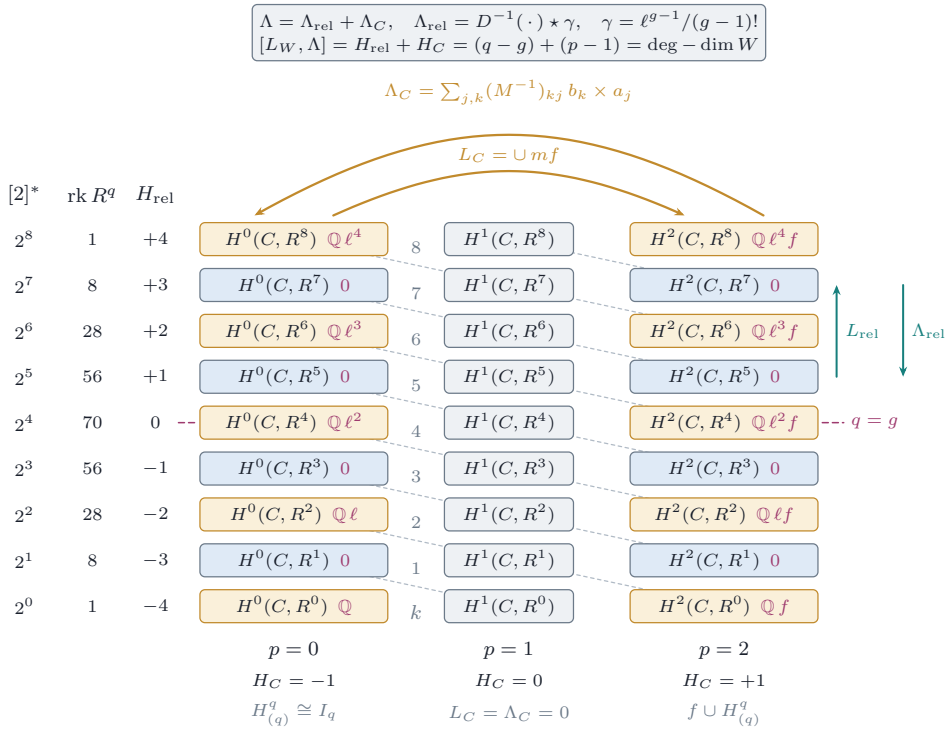


FIGURE 2. The proof of Theorem 2.21 for relative dimension four. The boxes are the summands $H^p(C, R^q\pi_*\mathbb{Q})$ of $H^*(W, \mathbb{Q})$, split off by the eigenvalue 2^q of $[2]^*$ shown on the left; the dotted diagonals join the summands of one $H^k(W, \mathbb{Q})$, $k = p + q$, and the dashed magenta line is the axis $q = g$ of hard Lefschetz on the fibres. The relative operators L_{rel} and Λ_{rel}, in teal, move within a column; Λ_{rel} is a Pontryagin product with an algebraic class on every abelian scheme (Proposition 2.20). The base operators L_C and Λ_C, in ochre, move along a row between the outer columns and kill the middle one; Λ_C is the displayed cycle built from bases (a_j), (b_k) of the invariant summands, and is algebraic exactly when the invariant cycles are. Below each column is the weight of $H = [L_W, \Lambda]$ on it, and the panel at the top is Step 6 of the proof. The ochre boxes carry the invariant cycles and their images under L_C; for a Mumford family they are the lines listed in magenta on the right, spanned by $\ell^i f$ (and by ℓ^i in the left column), and the other boxes of the outer columns are zero (Corollary 2.22).

Proof. By André's theorem [And92, Theorem 1] the connected monodromy group is a normal subgroup of the derived Mumford–Tate group of X, which is G. It is not trivial, because the family is not isotrivial, and G is $\mathbb{Q}$-simple, since over $\overline{\mathbb{Q}}$ its three factors are indexed by the embeddings of F and permuted transitively by the Galois group; so it is G. The monodromy invariants in $H^*(X, \mathbb{Q})$ are therefore contained in the invariants of G, which are the Hodge classes of X, and these are the powers of the polarisation by Proposition 2.74(i). The powers of ℓ restrict to the powers of the polarisation, so every invariant class is the restriction of an algebraic class, and Theorem 2.21 applies. ∎

Remark 2.23 (Scope of the argument). The total space W is a fivefold fibred in abelian fourfolds whose Hodge groups are not defined by endomorphisms; it is not a product and not an abelian variety. For abelian schemes over curves the mechanism of Theorem 2.21 goes back to Tankeev [Tan03], who proves B for the total space when certain canonically defined Hodge cycles are algebraic; the argument here does not use his hypotheses, and we make no claim of priority for Corollary 2.22. The difficulty lies in the hypothesis of Theorem 2.21. Applied to the fibre square $W \times_C W \to C$ the theorem gives $B(W \times_C W)$ as soon as every invariant class of $X \times X$ is the restriction of an algebraic class on $W \times_C W$; the invariant classes include the two exceptional classes of Proposition 2.74, so this is the smallest case of (F2) in a global form. Theorem 2.21 does not bear on (P2), on the Kuga–Satake class of Theorem 3.4, or on (L) for varieties that are not fibred in abelian varieties over a curve.

Proposition 2.24 (Powers of a Weil variety; cf. [Mil22]). *Let X be of $(K, -1, n)$-Weil type with $n \geq 2$ and $\mathrm{Hg}(X) = \mathrm{SU}(V, H)$. For every $k \geq 1$ the $\mathbb{Q}$-algebra $\mathrm{Hdg}^*(X^k)$ is generated by the divisor classes of X^k and the pull-backs $f_a^*\omega$ of the Weil classes ω of X along the homomorphisms $f_a = \sum_j a_j\, pr_j \colon X^k \to X$, $a \in \mathbb{Z}^k$. Consequently, if the Weil classes of X are algebraic, the Hodge conjecture holds for every power of X. By [Mar25a] this is so for every such X with $n = 2$, and with $n = 3$ and trivial discriminant.*

Proof. Write $H^1(X, \mathbb{C}) = V_+ \oplus V_-$ for the eigenspaces of K and $W = V_+$. The polarisation makes V_+ and V_- isotropic and pairs them perfectly, so $V_- = W^\vee$, and restriction to V_+ identifies $\mathrm{SU}(V, H)_{\mathbb{C}}$ with $\mathrm{SL}(W)$. As $\mathrm{Hg}(X^k)$ is $\mathrm{Hg}(X)$ acting diagonally on $H^1(X^k, \mathbb{C}) = (W \otimes U) \oplus (W^\vee \otimes U)$, $U = \mathbb{C}^k$, the Hodge classes of X^k, tensored with $\mathbb{C}$, are the $\mathrm{SL}(W)$-invariants in $\bigwedge^*((W \otimes U) \oplus (W^\vee \otimes U))$. That algebra is an $\mathrm{SL}(W)$-equivariant quotient of a tensor algebra, with an equivariant section given by antisymmetrisation, so Weyl's first fundamental theorem for $\mathrm{SL}(W)$, in its tensor form and applied before antisymmetrisation, shows that its invariants are generated by the images of the contractions and the determinant tensors: the classes $D(u, u') = \sum_i (e_i \otimes u) \wedge (e_i^\vee \otimes u')$ and the polarised determinants, which by polarisation are combinations of $\det_u = \bigwedge_i (e_i \otimes u)$ and $\det_u^\vee = \bigwedge_i (e_i^\vee \otimes u)$, $u, u' \in U$. For $2n \geq 3$ the $D(u, u')$ span the invariants of degree two, which are the divisor classes tensored with $\mathbb{C}$. With $\alpha_\pm$ spanning $\bigwedge^{2n} V_\pm$ ([I, Proposition 3.7]) one has $f_a^*\alpha_+ = \det_a$ and $f_a^*\alpha_- = \det_a^\vee$, and since $u \mapsto \det_u$ is polynomial and $\mathbb{Z}^k$ is Zariski dense in U, these span the same space as all $\det_u$. So the $\mathbb{Q}$-algebra generated by $\mathrm{NS}(X^k)_{\mathbb{Q}}$ and the $f_a^*\omega$ is contained in $\mathrm{Hdg}^*(X^k)$ and has the same complexification. The generators are algebraic when the Weil classes of X are, and they are for $n = 2$ and for $n = 3$ at trivial discriminant by [Mar25a]. ∎

The statement for powers is known (compare [Mil22], and [Li26] for the powers of CM fourfolds); we include the short proof because the next corollary uses it.

Corollary 2.25 (Abelian schemes of small relative dimension). *Let $\pi \colon W \to C$ be an abelian scheme of relative dimension g over a smooth projective connected curve, with W projective, and suppose that every monodromy invariant class in $H^*(W_c, \mathbb{Q})$ is a Hodge class, as happens when the identity component of the Zariski closure of the monodromy group is the Hodge group of a very general fibre.*

(i) *If $g \leq 4$, then $B(W)$ holds.*

(ii) *If the fibres are of $(K, -1, n)$-Weil type with connected monodromy group $\mathrm{SU}(V, H)$, and $n = 2$, or $n = 3$ with trivial discriminant, then B holds for every fibre power $W \times_C \cdots \times_C W$. In particular $B(W_C)$ of Corollary 2.31 holds in these cases.*

Proof. By the theorem of the fixed part the invariant classes form a sub-Hodge structure of every fibre, with a Hodge structure independent of the point, so they are Hodge classes on every fibre.

(i) The Hodge conjecture holds for abelian varieties of dimension at most four, by the Lefschetz theorem on $(1,1)$ classes and hard Lefschetz up to dimension three and by [Mar25a] in dimension four, so each invariant class is algebraic on every fibre; Lemma 2.27 makes it the restriction of an algebraic class on W, and Theorem 2.21 applies.

(ii) The fibre power is an abelian scheme with projective total space and the monodromy acting diagonally. Its invariant classes are $\mathrm{SU}(V,H)$-invariants, hence Hodge classes at every member with $\mathrm{Hg} = \mathrm{SU}(V,H)$, an uncountable set of points, and there the Hodge conjecture holds for the fibre by Proposition 2.24; Lemma 2.27 and Theorem 2.21 conclude as before. For W_C the connected monodromy group is $\mathrm{SU}(V,H)$ when it is Zariski dense, which is the hypothesis of Corollary 2.31(ii). ∎

The hypothesis on the monodromy in Corollary 2.25 can be checked with André's theorem [And92, Theorem 1]: the identity component of the Zariski closure of the monodromy group is always a normal subgroup of the derived Mumford–Tate group of a very general fibre, so the hypothesis holds whenever the family is not isotrivial and the Hodge group of a very general fibre is semisimple and $\mathbb{Q}$-simple, as for the Mumford families of Corollary 2.22.

Proposition 2.26 (The total space of a Mumford family)**.** *Let $W \to C$ be as in Corollary 2.22, with C the compact Shimura curve of* [Mum69] *of torsion-free level, or a finite étale cover of it. Then the Hodge conjecture holds for the fivefold W: its Hodge classes of degree $2p$ are spanned by ℓ^p and $f\ell^{p-1}$, with f the class of a fibre.*

Proof. By Step 1 of the proof of Theorem 2.21, $H^k(W,\mathbb{Q}) = \bigoplus_q H^k_{(q)}$ with $H^k_{(q)} \cong H^{k-q}(C, R^q\pi_*\mathbb{Q})$, the summands being sub-Hodge structures cut out by algebraic projectors. The summand with $k = q$ is the space of invariants, which by the proof of Corollary 2.22 is spanned by the powers of the polarisation, restrictions of the powers of ℓ. The summand with $k = q+2$ is $f \cup H^q_{(q)}$ by Step 5, spanned by the $f\ell^{q/2}$. It remains to see that $H^1(C, R^q\pi_*\mathbb{Q})$ carries no Hodge class. For q even its weight $q+1$ is odd. For q odd, over $\mathbb{C}$ the local system splits as a sum of $S^{k_1}V_1 \otimes U_\lambda \otimes M_\lambda$, where V_1 is the uniformising variation of the factor of G split at the real place, U_λ is a unitary variation of type $(0,0)$, M_λ is constant of type (a,a) with $2a + k_1 = q$, and $k_1 \equiv q \pmod 2$ because the central element $(-1,1,1)$ acts on V by -1. When C is étale over the Shimura curve the Kodaira–Spencer map of V_1 is an isomorphism, and then $H^1(C, S^{k_1}V_1 \otimes U_\lambda)$ has only the types $(k_1+1,0)$ and $(0,k_1+1)$ [Zuc79]; after the twist by M_λ these are never balanced, since $k_1 \geq 1$. ∎

Statements of the kind of Proposition 2.26 are studied for Kuga fibre varieties, and we make no claim of priority. The étale hypothesis is used only at the end of the proof, and it cannot be dropped from the argument: over a ramified cover the Kodaira–Spencer map of V_1 vanishes at the branch points, the Higgs field of $S^{k_1}V_1$ is no longer an isomorphism, and the torsion of its cokernel contributes to $H^1(C, R^q\pi_*\mathbb{Q})$ in balanced types.

2.8. **The two targets as cases of (L).** The argument of Section 2.7 does more than add a case of (L). For an abelian scheme over a curve it shows that (L) for the total space is exactly the statement that algebraicity propagates along the curve. The base points of this paper then turn each of the two targets of Remark 2.73, one Weil family and the first classes beyond the Weil lines, into (L) for one explicit variety. The reductions

are equivalences, so they locate the difficulty rather than remove it. Along the way the Hodge conjecture is proved for the self-product of a Mumford fourfold at every CM point (Theorem 2.29), and the two branches of [II, Remark 2.10] are shown not to be realisable by two objects (Theorem 2.32), an implication whose hypothesis [II, Theorem 3.5] shows is never met.

Lemma 2.27 (Spreading along a curve)**.** *Let $\pi\colon W \to C$ be a smooth projective morphism onto a smooth projective connected curve, W projective, and ξ a global section of $R^{2p}\pi_*\mathbb{Q}$. If ξ_t is algebraic for uncountably many $t \in C$, there are an algebraic class $\tilde{\xi}$ on W and an integer $d \geq 1$ with $\tilde{\xi}|_{W_t} = d\,\xi_t$ for every $t \in C$.*

Proof. The proof of [I, Theorem 16.4] applies verbatim with $(\mathcal{A}, \mathcal{S}, \omega)$ replaced by (W, C, ξ): the set of t at which ξ_t is algebraic is the union, over countably many tuples of Hilbert polynomials and rational coefficients, of the images of unions T of connected components of fibre products of relative Hilbert schemes, projective over C, on which $\sum_i q_i[Z_i] = \xi$. Uncountably many t force one image to be infinite, hence equal to C. Choose an irreducible component of that T surjecting onto C, two of its points over distinct points of C, and an irreducible curve through them; its normalisation C' maps finitely onto C, of some degree d. Pulling the universal subschemes back to $W' = W \times_C C'$ gives a cycle $z = \sum_i q_i[\mathcal{Z}_i]$ whose restriction to $W'_{c'} = W_{g(c')}$ is $\xi_{g(c')}$ for every $c' \in C'$. The projection $h\colon W' \to W$ is finite and flat of degree d, and by base change $h_*[z]$ restricts on W_t to the sum of the restrictions over the points of C' above t, counted with multiplicity, which is $d\,\xi_t$. ∎

Theorem 2.28 ((L) for an abelian scheme over a curve is propagation)**.** *Let $\pi\colon W \to C$ be as in Theorem 2.21, but with no hypothesis on its invariant cycles, and keep the notation of its proof. Let Λ_C be the operator that is zero on $H^p(C, R^q\pi_*\mathbb{Q})$ for $p \neq 2$ and inverts $L_C\colon H^q_{(q)} \to H^{q+2}_{(q)}$.*

(i) *The operator Λ of* [I, Definition 10.6] *for L_W is $\Lambda_{\mathrm{rel}} + \Lambda_C$. So $B(W)$ holds for L_W if and only if Λ_C is induced by an algebraic cycle.*

(ii) *If $B(W)$ holds for L_W, then for every $c \in C$ and every algebraic class y on W_c the class $\Lambda_C(i_{c*}y)$ is algebraic; up to the factor m it lies in $H^0(C, R^q\pi_*\mathbb{Q})$ and restricts on W_c to the unique monodromy invariant class v with $\int_{W_c} v \cup w = \int_{W_c} y \cup w$ for every invariant w of complementary degree. In particular every global section of $R^{2p}\pi_*\mathbb{Q}$ that is algebraic at one point is the restriction of an algebraic class on W, and is algebraic at every point.*

(iii) *If at one point c_0 every monodromy invariant class of $H^*(W_{c_0}, \mathbb{Q})$ is algebraic, then $B(W)$ holds for L_W if and only if every monodromy invariant class is the restriction of an algebraic class on W.*

Proof. (i) Steps 4 and 5 of the proof of Theorem 2.21 show that $L_C\colon H^q_{(q)} \to H^{q+2}_{(q)}$ is an isomorphism without using the hypothesis on invariant cycles: the nondegeneracy of $I_q \times I_{2g-q} \to \mathbb{Q}$ comes from the Hodge–Riemann relations alone. So Λ_C is defined, and Steps 5 and 6, which use only that Λ_C vanishes off $p = 2$ and inverts L_C, give $[L_W, \Lambda_{\mathrm{rel}} + \Lambda_C] = H$. By the uniqueness in [I, Definition 10.6] this is Λ. As Λ_{rel} is induced by an algebraic cycle (Step 3), Λ is if and only if Λ_C is.

(ii) The Gysin map i_{c*} commutes with $[2]_*$, which acts on $H^q(W_c)$ by 2^{2g-q}, so $i_{c*}y \in H^{q+2}_{(q)}$, the summand with $p = 2$. If y is algebraic so is $i_{c*}y$, and by (i) so is $\Lambda_C(i_{c*}y)$. Put $u = m\,\Lambda_C(i_{c*}y) \in H^q_{(q)}$, so $f \cup u = i_{c*}y$. For $w \in H^{2g-q}_{(2g-q)}$ the projection formula gives $\int_{W_c} u|_{W_c} \cup w|_{W_c} = \int_W (f \cup u) \cup w = \int_W i_{c*}y \cup w = \int_{W_c} y \cup w|_{W_c}$, and $w|_{W_c}$ runs over

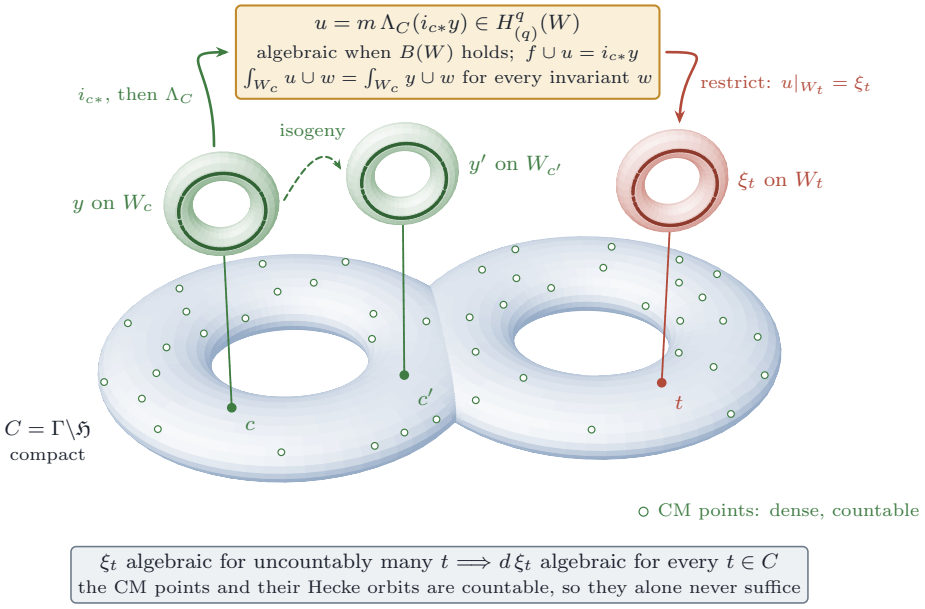


FIGURE 3. Theorem 2.28(ii) along a compact Shimura curve C. An algebraic class y on one fibre, at a base point of a Weil family or at a CM point of a Mumford family (hollow dots: the CM points, countable and dense), is pushed into W by the Gysin map and brought back to the invariant summand by the base operator Λ_C. When $B(W)$ holds the result u is an algebraic class on W, pinned down by the displayed pairing against the invariant classes, and its restriction to a very general fibre is the invariant part of y. The dashed arcs are isogenies between CM members along Hecke orbits: they carry algebraic classes to algebraic classes but reach only countably many points, and Lemma 2.27, in the box, needs uncountably many. This is how Corollary 2.31 and Corollary 2.30 turn each target into one case of (L).

the invariants of complementary degree (Step 4); by the nondegeneracy of Step 4 this determines $u|_{W_c}$. If $y = \xi_c$ for a global section ξ, then $v = \xi_c$ has the property, so $u|_{W_c} = \xi_c$, and u, being a global section of $R^q\pi_*\mathbb{Q}$ equal to ξ at c, restricts to ξ_t at every t.

(iii) If every invariant class is a restriction of an algebraic class, $B(W)$ holds by Theorem 2.21. Conversely, under $B(W)$, (ii) applied at c_0 to each invariant class makes it the restriction of an algebraic class. ∎

Figure 3 draws part (ii) along a compact Shimura curve.

Theorem 2.29 (The Mumford square at a CM point)**.** *Let $W \to C$ be a family as in Corollary 2.22 and $c \in C$ a CM point. Then every Hodge class on $X_c \times X_c$ is algebraic. In particular the two exceptional classes of Proposition 2.74 are algebraic on $X_c \times X_c$.*

Proof. The Hodge group T of X_c is a $\mathbb{Q}$-torus contained in G. It lies in a maximal $\mathbb{Q}$-torus of $G = \mathrm{Res}_{F/\mathbb{Q}}\mathrm{SL}_1(D)$, which is the restriction of scalars of the norm one torus of a quadratic extension of F inside D, of rank three, one factor for each embedding of F. The cocharacter of the Hodge structure lies in the factor of the embedding at which D splits, and the Galois group permutes the embeddings of F transitively, so the smallest

$\mathbb{Q}$-subtorus containing it has rank three: $T_{\mathbb{C}}$ is a maximal torus of $G_{\mathbb{C}} \cong \mathrm{SL}_2^3$. Its eight weights on $V = H^1(X_c, \mathbb{C}) = V_1 \otimes V_2 \otimes V_3$ are the vectors $(\pm 1, \pm 1, \pm 1)$, pairwise distinct. Hence $\mathrm{End}^0(X_c) \otimes \mathbb{C} = \mathrm{End}_T(V)$ is the algebra of diagonal matrices in a weight basis (x_w). For $n \geq 1$, $\mathrm{Hom}(X_c^n, X_c) \otimes \mathbb{C}$ is the space of n-tuples $(\varphi_1, \ldots, \varphi_n)$ of such matrices, acting by $x_w \mapsto x_w \otimes u_w$ with $u_w = (\varphi_{1,w}, \ldots, \varphi_{n,w}) \in \mathbb{C}^n$, so the vectors u_w can be prescribed independently. The Hodge classes of X_c^n, tensored with $\mathbb{C}$, are spanned by the products of vectors $x_w \otimes u$, $u \in \mathbb{C}^n$, in $\bigwedge^*(V \otimes \mathbb{C}^n)$ in which the weights w, counted with multiplicity, sum to zero.

We show that for every $n \geq 1$ the ring $\mathrm{Hdg}^*(X_c^n)$ is generated by pull-backs of Hodge classes of X_c along homomorphisms $X_c^n \to X_c$. Put $w_1 = (1,1,1)$, $w_2 = (1,-1,-1)$, $w_3 = (-1,1,-1)$ and $w_4 = (-1,-1,1)$; the weights are the $\pm w_i$, $w_1 + w_2 + w_3 + w_4 = 0$, and any three of the w_i are linearly independent, so the integer relations among them are the multiples of $(1,1,1,1)$. Let α_i and β_i be the multiplicities of w_i and $-w_i$ in a nonempty multiset of weights with sum zero. Then $\alpha_i - \beta_i$ is independent of i; so either some α_i and β_i are both positive, or all α_i are positive, or all β_i are. In each case the multiset contains one of the sets $\{w_i, -w_i\}$, $\{w_1, w_2, w_3, w_4\}$, $\{-w_1, -w_2, -w_3, -w_4\}$, each of sum zero, and by induction every product above is, up to sign, a product of monomials $\bigwedge_{w \in P}(x_w \otimes u_w)$ over such sets P, whose weights are pairwise distinct. Choosing $f \in \mathrm{Hom}(X_c^n, X_c) \otimes \mathbb{C}$ with $f^* x_w = x_w \otimes u_w$ for $w \in P$ gives $\bigwedge_{w \in P}(x_w \otimes u_w) = f^*(\bigwedge_{w \in P} x_w)$, with $\bigwedge_{w \in P} x_w$ a Hodge class of X_c tensored with $\mathbb{C}$. The map $f \mapsto f^*(t)$ is polynomial and the rational homomorphisms are Zariski dense in their complexification, so the same span is reached with f defined over $\mathbb{Q}$ and t rational. The Hodge classes of the abelian fourfold X_c are algebraic because the Hodge conjecture holds for abelian fourfolds [Mar25a], and pull-backs and products of algebraic classes are algebraic. The case $n = 2$ is the theorem. ∎

Corollary 2.30 (The first classes beyond the Weil lines are one case of (L))**.** *Let $W \to C$ be as in Corollary 2.22 and $W^{(2)} = W \times_C W \to C$, an abelian scheme of relative dimension eight with $W^{(2)}$ projective of dimension nine. The following are equivalent.*

(a) *$B(W^{(2)})$ holds for the class $L_{W^{(2)}}$.*
(b) *The two exceptional classes of $X_t \times X_t$ are algebraic for uncountably many $t \in C$.*
(c) *The Hodge conjecture holds for $X_t \times X_t$ for every $t \in C$.*
(d) *The Hodge conjecture holds for $W^{(2)}$; for this item C is taken to be the compact Shimura curve of torsion-free level, or a finite étale cover of it.*

Proof. As in the proof of Corollary 2.22, the connected monodromy group is G, so the monodromy invariants in $H^*(X_t \times X_t, \mathbb{Q})$ are contained in the G-invariants, which are the Hodge classes of $X_t \times X_t$ for t very general; conversely a G-invariant class is flat, being fixed by the monodromy. The Hodge group of every fibre lies in G, so the invariant classes are Hodge classes on every fibre. At a point t that is not a CM point the Hodge group H_t of X_t is G. Indeed, if H_t were smaller, the Hodge classes of the powers of X_t that are not G-invariant would stay Hodge only on a discrete subset of the period domain, the upper half plane; every point of the orbit of the Hodge structure at t under $H_t(\mathbb{R})^+$ carries them, so that orbit is a point, the cocharacter of the Hodge structure is central in H_t, and H_t, the smallest $\mathbb{Q}$-group containing it, is a torus: t would be a CM point. So at such t the Hodge classes of $X_t \times X_t$ are the invariant ones, and $\mathrm{End}^0(X_t) = \mathrm{End}_G(V) = \mathbb{Q}$, because $V \otimes \bar{\mathbb{Q}} = V_1 \otimes V_2 \otimes V_3$ is an irreducible representation of G; thus X_t is as in Setup 3.1.

(c) implies (b) trivially. (b) implies (c): the exceptional classes are invariant, hence flat, so by Lemma 2.27 a multiple of each is the restriction of an algebraic class on $W^{(2)}$, and they are algebraic on every fibre. At a point that is not a CM point, Theorem 3.7 with

$n = 2$ makes every Hodge class of $X_t \times X_t$ algebraic; at a CM point this is Theorem 2.29. (c) implies (a): by (c) every invariant class is algebraic on every fibre, so by Lemma 2.27 a multiple of it is the restriction of an algebraic class on $W^{(2)}$, and Theorem 2.21 gives $B(W^{(2)})$. (a) implies (c): at a CM point c_0 every Hodge class of $X_{c_0} \times X_{c_0}$ is algebraic by Theorem 2.29, so by Theorem 2.28(iii) every invariant class is the restriction of an algebraic class, and is algebraic on every fibre. This is (c) at the points that are not CM points, where the Hodge classes are the invariant ones; at a CM point (c) is Theorem 2.29.

For (d), split $H^*(W^{(2)}, \mathbb{Q})$ into the summands $H^p(C, R^q)$, $p \in \{0, 1, 2\}$, as in the proof of Proposition 2.26. The summands with $p = 1$ carry no Hodge class, by the argument given there, applied to the fibre square. Those with $p = 2$ are $f \cup H^q_{(q)}$, and $f \cup u = i_{c*}(u|_{W_c^{(2)}})$ for any c; taking c a CM point, where $u|_{W_c^{(2)}}$ is a Hodge class and so algebraic by Theorem 2.29, they are algebraic. The summands with $p = 0$ are the invariants, all of Hodge type. So (d) holds exactly when every invariant class is the restriction of an algebraic class, which by Theorem 2.28(iii), applied at a CM point, is (a). ∎

Corollary 2.31 (One Weil family is one case of (L))**.** *Keep* [I, Setup 16.1]*, let $s_0 \in \mathcal{S}$ be the image of a base point of* [I, Theorem 15.27]*, and let $t_1 \in \mathcal{S}$ lie outside every proper Zariski closed set $\Sigma_{\underline{P},\underline{q}}$ of* [I, Theorem 16.4]*. Let $\kappa\colon C \to \mathcal{S}$ be a morphism from a smooth projective connected curve whose image contains s_0 and t_1, and $W_C = \mathcal{A} \times_{\mathcal{S}} C$.*

(i) *If $B(W_C)$ holds for L_{W_C}, then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every member of the family with* $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$.

(ii) *Conversely, if $\Sigma = \mathcal{D}_{n,n}$ and the monodromy of W_C is Zariski dense in* $\mathrm{SU}(V, H)$*, then $B(W_C)$ holds for L_{W_C}.*

Proof. (i) The class $\kappa^*\omega$ is a global section of $R^{2n}\pi_*\mathbb{Q}$ on C, algebraic at a point over s_0 by [I, Theorem 15.27]. By Theorem 2.28(ii) it is algebraic at a point over t_1, so t_1 lies in some closed set $\Sigma_{\underline{P},\underline{q}}$, which is therefore all of $\mathcal{S}$. [I, Theorem 16.4] gives $\Sigma = \mathcal{D}_{n,n}$, and [I, Theorem 15.14(a)] the rest. (ii) With dense monodromy the invariants are the $\mathrm{SU}(V, H)$-invariants, which are the Hodge classes of a member with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$: the powers of the polarisation and the Weil classes, by [I, Theorem 4.14]. All are algebraic on every fibre when $\Sigma = \mathcal{D}_{n,n}$, so they are restrictions of algebraic classes by Lemma 2.27, and Theorem 2.21 applies. ∎

Curves κ as in Corollary 2.31 exist: the Baily–Borel compactification of $\mathcal{S}$ is projective [BB66], its boundary has codimension $2n - 1 \geq 3$, the rational boundary components of the domain of type $\mathrm{I}_{n,n}$ being of type $\mathrm{I}_{n-r,n-r}$ with $r \geq 1$, and a general complete intersection curve through s_0 and t_1 avoids it.

Theorem 2.32 (Two branches cannot be two objects)**.** *Let A be of $(K, -1, n)$-Weil type with $n \geq 2$, $\omega \neq 0$ a rational class on the Weil line, and E a perfect complex with* $\mathrm{ch}(E) = N\omega$*, $N \neq 0$, and σ_E injective. Let $E \cong \bigoplus_i F_i[k_i]$ be any decomposition. Then every summand F_i with* $\mathrm{ch}(F_i) \neq 0$ *has a Yoneda product $\bigwedge^2 \mathrm{Ext}^1(F_i, F_i) \to \mathrm{Ext}^2(F_i, F_i)$ that is not injective. In particular E is not a direct sum of shifts of line bundles, of skyscraper sheaves of points, or of their images under autoequivalences of $D^b(A)$, since the Ext algebra of each of these is the exterior algebra on its* Ext^1.

Proof. By [II, Theorem 2.11(ii)] the kernel of ev_E in degree two is $P \wedge Q$. The characteristic morphism is compatible with the decomposition, so $\mathrm{ev}_{F_i}(p \wedge q) = \mathrm{ev}_{F_i}(p)\,\mathrm{ev}_{F_i}(q) = 0$ for all $p \in P$ and $q \in Q$. Suppose the Yoneda product of $F = F_i$ is injective on $\bigwedge^2 \mathrm{Ext}^1$. Then the images $\bar{P}$ and $\bar{Q}$ of P and Q in $\mathrm{Ext}^1(F, F)$ satisfy $\bar{p} \wedge \bar{q} = 0$ for all $\bar{p}, \bar{q}$, so either $\bar{P} = 0$,

or $\bar{Q} = 0$, or $\bar{P} = \bar{Q}$ is a line. In the first case ev_F kills $P \wedge HH^1$, in the second $Q \wedge HH^1$, in the third all of $HH^2 = HH^1 \wedge HH^1$; by the compatibility $\sigma_F \circ \mathrm{ev}_F = (\,\cdot\,) \lrcorner \mathrm{ch}(F)$ of [II, Setup 2.7] the same subspace annihilates $x = \mathrm{ch}(F)$.

These annihilators are computed in the free module of rank one that $H^*(A, \mathbb{C})$ is over $HH^*(A) = \bigwedge^*(P \oplus Q)$, with generator the holomorphic volume form Ω: an element $y \cdot \Omega$ is killed by $P \wedge HH^1$ exactly when $p \wedge y$ has top degree for every $p \in P$, that is when $y \in \det P \wedge \bigwedge^* Q + \bigwedge^{\geq 4n-1}(P \oplus Q)$. Since $\det P \cdot \Omega = \pm\alpha_+$, the first summand gives the classes $\bigwedge^j Q \cdot \alpha_+$, of type $(n-b, n+a)$ with a wedges by $V_-^{0,1}$ and b contractions by T_+, and the second gives classes of type $(1, 2n)$, $(0, 2n-1)$ and $(0, 2n)$. The components of x are of type (k, k), so $x \in \mathbb{C}\alpha_+$. Symmetrically the second case gives $x \in \mathbb{C}\alpha_-$ and the third $x = 0$. Finally x is rational, complex conjugation carries V_+ to V_- and $H^{1,0}$ to $H^{0,1}$, so $\bar{\alpha}_+ \in \mathbb{C}\alpha_-$, and a rational multiple of α_+ or of α_- is zero. So $\mathrm{ch}(F) = 0$. As $\sum_i (-1)^{k_i} \mathrm{ch}(F_i) = N\omega \neq 0$, some summand has nonzero Chern character, and the first statement follows; the second is the special case in which every summand has an exterior Ext algebra. ∎

2.9. **Which hypotheses suffice.** The implications proved above, together with those of Section 2.6, determine exactly which combinations of statements yield the conjecture. The logical skeleton of the paper and of the literature it quotes is finite: a set of statements, each of them proved here, quoted from the literature, or taken as a hypothesis, and a set of inference rules, each of them one theorem. Consequences can then be taken.

The answer is only as strong as the rule set. A set of statements the rules do not carry to the conjecture is *insufficient* in the sense of the rules, and that means no more than that no rule derives the conjecture from it. An implication that is true and left out makes a set look necessary when it is not. One such implication is elementary, and it decides the shape of the answer, so we prove it first.

Proposition 2.33 (The conjecture for the varieties that are not abelian is the conjecture)**.** *Let A be a complex abelian variety of dimension $g \geq 1$ and let $X = A \times \mathbb{P}^1$. Then X is a smooth complex projective variety that is not an abelian variety, and if every Hodge class of degree $2p$ on X is algebraic, then so is every Hodge class of degree $2p$ on A. Consequently the statement*

> *the Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety*

implies the Hodge conjecture for every abelian variety, and it is equivalent to **HC**.

Proof. X is smooth and projective of dimension $g+1$. By the Künneth formula $h^{1,0}(X) = h^{1,0}(A) + h^{1,0}(\mathbb{P}^1) = g$, while an abelian variety of dimension $g+1$ has $h^{1,0} = g+1$; so X is not an abelian variety. (Equally, X contains the rational curves $\{a\} \times \mathbb{P}^1$, and every morphism from $\mathbb{P}^1$ to a complex torus is constant, since it lifts to the universal cover $\mathbb{C}^{g+1}$ [BL04, Chapter 1].)

Let $\mathrm{pr}\colon X \to A$ be the projection and $D = A \times \{0\}$, a smooth divisor that pr maps isomorphically onto A, so that $\mathrm{pr}_*[D] = 1$ in $H^0(A, \mathbb{Q})$. Let α be a Hodge class of degree $2p$ on A. Pull-back along a morphism is a morphism of Hodge structures, so $\mathrm{pr}^*\alpha$ is a Hodge class of degree $2p$ on X, and by hypothesis $\mathrm{pr}^*\alpha = [Z]$ for an algebraic cycle Z on X with rational coefficients. The projection formula gives

$$\mathrm{pr}_*\big([Z] \cup [D]\big) = \mathrm{pr}_*\big(\mathrm{pr}^*\alpha \cup [D]\big) = \alpha \cup \mathrm{pr}_*[D] = \alpha.$$

The cycle class map is compatible with intersection products and with proper push-forward [Ful98, Chapter 19], so the left side is the class of the cycle $\mathrm{pr}_*(Z \cdot D)$ and α is algebraic. A point, the abelian variety of dimension zero, satisfies the conjecture trivially.

So the displayed statement gives the conjecture for every abelian variety, and with it the conjecture for every smooth complex projective variety. The converse is immediate: the displayed statement is a special case of **HC**. ∎

Since the conjecture for the varieties that are not abelian is the whole conjecture, the statement that belongs on a frontier is weaker: it asks for algebraicity only modulo what abelian varieties supply.

Definition 2.34 (The conjecture modulo abelian varieties)**.** For a smooth complex projective variety X and $p \geq 0$ let $\mathrm{Ab}^p(X) \subseteq H^{2p}(X, \mathbb{Q})$ be the span of the classes $\Gamma_*\gamma = \mathrm{pr}_{X*}([\Gamma] \cup \mathrm{pr}_B^*\gamma)$ of degree $2p$, where B runs over the complex abelian varieties, the point included, γ over the Hodge classes on B, and Γ over the algebraic cycles on $B \times X$ with rational coefficients. The statement (F3′) is: for every X and every p,

$$\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X).$$

Each $\Gamma_*\gamma$ is a Hodge class, since cup product with an algebraic class and push-forward preserve Hodge classes, so $\mathrm{Ab}^p(X) \subseteq \mathrm{Hdg}^p(X)$; and taking B a point shows that $\mathrm{Ab}^p(X)$ contains every algebraic class. So (F3′) says that every Hodge class is algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences, which is the form of (F3) that Remark 2.72 asks for.

Proposition 2.35 (What is known of (F3′))**.**

(i) **HC** *implies* (F3′)*, and* (F3′) *together with the conjecture for abelian varieties implies* **HC**.

(ii) (F3′) *holds on every abelian variety and on every product* $A \times \mathbb{P}^1$ *with* A *abelian; the argument of Proposition 2.33 gives nothing for it.*

(iii) *Let* $\mathcal{A}$ *be the class of smooth complex projective varieties* X *for which* $H^*(X, \mathbb{Q})$ *is spanned by classes* $\Gamma_* y$ *with* B *abelian,* Γ *an algebraic cycle on* $B \times X$ *and* $y \in H^*(B, \mathbb{Q})$*. Every* $X \in \mathcal{A}$ *satisfies* (F3′)*. The class* $\mathcal{A}$ *contains every abelian variety and every smooth projective curve, and it is closed under products and under images of surjective morphisms. So it contains every product of curves, every symmetric product of a curve, and every smooth projective variety onto which a product of curves or an abelian variety maps surjectively.*

(iv) *Every* X *of dimension* n *satisfies* (F3′) *in the degrees* $2p$ *with* $p \in \{0, 1, n-1, n\}$*, so every* X *of dimension at most three satisfies it in every degree.*

Proof. (i) If **HC** holds, then $\mathrm{Hdg}^p(X)$ consists of algebraic classes, which lie in $\mathrm{Ab}^p(X)$. Conversely, if the conjecture holds for abelian varieties, every γ in Definition 2.34 is algebraic, so every $\Gamma_*\gamma$ is, since composition with a correspondence carries algebraic classes to algebraic classes [Ful98, Chapter 16]; then (F3′) says that every Hodge class is algebraic.

(ii) On an abelian variety A take $B = A$ and Γ the diagonal. On $X = A \times \mathbb{P}^1$ the Künneth formula gives $H^*(X, \mathbb{Q}) = \mathrm{pr}^*H^*(A, \mathbb{Q}) \oplus \mathrm{pr}^*H^*(A, \mathbb{Q}) \cup \eta$, with η the class of a point of $\mathbb{P}^1$ pulled back to X, which is algebraic. The first summand is $\Gamma_*H^*(A, \mathbb{Q})$ for Γ the transpose of the graph of pr, and the second is $\Gamma'_*H^*(A, \mathbb{Q})$ for $\Gamma' = \Gamma \cdot \mathrm{pr}_X^*\eta$. So $X \in \mathcal{A}$, and (iii) applies.

(iii) Let $X \in \mathcal{A}$ and fix p. Since $H^{2p}(X, \mathbb{Q})$ is finite dimensional, finitely many classes $\Gamma_{j*}y_j$ span it, and we may take each Γ_j of pure codimension c_j on $B_j \times X$ and each y_j of

pure degree k_j, with $k_j + 2c_j - 2\dim B_j = 2p$. Then $\Gamma_{j*}\colon H^{k_j}(B_j,\mathbb{Q}) \to H^{2p}(X,\mathbb{Q})$ is a morphism of Hodge structures up to a Tate twist, and the sum of these maps is a surjective morphism of polarisable Hodge structures onto $H^{2p}(X,\mathbb{Q})$. Polarisable Hodge structures are semisimple, so its kernel has a complement that is a sub-Hodge structure and maps isomorphically onto $H^{2p}(X,\mathbb{Q})$; hence every Hodge class of degree $2p$ is the image of a Hodge class, that is $\sum_j \Gamma_{j*}\gamma_j$ with each γ_j a Hodge class on B_j. So $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$.

An abelian variety lies in $\mathcal{A}$ through its diagonal. A curve C of genus $g \geq 1$ lies in $\mathcal{A}$ through an Abel–Jacobi embedding $\iota\colon C \to J(C)$: the transpose of its graph induces ι^*, which is an isomorphism on H^0 and H^1 and sends the class of the theta divisor to g times the class of a point, so it is surjective; the projective line is the image of an elliptic curve under a double cover, and is covered by the last closure property. For products, if $H^*(X_i,\mathbb{Q})$ is spanned by the $\Gamma_{i*}y$, then by the Künneth formula $H^*(X_1\times X_2,\mathbb{Q})$ is spanned by the $(\Gamma_1\times\Gamma_2)_*(y_1\otimes y_2)$, with $\Gamma_1\times\Gamma_2$ an algebraic cycle on $(B_1\times B_2)\times(X_1\times X_2)$. For images, let $f\colon Y\to X$ be a surjective morphism with $Y\in\mathcal{A}$, let $r = \dim Y - \dim X$ and let h be an ample class on Y. The correspondence $\Psi = [\Gamma_f]\cdot \mathrm{pr}_Y^* h^r$ on $Y\times X$ induces $\Psi_* y = f_*(y\cup h^r)$, and $\Psi_* f^* x = x\cup f_* h^r = c\,x$ with $c>0$ the degree of h^r on a general fibre. So Ψ_* is surjective, and $H^*(X,\mathbb{Q})$ is spanned by the classes $(\Psi\circ\Gamma)_* y$, with $\Psi\circ\Gamma$ algebraic [Ful98, Chapter 16]. A symmetric product of a curve is the image of a power of it.

(iv) Every class of degree 0 or $2n$ is algebraic, and every Hodge class of degree 2 is algebraic by the Lefschetz theorem on $(1,1)$ classes. For $n\geq 2$, the hard Lefschetz theorem makes cup product with h^{n-2}, for h an ample class, an isomorphism $H^2(X,\mathbb{Q})\to H^{2n-2}(X,\mathbb{Q})$ of Hodge structures up to a Tate twist, so every Hodge class of degree $2n-2$ is h^{n-2} times a Hodge class of degree 2, and is algebraic. Algebraic classes lie in $\mathrm{Ab}^p(X)$. When $n\leq 3$ every p between 0 and n is one of $0, 1, n-1, n$. ∎

Remark 2.36 (The reach of algebraic correspondences). (F3′) is the statement that the Hodge conjecture reduces to abelian varieties along algebraic correspondences, and Proposition 2.35 records the cases of such a reduction that are theorems: the varieties of $\mathcal{A}$, whose whole cohomology is reached from abelian varieties by algebraic correspondences, and the degrees in which the conjecture itself is a theorem. The next degree is degree four on fourfolds outside $\mathcal{A}$ and outside the scope of Proposition 2.43 and Proposition 2.44. The square $S\times S$ of a K3 surface is the standard example when $\mathrm{End}_{\mathrm{Hdg}}(T(S))$ is a totally real field of degree at least two, that is, when S has real multiplication. When that field is $\mathbb{Q}$ or a CM field, the Hodge conjecture holds on $S\times S$: by Proposition 2.38(i) in the first case, and in the second by [Bus19] and the fact that the elements of norm one span a CM field, as in [RM08] (Proposition 2.39). Its Hodge classes in $H^2(S)\otimes H^2(S)$ include the classes that induce the Hodge endomorphisms of the transcendental lattice $T(S)$, and $S\times S$ lies in $\mathcal{A}$ as soon as the Kuga–Satake embedding of $T(S)$ is induced by an algebraic cycle, the Kuga–Satake Hodge conjecture for S: the transpose of such a cycle maps the cohomology of the Kuga–Satake variety onto $T(S)$, and the rest of $H^*(S,\mathbb{Q})$ is algebraic. The rational Hodge isometries are algebraic [Bus19], but a real multiplication is not an isometry, and Remark 3.5 meets exactly this obstacle for the surfaces S_λ; the dihedral family of correspondences of [BB26b] crosses it for two cyclic cubic fields and reaches no other field.

For varieties whose cohomology is not of abelian type the position is starker. By Proposition 2.75 no correspondence moves the H^3 of a very general quintic threefold into the cohomology of an abelian variety, so such a variety does not lie in $\mathcal{A}$. That is no obstruction to (F3′) itself, since a Hodge class spans a Hodge structure of Tate type, which

is of abelian type, and nothing in Hodge theory forbids it from lying in $\mathrm{Ab}^p(X)$. Nor is it a route: to put a Hodge class into $\mathrm{Ab}^p(X)$ one must construct an algebraic cycle on $B \times X$ whose class carries it, which is the kind of construction the Hodge conjecture asks for, now on $B \times X$ instead of on X. The cases of (F3′) proved below outside $\mathcal{A}$ come from the Hodge conjecture itself, from hypotheses on zero-cycles, and, in degree four, from correspondences with members of $\mathcal{A}$ (Proposition 2.39 to Proposition 2.46); none of them reaches a variety in general position, and (F3′) enters Theorem 2.70 as a hypothesis.

Two further facts fix the position of (F3′). The first says how strong it is.

Proposition 2.37 (The strength of (F3′))**.** (F3′) *implies* (M)*: every Hodge class on every smooth complex projective variety is motivated in the sense of André. In particular it implies that every such class is absolutely Hodge. Adding the implication* (F3′) ⇒ (M) *to the rule set of Setup 2.69 leaves the five minimal sufficient sets of Theorem 2.70(ii) unchanged.*

Proof. Let $\gamma \in \mathrm{Hdg}^p(X)$. By (F3′), $\gamma = \sum_j \Gamma_{j*}\beta_j$ with β_j a Hodge class on an abelian variety B_j and Γ_j an algebraic cycle on $B_j \times X$. Every Hodge class on an abelian variety is motivated [And96b, Théorème 0.6.2], and the motivated classes form a graded $\mathbb{Q}$-algebra that contains the algebraic classes and is stable under pull-back and push-forward along morphisms [And96b, §2]. So $\mathrm{pr}_{B_j}^*\beta_j \cup [\Gamma_j]$ is motivated, and so is its push-forward $\Gamma_{j*}\beta_j$; hence γ is motivated. Motivated classes are absolutely Hodge [And96b, §2]. For the last assertion, the new rule has the premise (F3′) and the conclusion (M). The sets C_1 and C_2 of the proof of Theorem 2.70 omit (F3′), and C_3 and C_4 contain both statements, so the four sets stay closed and that proof applies without change; the five sets remain sufficient, since adding a rule only enlarges closures. ∎

So on the route $\{\mathrm{L}, \mathrm{F3}'\}$ the second member already contains (M), and that route is never easier than $\{\mathrm{L}, \mathrm{M}\}$. The second fact locates, on the powers of a variety of orthogonal type, the first class that (F3′) has to reach.

Proposition 2.38 (Powers of a variety of orthogonal type)**.** *Let Y be a smooth complex projective variety of even dimension m. Suppose that $H^k(Y,\mathbb{Q})$ is spanned by algebraic classes for $k \neq m$, that $H^m(Y,\mathbb{Q}) = N \oplus V$ with N spanned by algebraic classes, the intersection form ψ nondegenerate on N and $V = N^{\perp}$, and that the Hodge group of V is $\mathrm{SO}(V,\psi)$, with $t = \dim V \geq 3$. Let $\det(V) \in H^{mt}(Y^t,\mathbb{Q})$ be the image of a generator of $\bigwedge^t V \subseteq V^{\otimes t}$ under the Künneth embedding $V^{\otimes t} \subseteq H^{mt}(Y^t,\mathbb{Q})$.*

(i) *The Hodge conjecture holds for Y^n for every $n < t$.*

(ii) $\det(V)$ *is a Hodge class, and* $\mathrm{Hdg}^*(Y^t)$ *is spanned by algebraic classes and* $\det(V)$*.*

(iii) $\det(V)$ *does not lie in the $\mathbb{Q}$-algebra generated by the pull-backs along the projections of the Hodge classes of the Y^j, $j < t$.*

(iv) *The Hodge conjecture holds for every power of Y if and only if* $\det(V)$ *is algebraic, and* (F3′) *holds for every power of Y if and only if* $\det(V) \in \mathrm{Ab}(Y^t)$*.*

Proof. Algebraic classes have even degree, so $H^k(Y,\mathbb{Q}) = 0$ for odd k. Let $[\Delta] = \sum_k \pi_k$ be the Künneth decomposition of the class of the diagonal, $\pi_k \in H^{2m-k}(Y,\mathbb{Q}) \otimes H^k(Y,\mathbb{Q})$. For $k \neq m$, $\pi_k = \sum_i e_i^{\vee} \otimes e_i$ for a basis (e_i) of $H^k(Y,\mathbb{Q})$ and the Poincaré dual basis of $H^{2m-k}(Y,\mathbb{Q})$, both spanned by algebraic classes, so π_k is algebraic; likewise the part π_N of π_m in $N \otimes N$, since ψ is nondegenerate on N. So $\pi_V = [\Delta] - \sum_{k \neq m} \pi_k - \pi_N$, the class of ψ^{-1} in $V \otimes V \subseteq H^{2m}(Y \times Y,\mathbb{Q})$, is algebraic.

The Hodge group of Y^n is $\mathrm{SO}(V,\psi)$, acting on $H^*(Y^n,\mathbb{Q}) = \big(H^*(Y,\mathbb{Q})\big)^{\otimes n}$ through the factors V and trivially on the rest, which is spanned by algebraic classes; so the Hodge classes are the $\mathrm{SO}(V)$-invariants. By the first fundamental theorem for the special orthogonal group [GW09, Chapter 5], the invariants of $\mathrm{SO}(V)$ in $V^{\otimes j}$ are spanned by the complete contractions with ψ^{-1} when $j<t$, and, when $j=t$, by these and the determinant tensor; in general they are spanned by products of contractions and determinant tensors on disjoint sets of t factors. A contraction of the factors a and b of Y^n is the pull-back of π_V along the projection to those two factors, a determinant on a set I of t factors is the pull-back of $\det(V)$ along the projection to Y^I, and products of these with pull-backs of algebraic classes span the Hodge classes. This gives (i), (ii) and the first half of (iv). For (iii), the orthogonal group $\mathrm{O}(V,\psi)$ acts on $H^*(Y^j,\mathbb{Q})$ in the same way, and for $j<t$ every $\mathrm{SO}(V)$-invariant is $\mathrm{O}(V)$-invariant; pull-backs along projections and cup products commute with the diagonal action of $\mathrm{O}(V)$, so every element of the algebra in (iii) is $\mathrm{O}(V)$-invariant, while a reflection acts on $\det(V)$ by -1. For the second half of (iv), $\mathrm{Ab}^*(Y^n)$ contains the algebraic classes and is stable under pull-back along projections and under cup product: $\Gamma_*\gamma\cup\Gamma'_*\gamma'$ is the image of $\gamma\otimes\gamma'$ on $B\times B'$ under the correspondence $(\Gamma\times\Gamma')$ followed by the diagonal of Y^n. ∎

For a K3 surface S with $\mathrm{End}_{\mathrm{Hdg}}(T(S))=\mathbb{Q}$ the Hodge group of $T(S)$ is $\mathrm{SO}(T(S))$ [Zar83]; for a very general one, $t=21$. So the Hodge conjecture holds for S^n with $n\le 20$, and on S^{21} it is equivalent to the algebraicity of the single class $\det(T(S))$. The argument is the classical one from invariant theory, and we make no claim of priority for it. What it adds here is the location of the first instance of (F3′) on the powers of such a variety that invariant theory does not reach: one class, in one degree, on one power.

The same invariant theory, with the motivic decompositions of the literature and the theorems of Buskin and Markman on Hodge isometries, gives the Hodge conjecture itself, and so (F3′), on the other varieties built from a K3 surface and on varieties of $\mathrm{K3}^{[n]}$ type, up to a real multiplication and one determinant class.

Proposition 2.39 (Powers, Hilbert schemes and moduli spaces of a K3 surface)**.** *Let S be a projective K3 surface, $T=T(S)_{\mathbb{Q}}$, $t=\dim T$ and $E=\mathrm{End}_{\mathrm{Hdg}}(T)$, a totally real field or a CM field* [Zar83].

(i) *If E is a CM field, the Hodge conjecture holds for every power S^k* [RM08, Theorem 5.4], [Bus19]*, hence for every Hilbert scheme $S^{[n]}$* [dCM02]*, for every smooth projective moduli space of Gieseker stable (twisted) sheaves on S, or of σ-stable objects in $D^b(S,\alpha)$ for a generic stability condition σ and a Brauer class α* [Bül20, Theorem 0.1, Remark 2.1]*, and for the crepant resolutions of the ten-dimensional moduli spaces $M_\sigma(2v_0)$ of σ-semistable objects in $D^b(S,\alpha)$, with v_0 primitive, $v_0^2=2$ and σ v_0-generic* [FFZ21, Theorem 1.3].

(ii) *If $E=\mathbb{Q}$, the Hodge conjecture holds for $S^{[n]}$ for every $n<t(t+1)/2$, and for every moduli space M as in* (i) *with $\dim M<t$. For $N=t(t+1)/2$ the following are equivalent: the Hodge conjecture for $S^{[N]}$; the algebraicity of the class $\det(T)$ on S^t of Proposition 2.38; the Hodge conjecture for every power of S; and the Hodge conjecture for every $S^{[n]}$. For a very general S, the Hodge conjecture holds for $S^{[n]}$ with $n\le 230$, and for $S^{[231]}$ if and only if $\det(T(S))$ on S^{21} is algebraic.*

(iii) *If E is totally real of degree at least two, the Hodge conjecture for $S\times S$, the Hodge conjecture for $S^{[2]}$, and the statement that every element of E acts on T through an algebraic cycle on $S\times S$ are equivalent.*

Wherever the Hodge conjecture holds, so does (F3′). *This is a consequence of* [RM08, Bus19, dCM02, Bül20, FFZ21] *and Proposition 2.38; we make no claim of priority.*

Proof. We use three facts. First, $H^*(S,\mathbb{Q}) = A \oplus T$ with $A = H^0 \oplus \mathrm{NS}(S)_{\mathbb{Q}} \oplus H^4$ spanned by algebraic classes; the Künneth components of the diagonal of S are algebraic [Mur90], and so is its part π_T in $T \otimes T$, as in the proof of Proposition 2.38. The Hodge classes of S^k are the invariants in $(A \oplus T)^{\otimes k}$ of the Hodge group of T, which acts trivially on A. Second, if $H^*(X,\mathbb{Q})$ is spanned by the images of the $H^*(Y_i,\mathbb{Q})$ under algebraic correspondences and the Hodge conjecture holds for every Y_i, it holds for X: by the semisimplicity argument of the proof of Proposition 2.35(iii) every Hodge class of X is a sum of images of Hodge classes of the Y_i. Third, the correspondences Γ_ν of the proof of Proposition 2.42(ii) give an isomorphism of Chow motives between $h(S^{[n]})$ and $\bigoplus_\nu h(S^{(\nu)})(l(\nu)-n)$ [dCM02], so each $h(S^{(\nu)})(l(\nu)-n)$ is a direct summand of $h(S^{[n]})$, and the lift $\widetilde{\Gamma}_{\nu *}$ takes the same value on a class of $S^{l(\nu)}$ and on its average over G_ν. Hence the Hodge conjecture holds for $S^{[n]}$ if and only if, for every partition ν of n, every G_ν-invariant Hodge class of $S^{l(\nu)}$ is algebraic. By [Bül20, Theorem 0.1, Remark 2.1] the Chow motive of a moduli space M as in (i) is a direct summand of a sum of Tate twists of motives $h(S^{k_i})$ with $1 \le k_i \le \dim M$, and by [FFZ21, Corollary 4.5] the Chow motive of a crepant resolution as in (i) lies in the tensor subcategory generated by $h(S)$, whose objects are direct summands of sums of Tate twists of the $h(S^k)$, the dual of $h(S)$ being $h(S)(2)$.

(i) Over $\mathbb{C}$ the Hodge group of T is a product of groups $\mathrm{GL}(W_\sigma)$ acting on $T_{\mathbb{C}} = \bigoplus_\sigma (W_\sigma \oplus W_\sigma^\vee)$ [Zar83], and by the first fundamental theorem for the general linear group [GW09, Chapter 5] its invariants in $T_{\mathbb{C}}^{\otimes b}$ are spanned by products of contractions of one factor with another. Under the isomorphism $\mathrm{End}(T) \cong T \otimes T$, $\varphi \mapsto \tau_\varphi = \sum_i t_i^\vee \otimes \varphi(t_i)$, with $(t_i^\vee)$ the dual basis of a basis (t_i) for the intersection form, the Hodge classes of $T \otimes T$ are the τ_φ with $\varphi \in E$, and these contractions lie in their complex span. The rational span of the products of the τ_φ on pairs of factors has the same complexification as the space of invariants, so the Hodge classes of S^k are spanned by such products and pull-backs of algebraic classes. For $u \in E$ with $u\bar{u} = 1$ the map $\mathrm{id}_{\mathrm{NS}} \oplus u$ is a rational Hodge isometry of $H^2(S,\mathbb{Q})$, since the adjoint of u is $\bar{u}$ [Zar83]; it is induced by an algebraic cycle [Bus19], and composing that cycle with π_T on both sides gives τ_u, which is therefore algebraic. The elements of norm one span E. Indeed, write $E = E_0(y_0)$ with $y_0^2 = -\delta$, $\delta \in E_0$, and let $y \in E$ with $\bar{y} = -y$; for $s \in \mathbb{Q}$ the element $u_s = (1+sy)(1-sy)^{-1}$ has norm one. A linear form on E vanishing on all u_s gives a rational function of s vanishing on $\mathbb{Q}$, so it vanishes on the coefficients 1, $2y$, $2y^2$ of the expansion of u_s in powers of s. As y runs over $E_0 y_0$, the y span $E_0 y_0$ and the $y^2 = -a^2\delta$, $a \in E_0$, span E_0, since $2a = (a+1)^2 - a^2 - 1$. So every τ_u, $u \in E$, is algebraic, and the Hodge conjecture holds for every S^k; this is the argument of [RM08, Lemma 5.3, Theorem 5.4], which takes the algebraicity of the isometries from Mukai's announcement; [Bus19] proves it in general. We use [RM08] only when E is a CM field: for a totally real E the invariants of the Hodge group are not generated by those of degree two, since they include determinants, as Proposition 2.38 shows for $E = \mathbb{Q}$, and the powers do not reduce to the square. The rest of (i) follows from the second and third facts.

(ii) Here the Hodge group of T is $\mathrm{SO}(T)$ [Zar83], and $t \ge 3$, since a Hodge structure of K3 type of dimension two has complex multiplication. The Hodge conjecture holds for S^k, $k < t$, by Proposition 2.38(i), which gives the assertion on M by the second fact. Let ν be a partition of n with a_j parts equal to j, so that $G_\nu = \prod_j \mathfrak{S}_{a_j}$ permutes the factors of $S^{l(\nu)}$ within blocks of sizes a_j, one block for each distinct part size. By the proof of Proposition 2.38, the Hodge classes of $S^{l(\nu)}$ are spanned by algebraic classes and

by products $\beta = \det_I \cup \gamma$, where $\det_I$ is the pull-back of $\det(T)$ along the projection to a set I of t factors and γ is pulled back from the other factors. If two factors of I lie in one block, the transposition in G_ν exchanging them fixes γ and changes the sign of the alternating class $\det_I$, all classes involved being of even degree; so it takes β to $-\beta$, and the average of β over G_ν is zero. If ν has fewer than t distinct part sizes, every set of t factors meets some block twice, so a G_ν-invariant Hodge class, being its own average, is the average of an algebraic class, and is algebraic. A partition with t distinct part sizes has $n \geq 1+2+\cdots+t = t(t+1)/2$, so by the third fact the Hodge conjecture holds for $S^{[n]}$ when $n < t(t+1)/2$ (Figure 4). For $n = N$ the partition $(1, 2, \ldots, t)$ has $l(\nu) = t$ and $G_\nu = 1$, so the Hodge class $\det(T)$ of S^t is algebraic if the Hodge conjecture holds for $S^{[N]}$. If $\det(T)$ is algebraic, the Hodge conjecture holds for every power of S by Proposition 2.38(iv), and for every $S^{[n]}$ by the second fact; and the conjecture for S^t contains the algebraicity of $\det(T)$. A very general S has $E = \mathbb{Q}$ and $t = 21$, and $N = 231$.

(iii) The Hodge structure T has no nonzero Hodge class, so the Hodge classes of $A \otimes T$ and $T \otimes A$ vanish, and those of $S \times S$ are the algebraic classes and the τ_φ, $\varphi \in E$. A cycle inducing φ on T, composed with π_T on both sides, gives τ_φ, and τ_φ induces φ on T; this proves the equivalence for $S \times S$. By [dCM02], $h(S^{[2]}) = h(S^{(2)}) \oplus h(S)(-1)$, and the Hodge conjecture holds for S. The exchange of the two factors takes τ_φ to τ_{φ^*}, and the adjoint involution of a totally real E is trivial [Zar83], so every τ_φ is invariant under $\mathfrak{S}_2$ and the Hodge conjecture for $S^{[2]}$ asks for the same classes as that for $S \times S$. ■

Proposition 2.40 (Varieties of $\mathrm{K3}^{[n]}$ type)**.** *Let X be a smooth projective hyperkähler variety deformation equivalent to the Hilbert scheme of $n \geq 2$ points on a K3 surface, q the Beauville–Bogomolov form on $H^2 = H^2(X, \mathbb{Q})$, $\mathrm{NS} = \mathrm{NS}(X)_\mathbb{Q}$, $T = \mathrm{NS}^\perp$ and $E = \mathrm{End}_{\mathrm{Hdg}}(T)$, a totally real field or a CM field whose adjoint involution for q is complex conjugation* [Zar83]*. Let $\mathrm{SH}(X) \subseteq H^*(X, \mathbb{Q})$ be the subalgebra generated by H^2, and for $\varphi \in E$ with $\varphi^* = \varphi$ let $c_\varphi = \sum_i t_i^\vee \cup \varphi(t_i) \in H^4(X, \mathbb{Q})$, with (t_i) a basis of T and $(t_i^\vee)$ its dual basis for q.*

(i) *The Hodge classes of $\mathrm{SH}(X)$ are spanned by products of divisor classes and classes c_φ.*

(ii) *If E is $\mathbb{Q}$ or a CM field, every Hodge class of $\mathrm{SH}(X)$ is algebraic.*

(iii) *Let $n = 2$. Then $\mathrm{SH}(X) = H^*(X, \mathbb{Q})$, with $b_4(X) = 276 = \dim \mathrm{Sym}^2 H^2$, and $c_2(X) = \frac{6}{5} q^\vee$, where $q^\vee \in \mathrm{Sym}^2 H^2$ is dual to q. So the Hodge conjecture holds for X when E is $\mathbb{Q}$ or a CM field. When E is totally real of degree at least two, it holds for X if and only if every element of E, acting on T and by zero on NS, is induced by an algebraic cycle on $X \times X$.*

This is a consequence of [Mar24, CM13]*; we make no claim of priority.*

Proof. (i) Multiplication $\mathrm{Sym}^k H^2 \to \mathrm{SH}^{2k}(X)$ is a surjective morphism of polarisable Hodge structures, so by semisimplicity every Hodge class of $\mathrm{SH}^{2k}(X)$ is the image of one of $\mathrm{Sym}^k H^2 = \bigoplus_{a+b=k} \mathrm{Sym}^a \mathrm{NS} \otimes \mathrm{Sym}^b T$, whose Hodge classes are $\mathrm{Sym}^a \mathrm{NS} \otimes (\mathrm{Sym}^b T)^{\mathrm{Hdg}}$. The isomorphism $\mathrm{End}(T) \cong T \otimes T$ of the proof of Proposition 2.39, now for q, carries the adjoint to the exchange of the factors, so $(\mathrm{Sym}^2 T)^{\mathrm{Hdg}}$ consists of the tensors of the self-adjoint $\varphi \in E$, whose images in H^4 are the c_φ. Over $\mathbb{C}$ the Hodge group of T is a product of groups $\mathrm{SO}(T_\sigma)$ on $T_\mathbb{C} = \bigoplus_\sigma T_\sigma$, with $\dim T_\sigma = \dim_E T \geq 3$, when E is totally real, and of groups $\mathrm{GL}(W_\sigma)$ on $T_\mathbb{C} = \bigoplus_\sigma (W_\sigma \oplus W_\sigma^\vee)$ when E is a CM field [Zar83]. By the first fundamental theorems [GW09, Chapter 5], the invariant polynomials of one vector of T_σ, or of one vector of W_σ and one of $W_\sigma^\vee$, are generated by the quadratic form, or by the

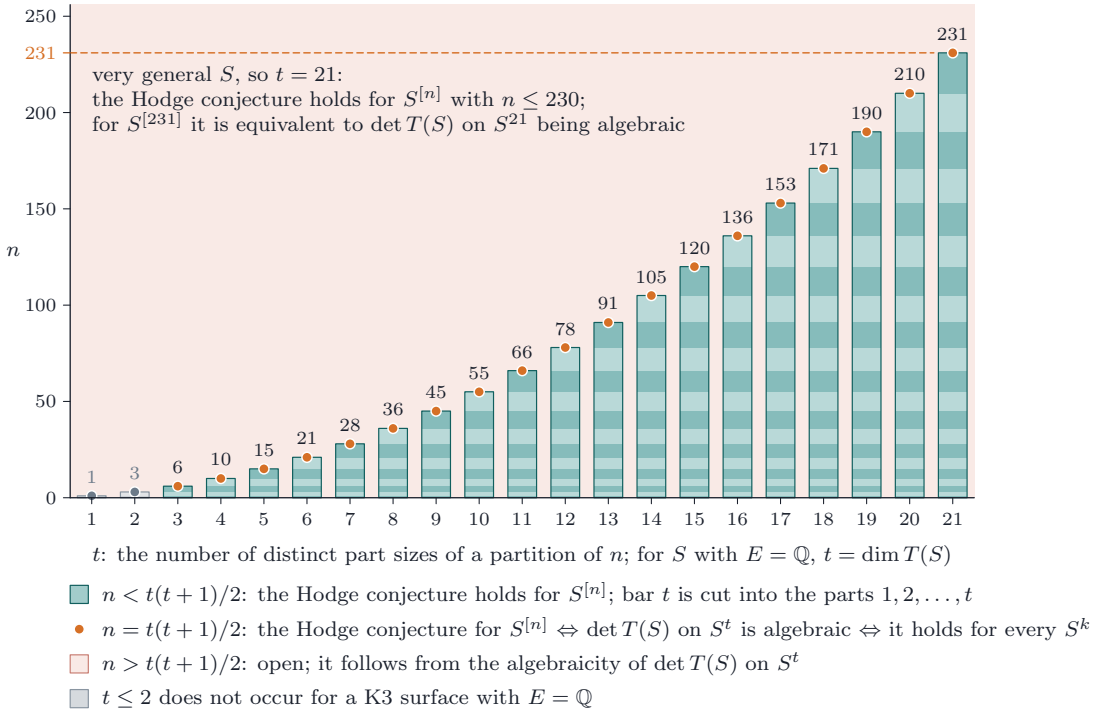


FIGURE 4. The threshold of Proposition 2.39(ii). For $t = 1, \dots, 21$ the bar rises to $t(t+1)/2$, the least n that has a partition with t distinct part sizes, and is cut into the parts $1, 2, \dots, t$ of the partition that attains it. For a K3 surface S with $E = \mathrm{End}_{\mathrm{Hdg}}(T(S)) = \mathbb{Q}$ and $t = \dim T(S)$, which is at least 3, the Hodge conjecture holds for $S^{[n]}$ with n below the bar; at its top it is equivalent to the algebraicity of the class $\det T(S)$ on S^t of Proposition 2.38, and to the Hodge conjecture for every power of S; above it, it follows from that algebraicity. A very general S has $t = 21$: the conjecture holds for $S^{[n]}$ with $n \le 230$, and for $S^{[231]}$ it is equivalent to the algebraicity of $\det T(S)$ on S^{21}, one of the frontier cases of (F3′) (Remark 2.68).

pairing: a determinant needs $\dim T_\sigma$ distinct vectors. So $(\mathrm{Sym}^b T)^{\mathrm{Hdg}} \otimes \mathbb{C}$ is spanned by products of elements of $(\mathrm{Sym}^2 T)^{\mathrm{Hdg}} \otimes \mathbb{C}$, and the rational span of the products of classes c_φ, which has the same complexification, is $(\mathrm{Sym}^b T)^{\mathrm{Hdg}}$.

(ii) Divisor classes are algebraic, so by (i) it suffices that every c_φ is. By [CM13, Theorem 1.1] the Lefschetz standard conjecture holds for X, so the Künneth projectors and the inverse of the isomorphism $L^{2n-2}\colon H^2 \to H^{4n-2}$, L the cup product with an ample class h, are induced by algebraic cycles [Kle68, Kle94]. Let $u \in E$ with $u\bar{u} = 1$, and $u = 1$ if $E = \mathbb{Q}$. The map $f_u = \mathrm{id}_{\mathrm{NS}} \oplus u$ is a rational Hodge isometry of H^2, so by [Mar24, Theorem 1.1] there is an algebraic class Z_u on $X \times X$ (the diagonal if $u = 1$) inducing on H^2 a nonzero rational multiple of f_u: the class of the correspondence $\tilde{f}$ of that theorem if it preserves degrees, and the class of $\tilde{f} \circ (c_2(X)^{n-1} \cup \cdot\,)$ if it reverses them. The class Z_u may act in other degrees too; its Künneth component in $H^{4n-2}(X) \otimes H^2(X)$ is algebraic and induces a nonzero multiple of f_u on H^2 and zero elsewhere. Applying $(L^{2n-2})^{-1} \otimes \mathrm{id}$ gives, up to a nonzero rational factor, the algebraic class $\sum_i b_i \otimes f_u(a_i)$, for a basis (a_i) of H^2 and

the dual basis (b_i) for $B(x,y)=\int_X xyh^{2n-2}$. Polarising the relation $\int_X \alpha^{2n}=c_X q(\alpha)^n$ of Fujiki and Beauville [Bea83, Fuj87] gives

$$B(x,y)=\frac{c_X}{2n-1}\,q(h)^{n-2}\big(q(h)q(x,y)+2(n-1)q(x,h)q(y,h)\big),$$

so NS and T are orthogonal for B, and $B=\kappa q$ on T with $\kappa=c_X q(h)^{n-1}/(2n-1)\neq 0$. With (a_i) adapted to $\mathrm{NS}\oplus T$, the pull-back along the diagonal is $\sum_i b_i\cup f_u(a_i)=\nu+\kappa^{-1}c_{(u+\bar u)/2}$ with $\nu\in\mathrm{Sym}^2\mathrm{NS}$ algebraic, because cup product on H^2 is commutative and the exchange of factors turns u into $\bar u$. So $c_{(u+\bar u)/2}$ is algebraic. When E is a CM field its elements of norm one span it (proof of Proposition 2.39(i)), so the $(u+\bar u)/2$ span E_0, the self-adjoint part of E; when $E=\mathbb{Q}$, $u=1$ suffices. So every c_φ is algebraic.

(iii) The variety X has the Betti numbers of $S^{[2]}$, which by [dCM02] are those of $\mathrm{Sym}^2H^*(S)\oplus H^*(S)(-1)$: $H^{\mathrm{odd}}(X)=0$ and $b_4(X)=253+1+22=276=\dim\mathrm{Sym}^2H^2$. Here $c_X=3$, and the relation of Fujiki and Beauville gives $\int_X abcd=q(a,b)q(c,d)+q(a,c)q(b,d)+q(a,d)q(b,c)$ for $a,b,c,d\in H^2$. This form on Sym^2H^2 is nondegenerate: in a q-orthogonal basis (e_i) with $q(e_i,e_i)=d_i$, the e_ie_j with $i<j$ are orthogonal to each other and to the squares, with $\int_X(e_ie_j)^2=d_id_j$, and on the squares the form has the matrix $D(J+2I)D$, with $D=\mathrm{diag}(d_i)$ and J the matrix of ones, of determinant $2^{22}\cdot 25\prod_i d_i^2$. So $\mathrm{Sym}^2H^2\to H^4$ is injective, hence bijective, and with $H^6=h^2\cup H^2$ by the hard Lefschetz theorem, $\mathrm{SH}(X)=H^*(X,\mathbb{Q})$. By [Fuj87], $\int_X c_2(X)xy=C\,q(x,y)$ for a constant C, and the Riemann–Roch polynomial $\chi(L)=\binom{q(L)/2+3}{2}$ [EGL01] gives $C=30$; the Fujiki relation gives $\int_X q^\vee xy=25\,q(x,y)$, so $c_2(X)=\frac{6}{5}q^\vee$ by nondegeneracy. For $E=\mathbb{Q}$ this makes $c_1=q^\vee-q^\vee_{\mathrm{NS}}$ algebraic without [CM13], with $q^\vee_{\mathrm{NS}}\in\mathrm{Sym}^2\mathrm{NS}$ the class dual to $q|_{\mathrm{NS}}$. With (i) and (ii) this proves the Hodge conjecture for X when E is $\mathbb{Q}$ or a CM field. Let E be totally real and $\varphi\in E$. If $\varphi\oplus 0$ is induced by an algebraic cycle, the computation of (ii) with $\varphi\oplus 0$ in place of f_u makes c_φ algebraic. Conversely, if c_φ is algebraic, so is the correspondence $x\mapsto(L^2)^{-1}(c_\varphi\cup x)$. The Fujiki relation gives $\int_X c_\varphi xy=\mathrm{Tr}(\varphi)q(x,y)+2q(\varphi x,y)$ for $x\in T$ and $y\in H^2$, and $B(z,y)=q(h)q(z,y)$ for $z\in T$, so that correspondence acts on T as $(\mathrm{Tr}(\varphi)+2\varphi)/q(h)$. The projector onto T, the Künneth projector π_2 minus $\sum_j n_j^\vee\otimes n_j$ over a basis (n_j) of NS and the Hodge classes $n_j^\vee\in H^6$ dual to it, is algebraic by Proposition 2.35(iv); composing with it, $\varphi\oplus 0$ is induced by an algebraic cycle. By (i) the Hodge conjecture for X is the algebraicity of the c_φ, which proves the equivalence. ∎

For varieties of generalised Kummer type every Hodge class in the subalgebra generated by H^2 is algebraic, whatever the field, and the Hodge conjecture holds in dimension four [FV25]. The preprint [O'G26] announces, with [Mar24], that rational Hodge isometries between varieties of the types $\mathrm{K3}^{[a_1^2+1]}$ and $\mathrm{K3}^{[a_2^2+1]}$ are induced by algebraic cycles; it is not used here.

Corollary 2.41 (Cubic fourfolds and their Fano varieties of lines)**.** *Let $Y\subset\mathbb{P}^5$ be a smooth cubic fourfold, $T(Y)\subset H^4(Y,\mathbb{Q})$ the orthogonal complement of the Hodge classes, $E(Y)=\mathrm{End}_{\mathrm{Hdg}}(T(Y))$, and $F=F(Y)$ the Fano variety of lines on Y. The Hodge conjecture for F, the Hodge conjecture for $Y\times Y$, and the statement that every element of $E(Y)$ acts on $T(Y)$ through an algebraic cycle on $Y\times Y$ are equivalent. They hold when $E(Y)$ is $\mathbb{Q}$ or a CM field, for instance when Y is very general.*

Proof. The Hodge conjecture holds for Y [Zuc77], and also by Proposition 2.43, since Y is covered by lines. So $H^*(Y,\mathbb{Q})$ is $T(Y)$ plus a space spanned by algebraic classes, on which

the intersection form is nondegenerate by the Hodge–Riemann relations, and as in the proof of Proposition 2.38 the part π_T of the diagonal in $T(Y)\otimes T(Y)$ is algebraic. The Hodge structure $T(Y)$ is of K3 type up to a Tate twist and has no nonzero Hodge class, so $E(Y)$ is a field [Zar83], and as in Proposition 2.39(iii) the Hodge conjecture for $Y\times Y$ is the third statement. By [BD85], F is a hyperkähler fourfold of $\mathrm{K3}^{[2]}$ type, and the universal line $P\subset F\times Y$ induces an isomorphism of Hodge structures $\alpha = P_*\colon H^4(Y,\mathbb{Q})\to H^2(F,\mathbb{Q})$ up to a Tate twist. Both $T(Y)$ and $T(F)$ are the sums of the isotypic components that are not of Tate type, so $\alpha(T(Y)) = T(F)$ and $E(F) = \alpha E(Y)\alpha^{-1}$. Let g be the Plücker class on F and $e = P_*^t\circ(g^2\cup\,\cdot\,)\circ\alpha$, an algebraic correspondence that maps $T(Y)$ to itself, so that $e|_{T(Y)}\in E(Y)$. For $0\neq\sigma\in H^{3,1}(Y)$, $\int_Y e(\sigma)\bar\sigma = \int_F g^2\alpha(\sigma)\alpha(\bar\sigma) = q(g)\,q(\alpha(\sigma),\alpha(\bar\sigma))\neq 0$ by the Fujiki relation, since $q(g)>0$ and $\alpha(\sigma)$ spans $H^{2,0}(F)$. So $e|_{T(Y)}$ is invertible, its inverse is a polynomial in it and is algebraic, and the inverse of α on $T(F)$, which is $e^{-1}\circ P_*^t\circ(g^2\cup\,\cdot\,)$, is induced by an algebraic cycle. Hence $\varphi\in E(Y)$ acts through an algebraic cycle on $Y\times Y$ if and only if $\alpha\varphi\alpha^{-1}$ acts through one on $F\times F$, and Proposition 2.40(iii) proves the equivalences and the case where $E(Y)=E(F)$ is $\mathbb{Q}$ or a CM field. A very general Y has $E(Y)=\mathbb{Q}$. ∎

The class $\mathcal{A}$ of Proposition 2.35(iii) is larger than the closure properties recorded there show. Four further operations preserve it, the last of them domination by correspondences, which brings in the varieties whose motives are known to be built from those of abelian varieties, and two classical families lie in it.

Proposition 2.42 (More varieties of $\mathcal{A}$)**.** *Let $\mathcal{A}$ be as in Proposition 2.35(iii).*

(i) *If $Y\in\mathcal{A}$ and $Z\subset Y$ is a smooth closed subvariety with $Z\in\mathcal{A}$, then the blow-up $\mathrm{Bl}_Z Y$ lies in $\mathcal{A}$. If $Y\in\mathcal{A}$ and V is a vector bundle on Y, then $\mathbb{P}(V)\in\mathcal{A}$.*

(ii) *If $S\in\mathcal{A}$ is a surface, then the Hilbert scheme $S^{[n]}$ lies in $\mathcal{A}$ for every $n\geq 1$.*

(iii) *Every Fermat hypersurface $X_m^r = \{x_0^m+\cdots+x_{r+1}^m=0\}\subset\mathbb{P}^{r+1}$, $r\geq 0$, $m\geq 1$, lies in $\mathcal{A}$.*

(iv) *A K3 surface S lies in $\mathcal{A}$ as soon as its Kuga–Satake correspondence is induced by an algebraic cycle, that is, as soon as some algebraic cycle Γ on $S\times B$, with B an abelian variety, induces a nonzero map $T(S)_{\mathbb{Q}}\to H^2(B,\mathbb{Q})$.*

(v) *If $H^*(X,\mathbb{Q})$ is the sum of the images of the $H^*(Y_i,\mathbb{Q})$, with $Y_i\in\mathcal{A}$, under algebraic correspondences, then $X\in\mathcal{A}$; in particular $X\in\mathcal{A}$ when its Chow motive is a direct summand of a sum of Tate twists of motives of members of $\mathcal{A}$, for instance when it is of abelian type. So $\mathcal{A}$ contains the $2n$-dimensional generalised Kummer varieties $K_n(A)$ of abelian surfaces A* [Xu15, Theorem 2.7, Corollary 2.8]*, the smooth projective moduli spaces of Gieseker stable (twisted) sheaves, or of σ-stable objects for a generic stability condition σ, on abelian surfaces and on K3 surfaces in $\mathcal{A}$* [Bül20, Theorem 0.1, Remark 2.1]*; O'Grady's six-dimensional varieties $\widetilde{K}_v(A,H)$ obtained from moduli spaces of sheaves on an abelian surface A, with $v=2v_0$, v_0 primitive, effective and of square 2, and H a v-generic polarisation* [Flo23, Theorem 1.1]*; the crepant resolutions of the ten-dimensional moduli spaces $M_\sigma(2v_0)$ of σ-semistable objects, with v_0 primitive, $v_0^2=2$ and σ v_0-generic, on abelian surfaces and on K3 surfaces in $\mathcal{A}$* [FFZ21, Theorem 1.3, Corollary 4.5]*; and the cubic fourfolds whose lattice of algebraic 2-cycles has rank greater than 19, with their Fano varieties of lines, the eightfolds of Lehn, Lehn, Sorger and van Straten of those containing no plane, and the tenfolds of Laza, Saccà and Voisin of infinitely*

many Pfaffian cubic fourfolds [ABLP26, Theorem 3.5 and its proof, Proposition 3.9, Theorem 4.2].

Consequently (F3′) *holds on every smooth projective variety obtained from abelian varieties, curves, Fermat hypersurfaces and K3 surfaces as in* (iv) *by products, surjective images, blow-ups along smooth centres in $\mathcal{A}$, projective bundles and Hilbert schemes of points on surfaces, and on every variety dominated by these as in* (v).

Proof. (i) Let c be the codimension of Z, $E=\mathbb{P}(N_{Z/Y})$ the exceptional divisor, $j\colon E\to\widetilde{Y}=\mathrm{Bl}_Z Y$ its inclusion, $\rho\colon E\to Z$ and $\pi\colon\widetilde{Y}\to Y$ the projections and $\xi=c_1(\mathcal{O}_E(1))$. By the blow-up formula [Voi02, §7.3.3],

$$H^*(\widetilde{Y},\mathbb{Q})=\pi^*H^*(Y,\mathbb{Q})\oplus\bigoplus_{i=0}^{c-2}j_*(\xi^i\cup\rho^*H^*(Z,\mathbb{Q})).$$

The first summand is the image of the transpose of the graph of π, and the i-th of the others is the image of the algebraic cycle $(\rho\times j)_*\xi^i$ on $Z\times\widetilde{Y}$. Composing with the correspondences that exhibit Y and Z as members of $\mathcal{A}$, which is allowed [Ful98, Chapter 16], shows that $H^*(\widetilde{Y},\mathbb{Q})$ is spanned by classes Γ_*y with B abelian. For $\mathbb{P}(V)$ the projective bundle formula $H^*(\mathbb{P}(V),\mathbb{Q})=\bigoplus_i\xi^i\cup\pi^*H^*(Y,\mathbb{Q})$ [Voi02, §7.3.3] does the same with the algebraic cycles $\Gamma_\pi^t\cdot\mathrm{pr}^*\xi^i$.

(ii) By de Cataldo and Migliorini [dCM02], for every partition ν of n with $l(\nu)$ parts the incidence correspondence $\Gamma_\nu\subset S^{(\nu)}\times S^{[n]}$ between the product $S^{(\nu)}$ of symmetric products attached to ν and the Hilbert scheme induces, after a Tate twist by $n-l(\nu)$, an isomorphism $\bigoplus_\nu H^*(S^{(\nu)},\mathbb{Q})\to H^*(S^{[n]},\mathbb{Q})$. Let $q_\nu\colon S^{l(\nu)}\to S^{(\nu)}$ be the quotient by the finite group G_ν. With rational coefficients the cycles and the cohomology of $S^{(\nu)}$ are the G_ν-invariants of those of $S^{l(\nu)}$ [Ful98, Example 1.7.6], so Γ_ν pulls back to a cycle $\widetilde{\Gamma}_\nu$ with rational coefficients on the smooth variety $S^{l(\nu)}\times S^{[n]}$, whose action on $H^*(S^{l(\nu)},\mathbb{Q})$ is $\Gamma_{\nu*}\circ q_{\nu*}$ up to a nonzero factor. As $H^*(S^{(\nu)},\mathbb{Q})=q_{\nu*}H^*(S^{l(\nu)},\mathbb{Q})$, the ν-th summand is the image of $H^*(S^{l(\nu)},\mathbb{Q})$ under $\widetilde{\Gamma}_\nu$. Products of members of $\mathcal{A}$ lie in $\mathcal{A}$, so $S^{[n]}\in\mathcal{A}$.

(iii) By induction on r, following the inductive structure of Shioda and Katsura [SK79]. For $r=0$, X_m^0 is a finite set of points, and X_m^1 is a curve. Let $r\geq 2$ and $r=a+b$ with $a,b\geq 1$; fix ϵ with $\epsilon^m=-1$ and put

$$\varphi\big((x),(y)\big)=\big(x_0y_{b+1}:\cdots:x_ay_{b+1}:\epsilon x_{a+1}y_0:\cdots:\epsilon x_{a+1}y_b\big),$$

a rational map $X_m^a\times X_m^b\dashrightarrow\mathbb{P}^{r+1}$; since $\sum_{i\leq a}x_i^m=-x_{a+1}^m$ and $\sum_{k\leq b}y_k^m=-y_{b+1}^m$, its image lies in X_m^r. It is defined by sections of $\mathcal{O}(1,1)$ whose common zero scheme is $\{x_{a+1}=0\}\times\{y_{b+1}=0\}$, the product $Z\cong X_m^{a-1}\times X_m^{b-1}$ of two smooth hyperplane sections, transversal because the Fermat hypersurfaces are smooth. Blowing up that reduced scheme resolves the map, so φ extends to a morphism $\widetilde{\varphi}\colon\mathrm{Bl}_Z(X_m^a\times X_m^b)\to X_m^r$. It is dominant: at a point z of X_m^r with $\sum_{i\leq a}z_i^m\neq 0$ the preimages with $x_{a+1}=y_{b+1}=1$ are the points $x_i=\lambda z_i$, $y_k=\lambda z_{a+1+k}/\epsilon$ with $\lambda^m\sum_{i\leq a}z_i^m=-1$, and both equations reduce to this one because $\sum_{\mathrm{all}}z_i^m=0$. So $\widetilde{\varphi}$ is surjective, being proper. By induction X_m^a, X_m^b, X_m^{a-1} and X_m^{b-1} lie in $\mathcal{A}$, hence so do their products, the blow-up by (i) and X_m^r as a surjective image; one may take $a=r-1$, $b=1$.

(iv) The transcendental part $T=T(S)_\mathbb{Q}$ is an irreducible Hodge structure, and $H^*(S,\mathbb{Q})=T\oplus T^\perp$ with $T^\perp$ spanned by algebraic classes. Let h be an ample class on B, $d=\dim B$, and let $\phi=\Gamma_*^t\circ(\cdot\cup h^{d-2})\circ\Gamma_*$, an algebraic correspondence from S to itself and a morphism of Hodge structures. It preserves T, because a morphism from the irreducible structure T to the algebraic part, which is of Tate type, vanishes; so $\phi|_T$ lies in

the field $\mathrm{End}_{\mathrm{Hdg}}(T)$. For $0 \neq \sigma \in H^{2,0}(S)$, $\Gamma_*\sigma \neq 0$ because T is irreducible and $\Gamma_*|_T \neq 0$; it is a class of type $(2,0)$ on B, hence primitive, and by the Hodge–Riemann relations $\int_S \phi(\sigma)\,\bar{\sigma} = \int_B \Gamma_*\sigma\,\overline{\Gamma_*\sigma}\,h^{d-2} \neq 0$. So $\phi|_T \neq 0$, it is invertible, and $T \subseteq \Gamma_*^t H^*(B,\mathbb{Q})$. With $T^\perp$ algebraic, $S \in \mathcal{A}$.

(v) If $H^*(X,\mathbb{Q}) = \sum_i \Psi_{i*}H^*(Y_i,\mathbb{Q})$ and each $H^*(Y_i,\mathbb{Q})$ is spanned by classes $\Gamma_{ij*}y$ with B_{ij} abelian, then $H^*(X,\mathbb{Q})$ is spanned by the classes $(\Psi_i \circ \Gamma_{ij})_*y$, and composites of algebraic correspondences are algebraic [Ful98, Chapter 16]. If $h(X)$ is a direct summand of $\bigoplus_i h(Y_i)(n_i)$, through α and β with $\beta\circ\alpha = \mathrm{id}$, then β_* is surjective on cohomology, and its components are the Ψ_i; a motive of abelian type is such a summand with abelian varieties as the Y_i. The objects of the tensor subcategory generated by $h(Y)$ are direct summands of sums of Tate twists of the motives $h(Y^k)$, since the dual of $h(Y)$ is a Tate twist of $h(Y)$, and $Y^k \in \mathcal{A}$ when $Y \in \mathcal{A}$. The Chow motive of $K_n(A)$ lies in the tensor subcategory generated by $h(A)$ [Xu15, Theorem 2.7, Corollary 2.8], see also [FTV19, Theorem 1.5]; that of a moduli space as in (v) is a direct summand of a sum of Tate twists of motives of powers of the surface [Bül20, Theorem 0.1]; those of the varieties $\widetilde{K}_v(A,H)$ and of the ten-dimensional crepant resolutions lie in the tensor subcategory generated by the motive of the surface [Flo23, Theorem 1.1], [FFZ21, Corollary 4.5]; and the cubic fourfolds and hyperkähler varieties of [ABLP26] listed in (v) have Chow motives of abelian type [ABLP26, Theorem 3.5 and its proof, Proposition 3.9, Theorem 4.2].

The last assertion is Proposition 2.35(iii). ∎

The known cases of (iv) are recalled in Remark 3.5; for the K3 surfaces among them whose powers are covered by [Var22] the Hodge conjecture itself holds on every power, which is stronger. What (iii) adds is a family on which (F3′) holds in every degree and dimension, while the Hodge conjecture for Fermat hypersurfaces is proved in a range of degrees and dimensions [Shi79]. Three recent preprints, if correct, extend that range to every Fermat variety of degree less than 65 other than 44, 51 and 52 [MMMV26], to the Fermat fourfolds of every odd degree at most 199, by a computer-assisted proof [Jum26], and to the Fermat fourfold of degree 33 [FK26]; (iii) is new for the degrees and dimensions outside these ranges. By (iii) and Proposition 2.35(i) the conjecture holds for X_m^r as soon as it holds for the powers of the Jacobian of the Fermat curve of degree m. The inductive structure is Shioda and Katsura's, and the consequence for (F3′) is immediate; we claim no priority for it.

Item (v) collects consequences of published motivic results, and we claim no priority for them either. For a generalised Kummer variety $K_n(A)$ it is needed in general. The restriction from $A^{[n+1]}$, which lies in $\mathcal{A}$ by (ii), reaches exactly the $A[n+1]$-invariant part of $H^*(K_n(A),\mathbb{Q})$, since $A \times K_n(A) \to A^{[n+1]}$ is an étale Galois cover with group $A[n+1]$. When $n+1$ is prime the rest is spanned by the classes of the punctual Hilbert schemes at the points of $A[n+1]$, which are algebraic, and (ii) suffices; when $n+1$ is composite the rest contains transcendental classes, and [Xu15, FTV19] are needed. These results concern the varieties $K_n(A)$ themselves, not every variety of generalised Kummer type; for products of varieties $K_n(A)$ the Hodge conjecture itself is known [Xu15, Theorem 3.3], and for the varieties $\widetilde{K}_v(A,H)$ it holds on all self-products [Flo23, Corollary 1.2]. If the preprint [Ped26] is correct, (v) also puts in $\mathcal{A}$ the tenfolds of Laza, Saccà and Voisin of every cubic fourfold in $\mathcal{A}$, whose motives it makes direct summands of Tate twists of the motive of the fifth power of the cubic fourfold. In low dimension the algebraicity results of Bloch–Srinivas and Conte–Murre give more.

Proposition 2.43 (Varieties whose zero-cycles are small)**.** *Let X be a smooth complex projective variety of dimension n whose group $\mathrm{CH}_0(X)_{\mathbb{Q}}$ is supported on a closed subvariety of dimension at most three. If $n \leq 5$, the Hodge conjecture holds for X, and so does* (F3′)*. This applies to every fourfold whose group of zero-cycles is supported on a proper closed subset, in particular every uniruled fourfold, and to every fivefold whose maximal rationally connected quotient has dimension at most three, in particular every rationally connected fivefold.*

Proof. By Bloch and Srinivas [BS83] every Hodge class of degree 4 on X is algebraic. By Proposition 2.35(iv) the Hodge classes of degree 0, 2, $2n-2$ and $2n$ are algebraic, which settles $n \leq 4$. For $n = 5$ the remaining degree is $6 = 2n-4$, and cup product with h, for h ample, is an isomorphism $H^4(X,\mathbb{Q}) \to H^6(X,\mathbb{Q})$ of Hodge structures up to a Tate twist by hard Lefschetz, so every Hodge class of degree 6 is h times one of degree 4. A uniruled fourfold has CH_0 supported on any ample divisor D, because a rational curve through a point meets D and any two points of $\mathbb{P}^1$ are rationally equivalent; this case is the theorem of Conte and Murre [CM78]. A proper closed subset of a fourfold has dimension at most three.

Let $\phi\colon X \dashrightarrow R$ be the maximal rationally connected fibration of Campana and of Kollár, Miyaoka and Mori [Cam92, KMM92]: R, the maximal rationally connected quotient, is smooth and projective and unique up to birational equivalence, and ϕ is dominant with connected fibres, the general one rationally connected. Resolving its indeterminacy [Hir64] gives a smooth projective variety $\widetilde{X}$, a birational morphism $\pi\colon \widetilde{X} \to X$, an isomorphism over the domain of definition of ϕ, and a surjective morphism $\widetilde{\phi}\colon \widetilde{X} \to R$ whose general fibre is smooth, by generic smoothness, and birational to a general fibre of ϕ, hence rationally connected. Suppose $\dim R \leq 3$, and let $W \subset \widetilde{X}$ be a subvariety that $\widetilde{\phi}$ maps generically finitely onto R, of degree e. Any two points of a general fibre F are joined by a rational curve, so every $x \in F$ satisfies $e[x] = [W \cap F]$ in $\mathrm{CH}_0(\widetilde{X})_{\mathbb{Q}}$. The general fibres cover a dense open subset of $\widetilde{X}$, and every zero-cycle is rationally equivalent to one supported on a given dense open subset, so $\mathrm{CH}_0(\widetilde{X})_{\mathbb{Q}}$ is supported on W and $\mathrm{CH}_0(X)_{\mathbb{Q}} = \pi_*\mathrm{CH}_0(\widetilde{X})_{\mathbb{Q}}$ on $\pi(W)$, of dimension $\dim R \leq 3$. A rationally connected fivefold is the case where R is a point. ∎

In degree four the hypothesis on zero-cycles can be relaxed to one on the fibres of a map.

Proposition 2.44 (Degree four along rationally connected fibrations)**.** *Let $f\colon X \to Y$ be a surjective morphism of smooth complex projective varieties whose general fibre is rationally connected. Then $\mathrm{Hdg}^2(X) = \mathrm{Ab}^2(X)$ if and only if $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, and every Hodge class of degree four on X is algebraic if and only if every one on Y is. Both properties are therefore birational invariants of smooth projective varieties, and they hold for X exactly when they hold for a smooth projective model of its maximal rationally connected quotient. Consequently* (F3′) *holds for every fourfold and every fivefold whose maximal rationally connected quotient is birational to a variety R with $\mathrm{Hdg}^2(R) = \mathrm{Ab}^2(R)$, for instance a member of $\mathcal{A}$. With the Hodge conjecture for abelian fourfolds* [Mar25a, Mar25b]*, the Hodge conjecture holds for every fourfold birational to an abelian fourfold, and for every fivefold admitting a dominant rational map with rationally connected general fibre onto an abelian fourfold.*

Proof. Let $n = \dim X$ and $m = \dim Y$. Let W be a component, dominating Y, of the intersection of X with a general linear space of codimension $n-m$, so that $W \to Y$ is generically finite, of degree $e \geq 1$; let $i_W\colon \widetilde{W} \to X$ be a resolution of W [Hir64] and

$g = f \circ i_W$. The correspondence $\Phi = \Gamma_f^t \circ \Gamma_g \circ \Gamma_{i_W}^t$ on $X \times X$ acts by $\Phi_* = f^* g_* i_W^*$ [Ful98, Chapter 16]. Let η be the generic point of X and K its field of functions. The restriction of Φ^t to $\{\eta\} \times X$ is the zero-cycle $W_{f(\eta)}$, of degree e, that W cuts on the fibre of X_K over the point $f(\eta)$, and that of $[\Delta_X]$ is the point η of the same fibre. Any two points of a general fibre of f are rationally equivalent, and the argument of Bloch and Srinivas [BS83] (a countable algebraically closed field of definition, an embedding of an algebraic closure of K into $\mathbb{C}$, specialisation and a norm) gives $e[\eta] = [W_{f(\eta)}]$ in $\mathrm{CH}_0(X_K)_{\mathbb{Q}}$. As $\mathrm{CH}_0(X_K)$ is the direct limit of the groups $\mathrm{CH}^n(U \times X)$ over the dense open subsets $U \subseteq X$, the cycle $e[\Delta_X] - \Phi^t$ vanishes on some $U \times X$, and by the localisation sequence [Ful98, Chapter 1] it is supported on $D \times X$ for a divisor $D \supseteq X \smallsetminus U$. The diagonal is symmetric, so $e[\Delta_X] - \Phi$ is supported on $X \times D$. Let $i\colon D' \to X$ be a resolution of D. Each component of a cycle supported on $X \times D$ is, up to a nonzero factor, the image of a subvariety of $X \times D'$ mapping generically finitely onto it, so $e[\Delta_X] - \Phi = (\mathrm{id} \times i)_* \Psi$ for a cycle Ψ with rational coefficients on $X \times D'$, and the projection formula gives, for every $a \in H^k(X, \mathbb{Q})$,

$$e\, a = f^* g_* i_W^* a + i_* \Psi_* a, \qquad \Psi_* a \in H^{k-2}(D', \mathbb{Q}).$$

Let $a \in \mathrm{Hdg}^2(X)$. Then $g_* i_W^* a \in \mathrm{Hdg}^2(Y)$, and $\Psi_* a$ is a Hodge class of degree two on D', algebraic by the Lefschetz theorem on $(1,1)$ classes. If $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, then $f^* g_* i_W^* a \in \mathrm{Ab}^2(X)$, because composites of algebraic correspondences are algebraic, and so $a \in \mathrm{Ab}^2(X)$; if every Hodge class of degree four on Y is algebraic, a is algebraic. Conversely, as in the proof of Proposition 2.35(iii), for h ample on X the correspondence $[\Gamma_f] \cdot \mathrm{pr}_X^* h^{n-m}$ sends $f^* b$ to $c\, b$, with $c > 0$ the degree of h^{n-m} on a general fibre; so $b \in \mathrm{Hdg}^2(Y)$ lies in $\mathrm{Ab}^2(Y)$, or is algebraic, as soon as $f^* b$ does, or is.

A birational morphism is surjective with general fibre a point, and two birational smooth projective varieties are dominated through birational morphisms by a resolution of the closure of the graph [Hir64]; so both properties are birational invariants. The morphism $\widetilde{\phi}\colon \widetilde{X} \to R$ of the proof of Proposition 2.43, from a variety birational to X onto its maximal rationally connected quotient, has rationally connected general fibre, which gives the statement on R. For $n \leq 5$ the degrees 0, 2, $2n-2$ and $2n$ are settled by Proposition 2.35(iv), and for $n = 5$ cup product with h is an isomorphism $H^4(X, \mathbb{Q}) \to H^6(X, \mathbb{Q})$ of Hodge structures up to a Tate twist, so that $\mathrm{Hdg}^3(X) = h \cup \mathrm{Hdg}^2(X)$; members of $\mathcal{A}$ satisfy (F3′) by Proposition 2.35(iii). An abelian fourfold B satisfies the Hodge conjecture [Mar25a, Mar25b]. A dominant rational map onto B with rationally connected general fibre becomes, after its indeterminacy is resolved, a morphism whose general fibre is smooth and rationally connected, as in the proof of Proposition 2.43; so the fourfolds and fivefolds of the last assertion have every Hodge class of degree four algebraic, and the other degrees are settled as before. ∎

Proposition 2.44 reaches varieties that Proposition 2.42 does not: conic bundles with singular fibres over members of $\mathcal{A}$ of dimension three or four, and fourfolds and fivefolds birational to members of $\mathcal{A}$ without being known to lie in $\mathcal{A}$. A projective bundle over a member of $\mathcal{A}$ already lies in $\mathcal{A}$ by Proposition 2.42(i), and so, by Proposition 2.42(v) and the Leray–Hirsch theorem, does the source of a smooth morphism onto a member of $\mathcal{A}$ on which algebraic classes restrict to a basis of the cohomology of a fibre, such as a Severi–Brauer bundle. Proposition 2.44 is the relative form of the argument of Bloch and Srinivas, and we claim no priority for it.

Remark 2.45 (Zero-cycles alone go no further). Hypotheses on zero-cycles cannot carry (F3′) further without the fourfolds. If $\mathrm{CH}_0(X)_{\mathbb{Q}}$ is supported on a closed subset Z, with

resolution $j\colon Z' \to X$, the proof of Proposition 2.44, with the closure in $X \times Z$ of a zero-cycle on Z_K rationally equivalent to the generic point in place of Φ^t, gives a divisor with resolution $i\colon D' \to X$ and cycles Φ on $Z' \times X$ and Ψ on $X \times D'$ with $a = \Phi_* j^* a + i_* \Psi_* a$ for every $a \in H^k(X, \mathbb{Q})$, where $\Psi_* a$ has degree $k-2$; this is the transpose of the decomposition of Bloch and Srinivas [BS83]. When $Z \neq X$ and a has type $(n,0)$, $n = \dim X$, the first term vanishes, as $\dim Z' < n$, and the second has no component of type $(n,0)$; so $H^{n,0}(X) = 0$. A uniruled fivefold has its zero-cycles supported on an ample divisor, and for a fourfold Y and $b \in \mathrm{Hdg}^2(Y)$ the class $\mathrm{pr}^* b$ is a Hodge class on $Y \times \mathbb{P}^1$ with $b = \mathrm{pr}_*(\mathrm{pr}^* b \cup \eta)$, for η the class of $Y \times \{0\}$. So (F3′), and likewise the Hodge conjecture, holds for every uniruled fivefold if and only if it holds in degree four for every fourfold.

It also holds for every rational sixfold if and only if it holds for every fourfold. A rational sixfold has its zero-cycles supported on a point, so the proof of Proposition 2.43 settles every degree but six; and it is the image under a birational morphism of a variety obtained from $\mathbb{P}^6$ by blow-ups along smooth centres [Hir64], on which, by the blow-up formula [Voi02, §7.3.3], the Hodge classes of degree six are images under algebraic correspondences of Hodge classes of degree at most four on the centres, of dimension at most four, and lie in Ab^3, or are algebraic, when the fourfolds satisfy (F3′), or the conjecture. Conversely, a fourfold Y is birational to a hypersurface in $\mathbb{P}^5$; place it in a hyperplane of $\mathbb{P}^6$, and let $\widetilde{Y}$ be its smooth strict transform under an embedded resolution $M \to \mathbb{P}^6$ [Hir64]. On the rational sixfold $X = \mathrm{Bl}_{\widetilde{Y}} M$, with exceptional divisor $j\colon E \to X$ and projection $\rho\colon E \to \widetilde{Y}$, the normal bundle of E is $\mathcal{O}_E(-1)$, so $a = -\rho_* j^* j_* \rho^* a$ for $a \in H^4(\widetilde{Y}, \mathbb{Q})$; this recovers $\mathrm{Hdg}^2(\widetilde{Y})$ from $\mathrm{Hdg}^3(X)$ through algebraic correspondences, and Proposition 2.44 passes from $\widetilde{Y}$ to Y.

Among smooth hypersurfaces, ordered by dimension and degree, the first family that these results do not cover is that of the sextic fourfolds: those of dimension at most three are covered by Proposition 2.35(iv), the fourfolds of degree at most five are Fano, hence rationally connected [Cam92, KMM92], and covered by Proposition 2.43, while a sextic fourfold has $h^{4,0} = 1$, so that its zero-cycles are not supported on a proper closed subset and it is its own maximal rationally connected quotient; the Fermat sextic lies in $\mathcal{A}$ by Proposition 2.42(iii). For a smooth sextic fourfold Y in a hyperplane of $\mathbb{P}^6$ no resolution is needed, and the rational sixfold $\mathrm{Bl}_Y \mathbb{P}^6$, with $h^{6,0} = 0$, $h^{5,1} = 1$, $h^{4,2} = 426$ and $h^{3,3} = 1753$ by the blow-up formula [Voi02, §7.3.3] and the Hodge numbers 1, 426, 1752 of Y, satisfies (F3′) if and only if Y does. For the Hodge conjecture these equivalences are folklore, and we claim no priority for them.

Surjective morphisms carry $\mathcal{A}$ to itself (Proposition 2.35(iii)). In degree four the morphism can be replaced by any dominant rational map, which reaches some of these sextic fourfolds.

Proposition 2.46 (Degree four along dominant rational maps)**.** *Let $\varphi\colon Y \dashrightarrow X$ be a dominant rational map of smooth complex projective varieties.*

(i) *If $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, then $\mathrm{Hdg}^2(X) = \mathrm{Ab}^2(X)$; if every Hodge class of degree four on Y is algebraic, so is every one on X.*

(ii) (F3′) *holds on every fourfold and every fivefold that admits a dominant rational map from a variety Y with $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, for instance from a member of $\mathcal{A}$. The Hodge conjecture holds on every fourfold that admits a dominant rational map from an abelian fourfold, such as a smooth projective model of the quotient of an abelian fourfold by a finite group, and on every fivefold that admits one from the product of an abelian fourfold and a projective space.*

Proof. (i) Let $\Gamma \subset Y \times X$ be the closure of the graph of φ and $\widetilde{Y} \to \Gamma$ a resolution of singularities [Hir64]. The projections give a birational morphism $\widetilde{Y} \to Y$ and a morphism $g\colon \widetilde{Y} \to X$, which is surjective, being proper and dominant. Both properties are birational invariants (Proposition 2.44), so they hold for $\widetilde{Y}$. Let $x \in \mathrm{Hdg}^2(X)$, let h be ample on $\widetilde{Y}$ and $r = \dim \widetilde{Y} - \dim X$, and let $\Psi = [\Gamma_g] \cdot \mathrm{pr}_{\widetilde{Y}}^* h^r$, so that $\Psi_* g^* x = c\,x$ with $c > 0$, as in the proof of Proposition 2.35(iii). The class g^*x is a Hodge class of degree four on $\widetilde{Y}$, hence a sum of classes $\Gamma_{j*}\gamma_j$ as in Definition 2.34, or algebraic. So $x = c^{-1}\sum_j (\Psi \circ \Gamma_j)_* \gamma_j$ lies in $\mathrm{Ab}^2(X)$, or x is algebraic, since composites of algebraic correspondences are algebraic [Ful98, Chapter 16].

(ii) On a fourfold or a fivefold the degrees other than four and six are settled by Proposition 2.35(iv), and the same argument, the Lefschetz theorem on $(1,1)$ classes and hard Lefschetz, makes the Hodge classes of those degrees algebraic; on a fivefold cup product with an ample class is an isomorphism from H^4 onto H^6 of Hodge structures up to a Tate twist, so degree six follows from degree four. Members of $\mathcal{A}$ satisfy (F3′) (Proposition 2.35(iii)). An abelian fourfold B satisfies the Hodge conjecture [Mar25a, Mar25b], and by the Künneth formula a Hodge class of degree four on $B \times \mathbb{P}^k$ is a sum of pull-backs of Hodge classes of degrees four, two and zero on B multiplied by powers of the hyperplane class, so it is algebraic. A quotient B/G by a finite group is dominated by B, and so is every smooth projective model of it. ∎

Lefschetz theory carries the two statements from a variety to its ample complete intersections, outside the middle degree always and in the middle degree for a very general member.

Proposition 2.47 (Ample complete intersections)**.** *Let X be a smooth complex projective variety of dimension N, L an ample line bundle on X, and $Y = Y_c \subset Y_{c-1} \subset \cdots \subset Y_0 = X$ with each Y_j a smooth divisor of Y_{j-1} in the linear system of $L^{m_j}|_{Y_{j-1}}$, $m_j \geq 1$; put $r = \dim Y = N - c \geq 1$ and let $i\colon Y \to X$ be the inclusion.*

(i) *If $2p \neq r$, every Hodge class of degree $2p$ on Y is $i^*\alpha$ for a Hodge class α of degree $2p$ on X. So if* (F3′)*, or the Hodge conjecture, holds for X in degree $2p$, it holds for Y in degree $2p$; if r is odd and it holds for X in every degree, it holds for Y in every degree.*

(ii) *Let r be even, $c = 1$ and L^{m_1} very ample, and let $V \subset H^r(Y, \mathbb{Q})$ be the vanishing cohomology, the kernel of the Gysin map i_*. If V is not of type $(r/2, r/2)$, then for a very general Y in the linear system every Hodge class of degree r on Y is $i^*\alpha$ for a Hodge class α on X, and the conclusion of (i) holds in degree r as well.*

So the Hodge conjecture holds for every smooth ample divisor of odd dimension in $B \times \mathbb{P}^k$, for B an abelian variety of dimension at most five, and for the very general member of every very ample linear system on an abelian fivefold; and (F3′) *holds for every smooth ample divisor of odd dimension in a member of $\mathcal{A}$.*

Proof. (i) By the Lefschetz hyperplane theorem [Voi03, Chapter 1], applied to each $Y_j \subset Y_{j-1}$, the restriction $i^*\colon H^k(X, \mathbb{Q}) \to H^k(Y, \mathbb{Q})$ is an isomorphism for $k < r$ and injective for $k = r$. Let $2p < r$. Then i^* is an isomorphism of Hodge structures in degree $2p$, so every Hodge class on Y is i^* of a Hodge class on X. Let $2p > r$. The Gysin map $i_*\colon H^{2p}(Y, \mathbb{Q}) \to H^{2p+2c}(X, \mathbb{Q})$ is Poincaré dual to i^* in degree $2r - 2p < r$, hence an isomorphism, and $i_* i^*$ is cup product with $[Y] = m_1 \cdots m_c\, c_1(L)^c$. By the hard Lefschetz theorem [Voi02, Chapter 6], cup product with $c_1(L)^c$ maps $H^{2p}(X, \mathbb{Q})$ onto $H^{2p+2c}(X, \mathbb{Q})$

when $2p \geq N - c$, so i^* is onto in degree $2p$. A surjective morphism of polarisable Hodge structures has a section that is a morphism of Hodge structures, so again every Hodge class on Y is i^* of a Hodge class on X. The restriction i^* is induced by the transpose of the graph of i, so it carries $\mathrm{Ab}^p(X)$ into $\mathrm{Ab}^p(Y)$ and algebraic classes to algebraic classes, which gives the rest of (i).

(ii) The projection formula shows that V is the orthogonal complement of $i^*H^r(X,\mathbb{Q})$ for the intersection form, and $H^r(Y,\mathbb{Q}) = i^*H^r(X,\mathbb{Q}) \oplus V$ as Hodge structures [Voi03, Chapter 2]; the Hodge classes in the first summand are the $i^*\alpha$ with α a Hodge class on X. Over the open set U of the linear system parametrising smooth members, V is a local system, and its monodromy group Γ acts irreducibly [Voi03, Chapter 3]. For each class v of the lattice of V at a point of the universal cover of U, the set where the flat continuation of v is of Hodge type is a closed analytic subset, either everything or nowhere dense; so at a very general point the Hodge classes of V are those whose continuation is of Hodge type everywhere. They form a subspace W stable under Γ, hence $W = 0$ or $W = V$, and $W = V$ would make V of type $(r/2, r/2)$. So $W = 0$.

For the consequences, $B \times \mathbb{P}^k$ satisfies the Hodge conjecture by the Künneth formula, since B does [Mar25a, Mar25b, MZ99]. For a very ample divisor Y of an abelian fivefold B, $r = 4$, and V is not of type $(2,2)$: $h^{4,0}(Y) = h^0(K_Y) = h^0(Y, \mathcal{O}_Y(Y)) = h^0(B, \mathcal{O}_B(Y)) + 4$, by the sequence $0 \to \mathcal{O}_B \to \mathcal{O}_B(Y) \to \mathcal{O}_Y(Y) \to 0$ and $h^1(\mathcal{O}_B(Y)) = 0$, exceeds $h^{4,0}(B) = 5$ because $h^0(\mathcal{O}_B(Y)) \geq 2$, so V has a nonzero part of type $(4,0)$. Members of $\mathcal{A}$ satisfy (F3′) by Proposition 2.35(iii). ∎

For the Hodge conjecture these transfers are classical, and we claim no priority. What they add to (F3′) is a large class outside $\mathcal{A}$: the vanishing cohomology of an ample divisor of odd dimension is not reached from abelian varieties in general, yet it carries no Hodge class in even degree, since its only degree is the odd middle one.

A hypersurface of Delsarte type is defined by as many monomials as variables. Shioda observed that such a surface is dominated by a Fermat surface [Shi86]; the same monomial substitution works in every dimension, and with Proposition 2.46 it gives the following.

Proposition 2.48 (Delsarte fourfolds)**.** *Let $A = (a_{ij})_{0 \leq i,j \leq r+1}$ be an invertible matrix of non-negative integers whose rows all sum to $m \geq 2$, let $F_A = \sum_i \prod_j x_j^{a_{ij}}$, suppose that $X_A = \{F_A = 0\} \subset \mathbb{P}^{r+1}$ is smooth, and let d be a positive integer with dA^{-1} integral, such as $|\det A|$.*

(i) *There is a dominant rational map from the Fermat hypersurface X_d^r onto X_A, given by monomials.*

(ii) *$\mathrm{Hdg}^2(X_A) = \mathrm{Ab}^2(X_A)$. For $r = 4$, X_A satisfies (F3′), and it satisfies the Hodge conjecture if X_d^4 does.*

(iii) *If $m \geq 3$ and F_A is a sum, in disjoint sets of variables, of terms x^m, chains $x_1^{m-1}x_2 + \cdots + x_{k-1}^{m-1}x_k + x_k^m$ and loops $x_1^{m-1}x_2 + \cdots + x_{k-1}^{m-1}x_k + x_k^{m-1}x_1$ with $k \geq 2$, then A is invertible and X_A is smooth. For $r = 4$ and $m = 6$ these polynomials have 29 shapes up to a renumbering of the variables, the Fermat sextic being one; each defines a smooth sextic fourfold, with $h^{4,0} = 1$, $h^{3,1} = 426$ and primitive $h^{2,2}$ equal to 1751. The least d is 6 for the Fermat sextic, 24 or 30 for six others, those whose terms other than x^m are all loops of length two or all chains of length two, at least 120 for the remaining twenty-two, and $5^6 - 1 = 15624$ for the loop $x_0^5x_1 + x_1^5x_2 + \cdots + x_5^5x_0$.*

Proof. (i) The matrix dA^{-1} is integral for $d = |\det A|$, by the formula for the inverse through the adjugate. Let $B = dA^{-1}$. From $A\mathbf{1} = m\mathbf{1}$, with $\mathbf{1}$ the vector of ones,

$B\mathbf{1} = (d/m)\mathbf{1}$, so the rows of the integral matrix B all sum to the integer d/m, and $AB = d \cdot I$. Choose $c \geq 0$ with every $b_{jk} + c \geq 0$ and let $\psi(y)_j = \prod_k y_k^{b_{jk}+c}$; the coordinates of ψ are monomials of the same degree, so ψ is a rational map $\mathbb{P}^{r+1} \dashrightarrow \mathbb{P}^{r+1}$, and

$$F_A(\psi(y)) = \sum_i \prod_k y_k^{\sum_j a_{ij}(b_{jk}+c)} = (y_0 \cdots y_{r+1})^{cm} \sum_i y_i^d.$$

On the torus T of points with all coordinates nonzero, ψ is induced by the homomorphism $y \mapsto (\prod_k y_k^{b_{jk}})_j$ of $\mathbb{G}_m^{r+2}$, which is surjective because $\det B \neq 0$. So every $z \in X_A \cap T$ is $\psi(y)$ for some $y \in T$, and the identity puts y on X_d^r. A smooth hypersurface of degree at least two and positive dimension is irreducible, so it is not contained in a coordinate hyperplane, and $X_A \cap T$ and $X_d^r \cap T$ are dense. Hence ψ restricts to a dominant rational map $X_d^r \dashrightarrow X_A$.

(ii) X_d^r lies in $\mathcal{A}$ by Proposition 2.42(iii), so (i) and Proposition 2.46 give both assertions, the degrees other than four being settled as in the proof of Proposition 2.46(ii).

(iii) The exponent matrix is block diagonal, with blocks m, $(m-1)I + N + E$ for a chain, N the nilpotent shift and E the matrix whose only nonzero entry is a 1 in the last diagonal place, and $(m-1)I + P$ for a loop, P the cyclic shift; their determinants are m, $m(m-1)^{k-1}$ and $(m-1)^k - (-1)^k$, which are nonzero. The partial derivatives of a sum in disjoint sets of variables are those of the summands, so it suffices that the partials of each summand vanish together only at 0. For x^m this is clear. For a chain, $\partial F/\partial x_1 = (m-1)x_1^{m-2}x_2$ vanishes only if $x_1 = 0$ or $x_2 = 0$, and if $x_2 = 0$ then $\partial F/\partial x_2 = x_1^{m-1}$; so $x_1 = 0$, the same argument applies to $x_2, \ldots, x_{k-1}$, and then $\partial F/\partial x_k = m x_k^{m-1}$ gives $x_k = 0$. For a loop, $\partial F/\partial x_j = (m-1)x_j^{m-2}x_{j+1} + x_{j-1}^{m-1}$, indices modulo k. If one coordinate vanishes, so does the one before it, hence all of them; if none vanishes, the product of the relations $x_{j-1}^{m-1} = -(m-1)x_j^{m-2}x_{j+1}$ gives $\prod_j x_j^{m-1} = (1-m)^k \prod_j x_j^{m-1}$, which is impossible because $|1-m| \geq 2$. The number of shapes, drawn in Figure 5, is the coefficient of t^6 in $(1-t)^{-1}\prod_{k\geq 2}(1-t^k)^{-2}$, which is 29, and the least d is the least common multiple of the orders of the cyclic groups attached to the terms in the proof of Corollary 2.53(iii): 6 for x^6, $6 \cdot 5^{k-1}$ for a chain and $5^k - (-1)^k$ for a loop of length k; so 24 or 30 when the terms other than x^6 are all loops or all chains of length two, and at least $\operatorname{lcm}(24, 30) = 120$, $5^3 + 1$ or $6 \cdot 5^2$ when both kinds of length two, or a loop or chain of length at least three, occur. The Hodge numbers are those of every smooth sextic fourfold, read off from the Jacobian ring [Voi03, Chapter 6]. ∎

A sextic fourfold has $h^{4,0} = 1$, so no hypothesis on zero-cycles or on rationally connected fibrations reaches it (Remark 2.45), and among the smooth sextic fourfolds only the Fermat one is covered by Proposition 2.42. The Delsarte sextics are finitely many up to isomorphism, since the torus rescales the coefficients of a Delsarte polynomial to 1, so they are isolated points in the 426-dimensional moduli of smooth sextic fourfolds: Proposition 2.48 adds points, not a family. By Proposition 2.48(ii) the Hodge conjecture for a Delsarte fourfold follows from that for one Fermat fourfold. For six of the sextic shapes that Fermat fourfold has degree 24 or 30, inside the range claimed in the preprint [MMMV26], so if it is correct the Hodge conjecture itself holds on these six; for the others the degree ranges from 120 to 18750, and from 120 to 3750 with the lattice degree of Corollary 2.53, beyond every known or claimed range [Shi79, MMMV26, Jum26]. The monomial cover is Shioda's, and we claim no priority for it.

Proposition 2.48 uses the covering map only in degree four, where algebraicity modulo abelian varieties descends along every dominant rational map. In higher degrees a birational

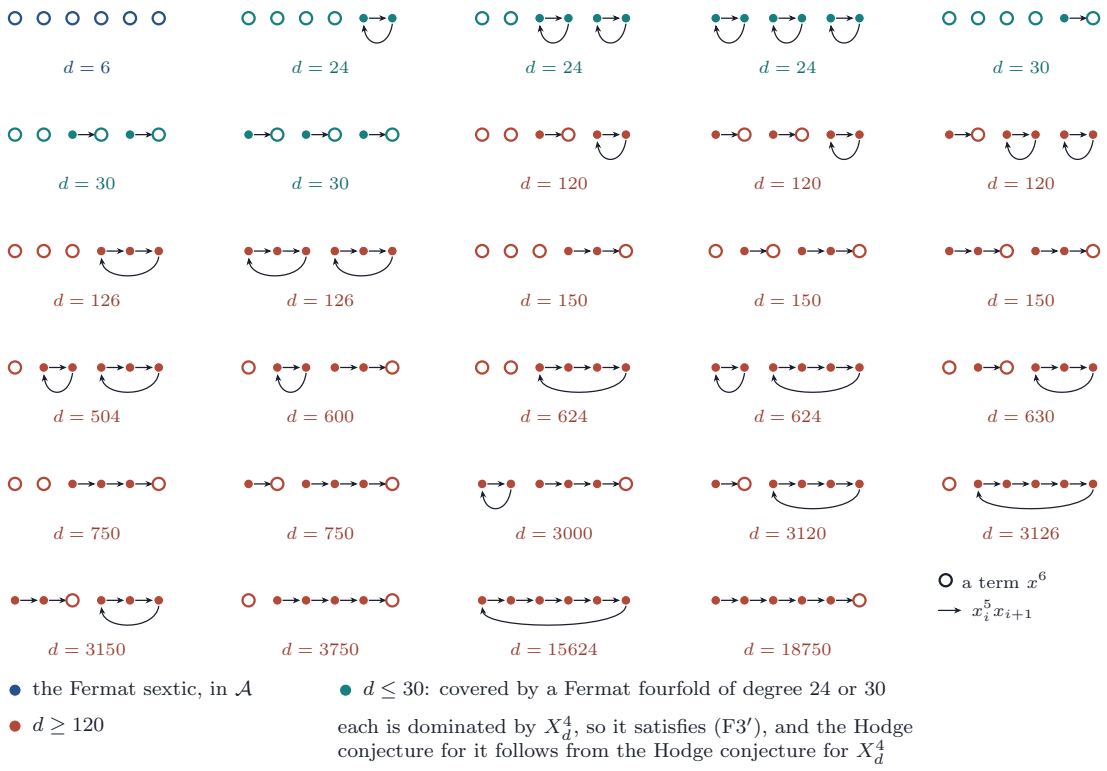


FIGURE 5. The 29 smooth sextic fourfolds of Proposition 2.48(iii), up to a renumbering of the variables. Each is drawn on its six variables: an arrow from x_i to x_j is a term $x_i^5 x_j$, and a ring is a term x^6, so a chain is a path of arrows ending at a ring and a loop is a cycle of arrows. Under each is the least d with dA^{-1} integral, the degree of the Fermat fourfold that covers it. All of them satisfy (F3′).

modification can add classes, so the boundary of a compactification has to be controlled. For a hypersurface of a torus defined by affinely independent monomials the boundary is again of the same kind, and the whole cohomology can be traced back to Fermat varieties of a single degree. We use toric varieties as in [CLS11]: a torus T with character lattice M and cocharacter lattice N, a fan Σ in $N_{\mathbb{R}}$ and its toric variety Y_Σ, and for a cone $\tau \in \Sigma$ the orbit $O(\tau)$, whose character lattice is $\tau^\perp \cap M$, and its closure $V(\tau)$, the toric variety of the fan formed by the images of the cones containing τ in $N_{\mathbb{R}}/\mathbb{R}\tau$ [CLS11, Chapter 3]; for a ray ρ we write $D_\rho = V(\rho)$. In what follows varieties need not be connected: a smooth projective variety lies in $\mathcal{A}$ when each of its connected components does, and the closure properties of Proposition 2.35(iii) and Proposition 2.42 hold component by component.

Definition 2.49 (Simplex type)**.** Let T be a torus of dimension n with character lattice M. A Laurent polynomial $f = \sum_{i=0}^{k} c_i \chi^{u_i}$ on T is of *simplex type* if every c_i is nonzero and $u_0, \dots, u_k \in M$ are affinely independent. Let $L' \subseteq M$ be the lattice spanned by the $u_i - u_0$, L its saturation, the set of $x \in M$ with a nonzero multiple in L', and $e(f)$ the exponent of the finite group L/L'; put $V_f = \{f = 0\} \subseteq T$. A fan Σ in $N_{\mathbb{R}}$ is *adapted* to f if for every cone $\sigma \in \Sigma$ there is an index i_σ with $\langle u_{i_\sigma}, v\rangle \leq \langle u_j, v\rangle$ for all j and all $v \in \sigma$, that is, if every cone of Σ lies in a cone of the normal fan of the simplex spanned by the

u_i. For a cone τ of an adapted fan let I_τ be the set of indices i with $\langle u_i, v\rangle = \min_j \langle u_j, v\rangle$ for every $v \in \tau$, fix $i_\tau \in I_\tau$, and let $f_\tau = \sum_{i\in I_\tau} c_i \chi^{u_i - u_{i_\tau}}$, a Laurent polynomial of simplex type on the torus $O(\tau)$, since its exponents lie in $\tau^\perp \cap M$.

The compactification is Khovanskii's [Kho77]; for a simplex the proof is short, and we give it.

Lemma 2.50 (The compactification)**.** *Let f be of simplex type on T, let Σ be a smooth fan adapted to f, and let Z be the closure of V_f in Y_Σ.*

(i) *V_f is empty if $k = 0$. If $k \geq 1$, it is a finite étale cover, of degree $[L : L']$, of the product of a torus of dimension $n - k$ and the complement of $k + 1$ hyperplanes in general position in $\mathbb{P}^{k-1}$; in particular it is smooth.*

(ii) *Z is smooth, and for every $\tau \in \Sigma$, $Z \cap O(\tau) = V_{f_\tau}$ and Z meets $V(\tau)$ transversally: $Z \cap V(\tau)$ is empty or smooth of codimension $\dim \tau$ in Z, and it is the closure of V_{f_τ} in $V(\tau)$. So the divisors $Z \cap D_\rho$ of Z are smooth and cross normally, and $Z \cap D_\rho$ is empty when I_ρ has one element.*

(iii) *The fan of $V(\tau)$ is smooth and adapted to f_τ.*

Proof. (i) If $k = 0$, f is a monomial and has no zero on T. Let $k \geq 1$, choose a complement C of the saturated lattice L in M and a basis $b_{k+1}, \dots, b_n$ of C. The characters $u_1 - u_0, \dots, u_k - u_0, b_{k+1}, \dots, b_n$ span a sublattice of index $[L : L']$ in M, so the map $T \to (\mathbb{C}^\times)^n$ that they define is an isogeny of that degree. The function $\chi^{-u_0} f$ is the pull-back of $c_0 + c_1 z_1 + \cdots + c_k z_k$, so V_f is the full preimage of the product of $(\mathbb{C}^\times)^{n-k}$ and the set H of points of $(\mathbb{C}^\times)^k$ with $c_0 + \sum_i c_i z_i = 0$. The projective closure of this affine hyperplane is a hyperplane $\mathbb{P}^{k-1}$ of $\mathbb{P}^k$, with homogeneous coordinates $w_0, \dots, w_k$, on which the $k + 1$ linear forms w_i satisfy the single relation $\sum_i c_i w_i = 0$, with every $c_i \neq 0$; so any k of them are independent, the hyperplanes $\{w_i = 0\}$ of $\mathbb{P}^{k-1}$ are in general position, and H is their complement.

(ii) Let $\sigma \in \Sigma$, spanned by part $v_1, \dots, v_s$ of a basis $v_1, \dots, v_n$ of N, let $w_1, \dots, w_n$ be the dual basis of M, and let $t_j = \chi^{w_j}$, the coordinates of $U_\sigma \cong \mathbb{C}^s \times (\mathbb{C}^\times)^{n-s}$. Put $i = i_\sigma$. Every $u_j - u_i$ lies in the dual cone of σ, so $g = \chi^{-u_i} f = \sum_j c_j \chi^{u_j - u_i}$ is a regular function on U_σ, and $Z \cap U_\sigma \subseteq \{g = 0\}$. A face τ of σ is spanned by the v_j with j in a subset J, and $O(\tau)$ is the locus where exactly the t_j with $j \in J$ vanish. A character χ^x regular on U_σ restricts to $O(\tau)$ as χ^x if $x \in \tau^\perp$ and as 0 otherwise [CLS11, Chapter 3]; since $\langle u_j - u_i, v\rangle \geq 0$ on σ, $u_j - u_i \in \tau^\perp$ exactly when $j \in I_\tau$. So $h = g|_{O(\tau)}$ is $\chi^{u_{i_\tau} - u_i} f_\tau$, a unit times f_τ. Let $p \in O(\tau)$ with $h(p) = 0$ and put $a_j = c_j \chi^{u_j - u_i}(p)$ for $j \in I_\tau$. The derivative of h at p along the one-parameter subgroup of $O(\tau)$ given by y in its cocharacter lattice is $\sum_j a_j \langle u_j - u_i, y\rangle$. If all of these vanished, then, with $\sum_j a_j = h(p) = 0$, the vector $\sum_j a_j (1, u_j - u_i)$ of $\mathbb{C} \oplus (\tau^\perp \cap M) \otimes \mathbb{C}$ would be zero, as the cocharacter lattice of $O(\tau)$ is dual to $\tau^\perp \cap M$; affine independence would give every $a_j = 0$, which is impossible at a point of the torus $O(\tau)$. So $dh(p) \neq 0$. Hence $\{g = 0\}$ is smooth at p and transversal to $O(\tau)$ and to every $\{t_j = 0,\ j \in J'\}$ with $J' \subseteq J$, whose tangent space at p contains that of $O(\tau)$. No component of $\{g = 0\}$ lies in a divisor $\{t_j = 0\}$, since g restricted to the orbit of the ray of v_j is a unit times a nonzero Laurent polynomial; so every component meets T, where $\{g = 0\}$ is V_f, and $\{g = 0\} = Z \cap U_\sigma$. This shows that Z is smooth, that $Z \cap O(\tau) = V_{f_\tau}$ and that $Z \cap V(\tau)$ is smooth of codimension $\dim \tau$. As $Z \cap O(\tau')$ has codimension at least $\dim \tau'$ in Z for every cone $\tau' \supsetneq \tau$, no component of $Z \cap V(\tau)$ lies in the smaller orbits, and $Z \cap V(\tau)$ is the closure of V_{f_τ}. If I_ρ has one element, f_ρ is a nonzero constant.

(iii) The fan of $V(\tau)$ is smooth because Σ is. Let $\sigma \supseteq \tau$. The index i_σ attains the minimum on σ, hence on τ, so $i_\sigma \in I_\tau$, and for $j \in I_\tau$ and $v \in \sigma$, $\langle u_{i_\sigma} - u_{i_\tau}, v\rangle \le \langle u_j - u_{i_\tau}, v\rangle$. Both sides depend only on the image of v in $N_{\mathbb{R}}/\mathbb{R}\tau$, because the exponents of f_τ lie in $\tau^\perp$; so the image of σ satisfies the condition of adaptedness for f_τ. ∎

Lemma 2.51 (Cohomology across a toric modification)**.** *Let f be of simplex type on T, let Σ_1 and Σ_0 be smooth projective fans adapted to f, with Σ_1 refining Σ_0, let $\varphi\colon Y_1 = Y_{\Sigma_1} \to Y_0 = Y_{\Sigma_0}$ be the toric morphism, and let Z_1 and Z_0 be the closures of V_f. For each ray ρ of Σ_1 that is not a ray of Σ_0 let $E_\rho = Z_1 \cap D_\rho$, with inclusion i_ρ. Then the kernel of $\varphi_*\colon H^k(Z_1,\mathbb{Q}) \to H^k(Z_0,\mathbb{Q})$ is contained in $\sum_\rho i_{\rho*}H^{k-2}(E_\rho,\mathbb{Q})$, and*

$$H^k(Z_1,\mathbb{Q}) = \varphi^*H^k(Z_0,\mathbb{Q}) + \sum_\rho i_{\rho*}H^{k-2}(E_\rho,\mathbb{Q}).$$

Proof. Let $W \subseteq Y_0$ be the union of the open sets U_τ with $\tau \in \Sigma_0 \cap \Sigma_1$. A cone of Σ_1 contained in such a τ is a face of it, since two cones of a fan meet in a common face; so $\varphi^{-1}(U_\tau) = U_\tau$, and φ is an isomorphism over W. A cone of Σ_1 whose rays are all rays of Σ_0 lies in a cone τ_0 of Σ_0, its rays are rays of τ_0, and τ_0 is simplicial, so it is a face of τ_0 and lies in Σ_0. Hence $Y_1 \smallsetminus \varphi^{-1}(W)$ is the union of the orbits of the cones with a ray not in Σ_0, that is, of the divisors D_ρ of those rays. So $U_1 = Z_1 \smallsetminus \bigcup_\rho E_\rho$ is the closure of V_f in $\varphi^{-1}(W)$, and φ maps it isomorphically onto $U_0 = Z_0 \cap W$, with $U_1 = Z_1 \cap \varphi^{-1}(U_0)$. The morphism $Z_1 \to Z_0$ is birational on each component, so $\varphi_*\varphi^* = \mathrm{id}$ by the projection formula, and every class is $\varphi^*\varphi_* a$ plus a class in the kernel of φ_*. Let $\varphi_* b = 0$. Push-forward along a proper map commutes with restriction to an open set and its preimage, so $b|_{U_1} = (\varphi|_{U_1})^*((\varphi_* b)|_{U_0}) = 0$. By Lemma 2.50(ii), $\bigcup_\rho E_\rho$ is a divisor with normal crossings and smooth components in the smooth projective variety Z_1, and by Deligne [Del71, §3.2] the weight spectral sequence of its complement U_1 degenerates at E_2; its term $E_2^{0,k}$, the cokernel of the Gysin map $\bigoplus_\rho H^{k-2}(E_\rho,\mathbb{Q}) \to H^k(Z_1,\mathbb{Q})$, is the image of $H^k(Z_1,\mathbb{Q})$ in $H^k(U_1,\mathbb{Q})$. So b lies in the image of the Gysin map. ∎

Theorem 2.52 (Hypersurfaces of simplex type)**.** *Let f be a Laurent polynomial of simplex type on a torus T of dimension n, with $e = e(f)$, let Y be a projective toric variety with torus T, not necessarily smooth, and let Z be the closure of V_f in Y. Suppose that Z is smooth.*

(i) *$Z \in \mathcal{A}$, so (F3′) holds on Z in every degree.*

(ii) *If the Hodge conjecture holds for the Fermat varieties X_e^j with $j \le \dim Z$, then it holds for Z.*

Proof. By induction on n. If $k = 0$, Z is empty; let $k \ge 1$.

The Fermat form. With C and b_j as in the proof of Lemma 2.50(i), let $\widetilde{M}$ be the lattice with basis $w_i = (u_i - u_0)/e$ for $1 \le i \le k$ and $w_j = b_j$ for $k < j \le n$. It contains M, since $eL \subseteq L'$ gives $L \subseteq \frac{1}{e}L'$. Let $\widetilde{T}$ be the torus with character lattice $\widetilde{M}$, $\widetilde{N} \subseteq N$ its cocharacter lattice and $\pi\colon \widetilde{T} \to T$ the isogeny dual to $M \subseteq \widetilde{M}$. In the coordinates $z_i = \chi^{w_i}$ of $\widetilde{T}$, $\pi^*(\chi^{-u_0} f) = c_0 + \sum_{i\le k} c_i z_i^e$; call it $\widetilde{f}$. It is of simplex type, $V_{\widetilde{f}} = \pi^{-1}(V_f)$, and after the z_i are rescaled it is $c_0(1 + \sum_{i\le k} z_i^e)$.

The model. Let Σ_0 be the fan in $\widetilde{N}_{\mathbb{R}}$ of $Y_0 = \mathbb{P}^k \times \mathbb{P}^{n-k}$, on whose torus $z_1, \dots, z_k$ are the affine coordinates of the first factor and $z_{k+1}, \dots, z_n$ those of the second. The closure of $V_{\widetilde{f}}$ in Y_0 is, after the rescaling, $Z_0 = X_e^{k-1} \times \mathbb{P}^{n-k}$, since the part of the Fermat variety in the torus is dense in it; $Z_0 \in \mathcal{A}$ by Proposition 2.42(iii) and Proposition 2.35(iii), $\mathbb{P}^{n-k}$

being the Fermat variety X_1^{n-k}. The fan Σ_0 is adapted to $\widetilde{f}$: $\min(0, e\langle w_1, v\rangle, \dots, e\langle w_k, v\rangle)$ depends only on the image of v in the fan of $\mathbb{P}^k$, the normal fan of the standard simplex, on whose cones it is linear.

The common refinement. Let Σ_Y be the fan of Y, the normal fan of a polytope P, as Y is projective [CLS11, Chapter 2], and Σ_0 that of a polytope Q. The face of $P+Q$ on which a vector v attains its minimum is the sum of those of P and of Q, so the normal fan of $P+Q$ consists of the intersections of cones of Σ_Y and Σ_0; it has a smooth projective refinement Σ_1 with respect to $\widetilde{N}$ [CLS11, Chapter 11]. It is adapted to $\widetilde{f}$, because Σ_0 is. Let $Y_1 = Y_{\Sigma_1,\widetilde{N}}$ and Z_1 the closure of $V_{\widetilde{f}}$ in it, smooth by Lemma 2.50(ii). The inclusion $\widetilde{N} \subseteq N$ maps each cone of Σ_1 into a cone of Σ_Y and defines a proper toric morphism $\psi\colon Y_1 \to Y$ extending π [CLS11, Chapter 3]; $\psi(Z_1)$ is closed, contains $\pi(V_{\widetilde{f}}) = V_f$ and lies in Z, so ψ maps Z_1 onto Z. The identity of $\widetilde{N}$ defines the toric morphism $\varphi\colon Y_1 \to Y_0$, which maps Z_1 onto Z_0 and is an isomorphism over $V_{\widetilde{f}}$.

The boundary. By Lemma 2.50, each $E_\rho = Z_1 \cap D_\rho$ is the smooth closure of $V_{\widetilde{f}_\rho}$ in the smooth projective toric variety D_ρ of dimension $n-1$. The exponents of $\widetilde{f}_\rho$ are the $e(w_i - w_{i_\rho})$ with $i \in I_\rho$, where $w_0 = 0$. The vectors $w_i - w_{i_\rho}$ with $i \le k$, $i \ne i_\rho$, together with $w_{k+1}, \dots, w_n$, form a basis of $\widetilde{M}$, so those with $i \in I_\rho$ span a saturated sublattice of $\widetilde{M}$, which lies in $\rho^\perp \cap \widetilde{M}$ and is saturated there. Hence $e(\widetilde{f}_\rho) = e$ if I_ρ has at least two elements, and E_ρ is empty otherwise. By induction every E_ρ lies in $\mathcal{A}$, and it satisfies the Hodge conjecture if the X_e^j with $j \le n-2$ do.

(i) By Lemma 2.51, $H^*(Z_1, \mathbb{Q})$ is spanned by the images of $H^*(Z_0, \mathbb{Q})$ and of the $H^*(E_\rho, \mathbb{Q})$ under φ^* and the $i_{\rho *}$, which are induced by algebraic cycles, the transposed graph of φ and the graphs of the i_ρ. So $Z_1 \in \mathcal{A}$ by Proposition 2.42(v), and $Z \in \mathcal{A}$ by Proposition 2.35(iii), as the image of Z_1 under ψ.

(ii) By the Künneth formula every Hodge class on $X_e^{k-1} \times \mathbb{P}^{n-k}$ is a sum of pull-backs of Hodge classes on X_e^{k-1} multiplied by powers of the hyperplane class, so the conjecture holds on Z_0, as $k-1 \le n-1 = \dim Z$; it holds on the E_ρ by induction. Let a be a Hodge class on Z_1. Then $\varphi_* a$ is a Hodge class, hence algebraic, and so is $\varphi^*\varphi_* a$. The class $b = a - \varphi^*\varphi_* a$ is a Hodge class in the kernel of φ_*, hence in the image of the Gysin map of Lemma 2.51, a morphism of polarisable Hodge structures after a Tate twist; as in the proof of Proposition 2.35(iii) it is the image of a Hodge class, which is algebraic, so b is algebraic. So the conjecture holds on Z_1, and it passes to Z along ψ: for a Hodge class x on Z, $\psi^* x$ is algebraic, and $\Psi_*\psi^* x = c\,x$ with $c > 0$ for the algebraic correspondence Ψ of the proof of Proposition 2.35(iii). ∎

Corollary 2.53 (Delsarte hypersurfaces of every dimension)**.** *Let $X_A \subset \mathbb{P}^{r+1}$, $r \ge 1$, be a smooth Delsarte hypersurface of degree m as in Proposition 2.48, let $M = \{x \in \mathbb{Z}^{r+2} : \sum_j x_j = 0\}$, and let e_A be the exponent of M/M', where M' is spanned by the differences $a_i - a_0$ of the rows of A.*

(i) *$X_A \in \mathcal{A}$, so* (F3′) *holds on X_A in every degree, and the Hodge conjecture holds on X_A if it holds on the Fermat varieties $X_{e_A}^j$ with $j \le r$. The number e_A divides every d of Proposition 2.48, and $[M : M'] = |\det A|/m$.*

(ii) *The same holds, with $j \le r+1$, for the cyclic cover $\{y^k = F_A(x)\}$ of $\mathbb{P}^{r+1}$ in the weighted projective space $\mathbb{P}(1, \dots, 1, m/k)$, for every k dividing m, with $M = \{x \in \mathbb{Z}^{r+3} : \sum_{j \le r+1} x_j + (m/k)x_{r+2} = 0\}$ and M' spanned by the differences of the exponents of $F_A - y^k$; it is smooth because X_A is.*

(iii) *For the* 29 *sextic shapes of Proposition 2.48(iii),* e_A *is the least* d *there, except for the chain* $x_0^5x_1 + x_1^5x_2 + \cdots + x_4^5x_5 + x_5^6$ *and the loop, for which it is* 3125 *and* 2604 *instead of* 18750 *and* 15624. *For the Klein quartic* $x^3y + y^3z + z^3x$ *it is* 7.

Proof. (i) On the torus of $\mathbb{P}^{r+1}$, with character lattice M, X_A is the closure of V_f for $f = \chi^{-a_0}F_A$, as in the proof of Proposition 2.48(i). The rows of A are linearly independent and lie on the affine hyperplane $\sum_j x_j = m$, which does not contain 0, so they are affinely independent, f is of simplex type, $L' = M'$ has rank $r+1$ and $L = M$; so $e(f) = e_A$, and Theorem 2.52 gives the first two assertions. If dA^{-1} is integral and $x \in M$, then $dx = \lambda A$ with $\lambda = dxA^{-1}$ integral and $\sum_i \lambda_i = d\sum_j x_j/m = 0$, as $A^{-1}\mathbf{1} = \mathbf{1}/m$; so $dx = \sum_i \lambda_i(a_i - a_0) \in M'$, and e_A divides d. The lattice M' is the intersection of M with the lattice R spanned by the rows, whose image under $x \mapsto \sum_j x_j$ is $m\mathbb{Z}$; the exact sequence $0 \to M/M' \to \mathbb{Z}^{r+2}/R \to \mathbb{Z}/m\mathbb{Z} \to 0$ gives $[M : M'] = |\det A|/m$.

(ii) The exponents of $F_A - y^k$ are linearly independent and lie on the hyperplane of weighted degree m, so the polynomial is of simplex type on the torus of $\mathbb{P}(1, \dots, 1, m/k)$, a projective toric variety whose character lattice is the given M. On the affine cone, the partial derivatives vanish together with $y^k - F_A$ only where $y = 0$ and x is a singular point of the cone over X_A, that is $x = 0$; so the cover is quasi-smooth, and it misses the only singular point $(0 : \cdots : 0 : 1)$ of the weighted projective space, where $y^k = F_A(0)$ would force $y = 0$. So it is smooth. It is irreducible, F_A being irreducible, and not contained in a coordinate hyperplane, so it is the closure of its part in the torus, and Theorem 2.52 applies.

(iii) The least d with dA^{-1} integral is the exponent of $\mathbb{Z}^{r+2}/R$, and the map $x \mapsto \sum_j x_j$ modulo m is well defined on it because the rows sum to m. For a polynomial as in Proposition 2.48(iii) the group $\mathbb{Z}^{r+2}/R$ is the direct sum of one group for each term. The relations of a chain are $x_{j+1} = (1-m)x_j$ and $mx_k = 0$, those of a loop are $x_{j+1} = (1-m)x_j$, indices modulo k, and that of x^m is $mx = 0$. So each summand is cyclic, generated by the class of its first variable, of order $N = m(m-1)^{k-1}$, $(m-1)^k - (-1)^k$ and m respectively, and the sum map sends that generator to 1, since $1 - m \equiv 1$ modulo m; each N is divisible by m. The least d is the least common multiple of these orders, which gives the values of Proposition 2.48(iii). The group M/M' is the kernel of the sum map. With one term only, the loop or the chain in six variables, the kernel is $m\mathbb{Z}/N\mathbb{Z}$, of exponent N/m, which is $15624/6 = 2604$ and $18750/6 = 3125$; for the Klein quartic, a loop with $m = 4$ and $k = 3$, it is $28/4 = 7$. With at least two terms, of orders $N_1, N_2, \dots$, the kernel contains the differences of the generators of the first and of the b-th summand, of order $\operatorname{lcm}(N_1, N_b)$; so its exponent is the least common multiple of all the N_b, which is the least d. ∎

For the Klein quartic the construction is classical: the isogeny has kernel of order $e_A^2/[M : M'] = 7$, and the Klein quartic is the quotient of the Fermat septic curve by a group of order seven; Figure 6 draws its Newton triangle and the smooth adapted fan of the proof. For $r = 4$, Corollary 2.53(i) recovers Proposition 2.48(ii), with the Fermat degree lowered to e_A. Its content lies in dimension five and more, where otherwise only the Fermat hypersurfaces are covered (Proposition 2.42(iii)), and in the cyclic covers, among them the Calabi–Yau varieties $\{y^2 = F_A(x)\}$ with F_A of degree $2r+4$ in $r+2$ variables. Khovanskii's compactification and Shioda's cover are classical, and we claim no priority for the two lemmas. What Theorem 2.52 adds is that the boundary of the compactification is again of simplex type with the same exponent e, so that the whole cohomology, not only its part of top Hodge level, comes from Fermat varieties of one degree.

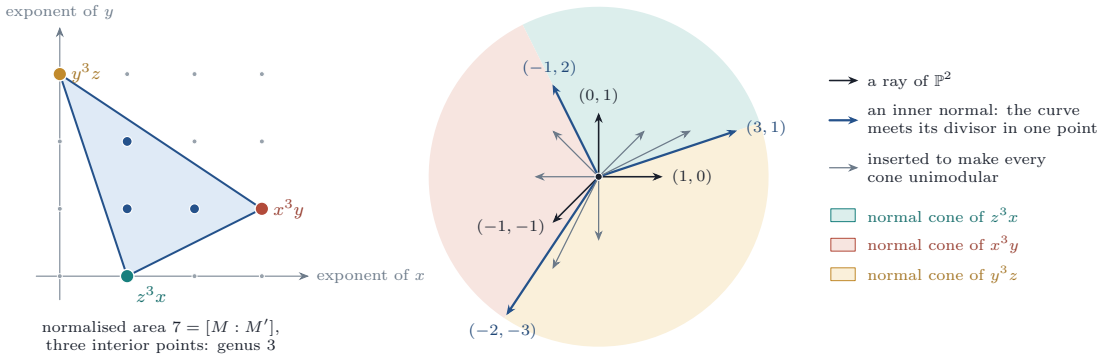


FIGURE 6. The Klein quartic $x^3y + y^3z + z^3x$ in the proof of Theorem 2.52. Left: its Newton triangle in the lattice M of the torus of $\mathbb{P}^2$, in the chart $z = 1$, with normalised area $7 = [M : M']$ and three interior points, the genus. Right: the rays of a smooth fan adapted to it, the three rays of $\mathbb{P}^2$, the three inner normals of the triangle and the six rays inserted to make every cone unimodular. The curve meets only the three divisors of the inner normals, in one point each, the lattice length of the edge, so its Euler number is $-7 + 3 = -4$. With $e_A = 7$ the isogeny of the proof makes it the quotient of the Fermat septic by a group of order seven.

Remark 2.54 (Quotients by diagonal automorphisms). Let $G \subset T$ be a finite subgroup of the torus of $\mathbb{P}^{r+1}$ on which every character $\chi^{a_i - a_0}$ is trivial, so that G acts on X_A; these G are the subgroups of the group G_A dual to M/M'. Then f descends to a Laurent polynomial of simplex type on T/G, whose character lattice M_G, the characters trivial on G, contains M'; so $L' = M'$, $L = M_G$, and $e(f)$ becomes the exponent e_G of M_G/M', a divisor of e_A. As $\mathbb{P}^{r+1}/G$ is a projective toric variety with torus T/G in which X_A/G is the closure of V_f/G, Theorem 2.52 applies to every projective model Z of X_A/G that is the closure of V_f/G in a projective toric variety with torus T/G and is smooth, for instance in one whose fan is smooth and adapted to f (Lemma 2.50): such a Z lies in $\mathcal{A}$, and the Hodge conjecture holds on it if it holds on the $X^j_{e_G}$ with $j \le r$. For $G = G_A$, $M_G = M'$ and $e_G = 1$, so the conjecture holds on Z. For the Fermat hypersurface of degree $m = r + 2$ and the group of diagonal automorphisms of determinant one modulo scalars, the starting point of the mirror construction for X_A, $M' = mM$ and M_G is the set of $x \in M$ whose coordinates are congruent modulo m, so $M_G/M' \cong \mathbb{Z}/m$ and $e_G = m = e_A$: the quotient needs the same Fermat varieties as X_A, and nothing is gained. For any finite group G of automorphisms of X_A, every smooth projective model X' of X_A/G has $\mathrm{Hdg}^2(X') = \mathrm{Ab}^2(X')$, so (F3′) holds on it when it is a fourfold or a fivefold, by Proposition 2.46 and Corollary 2.53(i).

The Fermat hypersurfaces are the complete intersections of one diagonal hypersurface. Complete intersections of several diagonal hypersurfaces of the same degree lie in $\mathcal{A}$ as well when their coefficients are of Vandermonde type, because they are quotients of powers of one curve. Fix $d \ge 2$, $N \ge 2$ and $1 \le c \le N - 1$, put $r = N - c$, and call a $c \times (N + 1)$ matrix (a_{ki}) *of Vandermonde type* if there are distinct $\lambda_0, \dots, \lambda_N \in \mathbb{C}$, nonzero $\mu_0, \dots, \mu_N$ and $g \in \mathrm{GL}_c(\mathbb{C})$ with $a_{ki} = \mu_i \sum_{l=1}^{c} g_{kl} \lambda_i^{l-1}$; for $c \ge 2$ this says that the columns are distinct points of a rational normal curve of degree $c - 1$ in $\mathbb{P}^{c-1}$. Put $w_i = \prod_{l \ne i} (\lambda_i - \lambda_l)^{-1}$

and

$$X_{d,r}(\lambda) = \Big\{ x \in \mathbb{P}^N : \sum_{i=0}^{N} w_i \lambda_i^k x_i^d = 0, \quad 0 \le k \le c-1 \Big\}.$$

For $r = 1$ this is the generalised Fermat curve of type (d, N) of González-Diez, Hidalgo and Leyton [GDHL09], the Galois cover of $\mathbb{P}^1$ with group $H = \mu_d^{N+1}/\mu_d$, acting by $u_i \mapsto h_i u_i$, branched over the λ_i. The characters of H are the $\chi_a(h) = \prod_i h_i^{a_i}$ with $a \in (\mathbb{Z}/d)^{N+1}$ and $\sum_i a_i = 0$; write $V_a \subset H^1(C, \mathbb{C})$ for the eigenspace on which h^* acts by $\chi_a(h)$, $[a]$ for the orbit of a under $(\mathbb{Z}/d)^\times$, and $B_{[a]}$ for the image of a multiple of the idempotent of $\mathbb{Q}[H]$ attached to $[a]$, an abelian subvariety of the Jacobian $J(C)$ whose first cohomology is the part $\bigoplus_{a' \in [a]} V_{a'}$ of $H^1(C, \mathbb{C})$.

Theorem 2.55 (Diagonal complete intersections of Vandermonde type)**.** *Let $X \subset \mathbb{P}^N$ be the complete intersection of c hypersurfaces $\sum_i a_{ki} x_i^d = 0$, $1 \le k \le c$, with (a_{ki}) of Vandermonde type, and let $C = X_{d,1}(\lambda)$ for the λ_i of (a_{ki}).*

(i) *X is smooth of dimension r and isomorphic to $X_{d,r}(\lambda)$, and C is a smooth curve of genus $1 + \frac{1}{2} d^{N-1}\big((N-1)(d-1) - 2\big)$.*

(ii) *The map $\Phi(u^{(1)}, \dots, u^{(r)}) = \big(\prod_j u_i^{(j)}\big)_i$ is a surjective morphism $C^r \to X_{d,r}(\lambda)$, and it induces an isomorphism $C^r/G \cong X_{d,r}(\lambda)$, where G is the group of order $r!\, d^{N(r-1)}$ generated by the permutations of the factors and the $(h_1, \dots, h_r) \in H^r$ with $h_1 \cdots h_r = 1$.*

(iii) *$X \in \mathcal{A}$, so (F3′) holds on X in every degree.*

(iv) *$\dim V_a = \#\{i : a_i \ne 0\} - 2$ for $a \ne 0$, and the Hodge conjecture holds on X as soon as it holds on the abelian varieties $B_{[a]}$ with $\dim V_a \ge r$, which have dimension $\frac{1}{2}|[a]| \dim V_a$.*

Proof. Everything rests on the Lagrange identity: interpolating a polynomial f of degree at most N at the λ_i and comparing the coefficients of t^N gives $\sum_i w_i f(\lambda_i) = 0$ when $\deg f \le N - 1$.[2]

(i) Replacing the equations by their combinations under g^{-1} and x_i by $\nu_i x_i$ with $\nu_i^d \mu_i = w_i$ turns them into those of $X_{d,r}(\lambda)$. Every $c \times c$ minor of (a_{ki}) is $\det g \prod \mu_i$ times a Vandermonde determinant in distinct λ_i, so any c columns are independent. At a point x of X the Jacobian matrix is $(d\, a_{ki} x_i^{d-1})$; if $S = \{i : x_i \ne 0\}$ had at most c elements, $\sum_{i \in S} x_i^d a_i = 0$ would be a relation between at most c columns, so $|S| > c$ and the Jacobian has rank c.[3] Hence X is a smooth complete intersection of dimension r, connected because $r \ge 1$, and the same holds for C. The genus follows from the Riemann–Hurwitz formula for the cover $C \to \mathbb{P}^1$ of degree d^N described below, which has $N + 1$ branch points of ramification index d.

[2] Lagrange interpolation writes $f(t) = \sum_i f(\lambda_i) \prod_{l \ne i} (t - \lambda_l)/(\lambda_i - \lambda_l)$ for $\deg f \le N$, and the coefficient of t^N on the right is $\sum_i w_i f(\lambda_i)$. For $N = 2$ and $f = 1$ it says that $\sum_i \prod_{l \ne i} (\lambda_i - \lambda_l)^{-1} = 0$ for any three distinct numbers. The identity says that the vector $(w_i \lambda_i^k)_i$ is orthogonal to the values of every polynomial of degree at most $N - 1 - k$, which is how the equations of $X_{d,r}(\lambda)$ become the condition that the d-th powers of the coordinates are the values of one polynomial of degree at most r.

[3] The columns of the Jacobian matrix with $i \in S$ are the columns of (a_{ki}) multiplied by the nonzero numbers $d\, x_i^{d-1}$, and any c of the columns of (a_{ki}) are independent; so c of them already give rank c. Without the Vandermonde condition this fails: for two diagonal quadrics in $\mathbb{P}^3$ with proportional columns $i = 0, 1$, the points with $x_2 = x_3 = 0$ and $a_0 x_0^2 + a_1 x_1^2 = 0$ are singular.

(ii) For $1 \le k \le N-1$ let Λ_k be the space of $y \in \mathbb{C}^{N+1}$ with $\sum_i w_i \lambda_i^l y_i = 0$ for $0 \le l \le N-k-1$. The vectors $(F(\lambda_i))_i$ with $\deg F \le k$ lie in Λ_k by the identity, since $\deg t^l F \le N-1$; they form a space of dimension $k+1$, since $k < N+1$; and Λ_k has dimension $k+1$, the $N-k$ equations having independent Vandermonde rows. So $\Lambda_k = \{(F(\lambda_i))_i : \deg F \le k\}$, and C and $X_{d,r}(\lambda)$ are the sets of points whose d-th powers lie in Λ_1 and in Λ_r. The point $u \in C$ determines $(s:t) \in \mathbb{P}^1$ with $u_i^d = \kappa(t - \lambda_i s)$ for all i; this is the quotient $C \to C/H = \mathbb{P}^1$, whose fibres are the H-orbits. If $(u_i^{(j)})^d = t_j - \lambda_i s_j$, then $\Phi(u)_i^d = F(\lambda_i)$ for $F(t) = \prod_j (t_j - t s_j)$, a nonzero polynomial of degree at most $r < N+1$; so $\Phi(u) \in X_{d,r}(\lambda)$ and $\Phi(u) \ne 0$, and Φ, given by forms of multidegree $(1,\dots,1)$ with no common zero, is a morphism. It is G-invariant. Conversely, if $x \in X_{d,r}(\lambda)$, then $(x_i^d)_i = (F(\lambda_i))_i$ for a unique nonzero F of degree at most r, which factors as $\prod_j (t_j - t s_j)$, uniquely up to order and to rescaling the factors. Choosing $u^{(j)}$ over $(s_j : t_j)$ gives $\Phi(u)_i = \epsilon_i x_i$ with $\epsilon_i \in \mu_d$, and multiplying $u^{(1)}$ by (ϵ_i^{-1}) gives a preimage. Two preimages have the same roots, so after a permutation $u'^{(j)} = h_j u^{(j)}$ with $h_j \in H$, and $\prod_j h_j$ acts trivially on the coordinates where $x_i \ne 0$; where $x_i = 0$ some $u_i^{(j)}$ vanishes and $h_{j,i}$ may be changed freely, so we may take $\prod_j h_j = 1$. Hence the fibres of Φ are the G-orbits, Φ induces a finite bijective morphism from the normal variety C^r/G to the smooth variety $X_{d,r}(\lambda)$, and by Zariski's main theorem this is an isomorphism.[4] The order of G is $r!\,|H|^{r-1}$.

(iii) C is a curve, so $C^r \in \mathcal{A}$, and X is the image of C^r under a surjective morphism (Proposition 2.35(iii)).

(iv) For $a \ne 0$, the forms $t^j \prod_i (t-\lambda_i)^{-b_i/d}\,dt$ with $b_i \in \{0,\dots,d-1\}$, $b_i \equiv -a_i$, and $0 \le j \le \sum_i b_i/d - 2$, are holomorphic on C and span the χ_a-eigenspace of $H^{1,0}(C)$;[5] its dimension $\sum_i \langle -a_i/d\rangle - 1$ and that of the χ_{-a}-eigenspace add up to $\#\{i : a_i \ne 0\} - 2$, and $H^1(C)^H = H^1(\mathbb{P}^1) = 0$. The group H acts trivially on $H^0(C)$ and $H^2(C)$, so a tensor in $H^r(C^r) = \bigoplus \bigotimes_j H^{k_j}(C)$ invariant under the (h_j) with $\prod_j h_j = 1$ has all its factors in H^1, with one character, or none: taking h in one factor and h^{-1} in another shows that two factors carry the same character, and a factor in H^0 or H^2 carries the trivial one. Taking the invariants of the permutations,

$$H^r(C^r,\mathbb{C})^G = \mathbb{C}\,\eta \;\oplus \bigoplus_{a \ne 0} \bigwedge^r V_a,$$

with η the symmetrised product of $r/2$ classes of points when r is even and $\eta = 0$ otherwise, and $W_{[a]} = \bigoplus_{a' \in [a]} \bigwedge^r V_{a'}$ is a sub-Hodge structure defined over $\mathbb{Q}$, zero unless $\dim V_a \ge r$. Let $\psi\colon C \to B_{[a]}$ be an Abel–Jacobi map followed by the projection, so that

[4] In characteristic zero a finite morphism that is bijective on closed points is generically étale of degree one, hence birational, and a finite birational morphism onto a normal variety is an isomorphism [Har77, Chapter III, Corollary 11.4]. The group G acts faithfully on C^r, since H is the group of deck transformations of $C \to \mathbb{P}^1$, so Φ has degree $|G|$.

[5] On the affine chart $s = 1$ put $u_i^d = t - \lambda_i$, so that the form is $t^j \prod_i u_i^{-b_i}\,dt$ and $h \in H$ multiplies it by $\prod_i h_i^{-b_i} = \chi_a(h)$; it is well defined on $H = \mu_d^{N+1}/\mu_d$ because $\sum_i b_i \equiv 0$. Over λ_i the cover is ramified of index d, with a local parameter z satisfying $z^d = t - \lambda_i$, and the form is a unit times $z^{-b_i}\,d(z^d) = d\,z^{d-1-b_i}\,dz$, holomorphic as $b_i \le d-1$. Over $t = \infty$, with $w = 1/t$ and $v^d = w$, it is a unit times $v^{\,d(\sum_i b_i/d - j - 2) + d - 1}\,dv$, holomorphic exactly when $j \le \sum_i b_i/d - 2$, the exponent of v being an integer. Forms with different characters, or with different j, are independent, and their number is $\frac12 \sum_{a \ne 0}\big(\#\{i : a_i \ne 0\} - 2\big) = \frac12\big((N+1)(d^N - d^{N-1}) - 2(d^N - 1)\big)$, which is the genus of (i); so they form a basis of $H^{1,0}(C)$, and each eigenspace is spanned by the forms of its character. For $d = 2$ and $N = 4$ they are the five forms $dt/\sqrt{(t-\lambda_{i_1})\cdots(t-\lambda_{i_4})}$, one for each choice of four of the five indices, on a curve of genus five.

ψ^* maps $H^1(B_{[a]},\mathbb{Q})$ isomorphically onto the $[a]$-part of $H^1(C,\mathbb{Q})$, and let $\sigma\colon C^r \to B_{[a]}$ be $\sum_j \psi\circ \mathrm{pr}_j$. Then $\sigma^*(\alpha_1\wedge\cdots\wedge\alpha_r) = \prod_k \sum_j \mathrm{pr}_j^*\psi^*\alpha_k$, whose component in $H^1(C)^{\otimes r}$ is the alternating sum of the $\psi^*\alpha_1\otimes\cdots\otimes\psi^*\alpha_r$. The Künneth projectors of a curve are algebraic,[6] and so is the average e_G over G, so $e_G\circ\pi_1^{\otimes r}\circ\sigma^*$ is induced by an algebraic cycle on $B_{[a]}\times C^r$, and it maps $H^r(B_{[a]},\mathbb{Q})$ onto $W_{[a]}$: the alternating tensors in the $[a]$-part are the image of $\pi_1^{\otimes r}\circ\sigma^*$, and e_G keeps those of one character. A surjective morphism of polarisable Hodge structures maps Hodge classes onto Hodge classes,[7] so the Hodge classes of $W_{[a]}$ are algebraic when those of $B_{[a]}$ are. Now let x be a Hodge class of degree r on $X = X_{d,r}(\lambda)$, the other degrees being multiples of powers of the hyperplane class by the Lefschetz hyperplane theorem. Then Φ^*x is a G-invariant Hodge class, its components in the $W_{[a]}$ and along η are Hodge classes, being cut out by the algebraic projectors attached to the action of H^r, and they are algebraic; so Φ^*x is algebraic, and so is $x = \Phi_*\Phi^*x/|G|$. Finally $\dim B_{[a]}$ is half of $\sum_{a'\in[a]}\dim V_{a'} = |[a]|\dim V_a$. ∎

Corollary 2.56 (Two diagonal hypersurfaces)**.**

(i) *Every smooth complete intersection of two diagonal hypersurfaces* $\sum_i a_i x_i^d = \sum_i b_i x_i^d = 0$ *of the same degree in* $\mathbb{P}^N$ *is of Vandermonde type, so it lies in* $\mathcal{A}$ *and satisfies* (F3′) *in every degree.*

(ii) *Let* $d\in\{3,4,6\}$ *and* $N\le 6$*, or* $d=2$ *and* $N\le 12$*. Then the Hodge conjecture holds on every complete intersection of Vandermonde type in* $\mathbb{P}^N$ *of* $N-r$ *diagonal hypersurfaces of degree* d*. In particular it holds on every smooth complete intersection of two diagonal hypersurfaces of degree* 3*,* 4 *or* 6 *in* $\mathbb{P}^6$*.*

(iii) *For* $d\in\{3,4,6\}$ *and* $N=6$*, each orbit* $[a]$ *of order at least three with six nonzero coordinates and* $\sum_i\langle a_i/d\rangle = 3$ *makes* $B_{[a]}$ *an abelian fourfold of Weil type for* $K = \mathbb{Q}(\zeta_d)$*, and* $W_{[a]}$ *is its space of Weil classes. On each of these fourfolds* X *the Weil classes of the* 70*,* 490 *and* 6125 *such orbits, for* $d=3,4,6$*, give Hodge classes of degree four outside* $\mathbb{Q}h^2$*, so the Hodge classes of* X *form a space of dimension at least* 141*,* 981 *and* 12251*, whereas the divisor classes of* X *are the multiples of* h*.*

Proof. (i) If two columns (a_i,b_i) and (a_j,b_j) were proportional, or one of them zero, X would contain a point supported on $\{i,j\}$, or the coordinate point e_i, at which the Jacobian has rank at most one; so the $N+1$ columns are distinct points of $\mathbb{P}^1$. Sending a point of $\mathbb{P}^1$ outside them to infinity writes them as $\mu_i(1,\lambda_i)$ with distinct λ_i, which is the Vandermonde form for $c=2$, and Theorem 2.55(iii) applies. (ii) For $d\in\{2,3,4,6\}$ every a has order 1, 2, 3, 4 or 6, so $|[a]|\le 2$, with $|[a]|=1$ for the a of order two, whose V_a is real, so that $\dim B_{[a]}\le\dim V_a\le N-1$. When $d=2$ every $a\ne 0$ has order two and an even number of nonzero coordinates, at most 12 when $N\le 12$, so $\dim B_{[a]} = \frac12\dim V_a\le 5$. So every $B_{[a]}$ has dimension at most five,[8] and the Hodge conjecture holds on abelian varieties of dimension at most five [Mar25a, Mar25b, MZ99]; Theorem 2.55(iv) gives the conclusion, and (i) the last statement. (iii) For such a the eigenspaces V_a and V_{-a} have dimension four

[6] For a curve C with a point o they are $\pi_0=[o\times C]$, $\pi_2=[C\times o]$ and $\pi_1=[\Delta_C]-\pi_0-\pi_2$, and $\pi_1^{\otimes r}$ is their exterior product on $C^r\times C^r$. The average $e_G = |G|^{-1}\sum_{g\in G}[\Gamma_g]$ of the graphs of the elements of G acts on $H^*(C^r)$ as the projector onto the invariants, and Φ^* identifies $H^*(X_{d,r}(\lambda),\mathbb{Q})$ with $H^*(C^r,\mathbb{Q})^G$.

[7] The category of polarisable rational Hodge structures is semisimple, so the kernel of a surjective morphism $f\colon V\to W$ has a complement V', a sub-Hodge structure that f maps isomorphically onto W; a Hodge class of W is then the image of a Hodge class of V'.

[8] Both bounds are sharp: for $d=3$ and $N=7$ a character with eight nonzero coordinates has $\dim V_a=6$ and $|[a]|=2$, so $\dim B_{[a]}=6$; for $d=2$ and $N=13$ a character with fourteen nonzero coordinates has $\dim B_{[a]}=6$.

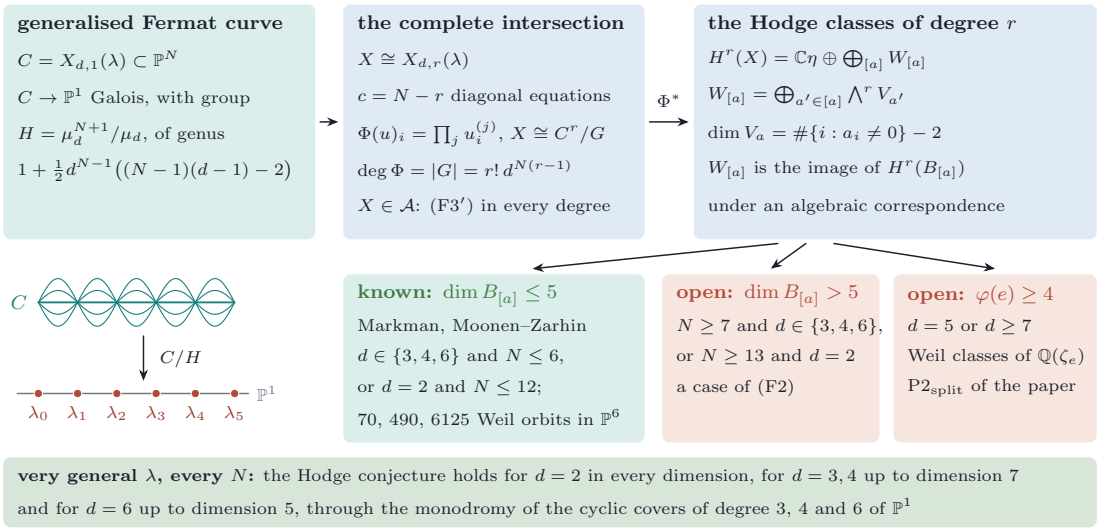


FIGURE 7. Theorem 2.55 and Corollary 2.56. Left: the generalised Fermat curve C over $\mathbb{P}^1$, its sheets meeting over the branch points $\lambda_0, \dots, \lambda_N$ (five sheets stand for the d^N of the cover). Top: C, the complete intersection $X \cong C^r/G$ with the map Φ of degree $|G|$, and the decomposition of $H^r(X)$ through Φ^*. Bottom: where the Hodge conjecture for X stands at an arbitrary configuration, according to the abelian varieties $B_{[a]}$ that carry its Hodge classes: known when they have dimension at most five, which happens for $d \in \{3,4,6\}$ and $N \leq 6$ and for $d = 2$ and $N \leq 12$, and open when they are larger, a case of (F2), or carry Weil classes of $\mathbb{Q}(\zeta_e)$ with $\varphi(e) \geq 4$, the case $\mathrm{P2}_{\mathrm{split}}$ of the paper. Foot: the very general configuration, Theorem 2.66.

and contain two-dimensional parts of type $(1,0)$, so K acts on $H^{1,0}(B_{[a]})$ with multiplicities $(2,2)$ and $B_{[a]}$, of dimension four, is of Weil type; $\bigwedge^4 V_a$ and $\bigwedge^4 V_{-a}$ are its Weil lines, of type $(2,2)$ for every λ, and they lie in $H^4(C^4)^G = \Phi^* H^4(X)$. The zero coordinate of a is one of seven, and the other six $a_i \in \{1, \dots, d-1\}$ add up to $3d$, which they do in 20, 141 and 1751 ways for $d = 3, 4, 6$, the coefficients of x^{3d-6} in $(1 + x + \cdots + x^{d-2})^6$; removing the character of order two with every $a_i = d/2$ leaves 20, 140 and 1750, and $(\mathbb{Z}/d)^\times = \{\pm 1\}$ pairs them off, since $-a$ again has coordinate sum $3d$; this gives $7 \cdot 20/2 = 70$, $7 \cdot 140/2 = 490$ and $7 \cdot 1750/2 = 6125$ orbits. Distinct orbits give independent classes, and the hyperplane class lies in the part of the trivial character, which gives the bounds; the Picard group of a complete intersection of dimension at least three is generated by h. ■

Figure 7 draws the construction. For $c = 1$ the theorem is Proposition 2.42(iii) again, by a different route, and for $r = 1$ it restates the definition of C.

At a very general configuration the abelian varieties $B_{[a]}$ carry only the Hodge classes that the monodromy of the curves allows, and the conjecture can then be decided in every dimension for $d = 2$, up to dimension seven for $d = 3$ and $d = 4$, and up to dimension five for $d = 6$. Write $U \subset \mathbb{C}^{N+1}$ for the space of $(N+1)$-tuples of distinct numbers. The varieties $X_{d,r}(\lambda)$ with $\lambda \in U$ form a smooth projective family over U, and so do the curves C and the maps Φ, with H acting on the family of curves.

Lemma 2.57 (Hodge classes of a very general member)**.** *Let $\mathbb{V}$ be a polarised variation of $\mathbb{Z}$-Hodge structure of weight $2p$ on a smooth connected quasi-projective variety S, and let $\Pi = \pi_1(S, s)$ act on $\mathbb{V}_s$. If s lies outside a countable union of proper Zariski closed subsets of S, the space $\mathrm{Hdg}(\mathbb{V}_s)$ of rational classes of type (p, p) is stable under Π, and a subgroup of finite index of Π fixes it pointwise.*

Proof. The locus of Hodge classes has countably many components ([I, Definition B.3], [I, Lemma B.4]), and the image in S of each is Zariski closed ([I, Theorem B.6]); let s avoid the images that are not all of S. For $\alpha \in \mathrm{Hdg}(\mathbb{V}_s)$ the component through (s, α) maps onto S, locally biholomorphically onto its image, so the flat section $\tilde{\alpha}$ through α on the universal cover $\tilde{S}$ is of type (p, p) on a nonempty open set. The set where it is of type (p, p) is a closed analytic subset of the connected manifold $\tilde{S}$, so it is all of $\tilde{S}$, and every transport of α along a loop at s is a Hodge class at s. The group Π therefore preserves $\mathrm{Hdg}(\mathbb{V}_s)$, the lattice $\mathbb{V}_{s,\mathbb{Z}} \cap \mathrm{Hdg}(\mathbb{V}_s)$ and the flat form $\tilde{Q}$ of the proof of [I, Lemma B.5], which is positive definite on the real classes of type (p, p); so it acts on $\mathrm{Hdg}(\mathbb{V}_s)$ through a finite group, whose kernel is the subgroup of the statement. ∎

For the characters of order three, four and six the monodromy of the curves D_a is determined by induction on the number of branch points. The step lets two branch points collide, and a lemma on two copies of SL of a hyperplane closes it. The base has four branch points, two of them with opposite exponents, whose full twist is a transvection (Figure 8).

Setup 2.58 (Cyclic covers)**.** Let $m \in \{3, 4, 6\}$, $\zeta = e^{2\pi i/m}$ and $K = \mathbb{Q}(\zeta)$, and let $a = (a_1, \dots, a_k)$ with $a_i \in \mathbb{Z}/m$ nonzero, $\sum_i a_i = 0$ and $\gcd(m, a_1, \dots, a_k) = 1$, so that a has order m. For λ in the space Conf_k of k-tuples of distinct complex numbers let D_a be the cyclic cover $y^m = \prod_i (t - \lambda_i)^{a_i}$ of $\mathbb{P}^1$, with representatives $0 < a_i < m$; it is unramified over ∞. Let $V_a \subset H^1(D_a, \mathbb{C})$ be the eigenspace of the deck group on which $H^{1,0}$ is spanned by the forms $t^j \prod_i (t - \lambda_i)^{-b_i/m}\, dt$, where $0 < b_i < m$, $b_i \equiv -a_i$ and $0 \le j \le \sum_i b_i/m - 2$ [DM86, §2], and put $n = \dim V_a$, $p = \dim V_a^{1,0} = \sum_i b_i/m - 1$ and $q = n - p$. Let $\Gamma_a \subset \mathrm{GL}(V_a)$ be the image of the pure braid group $P_k = \pi_1(\mathrm{Conf}_k)$ and G_a^0 the identity component of its Zariski closure.

Lemma 2.59 (A model of the monodromy)**.** *Let F_k be the free group on loops $x_1, \dots, x_k$ around $\lambda_1, \dots, \lambda_k$, put $t_i = \zeta^{a_i}$, let $\partial\colon K^k \to K$ map e_i to $t_i - 1$, and let $E_a = \ker \partial / K\ell$, where $\ell = \sum_i t_1 \cdots t_{i-1} e_i$. A pure braid β acts on F_k by Artin's automorphism and on K^k by $v \mapsto vJ_\beta$, where J_β is the matrix of Fox derivatives $\partial\beta(x_i)/\partial x_j$ evaluated at $x_j \mapsto t_j$. This action preserves $\ker \partial$ and fixes ℓ, and the representation of P_k on E_a, of dimension $k - 2$, is isomorphic to V_a, to its dual or to its complex conjugate. So $n = k - 2$, and $G_a^0 \supseteq \mathrm{SL}(V_a)$ if and only if the identity component of the Zariski closure of the image of P_k in $\mathrm{GL}(E_a)$ contains $\mathrm{SL}(E_a)$.*

Proof. Fox's formula $\sum_j (\partial w/\partial x_j)(x_j - 1) = w - 1$ [Fox53], evaluated at $x_j \mapsto t_j$, shows that the vector $\sum_j (\partial w/\partial x_j)(t)\, e_j$ of a word w with $t(w) = 1$ lies in $\ker \partial$. It is the class of the lift of the loop w in the summand of $H_1(D_a^\circ, K)$ on which the deck group acts through $x_i \mapsto t_i$, where D_a° is the unramified part of the cover, and that summand is $H_1(F_k, K_t) = \ker \partial$, computed from the free resolution of length one of the augmentation ideal of a free group.[9] Filling in the points over λ_i adds no relation: the loop around such

[9] By Shapiro's lemma $H_1(D_a^\circ, K) = H_1(F_k, K[\mu_m])$, and its summand for the character $x_i \mapsto t_i$ is $H_1(F_k, K_t)$. The augmentation ideal of $\mathbb{Z}[F_k]$ is free on the $x_i - 1$, so $H_1(F_k, K_t)$ is the kernel of $\bigoplus_i K e_i \to K$, $e_i \mapsto t_i - 1$, the chain e_i standing for the loop x_i and a word w for $\sum_j (\partial w/\partial x_j)(t) e_j$.

a point lifts x_i^e, e the order of t_i, whose vector $(1+t_i+\cdots+t_i^{e-1})e_i$ is zero. Filling in the m points over ∞, where the cover is unramified, kills the lift of $x_1\cdots x_k$, whose vector is $\ell\neq 0$. So E_a is the summand of $H_1(D_a,K)$ for the character $x_i\mapsto t_i$, of dimension $k-2$. A pure braid fixes $x_1\cdots x_k$ and maps each x_i to a conjugate of itself, so it preserves t, and the chain rule for Fox derivatives makes $\beta\mapsto J_\beta$ the action of P_k on this summand.[10] Poincaré duality and the choice of the generator of the deck group identify it with V_a, its dual or its conjugate, and the automorphisms $g\mapsto {}^tg^{-1}$ and $g\mapsto\bar g$ of GL preserve SL. ■

Lemma 2.60 (Collision of two branch points)**.** *In Lemma 2.59 suppose that $a_1+a_2\neq 0$, and let $a'=(a_1+a_2,a_3,\dots,a_k)$. Let $I=(1-t_2)e_1-(1-t_1)e_2$, the cycle of a disc around λ_1 and λ_2, and let $Y\subset E_a$ be the image of the cycles outside that disc, the vectors of $\ker\partial$ with $v_2=t_1v_1$. Then $E_a=Y\oplus KI$, and $\iota(v_1,v_3,\dots,v_k)=(v_1,t_1v_1,v_3,\dots,v_k)$ induces an isomorphism $E_{a'}\to Y$. If $\beta\in P_k$ is obtained from $\beta'\in P_{k-1}$ by doubling its first strand, then β preserves Y and KI, acts on Y as β' acts on $E_{a'}$, and multiplies I by a power of ζ.*

Proof. The vectors of Y have (v_1,v_2) proportional to $(1,t_1)$, and $(1-t_2,t_1-1)$ is proportional to $(1,t_1)$ only if $t_1t_2=1$; so $I\notin Y$. The map ι carries the kernel of ∂', where $\partial'(e_1)=t_1t_2-1$, onto the cycles outside the disc, because $v_1(t_1t_2-1)=v_1(t_1-1)+t_1v_1(t_2-1)$, and it carries the vector ℓ' of a' to ℓ; as $\dim E_{a'}+1=\dim E_a$, both assertions about Y follow. The doubled braid moves a disc containing λ_1,λ_2 rigidly. Its automorphism therefore maps $x_1\mapsto wx_1w^{-1}$, $x_2\mapsto wx_2w^{-1}$ and $x_j\mapsto u_j$ for $j\geq 3$, where $y\mapsto w'yw'^{-1}$, $x_j\mapsto u'_j$ is the automorphism of β' on the free group on $y,x_3,\dots,x_k$, and w,u_j are w',u'_j with y replaced by x_1x_2. For such a word u the chain rule gives $\partial u/\partial x_1=\partial u/\partial y$ and $\partial u/\partial x_2=(\partial u/\partial y)\,x_1$, which at t is the relation $v_2=t_1v_1$; together with $\partial(wx_iw^{-1})/\partial x_j=(1-t_i)\,\partial w/\partial x_j+t(w)\delta_{ij}$ at t, this shows that the rows of J_β for $j\geq 3$, and the first row plus t_1 times the second, are the images under ι of the rows of $J_{\beta'}$; this is the equivariance of ι. The same formula gives $IJ_\beta=(1-t_2)((1-t_1)\partial w+t(w)e_1)-(1-t_1)((1-t_2)\partial w+t(w)e_2)=t(w)I$, and $t(w)$ is a power of ζ. ■

Lemma 2.61 (Two hyperplanes)**.** *Let $E=Y\oplus L$ be a complex vector space with $\dim Y\geq 2$ and $\dim L=1$, let $S\subset\mathrm{SL}(E)$ be the subgroup acting on Y through $\mathrm{SL}(Y)$ and trivially on L, and let $\gamma\in\mathrm{GL}(E)$ satisfy $\gamma L\neq L$ and $\gamma Y\neq Y$. Then the group T generated by S and $\gamma S\gamma^{-1}$ acts irreducibly on E. If $\dim Y\geq 3$, then $T=\mathrm{SL}(E)$. If $\dim Y=2$, every connected semisimple subgroup of $\mathrm{SL}(E)$ that contains T is $\mathrm{SL}(E)$.*

Proof. The group T is connected. As a representation of S, E is the sum of the irreducible nontrivial Y and the trivial L, so its S-stable subspaces are 0, L, Y and E, and those of $\gamma S\gamma^{-1}$ are 0, γL, γY and E; by the hypotheses and the dimensions only 0 and E are in both lists, and T is irreducible.

Let $\dim Y\geq 3$. As a representation of $\mathfrak{sl}(Y)$,

$$\mathfrak{sl}(E)=\mathfrak{sl}(Y)\oplus\mathrm{Hom}(L,Y)\oplus\mathrm{Hom}(Y,L)\oplus\mathbb{C}z,$$

z acting by 1 on Y and by $-\dim Y$ on L, and the four summands, the adjoint representation, Y, Y^* and the trivial one, are pairwise non-isomorphic since $\dim Y\geq 3$. The Lie algebra $\mathfrak{t}$ of T contains $\mathfrak{sl}(Y)$ and is stable under it, so it is $\mathfrak{sl}(Y)$ plus some of the other three summands. Without $\mathrm{Hom}(L,Y)$ it preserves L, and without $\mathrm{Hom}(Y,L)$ it preserves Y;

[10] Artin's automorphism of the half twist σ_i of the i-th and $(i+1)$-st points is $x_i\mapsto x_ix_{i+1}x_i^{-1}$, $x_{i+1}\mapsto x_i$, the other generators being fixed; P_k is generated by the full twists $A_{st}=\sigma_{t-1}\cdots\sigma_{s+1}\sigma_s^2\sigma_{s+1}^{-1}\cdots\sigma_{t-1}^{-1}$. For a pure braid the chain rule reads $J_{\beta\beta'}=J_{\beta'}J_\beta$ or $J_\beta J_{\beta'}$, according to the convention for composing braids, with no twist of the coefficients, since $t\circ\beta=t$.

both are excluded by irreducibility. The bracket of a nonzero map $L \to Y$ with a map $Y \to L$ not vanishing on its image has a nonzero component along z, so $\mathfrak{t} = \mathfrak{sl}(E)$ and $T = \mathrm{SL}(E)$.

Let $\dim Y = 2$ and let $G \supseteq T$ be connected and semisimple. Then E is a faithful irreducible representation of dimension 3 of the Lie algebra $\mathfrak{g}$ of G. An irreducible representation of a product of simple Lie algebras is a tensor product of irreducible representations of the factors, and 3 is prime, so $\mathfrak{g}$ is simple. It lies in $\mathfrak{sl}(E)$, of dimension 8 and rank 2, and the simple Lie algebras of rank at most 2 are $\mathfrak{sl}_2$, $\mathfrak{sl}_3$, $\mathfrak{so}_5$ and $\mathfrak{g}_2$, of dimensions 3, 8, 10 and 14. If $\mathfrak{g} \cong \mathfrak{sl}_2$, then $\mathfrak{g}$ equals the Lie algebra of S, both being of dimension 3, so $G = S$, which preserves L, against the irreducibility of T. Hence $\mathfrak{g} = \mathfrak{sl}(E)$ and $G = \mathrm{SL}(E)$. ∎

Lemma 2.62 (Merging two branch points)**.** *Let m and a be as in Setup 2.58, with $k \geq 6$ and $p, q \geq 1$. Then there are $i \neq j$ with $a_i + a_j \neq 0$ such that the vector a' obtained by replacing a_i and a_j by $a_i + a_j$ has order m, and its numbers p' and q' are at least 1.*

Proof. Put $f_i = b_i/m \in [1/m, 1 - 1/m]$, so that $p + 1 = \sum_i f_i$ and $q + 1 = \sum_i (1 - f_i)$. Merging a_i and a_j keeps p and lowers q by one if $f_i + f_j < 1$, and lowers p by one and keeps q if $f_i + f_j > 1$. Since $a_i + a_j = -\sum_{l \neq i,j} a_l$, the order of a' is that of the subgroup generated by the a_l with $l \neq i, j$, the rest of a. Replacing a by $-a$ exchanges p and q, so we may assume $q \geq p$, hence $q \geq 2$ as $p + q = k - 2 \geq 4$; if $p \geq 2$, every merge keeps $p', q' \geq 1$ and only the order matters.

The order. For $m = 3$ two equal coordinates exist, their sum is not 0, and any nonzero rest generates $\mathbb{Z}/3$. For $m = 4$ the number of odd coordinates is even and positive; if it is at least 4, merge two equal odd ones, and if it is 2, merge an odd one with one of the at least four coordinates equal to 2. For $m = 6$ the rest generates $\mathbb{Z}/6$ exactly when it contains an odd coordinate and one prime to 3. The odd coordinates are even in number and the coordinates prime to 3 have sum divisible by 3, so each kind occurs at least twice. If exactly two coordinates are odd, the at least four others lie in $\{2, 4\}$; merge two equal ones among them. If exactly two are prime to 3, the others equal 3; merge a 3 with one of the two. Otherwise each kind occurs at least three times, and any two coordinates with nonzero sum can be merged; they exist, since pairwise sums 0 would force every coordinate to be 3.

The case $p = 1$. Then $\sum_i b_i = 2m$. Some b_i equals 1, since otherwise $\sum_i b_i \geq 2k \geq 12 \geq 2m$, with equality only if $m = 6$ and every coordinate is 4, when a has order three. Keep such a coordinate, $a_i = -1$, which generates $\mathbb{Z}/m$. The other $k - 1 \geq 5$ coordinates have b-sum $2m - 1$, so the two smallest values of b among them add up to at most $2(2m - 1)/5 < m$; merge these two coordinates. Their sum is not 0, $p' = p = 1$, $q' = q - 1 = k - 4 \geq 2$, and the rest contains a_i, so a' has order m. ∎

Lemma 2.63 (Five branch points)**.** *Let m and a be as in Setup 2.58, with $k = 5$ and $p, q \geq 1$. Then $a_i + a_j = 0$ for some $i \neq j$, and the other three coordinates can be labelled x, y, z so that $x + y \neq 0$ and $a' = (x + y, z, a_i, a_j)$ has order m and $p' = q' = 1$.*

Proof. Replacing a by $-a$ we may assume $p = 1$, so that $\sum_l b_l = 2m$ with five terms in $[1, m - 1]$; and $a_i + a_j = 0$ means $b_i + b_j = m$. For $m = 3$ the terms lie in $\{1, 2\}$ and exactly one is 2; take it and a 1. For $m = 4$, take a 3 and a 1 if both occur. If no term is 3, the terms lie in $\{1, 2\}$ and at least three are 2, which gives $2 + 2$; if no term is 1, they lie in $\{2, 3\}$ and add up to at least 10, which is impossible. For $m = 6$ suppose that no two terms add up to 6. Then at most one term is 3, and neither 1 and 5 nor 2 and 4 both occur. Terms in $\{1, 2, 3\}$ with at most one 3 add up to at most 11, and terms in $\{3, 4, 5\}$ to

at least 19. Terms in $\{1,3,4\}$, respectively $\{2,3,5\}$, cannot add up to 12, since the number y of terms 4, respectively 5, would satisfy $3y \in \{7,5\}$, respectively $3y \in \{2,1\}$. This proves the first assertion.

The other three coordinates add up to 0 and have b-sum m, so any two of them, x and y, have $x+y=-z\neq 0$ and b-sum less than m; merging them keeps $p'=1$, so $q'=1$, and a' contains $c=a_i$ and $a_j=-c$. The order of a' is that of the subgroup generated by z and c. If c generates $\mathbb{Z}/m$, any choice of z will do. Otherwise c is even, or $m=6$ and $c=3$. As a has order m, one of x,y,z is then odd, respectively prime to 3, and we take it as z; then z and c generate $\mathbb{Z}/m$. ∎

Proposition 2.64 (Monodromy of cyclic covers)**.** *In Setup 2.58, if $p=0$ or $q=0$ then Γ_a is finite, and if $n\geq 3$ and $p,q\geq 1$ then $G_a^0 \supseteq \mathrm{SL}(V_a)$.*

Proof. If $pq=0$, the polarisation is definite on V_a up to sign, and Γ_a preserves it and the $\mathbb{Z}[\zeta]$-lattice spanned by $H^1(D_a,\mathbb{Z})$; a discrete subgroup of a compact group is finite. Assume $p,q\geq 1$ and use the model of Lemma 2.59. Three facts are used throughout. If $u\in\Gamma_a$ is unipotent, the Zariski closure of its powers is the group $\exp(\tau\log u)$, so $\log u$ lies in the Lie algebra of G_a^0. The group Γ_a normalises G_a^0. And G_a^0 is semisimple: by André's theorem [And92, Theorem 1] the identity component of the Zariski closure of the monodromy group of $H^1(D_a,\mathbb{Q})$ is a normal subgroup of the derived group of the Mumford–Tate group of a very general fibre, hence semisimple, and G_a^0 is its image in $\mathrm{GL}(V_a)$, the monodromy commuting with the deck group.

Four points. Let $k=4$, so $n=2$, and suppose that two coordinates of a are opposite; relabel them a_1 and a_2, so that $t_1t_2=1$. The full twist A_{12} maps $x_i\mapsto wx_iw^{-1}$ for $i=1,2$, with $w=x_1x_2$, and fixes x_3,x_4. As $t(w)=1$, the formula for $\partial(wx_iw^{-1})/\partial x_j$ in the proof of Lemma 2.60 gives $vJ_{A_{12}}=v+\varphi(v)\,\delta$, with $\delta=e_1+t_1e_2$ and $\varphi(v)=(1-t_1)v_1+(1-t_2)v_2$, which equals $\sum_{j\geq 3}(t_j-1)v_j$ on $\ker\partial$. Here $\delta\in\ker\partial$, $\varphi(\delta)=0$, $\varphi(\ell)=t_1\cdots t_k-t_1t_2=0$, the vector δ is not a multiple of ℓ, whose third coordinate is $t_1t_2=1$, and $\varphi((t_3-1)e_1-(t_1-1)e_3)=-(t_1-1)(t_3-1)\neq 0$. So A_{12} acts on E_a as a unipotent $u\neq 1$, and G_a^0 is a connected semisimple subgroup of $\mathrm{SL}(E_a)\cong\mathrm{SL}_2$ containing $\exp(\tau\log u)$. The connected subgroups of SL_2 of dimension one or two are solvable, so $G_a^0=\mathrm{SL}(E_a)$.

The induction. Let $n\geq 3$. If $n=3$, Lemma 2.63 gives, after a relabelling, a vector $a'=(a_1+a_2,a_3,a_4,a_5)$ of order m with $a_1+a_2\neq 0$, $p'=q'=1$ and two opposite coordinates, so the identity component of the Zariski closure of the monodromy of $E_{a'}$ is $\mathrm{SL}(E_{a'})$ by the case of four points. If $n\geq 4$, Lemma 2.62 and a relabelling give $a_1+a_2\neq 0$ and $a'=(a_1+a_2,a_3,\ldots,a_k)$ of order m with $p',q'\geq 1$, and by induction on n that identity component contains $\mathrm{SL}(E_{a'})$. Let H be the identity component of the Zariski closure of the doubled braids of Lemma 2.60. Then $H\subseteq G_a^0$, H preserves Y and KI and acts trivially on KI, where the doubled braids act through a finite group, and its image in $\mathrm{GL}(Y)$ is the identity component of the Zariski closure of the monodromy of $E_{a'}$, which contains $\mathrm{SL}(Y)$. So G_a^0 contains the group S of Lemma 2.61 for $E_a=Y\oplus KI$. Let $\gamma=A_{23}$, the full twist of the second and third points; its automorphism fixes x_1 and maps $x_2\mapsto x_2x_3x_2x_3^{-1}x_2^{-1}$ and $x_3\mapsto x_2x_3x_2^{-1}$. Then $\gamma(I)=I+(1-t_1)t_2I_{23}$ with $I_{23}=(1-t_3)e_2-(1-t_2)e_3$, and $\gamma(KI)\neq KI$, since I, I_{23} and ℓ are independent, ℓ having a nonzero fourth coordinate. The functional $v\mapsto v_2-t_1v_1$ vanishes on ℓ and cuts out Y, and at $\gamma(v)$ for $v\in Y$ it equals $(t_3-1)(t_1t_2v_1-v_3)$, which is nonzero for the cycle $(1-t_4)e_3-(1-t_3)e_4$ of Y; so $\gamma Y\neq Y$. As γ normalises G_a^0, this group contains S and $\gamma S\gamma^{-1}$, and Lemma 2.61,

with $\dim Y = n-1$, gives $G_a^0 = \mathrm{SL}(E_a)$: for $n \geq 4$ directly, and for $n = 3$ because G_a^0 is connected and semisimple. ∎

For $m = 3$ the proposition also follows from the ℓ-adic theorem of Achter and Pries [AP07, Corollary 3.10], which identifies the monodromy of trielliptic curves with a special unitary group.

Lemma 2.65 (The discriminant of the new part)**.** *In Setup 2.58 let n be even with $p = q = n/2$, let $L = H^1(D_a, \mathbb{Z})$ with the cup product Q, and let $L_a = L \cap (V_a \oplus \overline{V_a})$. Then L_a is a free $\mathbb{Z}[\zeta]$-module of rank n, the K-hermitian form H with $Q = \frac{1}{2}\operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1}H)$ on it, $\delta = \sqrt{-1}$ for $m = 4$ and $\delta = \sqrt{-3}$ otherwise ([I, Lemma 4.5]), has signature $(n/2, n/2)$, and the class of $(-1)^{n/2}\det H$ in $\mathbb{Q}^\times/\mathrm{N}(K^\times)$ is trivial for $m = 3$ and $m = 4$; for $m = 6$ it is trivial if and only if the number of i with $a_i = 3$ is even.*

Proof. The deck transformation g is an isometry of the unimodular lattice L, so its eigenspaces for eigenvalues μ, ν are orthogonal unless $\mu\nu = 1$. Hence L_a, the part for the primitive m-th roots of unity, and the part L^{old} for the other eigenvalues are orthogonal primitive sublattices, each the orthogonal complement of the other, and $|\det(L_a)| = |\det(L^{\mathrm{old}})|$, which is 1 for $m = 3$, when $L^{\mathrm{old}} = 0$.[11] For a quotient map $\pi\colon D_a \to D'$ of prime degree e that is ramified, $\pi^* H^1(D', \mathbb{Z})$ is primitive in L and the cup product on it is e times that of D'.[12] For $m = 4$, $L^{\mathrm{old}} = \pi^* H^1(D', \mathbb{Z})$ for the quotient D' by g^2, ramified over the λ_i with a_i odd, so $|\det L^{\mathrm{old}}| = 2^{2g'}$, g' the genus of D'. For $m = 6$, the quotients D_2 by g^2 and D_3 by g^3 are ramified over the λ_i with $3 \nmid a_i$ and with a_i odd, which exist since a has order six, and L^{old} contains $A \oplus B$ with $A = \pi_2^* H^1(D_2, \mathbb{Z})$ and $B = \pi_3^* H^1(D_3, \mathbb{Z})$, of determinants 3^{2g_2} and 2^{2g_3}; the quotient $L^{\mathrm{old}}/(A \oplus B)$ embeds in $A^\vee/A$ and in $B^\vee/B$, of coprime orders, so $L^{\mathrm{old}} = A \oplus B$ and $|\det L^{\mathrm{old}}| = 3^{2g_2}2^{2g_3}$.

On L_a the deck transformation satisfies the cyclotomic equation of ζ, so L_a is a torsion-free module over the principal ideal domain $\mathbb{Z}[\zeta]$, free of rank n; the signature of H is (p, q) ([I, Remark 4.6]). The determinant of the trace form of a hermitian lattice gives $|\det(L_a)| = |d_K|^n \mathrm{N}(\frac{1}{2\delta})^n (\det H)^2$, that is $(\det H)^2$ for $K = \mathbb{Q}(i)$ and $(\det H)^2/4^n$ for $K = \mathbb{Q}(\sqrt{-3})$.[13] Hence $(-1)^{n/2}\det H$ equals 2^n for $m = 3$, $2^{g'}$ for $m = 4$ and $2^{n+g_3}3^{g_2}$ for $m = 6$, the sign being that of a form of signature $(n/2, n/2)$. For $m = 3$ this is $\mathrm{N}(2^{n/2})$, and for $m = 4$ a norm as $2 = \mathrm{N}(1+i)$. For $m = 6$, $3 = \mathrm{N}(\sqrt{-3})$ and $4 = \mathrm{N}(2)$ are norms and 2 is not, since $x^2 + xy + y^2$ is even only when x and y are, so that every norm has even 2-adic valuation; so the class is trivial exactly when g_3 is even. By the Riemann–Hurwitz formula D_3, the triple cover $y^3 = \prod_i (t - \lambda_i)^{a_i}$ branched over the i with $3 \nmid a_i$, has genus $k - c_3 - 2 = n - c_3$, where c_3 is the number of i with $a_i = 3$; as n is even, this is the assertion. ∎

[11] If $M \subset L$ is primitive and L is unimodular, restriction $L \to M^\vee$ is onto with kernel $M^\perp$, so $L/(M \oplus M^\perp) \cong M^\vee/M$, and likewise $\cong (M^\perp)^\vee/M^\perp$; the order of $M^\vee/M$ is $|\det M|$.

[12] A loop in D' avoiding the branch points, composed with a loop around a branch point, whose monodromy generates the group of order e, lifts to a closed loop with the same image in $H_1(D', \mathbb{Z})$; so $\pi_*\colon H_1(D_a, \mathbb{Z}) \to H_1(D', \mathbb{Z})$ is onto, and its transpose π^* is a split injection. The factor e is $\pi_*\pi^* = e$.

[13] For a free $\mathcal{O}_K$-module with hermitian Gram matrix M and $c \in K^\times$, the $\mathbb{Z}$-bilinear form $\operatorname{Tr}_{K/\mathbb{Q}}(c\, v^T M \bar{w})$ has determinant of absolute value $|d_K|^n \mathrm{N}(c)^n (\det M)^2$, where d_K is the discriminant of K: for $n = 1$ this is the discriminant of the trace form of $\mathcal{O}_K$ scaled by $\mathrm{N}(cM)$. Changing the basis by $P \in \mathrm{GL}_n(K)$ multiplies both sides by $\mathrm{N}_{K/\mathbb{Q}}(\det P)^2$, and a basis in which M is diagonal makes the form an orthogonal sum of forms of rank one. Here $d_K = -4$ and $\mathrm{N}(\frac{1}{2i}) = \frac{1}{4}$, or $d_K = -3$ and $\mathrm{N}(\frac{1}{2\sqrt{-3}}) = \frac{1}{12}$.

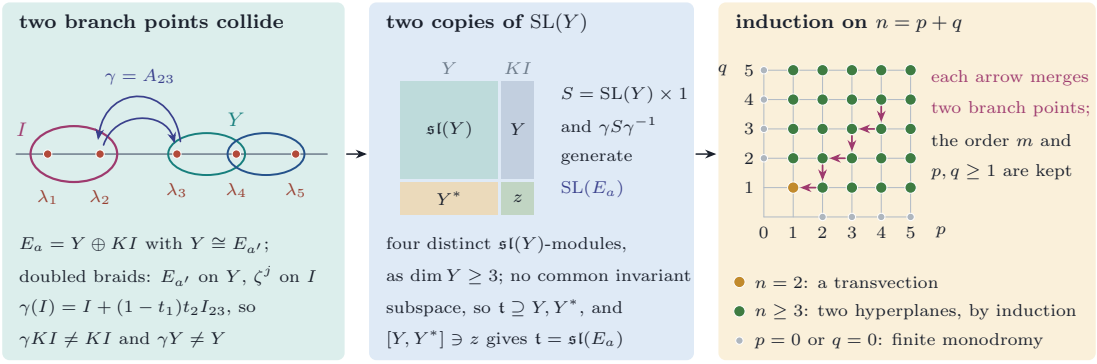


FIGURE 8. The proof of Proposition 2.64. Left: two branch points λ_1, λ_2 collide inside a disc (Lemma 2.60); the cycles outside the disc form the hyperplane $Y \cong E_{a'}$, the cycle I of the disc spans the complementary line, and the twist γ of the second and third points moves both. Middle: the two copies of $\mathrm{SL}(Y)$ of Lemma 2.61, which generate $\mathrm{SL}(E_a)$ when $\dim Y \geq 3$. Right: the induction on $n = k - 2$, each merge lowering p or q by one (Lemma 2.62, Lemma 2.63), down to $n = 2$, where the full twist of two points with opposite exponents is a transvection.

Theorem 2.66 (Very general diagonal complete intersections)**.** *Keep the notation of Theorem 2.55, let $r \geq 2$ be even, and let $\lambda \in U$ lie outside a countable union of proper Zariski closed subsets of U. For a Hodge class x of degree r on $X = X_{d,r}(\lambda)$, write $x_{[a]}$ for the component of $\Phi^* x$ in $W_{[a]}$.*

(i) *If a has order two, then $x_{[a]}$ is a rational multiple of the image of $\theta_a^{r/2}$, where θ_a is the class of a polarisation of $B_{[a]}$, under the correspondence of the proof of Theorem 2.55(iv); in particular $x_{[a]}$ is algebraic.*

(ii) *If a has order $m \in \{3, 4, 6\}$ and $x_{[a]} \neq 0$, then $\dim V_a = r$ and $\dim V_a^{1,0} = r/2$. In that case $B_{[a]}$ is an abelian variety of dimension r of Weil type for $\mathbb{Q}(\zeta_m)$, and $W_{[a]}$ is the image of its space of Weil classes. Its discriminant is trivial for $m = 3, 4$, and for $m = 6$ exactly when an even number of the coordinates of a equal $d/2$.*

(iii) *The Hodge conjecture holds on $X_{d,r}(\lambda)$ for every N when $d = 2$, when $d \in \{3, 4\}$ and $r \leq 6$, and when $d = 6$ and $r \leq 4$. For $d = 6$ and $r = 6$ the Hodge classes are algebraic modulo the spaces $W_{[a]}$ of the characters of order six of (ii) with an odd number of coordinates 3, which are spaces of Weil classes of abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$ and occur for every N. For $d \in \{2, 3, 4, 6\}$ the Hodge classes of degree r form a space of dimension*

$$1 + \binom{N+1}{r+2} T_d(r+2) + \epsilon_d \sum_{j \geq r/2+2} \binom{N+1}{2j},$$

where $T_d(k)$ is the number of $(s_1, \ldots, s_k) \in \{1, \ldots, d-1\}^k$ with $\sum_i s_i = dk/2$, and ϵ_d is 1 for d even and 0 for d odd; thus $T_2(k) = 1$, $T_3(k) = \binom{k}{k/2}$, $T_4(6) = 141$ and $T_6(6) = 1751$.

For odd r there is nothing to prove, since every Hodge class of X is then a multiple of a power of the hyperplane class. So the theorem gives the Hodge conjecture for the

very general complete intersection of Vandermonde type of diagonal quadrics in every dimension, of diagonal cubics or quartics up to dimension seven, and of diagonal sextics up to dimension five, in every $\mathbb{P}^N$.

Proof. Apply Lemma 2.57 to $H^r(X_{d,r}(\lambda),\mathbb{Z})$ over U, polarised by the hyperplane class, and let $\Pi'\subset\pi_1(U,\lambda)$ be the subgroup of finite index that fixes the Hodge classes. The monodromy commutes with H and with Φ^*, so Π' fixes $x_{[a]}$ and each of its parts in the spaces $\bigwedge^r V_{a'}$. Let m be the order of a, take representatives $0\le a_i<d$, and put $a_i'=ma_i/d$. The function $y=\prod_i u_i^{a_i}$ is invariant under the kernel of χ_a and satisfies $y^m=\prod_i(t-\lambda_i)^{a_i'}$, so the quotient of C by that kernel is the cyclic cover D_a of $\mathbb{P}^1$ of degree m given by this equation, and pull-back identifies $H^1(D_a,\mathbb{Q})$ with the part $\bigoplus_{j\neq0}V_{ja}$ of $H^1(C,\mathbb{Q})$, compatibly with the monodromy. The image $B_{[a]}$ is isogenous to the abelian subvariety of $J(D_a)$ whose first cohomology is $\bigoplus_{j\in(\mathbb{Z}/m)^\times}V_{ja}$, all of $J(D_a)$ for $m\le3$, its polarisation being a rational multiple of the restriction of the principal one, since pull-back along a map of degree e multiplies the cup product by e.

(i) For $m=2$, D_a is the hyperelliptic curve $y^2=\prod_{i\in S}(t-\lambda_i)$ over the support S of a, of genus $g=|S|/2-1$, and $H^1(D_a,\mathbb{Q})=V_a$. Move the base point so that the λ_i with $i\in S$ are real and consecutive, $\lambda_{s_1}<\cdots<\lambda_{s_{2g+2}}$, the others lying away from the segment they span. The loop in which λ_{s_k} and $\lambda_{s_{k+1}}$ turn once around each other, inside a disc that contains no other λ_i, makes D_a acquire a node with vanishing cycle c_k, the lift of the segment $[\lambda_{s_k},\lambda_{s_{k+1}}]$, and acts by the square of the Picard–Lefschetz transvection, $T_k^2(z)=z+2\langle z,c_k\rangle c_k$ up to orientation [Voi03, Chapter 3].[14] The classes $c_1,\dots,c_{2g}$ satisfy $\langle c_k,c_{k+1}\rangle=\pm1$ and $\langle c_k,c_l\rangle=0$ for $|k-l|\ge2$, so their intersection matrix has determinant 1 and they form a basis of $H_1(D_a,\mathbb{Q})$; they are the distinguished basis of type A of [ACa79]. Some power T_k^{2e} with $e\ge1$ lies in the image of Π'; it is unipotent, so the Zariski closure of that image contains the one-parameter group $\exp(\tau N_k)$, $N_k(z)=\langle z,c_k\rangle c_k$, and its Lie algebra contains $N_1,\dots,N_{2g}$. These generate the Lie algebra $\mathfrak{sp}(V_a)$.[15] So the Zariski closure contains $\mathrm{Sp}(V_a)$, whose invariants in $\bigwedge^r V_a$ are the multiples of $\omega_a^{r/2}$, ω_a the polarisation form [FH91, Theorem 17.5].[16] The correspondence of the proof of Theorem 2.55(iv) maps $H^r(B_{[a]},\mathbb{Q})=\bigwedge^r H^1(B_{[a]},\mathbb{Q})$ isomorphically and equivariantly onto $\bigwedge^r V_a$; the class θ_a goes to an invariant nondegenerate form, a nonzero multiple of ω_a, and $\theta_a^{r/2}$ to a nonzero multiple of $\omega_a^{r/2}$, which is algebraic.

(ii) For $m\in\{3,4,6\}$, D_a is the cyclic cover of Setup 2.58 for the exponents $a_i'\in\{1,\dots,m-1\}$ on the support S of a, and V_a is its eigenspace, of dimension $n=|S|-2$ (Lemma 2.59). The map from U to $\mathrm{Conf}_{|S|}$ that forgets the λ_i outside S is a fibration with connected fibres, so it is onto on fundamental groups, and Π' maps onto a subgroup of finite index of $P_{|S|}$; the Zariski closure of its image in $\mathrm{GL}(V_a)$ therefore contains G_a^0. If $n\le2$, then $\bigwedge^r V_a$ is zero unless $r=n=2$. If $V_a^{1,0}$ or $V_a^{0,1}$ is zero, then $\bigwedge^r V_a$ has type $(r,0)$ or

[14] Near the collision the curve is $y^2=(t-\lambda_{s_k})(t-\lambda_{s_{k+1}})\,v$ with v a unit, analytically $y^2=t^2-\epsilon$ with ϵ a multiple of $(\lambda_{s_{k+1}}-\lambda_{s_k})^2$; one turn of the two points around each other is two turns of ϵ around 0, and one turn of ϵ acts by the Dehn twist along the vanishing cycle, whose action on H_1 is the transvection T_k.

[15] Identify $\mathfrak{sp}(V)$ with Sym^2V, the product uv acting by $z\mapsto\langle z,u\rangle v+\langle z,v\rangle u$, so that $N_k=\frac12c_k^2$ and $[uv,wz]=\langle u,w\rangle vz+\langle u,z\rangle vw+\langle v,w\rangle uz+\langle v,z\rangle uw$ up to an overall sign. Then $[c_k^2,c_{k+1}^2]=\pm4c_kc_{k+1}$, and if all products c_ic_j with $i,j\le k$ lie in the generated algebra, so do the $c_ic_{k+1}=\pm\frac12[c_ic_k,c_{k+1}^2]$ for $i<k$, because c_{k+1} meets only c_k among $c_1,\dots,c_k$. By induction every c_ic_j lies in it, and these span Sym^2V.

[16] The exterior algebra of V is a representation of $\mathrm{Sp}(V)$ and of the $\mathfrak{sl}_2$ spanned by $\omega\wedge$ and its adjoint, and $\bigwedge^r V=\bigoplus_j\omega^j\wedge P^{r-2j}$, where P^k, the kernel of the contraction with ω in degree $k\le g$, is irreducible and nontrivial for $k\ge1$, so the invariants are the multiples of $\omega^{r/2}$, for r even.

$(0,r)$ and contains no class of type $(r/2,r/2)$. Otherwise $n\geq 3$, and Proposition 2.64 gives $G_a^0\supseteq \mathrm{SL}(V_a)$, whose invariants in $\bigwedge^r V_a$ vanish for $0<r<n$. In every case $x_{[a]}\neq 0$ forces $r=n$, and then $\bigwedge^r V_a$ is a line of type $(p_a,r-p_a)$ with $p_a=\dim V_a^{1,0}$, which contains a class of type $(r/2,r/2)$ only if $p_a=r/2$. The field $\mathbb{Q}(\zeta_m)$ then acts on $B_{[a]}$, of dimension $|[a]|\dim V_a/2=r$, with multiplicities $(r/2,r/2)$ on the tangent space and compatibly with the polarisation, the automorphism of D_a preserving the cup product; so $B_{[a]}$ is of Weil type ([I, Definition 3.1]), and $W_{[a]}=\bigwedge^r V_a\oplus\bigwedge^r V_{-a}$ is the image of $\bigwedge^r_{\mathbb{Q}(\zeta_m)}H^1(B_{[a]},\mathbb{Q})$, its space of Weil classes. Its discriminant ([I, Definition 15.22]) is that of the lattice L_a of Lemma 2.65: the isogeny with the abelian subvariety of $J(D_a)$ whose first cohomology is $L_a\otimes\mathbb{Q}$ scales the polarisation by a rational number and $\det H$ by its r-th power, a norm. The coordinates of a equal to $d/2$ are those with $a_i'=3$ when $m=6$, and Lemma 2.65 gives the assertion.

(iii) Let x be a Hodge class of degree r. The component of Φ^*x along η is algebraic. For $d\in\{2,3,4,6\}$ every $a\neq 0$ has order 2, 3, 4 or 6, and by (i) and (ii) the components $x_{[a]}$ are images of powers of polarisations or of Weil classes of abelian r-folds of Weil type for $\mathbb{Q}(\sqrt{-3})$ or $\mathbb{Q}(i)$. These are algebraic for $r=2$ by the theorem of Lefschetz, for $r=4$ by Markman's theorem on abelian fourfolds of Weil type, and for $r=6$ by Markman's theorem on abelian sixfolds of split Weil type [Mar25a, Mar25b] when the discriminant is trivial: always for $d=3,4$, and for $d=6$ except along the characters of order six with an odd number of coordinates 3. So Φ^*x is algebraic in the cases stated, and so is $x=\Phi_*\Phi^*x/|G|$. For the dimension, the classes found are Hodge at every λ, and by (i) and (ii) there are no others. Each character of order two with $|S|\geq r+2$ gives one class, the image of $\theta_a^{r/2}$, nonzero since $r\leq\dim V_a$. A character of order at least three gives a class only if $|S|=r+2$ and $\dim V_a^{1,0}=\sum_{i\in S}(d-a_i)/d-1=r/2$, that is $\sum_i a_i=d(r+2)/2$ with representatives $0\leq a_i<d$, and then exactly one, in $\bigwedge^r V_a$. The characters with $|S|=r+2$ whose coordinates on S average $d/2$ are counted by $T_d(r+2)$; for d even they include the one of order two with all coordinates $d/2$, and the characters of order two with $|S|\geq r+4$ give the last sum. ∎

Figure 9 draws the steps of the proof. The dimensions of (iii) never exceed $h^{r/2,r/2}(X)$, and they equal it for a quadric, for the Fermat hypersurfaces (for cubics 7, 21 and 71 in dimensions 2, 4 and 6, for quartics 142 and 1108 in dimensions 4 and 6, for sextics 1752 in dimension 4) and for two quadrics in $\mathbb{P}^N$ with N even $(N+2)$, where every class of type $(r/2,r/2)$ is algebraic. For two cubics in $\mathbb{P}^6$ the very general member has 141 Hodge classes of degree four against $h^{2,2}=267$, and for three quartics in $\mathbb{P}^7$ it has 3950 against $h^{2,2}=29872$ (Figure 10). Figure 11 shows, for each degree d and dimension r, what is known for the very general member.

Remark 2.67 (What the diagonal complete intersections give). Theorem 2.55 adds to the domain of (F3′) a family of complete intersections of every dimension and every degree, parametrised by configurations of $N+1$ points on the line, among them all smooth complete intersections of two diagonal hypersurfaces; for $c\geq 3$ and $r\geq 2$ the condition on the columns is a real restriction. Corollary 2.56 shows on them how (F3′) and (F2) together give the conjecture: once the Hodge classes of X are carried to the abelian varieties $B_{[a]}$, the conjecture for X is the algebraicity of their Hodge classes, a question of type (F2), and for the fourfolds in $\mathbb{P}^6$ of degrees 3, 4 and 6 it is answered by Markman's theorem: their exceptional classes are Weil classes of abelian fourfolds of Weil type, 490 families of them for $d=4$. For $d=3$ these fourfolds are Fano and the conjecture also follows from the theorem of Conte and Murre on uniruled fourfolds [CM78]; for $d=4$ and $d=6$

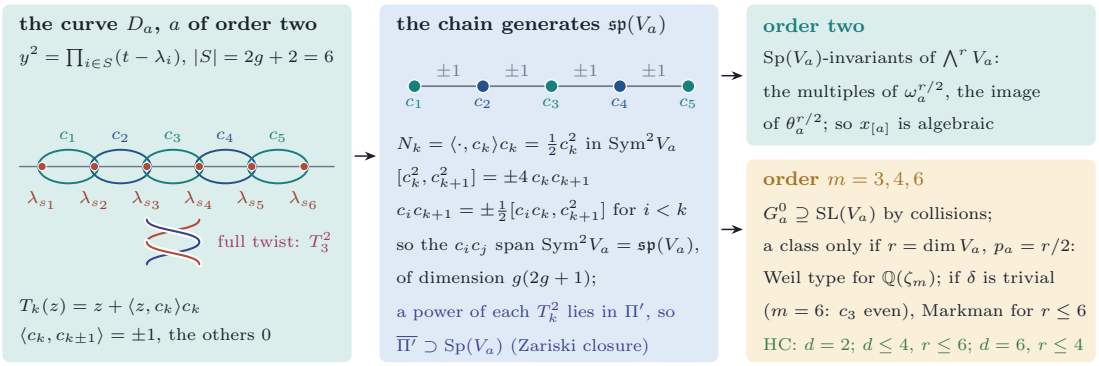


FIGURE 9. The proof of Theorem 2.66. Left: the hyperelliptic curve D_a of a character of order two, drawn over the six points of its support for $g = 2$, with the chain of vanishing cycles $c_1, \dots, c_5$; a full twist of two neighbouring branch points, drawn as a braid, acts by the square of the transvection T_k. Middle: the chain as a graph, and the induction in $\mathrm{Sym}^2 V_a$ by which the logarithms N_k generate $\mathfrak{sp}(V_a)$. Right: what the monodromy leaves of the Hodge classes, the powers of the polarisation for characters of order two, and for characters of order three, four and six the Weil classes of an abelian variety of Weil type for $\mathbb{Q}(\sqrt{-3})$ or $\mathbb{Q}(i)$.

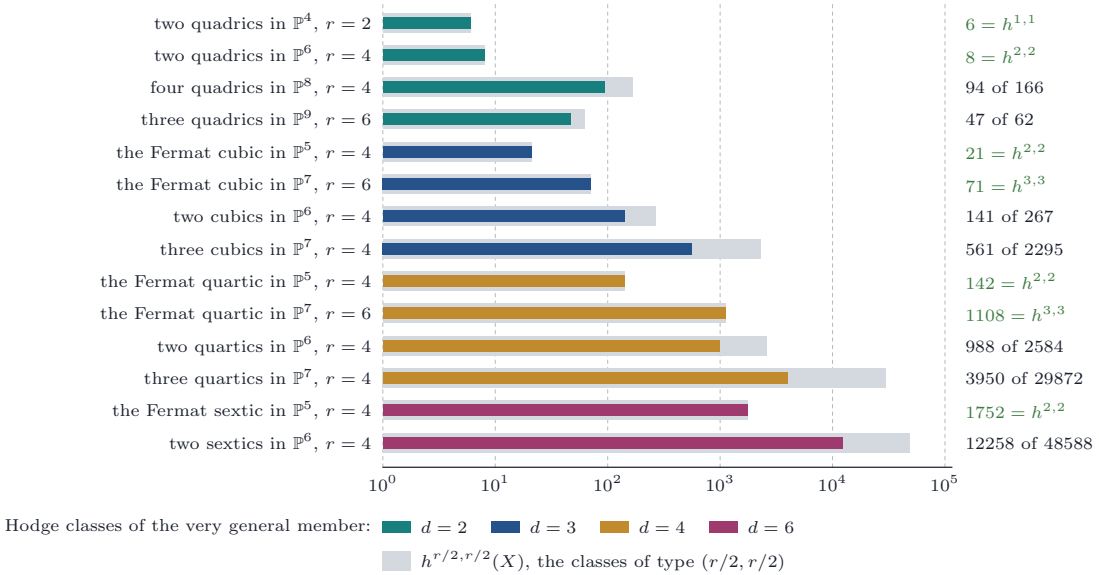


FIGURE 10. The dimension of the space of Hodge classes of degree r on the very general member, from Theorem 2.66(iii), against the Hodge number $h^{r/2,r/2}$, on a logarithmic scale. Where the two agree, the space $H^{r/2,r/2}$ is spanned by Hodge classes at every configuration; elsewhere the very general member has fewer Hodge classes, and the members with more lie over a countable union of proper closed subsets of the space of configurations.

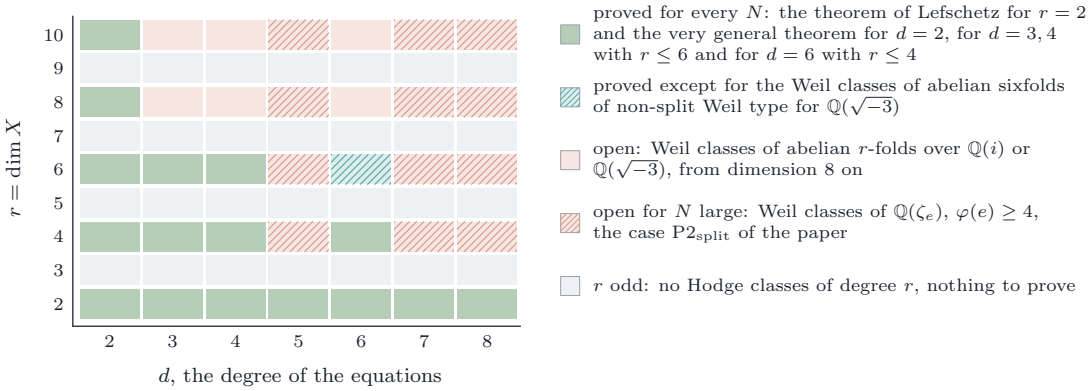


FIGURE 11. The Hodge conjecture for the very general complete intersection $X_{d,r}(\lambda)$ of Vandermonde type, for every N. Grass: proved, by the theorem of Lefschetz for $r = 2$ and by Theorem 2.66 for $d = 2$, for $d = 3, 4$ with $r \leq 6$ and for $d = 6$ with $r \leq 4$. Teal hatching: proved except for the Weil classes of abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$. Clay: open, the Weil classes of abelian varieties of Weil type over $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$ in dimension at least eight, or, hatched, of a CM field of degree at least four (Remark 2.67). For odd r there is nothing to prove.

they are of general type, with $h^{2,2} = 2584$ and 48588. Complete intersections of diagonal hypersurfaces, and of quadrics, are also the subject of Terasoma's paper [Ter88]. At a very general configuration Theorem 2.66 goes further, in every $\mathbb{P}^N$: for $d = 2$ in every dimension, for $d = 3$ and $d = 4$ up to dimension seven, and for $d = 6$ up to dimension five, through the monodromy of cyclic covers of degree three, four and six (Proposition 2.64). For $d = 3$ and $r \geq 8$ the conjecture for the very general member is the algebraicity of the Weil classes of the abelian r-folds $B_{[a]}$ of split Weil type for $\mathbb{Q}(\sqrt{-3})$, which is open from dimension eight on, and the same holds for $d = 4$ with $\mathbb{Q}(i)$. For $d = 6$ and $r = 6$ the open classes are those of the abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$ of Theorem 2.66(iii), such as the one of the character $(1, 1, 2, 2, 3, 5, 5, 5)$ on eight points. Beyond these cases the same reduction meets (F2) itself: for $d = 5$ or $d \geq 7$ and N large enough, some $B_{[a]}$ with $\dim V_a = r$ carries an action of $\mathbb{Q}(\zeta_e)$, $e \mid d$, with $\varphi(e) \geq 4$, and $W_{[a]}$ is the top exterior power of its first cohomology over that CM field, which consists of Hodge classes when the types are balanced: Weil classes for a CM field of degree at least four, of the kind treated in [I, Theorem 1.1]; and for larger N the $B_{[a]}$ have dimension above five.

Remark 2.68 (Where (F3′) stops). With Proposition 2.35, Proposition 2.42 and Proposition 2.43, (F3′) holds on every variety of dimension at most three, on every fourfold whose zero-cycles are supported in dimension three (in particular every uniruled fourfold), on every fivefold whose zero-cycles are supported in dimension at most three, on every fourfold and fivefold whose maximal rationally connected quotient is birational to a variety satisfying (F3′) in degree four (Proposition 2.44), on every fourfold and fivefold dominated by a rational map from such a variety (Proposition 2.46), and on every member of $\mathcal{A}$, a class containing the Fermat hypersurfaces, the smooth Delsarte hypersurfaces with the cyclic covers branched along them (Corollary 2.53), and the complete intersections of diagonal hypersurfaces of Vandermonde type, among them every smooth complete intersection of two

diagonal hypersurfaces (Theorem 2.55, Corollary 2.56), of every degree and dimension, the Hilbert schemes of points on abelian surfaces, on products of curves and on the K3 surfaces of Proposition 2.42(iv), the generalised Kummer varieties $K_n(A)$ and the other varieties of Proposition 2.42(v), and every blow-up and projective bundle built from these. Through the Hodge conjecture itself it holds on the complete intersections of Corollary 2.56(ii), among them the fourfolds of general type cut out by two diagonal quartics or sextics in $\mathbb{P}^6$, on the very general complete intersections of diagonal quadrics, of diagonal cubics and quartics up to dimension seven and of diagonal sextics up to dimension five, of Theorem 2.66, and on the varieties of Proposition 2.39, Proposition 2.40 and Corollary 2.41: for a K3 surface S with $E = \mathrm{End}_{\mathrm{Hdg}}(T(S))$ a CM field, on every power, Hilbert scheme and moduli space of S; for $E = \mathbb{Q}$ and $t = \dim T(S)$, on S^k for $k < t$, on $S^{[n]}$ for $n < t(t+1)/2$ and on the moduli spaces of dimension less than t; on every fourfold of $\mathrm{K3}^{[2]}$ type whose field E is $\mathbb{Q}$ or a CM field; on $F(Y)$ and $Y \times Y$ for every cubic fourfold Y with such a field; and on every fourfold of generalised Kummer type [FV25]. Its frontier cases are these (Figure 12), the first two in degree four on fourfolds whose zero-cycles are not supported on a proper closed subset and which are not birational to a member of $\mathcal{A}$ by any construction above: $S \times S$ and $S^{[2]}$ for a K3 surface whose real multiplication is not induced by a cycle constructed so far, such as S_λ (the known cases are recalled in Remark 3.5); the fourfolds of $\mathrm{K3}^{[2]}$ type with real multiplication, among them $F(Y)$ for a cubic fourfold Y with real multiplication, and with it $Y \times Y$; and, for a very general K3 surface, the single class $\det T(S)$ on S^{21} of Proposition 2.38, equivalently the Hodge conjecture for $S^{[231]}$. In the first two the rational Hodge isometries [Bus19, Mar24] act on T by ± 1, and a composite of Hodge similarities that returns to T is an element $g \in E$ with $g^2 \in \mathbb{Q}$, the adjoint involution of E being trivial; such elements span only the compositum of the quadratic subfields of E, which is $\mathbb{Q}$ for the cubic field of S_λ. Apart from proving the conjecture itself where zero-cycles are small, none of these results touches the step that Remark 2.36 identifies: putting a Hodge class into $\mathrm{Ab}^p(X)$ for a variety that no correspondence connects to an abelian variety. Theorem 2.70 is unchanged, since every statement proved here is a special case of one of the hypotheses of its rule set, not a new rule.

Setup 2.69 (The rule set)**.** The rule set has forty-two statements and twenty-two rules. Ten statements are proved in this paper: the base point with an explicit cycle in every Weil family of an imaginary quadratic field ([I, Theorem 15.27]) and of a CM field of degree at least four ([I, Theorem 17.5]); for each of the two kinds of field, the reduction of the Weil classes of a family to one class at every point of one connected domain, and the stability of the algebraic locus under the rational points, with dense orbits ([I, Section 12], [I, Proposition 16.7], [I, Theorem 17.7]); and four statements on secant objects: the form of the weakened criterion for the transform ([II, Theorem 4.56]), the conditions of Markman's strategy that bear on the Chern character, the description of the local obstruction as a failure of Cohen–Macaulayness ([II, Corollary 4.77]), and three of the four ingredients of the construction in every dimension. Four are quoted: $\mathbf{W}(K, 2, \delta)$ for every K and δ, $\mathbf{W}(K, 3, \delta_0)$ [Mar25a, Mar25b], the deformation theorem for motivated classes [And96b, Théorème 0.5], and the accessibility of Hodge classes on abelian varieties [Mil20, §5.1, Theorem 3]. Sixteen are hypotheses: (F3), (F2) of Corollary 2.71 and (F3′) of Definition 2.34; (L), (M), (V) of Definition 2.14; (P2), in the polarised form of [II, Remark 3.6], for four groups of families, the split and the other families of the imaginary quadratic fields and of the CM fields of degree at least four, the third group giving $\mathrm{P2}_{\mathrm{split}}$; secant objects meeting the weakened criterion in every dimension, and an affirmative

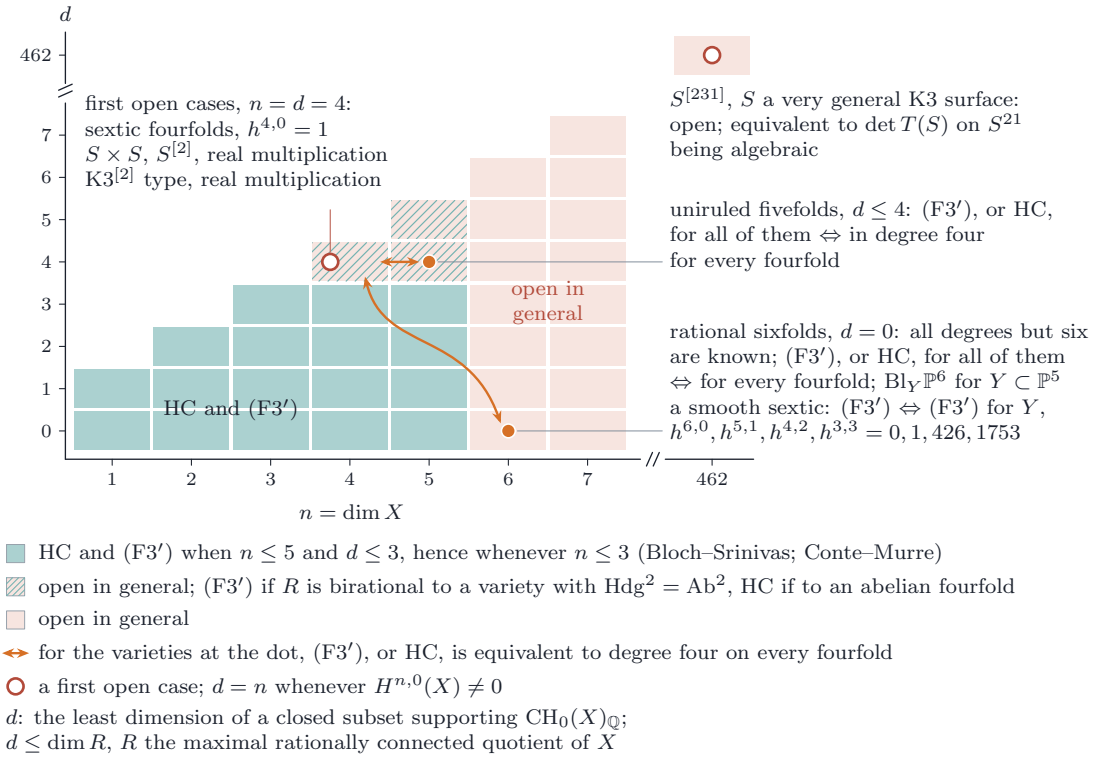


FIGURE 12. What arguments on zero-cycles and on rationally connected fibrations give for (F3′) and for the Hodge conjecture (HC in the figure) on a variety X, over the plane of $n = \dim X$ and the least dimension d of a closed subset supporting $\mathrm{CH}_0(X)_{\mathbb{Q}}$; d is at most the dimension of the maximal rationally connected quotient R of X, and $d = n$ whenever $H^{n,0}(X) \neq 0$ (Remark 2.45). Both hold when $n \leq 5$ and $d \leq 3$ (Proposition 2.43), hence on every variety of dimension at most three. In the hatched cells (F3′) holds when R is birational to a variety with $\mathrm{Hdg}^2 = \mathrm{Ab}^2$, for instance a member of $\mathcal{A}$, and the Hodge conjecture when R is birational to an abelian fourfold (Proposition 2.44). The double arrows are the equivalences of Remark 2.45: (F3′), and the Hodge conjecture, hold for every uniruled fivefold, and for every rational sixfold, if and only if they hold in degree four for every fourfold; the sixfold $\mathrm{Bl}_Y\mathbb{P}^6$ of a smooth sextic fourfold Y in a hyperplane carries the Hodge numbers listed. The rings are the frontier cases (Remark 2.68): the sextic fourfolds, the first family of smooth hypersurfaces beyond these results (the 29 Delsarte sextics lie in $\mathcal{A}$ by Corollary 2.53); $S \times S$ and $S^{[2]}$ for a K3 surface S with real multiplication, such as S_λ; the fourfolds of $\mathrm{K3}^{[2]}$ type with real multiplication, among them $F(Y)$; and, beyond the breaks of the axes, $S^{[231]}$ for a very general K3 surface, which is equivalent to the algebraicity of $\det T(S)$ on S^{21} (Proposition 2.39, Figure 4).

answer to [Mar25b, Question 11.4]; and, for a smooth and for a singular Cohen–Macaulay support on an abelian fourfold, its existence with the required extension property and the vanishing of the image of the product of translation classes. The other twelve are derived: **HC**, the conjecture for every abelian variety, written **AB**, and the Weil classes of every CM field, of the imaginary quadratic fields, of their families of trivial and of nontrivial discriminant, of the CM fields of degree at least four and of their split and their other families, together with one secant object on an abelian fourfold, $\mathbf{W}(K, 4, \delta_0)$, and the range settled in the literature. Each rule is a finite set of premises with one conclusion, and each is a theorem of this paper: Table 1 lists twenty-one of them, and the twenty-second derives the range settled in the literature from the two quoted Weil statements. For a set T of statements write $\mathrm{Cn}(T)$ for the least set that contains T and is closed under the rules, which exists because each rule has a single conclusion and the consequence operator is therefore monotone. Write $\mathbf{T}$ for the set of statements proved or quoted. Call a set S of hypotheses *sufficient* if $\mathbf{HC} \in \mathrm{Cn}(\mathbf{T} \cup S)$, and *minimal* if no proper subset of S is sufficient.

Theorem 2.70 (The closure of what is proved here)**.** *With the notation of Setup 2.69:*

(i) $\mathbf{HC} \notin \mathrm{Cn}(\mathbf{T})$. *Nothing in this paper, combined with the theorems it quotes, implies the Hodge conjecture.*

(ii) *There are exactly five minimal sufficient sets:*

$$\{\mathrm{F3}\}, \quad \{\mathrm{L}, \mathrm{M}\}, \quad \{\mathrm{L}, \mathrm{F3}'\}, \quad \{\mathrm{V}, \mathrm{F3}'\}, \quad \{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\},$$

with (L), (M), (V) *as in Definition 2.14,* (F2), (F3) *as in Corollary 2.71, (F3′) as in Definition 2.34, and* $\mathrm{P2}_{\mathrm{split}}$ *the propagation statement* (P2)*, read in the polarised form of* [II, Remark 3.6]*, for the split families of the CM fields of degree at least four. No minimal set contains* (P2) *for an imaginary quadratic field or for a family of nontrivial discriminant: Proposition 2.13 derives those. The single hypothesis (F3) is sufficient, and it is equivalent to* **HC** *(Proposition 2.33); no other single hypothesis is. The sets* $\{\mathrm{L}, \mathrm{F3}\}$, $\{\mathrm{V}, \mathrm{F3}\}$ *and* $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}\}$ *are sufficient and not minimal. The last of the five is the only one with three elements and the only one whose derivation of* **HC** *uses a statement proved in this paper.*

(iii) *In each of the five, every element is necessary: removing any one leaves* **HC** *outside the closure.*

(iv) *The secant route lies in no minimal sufficient set. Granting both of its demands, a secant object satisfying the weakened criterion in every dimension and an affirmative answer to* [Mar25b, Question 11.4]*, puts* $\mathbf{W}(K, n, \delta_0)$ *in the closure, and with it, by Proposition 2.13,* $\mathbf{W}(K, n, \delta)$ *for every* δ*; it leaves the Weil classes of the CM fields of degree at least four outside.*

(v) *No statement proved in this paper rests, at any depth, on a hypothesis, and the rule set is acyclic.*

(vi) *Every element of a minimal sufficient set other than* $\mathrm{P2}_{\mathrm{split}}$ *is a consequence of* **HC**, *and* $\{\mathrm{F3}\}$ *and* $\{\mathrm{L}, \mathrm{M}\}$ *are each equivalent to it; no implication from* **HC** *to* (P2) *is known.*

Proof. (i) and (v). No rule has a proved or quoted statement as its conclusion, so each of them lies in $\mathrm{Cn}(\mathbf{T})$ by itself, on the strength of its proof in this paper or in the literature; this is the first assertion of (v). The twenty-second rule is the only one whose premises all lie in $\mathbf{T}$: every other rule has among its premises a hypothesis or a derived statement other than the range settled in the literature. Hence $\mathrm{Cn}(\mathbf{T})$ is $\mathbf{T}$ together with that range,

premises	conclusion	by
the conjecture for abelian varieties, (F3)	**HC**	Section 2.5
(F3)	abelian varieties	Proposition 2.33
the conjecture for abelian varieties, (F3′)	**HC**	Proposition 2.35
the Weil classes of every CM field, (F2)	abelian varieties	Section 2.5
K imaginary quadratic, the fields of degree at least four	every CM field	[I, Appendix C]
$\delta = \delta_0$, $\delta \neq \delta_0$	every discriminant	Proposition 2.12
a base point, the reduction, the orbit, (P2) for the family	that family	[II, Theorem 2.3]
the same three for a CM field of degree at least four, (P2) for the family	that family	[I, Theorem 17.7]
its split families, its other families	the fields of degree at least four	[I, Proposition 17.3]
$\delta = \delta_0$	$\delta \neq \delta_0$	Proposition 2.13
the split families of the fields of degree at least four	their other families; $\delta = \delta_0$	Proposition 2.13
a secant object in every dimension, Question 11.4	$\delta = \delta_0$	[I, Proposition 15.19]
a Cohen–Macaulay support, smooth or singular	one secant object	[II, Corollary 4.77]
one secant object, Question 11.4	$\mathbf{W}(K, 4, \delta_0)$	[II, Corollary 4.57]
(L), (M)	**HC**	Proposition 2.16
(L), the deformation theorem for motivated classes	(V)	Proposition 2.17
(V), the accessibility of Hodge classes on abelian varieties	abelian varieties	Theorem 2.18

TABLE 1. The rules of Setup 2.69 whose conclusion is derived, with the theorem that proves each; the rows for (P2) and for the Cohen–Macaulay support each stand for two rules, the row for descent over the fields of degree at least four for two, descent within a field and scalar extension down to the imaginary quadratic fields, the second row is Proposition 2.33, the third is Proposition 2.35(i) for the weaker statement (F3′) of Definition 2.34, the last three rows are the rules of Section 2.6, one of whose conclusions, (V), is itself a hypothesis of the rule set, and the one remaining rule records what the literature supplies on Weil classes. The bounded criterion (P1) is absent: by [I, Theorem 16.17] it is equivalent to its own conclusion, so it is a restatement and not a premise.

and it does not contain **HC**, which is (i). For acyclicity, list first the thirty statements that are proved, quoted or hypotheses other than (V); then one secant object, $\mathbf{W}(K, 4, \delta_0)$ and the range settled in the literature; the split and then the other families of the CM fields of degree at least four; the families of trivial and then of nontrivial discriminant of the imaginary quadratic fields; the imaginary quadratic fields, the CM fields of degree at least four and every CM field; and last (V), **AB** and **HC**. The premises of every rule come before its conclusion in this list.

(ii) and (iii). The five sets are sufficient: F3 $\Rightarrow$ **AB** (Proposition 2.33) and **AB** $\wedge$ F3 $\Rightarrow$ **HC**; L $\wedge$ M $\Rightarrow$ **HC** (Proposition 2.16); (L) and the deformation theorem give (V) (Proposition 2.17), (V) and accessibility give **AB** (Theorem 2.18), and **AB** $\wedge$ F3$'$ $\Rightarrow$ **HC** (Proposition 2.35(i)), which settles $\{\mathrm{L}, \mathrm{F3}'\}$ and $\{\mathrm{V}, \mathrm{F3}'\}$; and $\mathrm{P2}_{\mathrm{split}}$ with the CM base point, the reduction and the orbit gives the split families of the CM fields of degree at least four ([I, Theorem 17.7]), descent gives their other families and the families of trivial discriminant of the imaginary quadratic fields (Proposition 2.13), hence the fields of degree at least four ([I, Proposition 17.3]), the nontrivial discriminant (Proposition 2.13), the imaginary quadratic fields (Proposition 2.12) and every CM field ([I, Appendix C]), and with (F2) this is **AB** and with (F3$'$) it is **HC**.

Let C_1, respectively C_2, be the set of all statements other than **HC**, (F3), (F3$'$) and (L), respectively (M). Let C_3 be the set of all statements other than **HC**, **AB**, the Weil classes of every CM field, of the fields of degree at least four and of their split families, (F3), (L), (V) and $\mathrm{P2}_{\mathrm{split}}$; and let C_4 be the set of all statements other than **HC**, **AB**, (F3), (L), (V) and (F2). Each C_j contains **T** and not **HC**, and each is closed under the rules: every rule whose conclusion it omits has a premise it omits. For C_1 and C_2 the only omitted conclusion is **HC**, and the three rules with that conclusion have the premises (F3), (F3$'$), and (L) and (M). For C_3 and C_4 the rules with conclusion **HC** have the premises **AB** or (L); those with conclusion **AB** have the premises (F3), the Weil classes of every CM field together with (F2), and (V); the only rule with conclusion (V) has the premise (L); and for C_3 the rule for every CM field needs the fields of degree at least four, the rule for these needs their split families, and the rule for the split families needs $\mathrm{P2}_{\mathrm{split}}$. Since Cn is monotone, a set of hypotheses contained in some C_j has its closure, together with **T**, inside C_j, and is not sufficient. So a sufficient set meets each of

$$H_1 = \{\mathrm{F3}, \mathrm{F3}', \mathrm{L}\}, \qquad H_2 = \{\mathrm{F3}, \mathrm{F3}', \mathrm{M}\},$$
$$H_3 = \{\mathrm{F3}, \mathrm{L}, \mathrm{V}, \mathrm{P2}_{\mathrm{split}}\}, \qquad H_4 = \{\mathrm{F3}, \mathrm{L}, \mathrm{V}, \mathrm{F2}\},$$

the hypotheses omitted from $C_1, \ldots, C_4$. Let S meet all four. If F3 $\in S$, then $S \supseteq \{\mathrm{F3}\}$. If F3 $\notin S$ and L $\in S$, then S meets H_2 in (F3$'$) or (M). If neither is in S, then S contains (F3$'$), from H_1, and (V) or both $\mathrm{P2}_{\mathrm{split}}$ and (F2), from H_3 and H_4. So a set of hypotheses is sufficient exactly when it contains one of the five sets, and as none of the five contains another, they are the minimal sufficient sets. The other assertions of (ii) can be read off the list, and the derivations above show that only the last uses a statement proved here; (iii) is the minimality.

(iv) Granting the two demands of the secant route, the rule of [I, Proposition 15.19] gives the families of trivial discriminant, descent those of nontrivial discriminant, and Proposition 2.12 the imaginary quadratic fields. The set formed by **T**, the range settled in the literature, the two demands and these three statements is closed, since every other rule has a premise outside it: a hypothesis not granted, one secant object, or a Weil statement about the fields of degree at least four or about every CM field. Every base point the secant route produces is a member $X \times \widehat{X}$, whose discriminant is trivial whatever β is

chosen ([I, Proposition 15.19(ii)]); the deformation argument stays inside the family of trivial discriminant, and it is Proposition 2.13(i), not the deformation, that carries the result to the other discriminants. The route has no base point over a field of degree at least four, so it reaches none of them, and none of its hypotheses lies in one of the five sets.

(vi) (L) follows from **HC** by [I, Theorem 12.2], (M) by Proposition 2.16, (V) by Proposition 2.17, (F3′) by Proposition 2.35(i), and (F2) and (F3) are special cases of **HC**. The equivalence of {F3} with **HC** is Proposition 2.33, and {L, M} is equivalent to it by Proposition 2.16 and the first two implications. ■

Corollary 2.71 (The three statements, one of which is the conjecture)**.** *The Hodge conjecture follows from the results of this paper together with the following three statements. No rule of Setup 2.69 derives it from (F1) and (F2) together, and it follows from (F3) alone, which is equivalent to it (Proposition 2.33); so the list is not a reduction of the conjecture to three smaller problems. With (F3) replaced by the weaker statement* (F3′) *of Definition 2.34, the conjecture modulo abelian varieties, the three statements (F1), (F2),* (F3′) *still suffice, and no rule of Setup 2.69 derives the conjecture from a proper subset of them. With the rule of Proposition 3.49, which the rule set does not contain, (F2) and* (F3′) *already suffice (Remark 3.50).*

(F1) One propagation statement, for every CM field. (P2)*, in the polarised form of* [II, Remark 3.6]*: at one base point of each Weil family of each CM field some perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the polarisation has injective semiregularity map. Its first form, with Chern character exactly a multiple of the Weil class, is false for every imaginary quadratic field and every* $n \geq 2$ *(*[II, Theorem 3.5]*): such a complex would have to deform to non-algebraic Weil tori, whose coherent sheaves have no Chern character in degrees* 1 *to* $2n-1$*. The polarised form closes* $\mathbf{W}(K, n, \delta)$ *for every imaginary quadratic* K*, every* n *and every* δ *at once, by* [II, Theorem 2.3]*, and* $\mathbf{W}(F, n, \delta)$ *for every CM field* F *of higher degree, by* [I, Theorem 17.7]*, because those theorems put* $\Sigma = \mathcal{D}$*, so the Weil class is algebraic at every member of the family and not only at the general one, and because* [I, Theorem 15.27] *and* [I, Theorem 17.5] *supply the base point in every one of those families. By Proposition 2.13 it is enough to have it for the split families of the CM fields of degree at least four, in infinitely many dimensions.* [II, Theorem 2.5], [II, Theorem 2.13]*, Theorem 2.32,* [II, Theorem 2.11] *and* [II, Theorem 2.14] *describe the object of the first form and are vacuous by* [II, Theorem 3.5]*; for the polarised form* [II, Theorem 3.9] *computes the number that* $\dim \mathrm{Ext}^2(E, E)$ *has to reach,* $7n^2 - 2n$ *for a general shape. For one family its conclusion is equivalent to the Lefschetz standard conjecture for one variety (Corollary 2.31). The bounded criterion* (P1) *is not a second form of the same demand: by* [I, Theorem 16.17] *it is equivalent to the conclusion it was to imply, and* [I, Theorem 16.15] *shows that the one cycle it could have been tested on, the transport of the base cycle, does not satisfy it.*

(F2) The classes beyond the Weil lines. *Every Hodge class on an abelian variety that lies outside the subring generated by divisor classes and by Weil classes of quotients is algebraic. Through dimension five there is no such class* [MZ99]*; in dimension eight there are (Proposition 2.74), so this is not a reduction of the conjecture for abelian varieties to Weil classes, which is false as a statement about Hodge rings: it is the conjecture for those classes, and by Proposition 3.49 it is equivalent to the conjecture for every abelian variety. Its smallest case is brought down to one*

Kuga–Satake class in Theorem 3.4; it holds at every CM point (Theorem 2.29), and on all members of a Mumford family it is equivalent to the Lefschetz standard conjecture for one ninefold (Corollary 2.30). Its classes are rigid along the base curve (Theorem 3.21), and one complex at one CM point with Ext^2 *of dimension* 119 *would close it (Theorem 3.27), as would a bound on the cycles that represent them at the closed points of the reductions of the curve modulo primes (Theorem 3.43).*

(F3) The varieties that are not abelian. *The Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety. This is the Hodge conjecture itself. The varieties it covers include* $A \times \mathbb{P}^1$ *for every abelian variety* A*, and the conjecture for* $A \times \mathbb{P}^1$ *gives it for* A *(Proposition 2.33); so (F3) implies the conjecture for abelian varieties, hence (F1) and (F2), and hence* **HC***. Nor is it a reduction to abelian varieties in the other direction: no realisation of the cohomology of an arbitrary variety inside that of abelian varieties exists (Proposition 2.75), and* [I, Theorem 10.3] *and* [I, Theorem 10.8] *close the two routes through hyperplane sections that would be the natural candidates for one.*

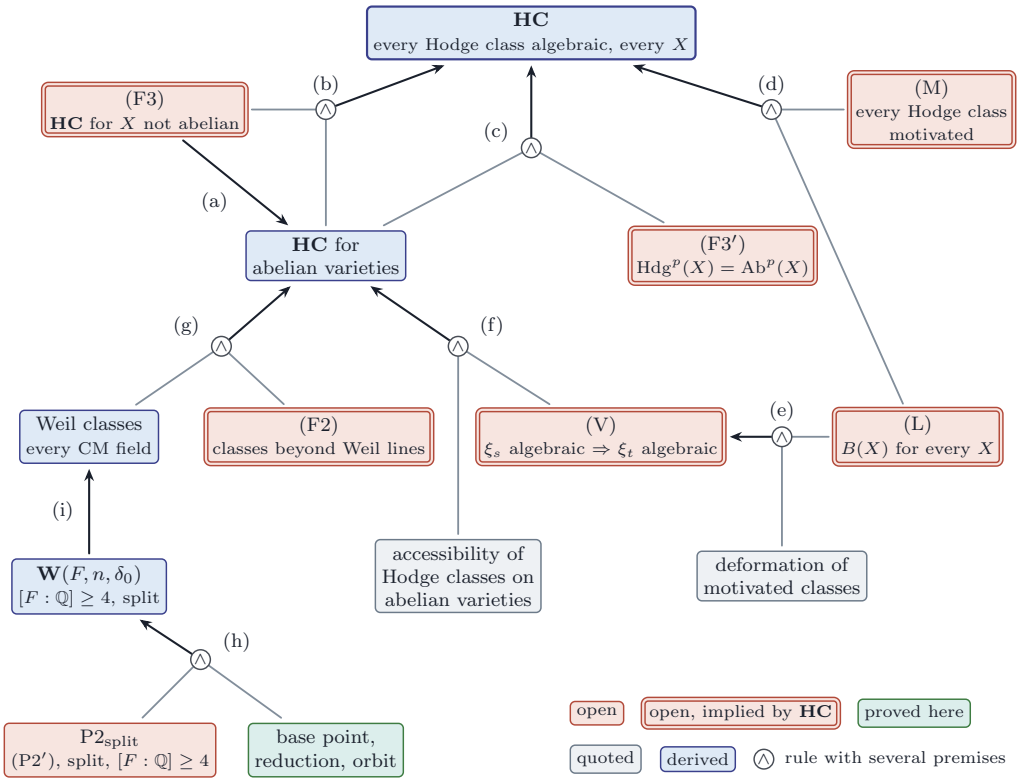
Proof. The three together are sufficient by the derivation of Theorem 2.70, and (F3) alone by Proposition 2.33. That (F1) and (F2) together are not sufficient in the rule set is Theorem 2.70(ii), since every minimal sufficient set contains (F3), (F3′) or (M); the statement about (F1), (F2), (F3′) is Theorem 2.70(ii) and (iii) for the last of the five minimal sets. The Weil classes of the CM fields of degree at least four are not a fourth statement: they are derived from (F1) by [I, Theorem 17.7]. Nor is the bounded criterion a premise, since by [I, Theorem 16.17] it is a restatement of its own conclusion. ∎

Remark 2.72 (Necessity relative to a rule set, and the wording of (F2))*.* A rule set can show that a set of statements suffices; it can show that a set is needed only relative to the implications written into it. A rule set that used (F3) only through (abelian varieties) ∧ (F3) ⇒ **HC**, without the elementary rule (F3) ⇒ (abelian varieties) of Proposition 2.33, would make the three statements of Corollary 2.71 look jointly necessary, although (F3) alone suffices.

The wording of (F2) has the same feature, one degree at a time. Let $R(A) \subseteq H^*(A, \mathbb{Q})$ be the subring generated by divisor classes and Weil classes of quotients. Suppose some Hodge class β of degree $2p$ on A lies outside $R(A)$, and let α be any Hodge class of degree $2p$. If $\alpha \notin R(A)$, (F2) makes it algebraic. If $\alpha \in R(A)$, then $\alpha + \beta \notin R(A)$, because $R(A)$ is closed under subtraction, so (F2) makes $\alpha + \beta$ and β algebraic, and with them α. So in every degree in which (F2) says anything it is the whole conjecture for that degree: on the square of a Mumford fourfold it gives every Hodge class of H^4, the six products of divisor classes included. Multiplying by an exceptional class of that square moves every Hodge class of every abelian variety outside the subring, and Proposition 3.49 deduces from this that (F2) is the whole conjecture for abelian varieties. The rule set does not contain that implication; Remark 3.50 says how Theorem 2.70(ii) changes when it is adjoined.

The form of (F3) that is not the conjecture itself asks only for algebraicity modulo classes coming from abelian varieties. It is (F3′) of Definition 2.34; Proposition 2.35 says what is known of it, and Remark 2.36 why it is not closed.

Remark 2.73 (The target, and what it would take)*.* With the rules of Section 2.6, Proposition 2.33 and Proposition 2.35 in place, Theorem 2.70 changes the reading of Corollary 2.71 in three ways. First, the route through this paper is a route only with (F3) replaced by (F3′). As stated, its last statement (F3) is the conjecture (Proposition 2.33), so (F1) and



minimal set	(F3)	(L)	(M)	(V)	(F3′)	(F2)	$\mathrm{P2}_{\mathrm{split}}$	rules used
S_1	●	·	·	·	·	·	·	(a), (b)
S_2	·	●	●	·	·	·	·	(d)
S_3	·	●	·	·	●	·	·	(c), (e), (f)
S_4	·	·	·	●	●	·	·	(c), (f)
S_5	·	·	·	·	●	●	●	(c), (g), (h), (i)
implied by **HC**	yes	yes	yes	yes	yes	yes	not known	

FIGURE 13. The rule set of Setup 2.69 with the rules of Section 2.6, drawn as a hypergraph. A rule with several premises is a junction marked ∧, joined to its premises and with one arrow to its conclusion. The letters name the results that prove the rules: (a) is Proposition 2.33; (b) and (g) are the closure map of Section 2.5; (c) is Proposition 2.35(i); (d) is Proposition 2.16; (e) is Proposition 2.17; (f) is Theorem 2.18; (h) is [I, Theorem 17.7]; and (i) is the chain of Proposition 2.13, Proposition 2.12, [I, Proposition 17.3] and [I, Appendix C]. Clay marks a hypothesis, a double frame a hypothesis that the conjecture implies, grass what this paper proves, slate what it quotes, and indigo a derived statement. The table lists the five minimal sufficient sets of Theorem 2.70(ii), the rules each of them uses, and which of the hypotheses the conjecture implies; only $\mathrm{P2}_{\mathrm{split}}$, which asks for a particular complex, is drawn without that frame. The secant route, which reaches the imaginary quadratic families only, is drawn in Figure 14.

(F2), and everything this paper proves about Weil classes, drop out of every minimal derivation that uses (F3). With (F3′) in its place the route $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\}$ is minimal, and it is not the shortest: (F1) and (F2), which between them give the conjecture for abelian varieties along this route, are replaced on the routes of the literature by the single statement (V) (Theorem 2.18), or by (L), which implies it (Proposition 2.17). Second, every element of every minimal sufficient set except $\mathrm{P2}_{\mathrm{split}}$ is a consequence of the conjecture (Theorem 2.70(vi)), and two of the sets, $\{\mathrm{F3}\}$ and $\{\mathrm{L}, \mathrm{M}\}$, are each equivalent to it. (P2) is a statement about semiregularity maps, and a proof of the conjecture need not prove it; in its first form, with a Chern character that is exactly a Weil class, it is false ([II, Theorem 3.5]), and the route runs through the polarised form of [II, Remark 3.6]. Third, every minimal sufficient set contains (F3), (F3′) or (M): the varieties that are not abelian are reached only through the conjecture for them, which is the whole conjecture, through the conjecture modulo abelian varieties (Remark 2.36), or through motivated classes.

So the target at the centre is (L). It lies in two of the five minimal sets, it gives (V) and with it the whole abelian half, the conjecture implies it, and with (M) it is the conjecture. For each X it asks for one explicit Hodge class on $X \times X$ to be algebraic, the class inducing Λ. This paper proves (L), (M), (V) and (F3′) on the classes of varieties named here: Corollary 2.22 adds the total spaces of Mumford families to the known cases of (L), Proposition 2.35 records the known cases of (F3′), and the routes of the literature use no other theorem of this paper.

Two of the three are not reductions. One could hope that (F2) and (F3) are reductions in the ordinary sense, statements to the effect that every Hodge class is carried, by some correspondence, to a class on a Weil type variety or on an abelian variety, so that the whole conjecture would follow from the Weil classes alone. Both hopes are refutable, and we refute them, because a frontier that contains a false statement is not a frontier.

Proposition 2.74 (The Hodge ring is not generated by divisor and Weil classes)**.** *Let X be a complex abelian fourfold whose Hodge group is a $\mathbb{Q}$-form of an almost direct product of three copies of SL_2, acting on $V = H^1(X, \mathbb{Q})$ through the tensor product $V_1 \otimes V_2 \otimes V_3$ of the three standard representations, with $\mathrm{End}^0(X) = \mathbb{Q}$; such fourfolds exist and form families of dimension one, by Mumford* [Mum69]*, and by* [MZ99, 2.5(i)] *they are exactly the fourfolds with $\mathrm{End}^0(X) = \mathbb{Q}$ and Hodge group smaller than Sp_8. Then:*

(i) *the Hodge ring of X is generated by the polarisation:* $\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X) = \dim_{\mathbb{Q}} \mathrm{Hdg}^2(X) = 1$*;*

(ii) $\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X \times X) = 3$ *and* $\dim_{\mathbb{Q}} \mathrm{Hdg}^2(X \times X) = 8$*, while the products of divisor classes span a subspace of dimension* 6 *in $H^4(X \times X, \mathbb{Q})$;*

(iii) *the two remaining classes lie in $H^4(X \times X, \mathbb{Q})$ and are Weil classes of no CM field acting on $X \times X$, on a quotient of it, or on an abelian subvariety of it.*

In particular the statement that every Hodge class on an abelian variety lies in the subring generated by divisor classes and Weil classes of quotients is false in dimension eight, though true through dimension five [MZ99].

Proof. Hodge classes are the invariants of the Hodge group, so $\mathrm{Hdg}^k(Y) \otimes \mathbb{C}$ is the space of $\mathrm{Hg}(Y)_{\mathbb{C}}$-invariants in $\bigwedge^{2k} H^1(Y, \mathbb{C})$, and $\mathrm{Hg}(X \times X) = \mathrm{Hg}(X)$ acting diagonally on $V \oplus V$. Over $\mathbb{C}$ the group is SL_2^3 and the weights of V under its maximal torus are the eight sign vectors $(\pm 1, \pm 1, \pm 1)$. Write $V = V_1 \otimes W$ with $W = V_2 \otimes V_3$, on which $\mathrm{SL}_2^2 = \mathrm{Spin}(4)$ acts orthogonally. The Cauchy formula $\bigwedge^k (V_1 \otimes W) = \bigoplus_{|\lambda|=k} S_\lambda V_1 \otimes S_{\lambda'} W$ [FH91, §6.1], with $S^2 V_1 = S_{(3,1)} V_1 = V(2)$, $\bigwedge^2 V_1 = S_{(2,2)} V_1 = \mathbf{1}$ and $S^4 V_1 = V(4)$, gives

$\bigwedge^2 V = V(2)\otimes\bigwedge^2 W \oplus S^2W$ and $\bigwedge^4 V = V(4)\otimes\bigwedge^4 W \oplus V(2)\otimes S_{(2,1,1)}W \oplus S_{(2,2)}W$. Under $\mathrm{Spin}(4)$, $\bigwedge^2 W = V(2,0)\oplus V(0,2)$, $S^2W = V(2,2)\oplus \mathbf{1}$, $\bigwedge^4 W = \mathbf{1}$, $S_{(2,1,1)}W \cong \mathfrak{sl}(W) = \bigwedge^2 W \oplus S^2_0 W$, and $S_{(2,2)}W = V(4,0)\oplus V(0,4)\oplus V(2,2)\oplus \mathbf{1}$, the decomposition of the algebraic curvature tensors in dimension four into the self-dual and anti-self-dual Weyl parts, the traceless Ricci part and the scalar part. Hence

$$\bigwedge^2 V = V(2,2,0)\oplus V(2,0,2)\oplus V(0,2,2)\oplus \mathbf{1},$$
$$\bigwedge^4 V = V(2,2,2)\oplus \bigoplus\nolimits_{\mathrm{cyc}} V(4,0,0)\oplus \bigoplus\nolimits_{\mathrm{cyc}} V(2,2,0)\oplus \mathbf{1},$$

each with one trivial summand, which is (i). For (ii), $\bigwedge^4(V\oplus V) = \bigoplus_a \bigwedge^a V\otimes\bigwedge^{4-a}V$ and the invariants in $\bigwedge^a V\otimes\bigwedge^{4-a}V$ are $\mathrm{Hom}_{\mathrm{SL}^3_2}(\bigwedge^a V, \bigwedge^{4-a}V)$, since every summand is self-dual; the counts are $1,1,4,1,1$, total 8, the 4 coming from the four distinct irreducible summands of $\bigwedge^2 V$. Likewise the invariants in $\bigwedge^2(V\oplus V)$ number $1+1+1=3$. The divisor classes of $X\times X$ are the three symplectic forms ψ_{11}, ψ_{12}, ψ_{22} on $V\oplus V$ built from the polarisation ψ of V. Their six pairwise products are linearly independent: they lie in the summands $\bigwedge^a V\otimes\bigwedge^{4-a}V$ with $a = 4,3,2,2,1,0$, each is nonzero, and in the summand $a=2$ the classes $\psi_{11}\psi_{22}$ and ψ_{12}^2 correspond, as maps $\bigwedge^2 V\to\bigwedge^2 V$, to a multiple of the projection onto $\mathbb{C}\psi$ and to a multiple of the identity. The Sp_8-invariants of $\bigwedge^4(V\oplus V)$ number $1+1+2+1+1=6$ by the same count, since $\bigwedge^1 V = V$, $\bigwedge^2 V = \bigwedge^2_0 V\oplus\mathbf{1}$ and $\bigwedge^3 V = \bigwedge^3_0 V\oplus V$ under Sp_8; so the six products are all the classes a Hodge group equal to Sp_8 would allow. For (iii), the Weil classes of an abelian variety Y with respect to a CM field $F\subseteq \mathrm{End}^0(Y)$ are, by definition, the classes in $\bigwedge^{2k}_F H^1(Y,\mathbb{Q})\subseteq H^{2k}(Y,\mathbb{Q})$ with $2k = \dim_F H^1(Y,\mathbb{Q})$ ([I, Setup 17.1]). A CM field acting on $X\times X$ lies in $\mathrm{End}^0(X\times X) = M_2(\mathbb{Q})$, so it is imaginary quadratic, $2k=8$, and its Weil classes lie in H^8, while the two classes of (ii) have degree four. Every proper abelian subvariety and every proper quotient of $X\times X$ is isogenous to X, because X is simple, and $\mathrm{End}^0(X)=\mathbb{Q}$ contains no CM field, so X carries no Weil classes at all. This is the statement of [MZ99, 0.1(iii)] that the exceptional classes of $X\times X$ are not of Weil type. The last sentence is (ii) and (iii) together with the theorem of [MZ99] for dimension at most five. ∎

Proposition 2.75 (No Hodge-theoretic reduction to abelian varieties)**.** *Let Y be a very general smooth quintic threefold in $\mathbb{P}^4$, and let $V = H^3(Y,\mathbb{Q})$ with its polarised Hodge structure of weight three. Then V is not isomorphic to a subquotient of any Hodge structure of the form $\bigoplus_i T_i$, where each T_i is obtained from the H^1 of finitely many abelian varieties by tensor products, duals and Tate twists. In particular the Hodge structure $H^*(Y,\mathbb{Q})$ is not of abelian type, and no correspondence between Y and an abelian variety, or a product of abelian varieties, induces an injection of V into the cohomology of the latter.*

Proof. Write $\mathrm{MT}(W)$ for the Mumford–Tate group of a rational Hodge structure W and $\mu_W\colon \mathbb{G}_m\to \mathrm{MT}(W)_{\mathbb{C}}$ for its Hodge cocharacter, acting on $W^{p,q}$ by z^p. The adjoint action of μ_W on $\mathrm{Lie}\,\mathrm{MT}(W)_{\mathbb{C}}\subseteq\mathrm{End}(W_{\mathbb{C}})$ has weights among the differences $p-p'$ of two Hodge indices of W.

If W has weight one, these differences lie in $\{-1,0,1\}$. If W' is a subquotient of a tensor construction T on W, then $\mathrm{MT}(W')$ is the image of $\mathrm{MT}(W)$ acting on W', and $\mu_{W'}$ is the image of μ_W; so $\mathrm{Lie}\,\mathrm{MT}(W')$ is a quotient of $\mathrm{Lie}\,\mathrm{MT}(W)$ as a representation of $\mathbb{G}_m$ through μ, and its weights are a subset of $\{-1,0,1\}$. The same holds with $W=\bigoplus H^1(A_i)$. Hence for every Hodge structure of abelian type the adjoint weights of the Hodge cocharacter lie in $\{-1,0,1\}$.

For $V = H^3(Y,\mathbb{Q})$ we show that the adjoint weight 3 occurs. The canonical bundle of a quintic threefold is trivial by adjunction, so $h^{3,0}(Y) = 1$; pick $0 \neq e \in H^{3,0}$. The intersection form ψ on V is symplectic and pairs $H^{3,0}$ with $H^{0,3}$ perfectly, so the rank one endomorphism $x \mapsto \psi(x,e)\,e$ of $V_{\mathbb{C}}$ is nonzero, sends $H^{0,3}$ to $H^{3,0}$ and kills the rest, and lies in $\mathfrak{sp}(V_{\mathbb{C}},\psi)$ because $\psi(\psi(x,e)e,y) + \psi(x,\psi(y,e)e) = \psi(x,e)\big(\psi(e,y)+\psi(y,e)\big) = 0$; its adjoint weight is $3-0=3$. It remains to see that $\mathfrak{sp}(V_{\mathbb{C}},\psi) \subseteq \operatorname{Lie}\mathrm{MT}(V)_{\mathbb{C}}$. Let S be the parameter space of smooth quintic threefolds, connected and nonsingular, and $\mathbb{V}$ the polarisable variation of Hodge structure R^3 of the universal family. The connected algebraic monodromy group G^0 of $\mathbb{V}$ contains the connected monodromy group of any Lefschetz pencil of quintics, that is of hyperplane sections of the fifth Veronese image of $\mathbb{P}^4$; for a Lefschetz pencil of odd fibre dimension the image of the monodromy representation on the ℓ-adic vanishing cohomology $E\otimes\mathbb{Q}_\ell$ is open in $\mathrm{Sp}(E\otimes\mathbb{Q}_\ell,\psi)$ [Del74, Theorem 5.10], so the Zariski closure of the monodromy group over $\mathbb{Q}$ is $\mathrm{Sp}(E,\psi)$; here $E = H^3(Y_t,\mathbb{Q})$, the vanishing cohomology being the orthogonal complement of the image of $H^3(\mathbb{P}^4,\mathbb{Q}) = 0$. Hence $G^0 \supseteq \mathrm{Sp}(V,\psi)$, and $G^0 \subseteq \mathrm{Sp}(V,\psi)$ since the monodromy preserves ψ. At a very general point Y the group G^0 is a normal subgroup of the derived group of $\mathrm{MT}(V)$ [And92, Theorem 1], so $\mathrm{Sp}(V,\psi) \subseteq \mathrm{MT}(V)^{\mathrm{der}} \subseteq \mathrm{Sp}(V,\psi)$, and the adjoint weight 3 occurs in $\operatorname{Lie}\mathrm{MT}(V)$. So V is not of abelian type, and neither is any Hodge structure containing it as a subquotient, in particular $H^*(Y,\mathbb{Q})$. A correspondence inducing an injection of V into the cohomology of an abelian variety would exhibit V as a sub-Hodge structure of a Tate twist of that cohomology, which is of abelian type. ∎

2.10. **Routes from outside algebraic geometry.** By Proposition 3.49 and Proposition 2.35(i), the two statements (F2) and (F3′) are the two halves of the conjecture: the conjecture for abelian varieties, and the conjecture modulo abelian varieties. A route from another domain must therefore either produce algebraic cycles or arrive at one of the hypotheses of Setup 2.69. This subsection records what the routes most often proposed do. One of them, the analytic route through positive currents, can be made exact: the conjecture is the vanishing of an integrality gap in a mass minimisation problem whose real version is solved, and positivity, which is what analysis sees first, cannot distinguish an algebraic class from any other.

Throughout, X is a smooth complex projective variety of dimension n, $h \in H^2(X,\mathbb{Z})$ is the class of an ample line bundle, ω_h is a Kähler form with class h, and for $c \in H^{2p}(X,\mathbb{Q})$ we put

$$L_h(c) = \int_X c \cup \frac{h^{n-p}}{(n-p)!} \in \mathbb{Q}.$$

A real (p,p)-form is *strongly positive* if at every point it is a nonnegative combination of forms $\mathrm{i}\beta_1\wedge\overline{\beta}_1\wedge\cdots\wedge\mathrm{i}\beta_p\wedge\overline{\beta}_p$ with β_j of type $(1,0)$; the closed positive currents of bidegree (p,p) are the currents nonnegative on such forms of the complementary degree, and every smooth closed strongly positive form is one [Dem12, Chapter III].

Proposition 2.76 (Positivity is no test)**.** *For every class $\alpha \in H^{p,p}(X,\mathbb{R})$, $0 \le p \le n$, there is an integer N such that $\alpha + Nh^p$ is the class of a smooth closed strongly positive (p,p)-form. Consequently every real (p,p)-class becomes the class of a closed positive current after the addition of a multiple of h^p, and, since h^p is algebraic, a Hodge class is algebraic if and only if the positive class $\alpha + Nh^p$ is: no property of a class implied by positivity, or by its being a limit of classes of effective cycles, separates the algebraic Hodge classes from the others.*

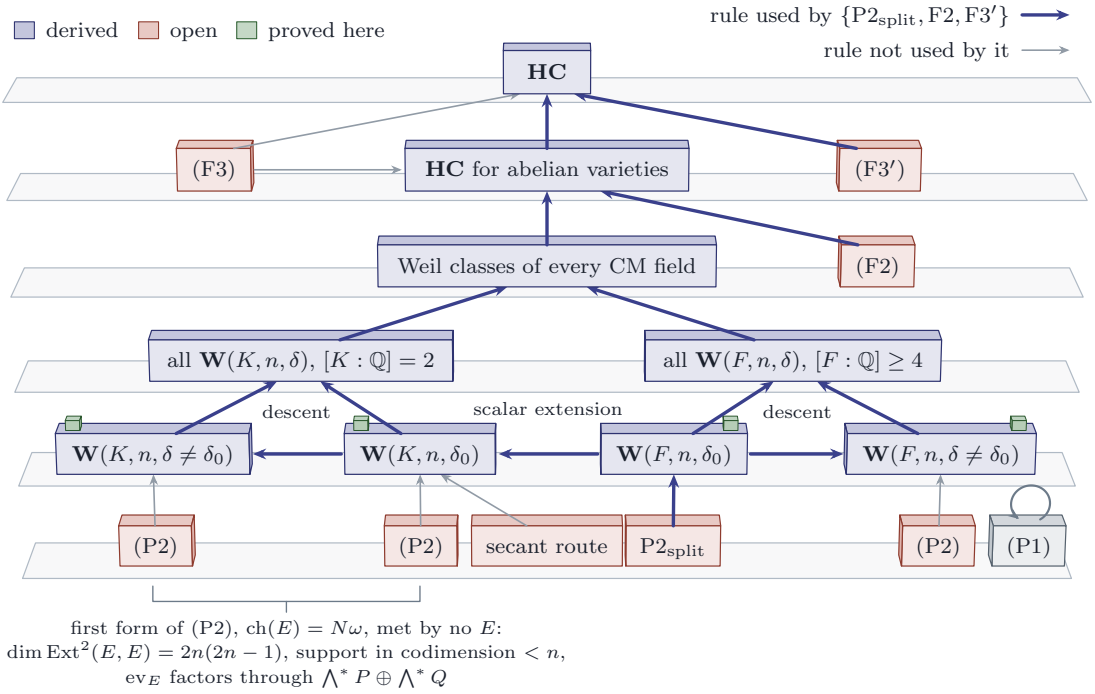


FIGURE 14. The route of this paper through the rule set of Theorem 2.70, drawn as a stack of plates, one per level of the derivation; the other routes are drawn in Figure 13. Heavy arrows are the rules used by the minimal set $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\}$, light arrows the rules it does not use. The bottom plate carries the propagation statement, read in the polarised form of [II, Remark 3.6], once for each of the four kinds of Weil family, with the secant route beside it. The small grass blocks on the next plate are the base points of [I, Theorem 15.27] and [I, Theorem 17.5], one in every family. The three arrows inside that plate are Proposition 2.13: descent from the trivial discriminant to the others, and scalar extension from the fields of degree at least four to the quadratic fields. With them (P2) is needed only for the split families of the fields of higher degree, and the other three (P2) blocks are not used. (F2) is the conjecture for the Hodge classes that no Weil line reaches, and (F3) is the whole conjecture (Proposition 2.33), so the route is minimal only with (F3) replaced by (F3′) of Definition 2.34. The bracket under the two quadratic (P2) blocks records the first form of the criterion, in which the Chern character is exactly a Weil class, as [II, Theorem 2.11], [II, Theorem 2.13] and [II, Corollary 2.9] describe it; by [II, Theorem 3.5] that form is met by no complex. The secant route reaches the trivial discriminant, since the member it is built on is split, and through descent the rest of the quadratic case; it has no edge to the fields of higher degree. The bounded criterion (P1) stands apart, closed on itself: it is equivalent to its own conclusion.

Proof. Let $V = T_xX$ with the hermitian form $\omega_h(x)$, and let $\mathrm{SP}_p \subseteq \Lambda^{p,p}_{\mathbb{R}}V^\vee$ be the closed convex cone generated by the forms $\mathrm{i}^p\beta_1\wedge\overline{\beta}_1\wedge\cdots\wedge\beta_p\wedge\overline{\beta}_p$. These forms span $\Lambda^{p,p}_{\mathbb{R}}V^\vee$, so SP_p has nonempty interior, and $\omega_h(x)^p$ lies in that interior: in a unitary basis $dz_1,\dots,dz_n$ of $V^\vee$,

$$\omega_h^p = p!\sum_{|J|=p}\Big(\frac{\mathrm{i}}{2}\Big)^p dz_J\wedge d\overline{z}_J,$$

and a generator $D = \mathrm{i}^p\beta_1\wedge\overline{\beta}_1\wedge\cdots\wedge\beta_p\wedge\overline{\beta}_p$ equals $|\det B|^2\,2^p(\mathrm{i}/2)^p dz_1\wedge d\overline{z}_1\wedge\cdots\wedge dz_p\wedge d\overline{z}_p$ after a unitary change of coordinates taking the span of the β_j to that of $dz_1,\dots,dz_p$, with B the matrix of the β_j in that basis; so $\omega_h^p - \delta D \in \mathrm{SP}_p$ for $0<\delta\le p!/(2^p|\det B|^2)$. As every real (p,p)-form is a finite real combination of generators, $\omega_h^p+\varepsilon\gamma\in\mathrm{SP}_p$ for every γ and all small ε, which is interiority. The cone, the form $\omega_h(x)^p$ and the norm defined by $\omega_h(x)$ are carried into each other by unitary frames, so the radius $r>0$ of a ball around $\omega_h(x)^p$ contained in SP_p does not depend on x. Represent α by a smooth closed real (p,p)-form a, for instance its harmonic representative, and let $N\ge\max_X|a|/r$. Then $a+N\omega_h^p = N(\omega_h^p + a/N)$ is strongly positive at every point, closed, and represents $\alpha+Nh^p$. The remaining assertions follow, the last because a class of the form $\alpha+Nh^p$ is positive for every α. ∎

The proposition says what positivity cannot do. Demailly studied the cone of closed positive currents against the conjecture and constructed extremal closed strongly positive currents that are not integration currents on analytic sets [Dem82], so extremality does not single out the algebraic cycles either. What positivity can do is turn the conjecture into a variational statement, and the statement is exact. It is the Kähler case of a formulation that Harvey and Lawson give for every calibrated manifold [HL09, Section 6, Proposition 6.13 and Remarks 6.9 and 6.11]; we state it for the classes of this paper, with the effective representative that Lemma 2.77 supplies, and then apply it to the Mumford square. For a closed integral current T of dimension $2(n-p)$ on X, Wirtinger's inequality [Fed69, 1.8.2], [HK74], [HL82, II.3] gives

$$\mathbf{M}(T)\;\ge\;T\Big(\frac{\omega_h^{n-p}}{(n-p)!}\Big)\;=\;L_h(c), \tag{2}$$

where $\mathbf{M}$ is the mass and $c\in H^{2p}(X,\mathbb{Z})$ is the Poincaré dual of the homology class of T, with equality if and only if the approximate tangent plane of T is a complex subspace, positively oriented, at $\|T\|$-almost every point. The right side depends on c alone.

Lemma 2.77 (Complete intersection classes are big)**.** *For every algebraic class $\beta\in H^{2p}(X,\mathbb{Q})$ there is an integer M such that $\beta+Mh^p$ is a positive rational combination of classes of irreducible subvarieties of codimension p; so some positive integer multiple of $\beta+Mh^p$ is the class of an effective algebraic cycle.*

Proof. Replacing h by a multiple, which changes M, we may assume h very ample. Let $Z\subseteq X$ be irreducible of codimension p, and let d be such that the ideal sheaf $I_Z(d)$ is globally generated. The base locus of the linear system of divisors of degree d containing Z is exactly Z, so by Bertini p general members $D_1,\dots,D_p$ of it meet properly away from Z. Every component of $D_1\cap\cdots\cap D_p$ has codimension at most p; a component not contained in Z has codimension exactly p, because the D_i meet properly off Z, and a component contained in Z has codimension at least p and so equals Z. The intersection is therefore the union of Z and a closed set W' of pure codimension p. Hence $d^ph^p = [D_1]\cdots[D_p] = m[Z]+[W']$ with $m\ge1$ and W' effective [Ful98, Chapter 7], and

$h^p - \varepsilon[Z]$ is an effective rational class for $0 < \varepsilon \leq m/d^p$. Write $\beta = \sum_i n_i[Z_i]$ with $n_i \in \mathbb{Q}$ and choose ε_i for each Z_i with $n_i < 0$; then for $M \geq \sum_{n_i<0} |n_i|/\varepsilon_i$ the class $\beta + Mh^p$ is $\sum_{n_i>0} n_i[Z_i] + \sum_{n_i<0}(|n_i|/\varepsilon_i)(h^p - \varepsilon_i[Z_i]) + (M - \sum_{n_i<0}|n_i|/\varepsilon_i)h^p$, a positive rational combination of effective classes, h^p itself being the class of a complete intersection. ■

Theorem 2.78 (The conjecture as a mass gap)**.** *Let $\alpha \in \mathrm{Hdg}^p(X)$ be a rational Hodge class.*

(i) *Let $N \geq 1$ be an integer with $N\alpha$ integral, let M be an integer for which $\alpha + Mh^p$ is the class of a smooth closed strongly positive form, as every large M is by Proposition 2.76, and put $c = N(\alpha + Mh^p)$. The minimum of the mass over closed real normal currents of dimension $2(n-p)$ in the class dual to c is $L_h(c)$, and it is attained by a smooth strongly positive form. The minimum over closed integral currents in the same class exists and is at least $L_h(c)$.*

(ii) *The class α is algebraic if and only if there are integers $N \geq 1$ and $M \geq 0$, with $N\alpha$ integral, for which the minimum of the mass over closed integral currents in the class dual to $N(\alpha + Mh^p)$ equals $L_h(N(\alpha + Mh^p))$; and then every mass minimising integral current in that class is the integration current of an effective algebraic cycle.*

(iii) *Write $\|c\|_{\mathbb{Z}}$ for the minimum in* (i) *over integral currents. By a theorem of Federer, recorded in this form by Harvey and Lawson* [HL09, Section 6], [Fed69, Fed74], *$\lim_{k\to\infty} \|kc\|_{\mathbb{Z}}/k = L_h(c)$: the integrality gap closes asymptotically for every class. The class α is algebraic if and only if, for every sufficiently large M, the gap is closed at some finite k, that is, $\|kc\|_{\mathbb{Z}} = kL_h(c)$ for some $k \geq 1$.*

In particular the conjecture for X is the statement that for every rational Hodge class the integrality gap of this problem vanishes for some N and M; (F2) *is that statement for abelian varieties. Teh and Yang give a different sufficient condition in the same setting, through Lipschitz chains that are closed for d and d^c with positive (k,k)-part* [TY21, Theorem 4.2].

Proof. (i) The form $\omega_h^{n-p}/(n-p)!$ has comass one and calibrates exactly the complex $2(n-p)$-planes, which is Wirtinger's inequality [Fed69, 1.8.2]; so every closed real normal current S in the class has $\mathbf{M}(S) \geq S(\omega_h^{n-p}/(n-p)!) = L_h(c)$. By Proposition 2.76 the class is represented by a smooth closed strongly positive form a, and for a strongly positive (p,p)-form the mass of the current it defines is $\int_X a \wedge \omega_h^{n-p}/(n-p)!$, because a strongly positive $2(n-p)$-vector is a nonnegative combination of complex unit planes, each of mass one and calibrated. So the minimum over real currents is $L_h(c)$ and a attains it. Closed normal currents represent every real homology class and closed integral currents every integral one [FF60, 5.11], and by the compactness theorem of Federer and Fleming the mass attains its minimum on the closed integral currents in a given class [FF60, 9.6]; the lower bound is (2).

(ii) If α is algebraic, Lemma 2.77 gives M and N with $N(\alpha + Mh^p) = \sum_j n_j[V_j]$, $n_j \in \mathbb{Z}_{>0}$, V_j irreducible of codimension p. The integration current $T = \sum_j n_j[V_j]$ is a closed locally integral current [Kin71, Theorem 3.1.1], its tangent planes are complex and positively oriented, so equality holds in (2) and $\mathbf{M}(T) = L_h(c)$ with $c = N(\alpha + Mh^p)$. Conversely let T be a closed integral current in the class dual to c with $\mathbf{M}(T) = L_h(c)$, for instance a mass minimiser when the minimum is $L_h(c)$. Equality in (2) makes T a closed locally rectifiable current of bidimension $(n-p, n-p)$ which is positive, and by King's theorem [Kin71, Theorem 5.2.1] such a current is a holomorphic cycle $\sum_j n_j[V_j]$ with n_j nonzero integers and V_j irreducible analytic subvarieties of dimension $n-p$; positivity makes every n_j positive. (Harvey and Shiffman characterised the holomorphic chains

among closed rectifiable currents of bidimension $(n-p, n-p)$ without positivity, under a hypothesis on the support that Alexander removed [HS74, Ale97].) By Chow's theorem [GH78, p. 167] the V_j are algebraic, so c is the class of an effective algebraic cycle and α is algebraic. This is the argument of [HL09, Proposition 6.13].

(iii) The limit is Federer's theorem on the real and integral masses of a class, as stated in the final note of [HL09, Section 6]; with (i) the limit is $L_h(c)$, since the real minimum is attained by a smooth form. If the gap is closed at k, then (ii), applied with kN in place of N, makes α algebraic. Conversely, if α is algebraic, Lemma 2.77 gives M_0 such that $\alpha + M_0 h^p$ is a positive rational combination of classes of subvarieties; for $M \geq M_0$ so is $\alpha + M h^p$, since h^p is the class of a complete intersection, and some multiple kc is the class of an effective cycle, whose integration current attains the bound. For a fixed small M the equivalence is not asserted, since it would need an effective multiple of every algebraic class represented by a strongly positive form, in codimension at least two. The last sentence of the theorem is the definition of the conjecture for X, and Proposition 3.49 for (F2). ■

Example 2.79 (The Mumford square)**.** Let $Y = X \times X$ be the square of a Mumford fourfold at a very general point of its curve, with the product principal polarisation θ_Y, and let ω be one of the two exceptional classes of Proposition 2.74, normalised in $H^2(X) \otimes H^2(X)$ with $a_0 = 0$ in the notation of Setup 3.18, so that it corresponds to an endomorphism of $H^2(X)$ killing θ with image in the primitive part. Then $L_{\theta_Y}(\omega) = 0$: the pairing of ω with $\theta_Y^6 = \sum_k \binom{6}{k} \theta^k \otimes \theta^{6-k}$ sees only the term $20\,\theta^3 \otimes \theta^3$, and $\int_X x \cup \theta^3 = 0$ for every primitive $x \in H^2(X)$. Also $L_{\theta_Y}(\theta_Y^2) = 8!/6! = 56$, and $4\theta_Y^2$ is the class of the complete intersection $D \cap D'$ of two general members of $|2\theta_Y|$, a base point free system, whose integration current has mass 224. By Theorem 2.78 and Corollary 3.45, the Hodge conjecture for every member of the Mumford family, and with it $B(W \times_C W)$, is equivalent to the following: for some integers $N \geq 1$ and $M \geq 1$ with $N\omega$ integral, the class dual to $N(M\theta_Y^2 + \omega)$ contains an integral current of mass exactly $56NM$; and any such current is an effective algebraic cycle representing the class. For M divisible by four the class $NM\theta_Y^2$ contains one, $NM/4$ copies of $D \cap D'$, of mass $56NM$; so the question is whether the perturbation by $N\omega$, which changes the Wirtinger bound by nothing, can be absorbed without mass. By Theorem 2.29 the answer is yes at every CM point of the curve, with integers N and M depending on the point; and if the same N and M served at infinitely many CM points the answer would be yes everywhere, by Corollary 3.28, because the components of an effective cycle of degree $56NM$ have degree at most $56NM$ and the Hilbert polynomials of subvarieties of bounded degree form a finite set.

Figure 15 draws the theorem and the example.

2.11. **The routes at a glance.**

Table 2: The routes considered, and the boundary of each result.

Route	**Result**	**Beyond the result**
Ambient object, restricted to $A^\vee$, transported by $\Phi_{\mathcal{P}}$	closed by [I, Theorem 7.3]	ambient varieties with exceptional classes

continued on the next page

Table 2, continued.

Route	**Result**	**Beyond the result**
Any K-equivariant construction with the norm character	closed by [I, Theorem 5.6]	constructions that are not eigenclasses
Secant dimension count	no obstruction, [I, Corollary 8.4]	none
Semiregularity dimension count	no obstruction, [I, Corollary 9.3]	none
Semiregularity of an object whose Chern character sits on the Weil line	closed by [I, Theorem 9.7] and [I, Corollary 9.18] for direct sums, and for every object by [II, Theorem 3.5], for every CM field by [II, Theorem 5.4]	objects of positive rank; the weaker hypothesis of [Mar25b, Question 11.4]
The polarised criterion at a quartic CM field, $n = 2$	computed for every shape, [II, Theorem 5.14]: a complex meeting it has $\dim \mathrm{Ext}^2(E, E) \geq 88$, [II, Corollary 5.15]	existence of such a complex
Induction through hyperplane sections	closed by [I, Theorem 10.3], [I, Theorem 10.8]	correspondences, degenerations, coniveau
Kuga–Satake, from a weight two structure of K3 type	not obstructed, [I, Proposition 11.9]	available only at $n = 1$, [I, Corollary 11.12]
Divisors, on a member with quaternionic multiplication	succeeds, [I, Theorem 15.25]	fails at full Hodge group, [I, Proposition 15.18]
Propagation off the base locus	reduced to one bound or one rank, [I, Theorem 16.9], [II, Theorem 2.3]	the bound or the rank, [II, Remark 3.7]
Objects with effective Chern character at a base point	closed by [II, Theorem 4.5]	objects indecomposable in D^b, [II, Remark 4.8]

continued on the next page

Table 2, continued.

Route	Result	Beyond the result
The weakened criterion for a secant object of positive rank	reduced to two conditions on the factor, [II, Theorem 4.56], the first automatic for Markman's construction, [II, Corollary 4.57]; the identity fails for Markman's candidate, [II, Theorem 4.61], and at every transversal double point, [II, Corollary 4.63]	objects whose support has no such point, [II, Remark 4.64]
A secant object $\mathcal{I}_S(b\Theta)$ with smooth support	invariants forced and $d \leq b^2$, so $d \leq 9$ at $b = 3$, [II, Theorem 4.67], [II, Corollary 4.68]	existence for $d \in \{1, 3, 5, 7\}$ at $b = 3$, one vanishing in $H^1(S, N_{S/X})$ and first-order extension, [II, Remark 4.69]
The local part of the weakened criterion	it is a condition on depth: no obstruction at a Cohen–Macaulay point, [II, Theorem 4.75], [II, Corollary 4.77]	Cohen–Macaulay supports, where the identity becomes one global vanishing beside first-order extension
A support that is a complete intersection	closed for every b and every d, [II, Theorem 4.81]	supports of higher Hilbert–Burch rank, [II, Proposition 4.86]
A support with a Hilbert–Burch resolution by line bundles or by simple semi-homogeneous bundles	closed for every rank, twist and d, [II, Theorem 4.89], [II, Theorem 4.92]	resolutions over bundles that are not projectively flat
The secant route, granted every demand it makes	reaches $\delta = \delta_0$, and every δ by Proposition 2.13, Theorem 2.70(iv)	the CM fields of degree at least four, which give the quadratic ones by Proposition 2.13(ii)
Propagation from a base point	closes every imaginary quadratic family at once, [I, Theorem 16.9], [II, Theorem 2.3]	(P2) for the family, Corollary 2.71

continued on the next page

Table 2, continued.

Route	**Result**	**Beyond the result**
The bounded form (P1), read on the transported cycle	closed: the transport spreads each component by c^{2n}/m_i and the data is unbounded whenever a component has finite stabiliser, [I, Lemma 16.14], [I, Theorem 16.15]	nothing; the other reading is closed too
The bounded form (P1), read on the minimal representative	closed: equivalent to the conjecture for the family, [I, Theorem 16.17]	nothing; it is a restatement, so it is not a route
The pure form of (P2), for a given representative	forces the evaluation map to vanish on the $4n^2$-dimensional annihilator, and excludes every representative that is a sheaf on a smooth germ of codimension two of its support, [II, Theorem 2.5], [II, Corollary 2.6]	nothing: no representative exists, [II, Theorem 3.5]
The same form, tested by a number	$\dim \operatorname{Ext}^2(E,E) = 2n(2n-1)$ at one base point would suffice, [II, Theorem 2.11], and no representative attains it, [II, Theorem 3.5]	nothing
The same form, for a representative in codimension n	closed at every member of every family: the support must have a component of codimension less than n, and at a general member every component of codimension at least two has multiplicity zero, [II, Lemma 2.12], [II, Theorem 2.13]	nothing: no representative exists, [II, Theorem 3.5]

continued on the next page

Table 2, continued.

Route	Result	Beyond the result
The Hochschild action on such a representative	the annihilator is the mixed part of a splitting of HH^1 and the action factors through a ring of dimension $2^{2n+1}-1$ in one of dimension 2^{4n}, [II, Theorem 2.8], [II, Corollary 2.9]	nothing: no such complex exists, [II, Theorem 3.5]
The shape of such a representative	an indecomposable summand carries the class; at $n=2$, and at $n=3$ granting the compatibility of [II, Corollary 2.9(iii)], its endomorphism algebra has dimension at least three, so no simple object serves, [II, Theorem 2.14]	nothing: no object of any shape meets the pure form, [II, Theorem 3.5]
The pure form of (P2), Chern character exactly a Weil class	false for every imaginary quadratic field and every $n \geq 2$: a semiregular such complex would deform to very general non-algebraic Weil tori, whose coherent sheaves have no Chern character below the top degree, [II, Proposition 3.4], [II, Theorem 3.5]	the polarised form (P2′), whose Chern character may carry powers of the polarisation, [II, Remark 3.6]
The reduction to Weil classes, and to abelian varieties	false as a statement about Hodge rings in dimension eight, Proposition 2.74; impossible Hodge-theoretically, Proposition 2.75	the algebraicity of the classes beyond the Weil lines, which is (F2), and (F3), which is the whole conjecture, Proposition 2.33
The two classes on the self-product of a Mumford fourfold	a real multiplication by a cubic field on the primitive part of H^2, algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen is, Proposition 3.2, Theorem 3.4	that Kuga–Satake class, Remark 3.5

continued on the next page

Table 2, continued.

Route	Result	Beyond the result
Other routes to the same two classes	degeneration and larger families closed off by rigidity; semiregularity reduced to $\dim \mathrm{Ext}^2 = 119$ at one CM point; specialisation reduced to a degree bound, Theorem 3.21, Theorem 3.27, Corollary 3.28	an object with $\dim \mathrm{Ext}^2 = 119$, or the degree bound, Remark 3.32
The cycles at a CM point, for the same two classes	the fourfold is of Weil type for $k_c \subset \mathrm{End}^0(X_c)$ with the two exotic classes as its Weil line; pull-backs of that line along homomorphisms with components in k_c span, with divisor products, 110 of the 132 Hodge classes of the square and no exceptional class, and an exceptional class is reached exactly by components generating the endomorphism algebra over k_c, Proposition 3.29	a degree bound for cycles using the real multiplication, at infinitely many CM points, Remark 3.31
Objects built from line bundles, for the same two classes	closed at every point of the curve: the Chern characters stay invariant under the Lefschetz group, Theorem 3.36	steps that break that symmetry, Remark 3.37
Twistor deformation of the square, for the same two classes	closed: the component of the Hodge locus among all complex tori is the Mumford curve, which contains no twistor line, Proposition 3.24	the Kuga–Satake class, which moves in a seven-dimensional family
Five further routes, for the same two classes	one statement: three of them are equivalent to the target and two imply it, so any one of them suffices, Corollary 3.45	the target itself, which is only the first case of (F2), Remark 3.46

continued on the next page

Table 2, continued.

Route	Result	Beyond the result
Pull-back from a hyperkähler manifold, for the same two classes	the square is holomorphic symplectic and the classes are the twisted dual forms of the part of H^2 its symplectic class generates; a symplectic rational map to some $S_\mu^{[n]}$, $n \geq 5$, suffices, and no K3 surface or hyperkähler eightfold serves, Proposition 3.25	such a map, Remark 3.26
Reduction modulo p, for the same two classes	represented by cycles at every closed point of every reduction; the target is equivalent to a bound on their Hilbert data on a Zariski dense set, Proposition 3.42, Theorem 3.43	the bound, Remark 3.44
The Weil classes of every CM field	the same propagation problem, with a CM base point in every family, [I, Theorem 17.5], [I, Theorem 17.7]	(P2) for the split families
Routes from outside algebraic geometry	positivity of currents sees nothing; the conjecture is the vanishing of the integrality gap of a mass minimisation whose real version is solved, Proposition 2.76, Theorem 2.78	a calibrated integral current in one class, Example 2.79

continued on the next page

Table 2, continued.

Route	**Result**	**Beyond the result**
The closure, as a whole	five minimal sets of hypotheses: {F3} and {L, M}, each equivalent to the conjecture, and three through the conjecture modulo abelian varieties, {L, F3′}, {V, F3′} and this route, {$P2_{split}$, F2, F3′}, Theorem 2.70, Proposition 2.33, Corollary 2.71, Remark 2.73	the Lefschetz standard conjecture, (M), (V), (F2), (F3′)
Convolutions of the Weil pieces on E_0^{2n}	at $n = 3$ every convolution has $\dim \mathrm{Ext}^2 \geq 69 > 57$, [II, Theorem 4.35]; at $n = 4$, with the pieces at two shifts, $\dim \mathrm{Ext}^2 \geq 40960 > 104$, [II, Theorem 4.37], at three consecutive shifts $\dim \mathrm{Ext}^2 \geq 22528$, [II, Theorem 4.38], with the shifts within six consecutive values $\dim \mathrm{Ext}^2 \geq 640$, [II, Corollary 4.48], and with each parity at one shift, at any odd distance, $\dim \mathrm{Ext}^2 \geq 1792$, [II, Proposition 4.46]	the other arrangements of the shifts at $n = 4$, [II, Remark 4.50]

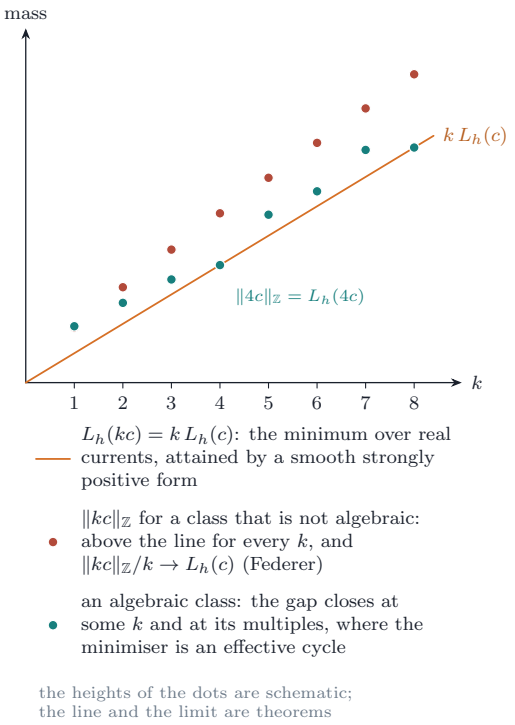
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Table 2, continued.

Route	Result	Beyond the result
Diagonal complete intersections of Vandermonde type	quotients of powers of a generalised Fermat curve, so in $\mathcal{A}$; the conjecture holds when it holds on the abelian varieties $B_{[a]}$, and holds for two diagonal hypersurfaces of degree 3, 4, 6 in $\mathbb{P}^6$, Theorem 2.55, Corollary 2.56; it holds on the very general member for $d = 2$ in every dimension, for $d = 3, 4$ up to dimension seven and for $d = 6$ up to dimension five, Theorem 2.66	$B_{[a]}$ of dimension above five, the Weil classes of $\mathbb{Q}(\zeta_e)$ with $\varphi(e) \geq 4$, and the very general member for $d = 3, 4, 6$ from dimension eight on and for $d = 6$ in dimension six, Remark 2.67
The conjecture modulo abelian varieties, (F3′)	holds on the class $\mathcal{A}$ of varieties whose cohomology is reached from abelian varieties by algebraic correspondences (curves, abelian varieties, products, surjective images), in degrees 0, 2, $2n-2$, $2n$, and in dimension at most three, Proposition 2.35; on the Delsarte hypersurfaces and the diagonal complete intersections of Vandermonde type, Corollary 2.53, Theorem 2.55; on the fourfolds and fivefolds of Proposition 2.43, Proposition 2.44 and Proposition 2.46; through the Hodge conjecture on the varieties of Proposition 2.39 and Proposition 2.40	degree four on fourfolds outside $\mathcal{A}$ not covered by these results, and beyond, for instance squares of K3 surfaces with real multiplication, such as S_λ, Remark 2.36, Remark 2.68

3. Hodge classes beyond the Weil lines

3.1. **The first classes beyond the Weil lines.** Statement (F2) is the Hodge conjecture for the Hodge classes on abelian varieties that no Weil line reaches, and Proposition 2.74 exhibits the first of them: two classes in $H^4(X \times X, \mathbb{Q})$ for an abelian fourfold X of



The Mumford square $Y = X \times X$, $\theta_Y = \theta \otimes 1 + 1 \otimes \theta$

$\theta_Y^6 = \sum_k \binom{6}{k}\, \theta^k \otimes \theta^{6-k}$

k	0	1	2	3	4	5	6
$\binom{6}{k}$	1	6	15	20	15	6	1
$\int_Y \pi_0\, \theta^k \otimes \theta^{6-k}$	0	0	0	144	0	0	0
$\int_Y \pi_{ij}\, \theta^k \otimes \theta^{6-k}$	0	0	0	0	0	0	0

$\int_Y \pi_0\, \theta_Y^6 = 20 \cdot 144 = 2880$, so
$L_{\theta_Y}(\omega_a) = 2880\, a_0/6! = 4a_0$

$L_{\theta_Y}(\omega) = 0$ for an exceptional class with $a_0 = 0$

$L_{\theta_Y}(\theta_Y^2) = 8!/6! = 56$; $4\theta_Y^2 = [D \cap D']$ for $D, D' \in |2\theta_Y|$, of mass 224

open: an integral current of mass $56NM$ in the class dual to $N(M\theta_Y^2 + \omega)$; known at every CM point with N, M depending on the point; the same N, M at infinitely many CM points would give every member

FIGURE 15. Theorem 2.78 and Example 2.79. Left: the mass of a current in the class dual to kc against k. The line $k\,L_h(c)$ is the minimum over closed real normal currents, attained by a smooth strongly positive form; the minima over closed integral currents lie on or above it, with $\|kc\|_{\mathbb{Z}}/k \to L_h(c)$ by Federer's theorem, and for M large the class is algebraic exactly when one of them lies on the line, where the minimiser is an effective algebraic cycle. The heights of the dots are schematic. Right: the numbers of the Mumford square, in the split model. Of the seven terms $\binom{6}{k}\theta^k \otimes \theta^{6-k}$ of θ_Y^6 only $k = 3$ pairs with $\pi_0 = \frac{1}{4}\theta \otimes \theta$, giving $20 \cdot 144 = 2880$, and none pairs with the primitive summands; so $L_{\theta_Y}(\omega_a) = 4a_0$, the Wirtinger bound of an exceptional class with $a_0 = 0$ is zero, and the perturbation of $NM\theta_Y^2$ by $N\omega$ changes the bound $56NM$ by nothing.

Mumford type. None of the constructions of the preceding sections acts on them. In this subsection we identify them as a real multiplication on the primitive part of $H^2(X)$, and we prove that they are algebraic as soon as the Kuga–Satake class of one K3 surface of Picard number thirteen attached to X is algebraic. We do not prove that the Kuga–Satake class is algebraic, and we know of no proof. The reduction changes the kind of question: two classes on an eightfold are traded for one correspondence of a well-studied type, between a surface and its Kuga–Satake variety. The cubic field of the real multiplication enters through the Clifford relations, which force the blocks of the Kuga–Satake embedding to span it.

Setup 3.1 (Mumford type)**.** Let X be as in Proposition 2.74, with $V = H^1(X, \mathbb{Q})$, polarisation ψ and polarisation class $\theta \in H^2(X, \mathbb{Q})$, and let $G = \mathrm{Hg}(X)$. Over $\bar{\mathbb{Q}}$ the Hodge cocharacter is nontrivial on exactly one of the three factors of G: on V_i it has weights

$\pm a_i/2$ with $a_i \geq 0$, and V has only the two weights $\pm\frac{1}{2}$, so exactly one a_i is nonzero. Since $\mathrm{End}^0(X) = \mathbb{Q}$, it follows that G is $\mathbb{Q}$-simple: the Lie algebra of a $\mathbb{Q}$-factor not containing that geometric factor would consist of Hodge classes in $\mathrm{End}(V)$, that is, of endomorphisms of X. So G is the image in $\mathrm{GL}(V)$ of $\mathrm{Res}_{F/\mathbb{Q}} D^1$, where F is a totally real cubic field, D is a quaternion algebra over F split at one real place σ_3 and definite at the other two, σ_1 and σ_2, with $\mathrm{Cor}_{F/\mathbb{Q}} D \cong M_8(\mathbb{Q})$, and D^1 is the group of elements of reduced norm one [Mum69, Gal00]; the kernel is the central subgroup of triples of signs with product one. Over $\bar{\mathbb{Q}}$, $V = V_1 \otimes V_2 \otimes V_3$, where V_i is the standard representation of $D \otimes_{F,\sigma_i} \bar{\mathbb{Q}} = M_2(\bar{\mathbb{Q}})$; the Galois group permutes the three factors as it permutes the three embeddings, and $H^{1,0}(X) = V_1 \otimes V_2 \otimes V_3^{1,0}$. Put

$$T = \mathrm{Lie}\, G = D^0, \qquad q_\lambda(x,y) = \mathrm{Tr}_{F/\mathbb{Q}}(\lambda\, \mathrm{trd}(xy)) \quad (\lambda \in F^\times),$$

where D^0 is the space of quaternions of reduced trace zero, regarded as a $\mathbb{Q}$-vector space of dimension nine, and trd is the reduced trace. Over $\bar{\mathbb{Q}}$, $T = T_1 \oplus T_2 \oplus T_3$ with $T_i = \mathfrak{sl}(V_i)$, and $a \in F$ acts on T_i by $\sigma_i(a)$. Write m_a for multiplication by a. The space $T \subset \mathrm{End}(V)$ is a sub-Hodge structure of weight zero.

Proposition 3.2 (The two classes are a real multiplication)**.** *In Setup 3.1:*

(i) $H^2(X,\mathbb{Q}) = \mathbb{Q}\theta \oplus U$, *where* U *is the primitive part, of dimension* 27, *and over* $\bar{\mathbb{Q}}$

$$U = U_{12} \oplus U_{13} \oplus U_{23}, \qquad U_{ij} = S^2 V_i \otimes S^2 V_j \otimes \textstyle\bigwedge^2 V_k, \quad \{i,j,k\} = \{1,2,3\},$$

each of dimension nine. The summand U_{12} *is of type* $(1,1)$, *and* U_{13} *and* U_{23} *have Hodge numbers* $(3,3,3)$.

(ii) *The algebra of Hodge endomorphisms of* $H^2(X,\mathbb{Q})$ *is* $\mathbb{Q} \times F$, *acting by scalars on* $\mathbb{Q}\theta$, *with* $a \in F$ *acting on* U_{ij} *by* $\sigma_k(a)$. *Through Poincaré duality and the Lefschetz isomorphism* $\theta^2 \colon H^2 \to H^6$ *this algebra is the space of Hodge classes in the Künneth component* $H^2(X) \otimes H^2(X)$ *of* $H^4(X \times X, \mathbb{Q})$; *the subalgebra* $\mathbb{Q} \times \mathbb{Q}$ *is spanned by the products of divisor classes that lie there, and the two classes of Proposition 2.74(ii) may be taken to be the elements of* F *of trace zero.*

(iii) $T(-1)$ *is a Hodge structure of K3 type with Hodge numbers* $(1,7,1)$, *and its Hodge endomorphisms are the* m_a, $a \in F$. *The* G*-invariant symmetric forms on* T *are the* q_λ, *and* q_λ *polarises* $T(-1)$ *if and only if* λ *is totally positive.*

Proof. (i) is the decomposition of $\bigwedge^2(V_1 \otimes V_2 \otimes V_3)$ in the proof of Proposition 2.74, where U_{ij} appears as the summand of highest weight 2 on the i-th and j-th factors. The Hodge cocharacter acts trivially on V_1 and V_2 and splits $V_3 = V_3^{1,0} \oplus V_3^{0,1}$, so $\bigwedge^2 V_3$ has type $(1,1)$ and $S^2 V_3$ has one dimension of each of the types $(2,0)$, $(1,1)$, $(0,2)$.

(ii) A Hodge endomorphism commutes with the Mumford–Tate group $G \cdot \mathbb{G}_m$, and $\mathbb{G}_m$ acts on H^2 by scalars, so the Hodge endomorphisms are the G-endomorphisms. Over $\bar{\mathbb{Q}}$ the four summands $\mathbb{Q}\theta$, U_{12}, U_{13}, U_{23} are irreducible and pairwise non-isomorphic, so the G-endomorphisms are the combinations of the four projections. The Galois group fixes the projection onto $\mathbb{Q}\theta$ and permutes the other three as it permutes the indices k, that is, as it permutes the embeddings of F. For $a \in F$ an endomorphism acting on U_{ij} by $\sigma_k(a)$ is therefore Galois invariant, hence rational, and with the scalars on $\mathbb{Q}\theta$ these exhaust the rational points, which form $\mathbb{Q} \times F$. Poincaré duality identifies $\mathrm{End}\, H^2(X,\mathbb{Q})$ with $H^6(X) \otimes H^2(X)$, and $\theta^2 \otimes 1$ carries $H^2 \otimes H^2$ onto it; both are given by algebraic classes, and on an abelian variety so is the inverse of the Lefschetz isomorphism [Kle68, §2]. The products of divisor classes in $H^2 \otimes H^2$ are $\psi_{11}\psi_{22}$ and ψ_{12}^2 in the notation of the proof of Proposition 2.74. They are invariant under the whole of $\mathrm{Sp}(V,\psi)$, whose endomorphisms

of $H^2 = \mathbb{Q}\theta \oplus U$ are spanned by the identity and the projection onto $\mathbb{Q}\theta$, and they are linearly independent, being a multiple of the projection and a multiple of the identity (proof of Proposition 2.74); so they span $\mathbb{Q} \times \mathbb{Q}$, and any complement of $\mathbb{Q} \times \mathbb{Q}$ in $\mathbb{Q} \times F$ may serve as the two remaining classes.

(iii) The adjoint action of the Hodge cocharacter is trivial on T_1 and T_2 and has weights -1, 0, 1 on T_3, so T has Hodge numbers $(1, 7, 1)$ in the types $(-1, 1)$, $(0, 0)$, $(1, -1)$, and $T(-1)$ is of K3 type. Its Hodge endomorphisms are again the G-endomorphisms, and the argument of (ii), applied to the three pairwise non-isomorphic summands T_i, gives F acting by multiplication. On each T_i the invariant symmetric forms are the multiples of $\operatorname{tr}(xy)$, and $\operatorname{tr}(x_i y_i) = \sigma_i(\operatorname{trd}(xy))$ for $x, y \in D^0$, so the Galois-invariant ones are the q_λ. At σ_1 and σ_2 the algebra $D \otimes \mathbb{R}$ is Hamilton's quaternions, T_i is of type $(1, 1)$, and $\operatorname{trd}(x^2) = -2\operatorname{nrd}(x)$ is negative on nonzero pure quaternions. At σ_3 the form $\operatorname{tr}(xy)$ on $\mathfrak{sl}_2(\mathbb{R})$ is negative on the line of type $(1, 1)$, the Lie algebra of the rotations fixed by the Hodge cocharacter, and positive on its orthogonal complement, the real points of the types $(2, 0)$ and $(0, 2)$. The Hodge–Riemann relations for a structure of K3 type ask for exactly these signs after multiplication by $\sigma_i(\lambda)$, so q_λ polarises $T(-1)$ precisely when every $\sigma_i(\lambda)$ is positive. ∎

The two classes are thus the endomorphisms of U given by the elements of F of trace zero. A Hodge structure of K3 type with the same field sits beside them, and the next lemma connects the two.

Lemma 3.3 (The product map)**.** *For $x, y \in T$ let $\mu(x \otimes y) \in H^2(X, \mathbb{Q})$ be the class that corresponds under ψ to the alternating form $(v, w) \mapsto \psi(xv, yw) + \psi(yv, xw)$ on V. Then μ is a G-equivariant map $T \otimes T \to H^2(X, \mathbb{Q})$, and:*

(i) *for $i \neq j$ it restricts to an isomorphism $T_i \otimes T_j \to U_{ij}$, and it carries $T_i \otimes T_i$ onto $\mathbb{Q}\theta$;*

(ii) *for $e, \lambda \in F^\times$ put $\mu_e = \mu \circ (m_e \otimes m_e)$ and let $\mu_e^\dagger$ be its adjoint for the form $q_\lambda \otimes q_\lambda$ on $T \otimes T$ and the Lefschetz pairing $B(\alpha, \beta) = \int_X \alpha\beta\,\theta^2$ on $H^2(X, \mathbb{Q})$. Then $\mu_e\mu_e^\dagger$ preserves $\mathbb{Q}\theta$ and acts on U as the element*

$$\nu\, \mathrm{N}_{F/\mathbb{Q}}(g)\, g^{-1} \in F, \qquad g = e^2/\lambda,$$

where ν is a nonzero rational number independent of e and λ.

Proof. Equivariance is clear, since μ is built from ψ and the action of $\operatorname{End}(V)$ on V. An element $x \in T_i$ acts on V through the i-th factor and lies in the symplectic Lie algebra of $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$, so $\psi(xv, yw) = -\psi(v, xyw)$ and $\mu(x \otimes y)$ is the form $-\psi(v, (xy + yx)w)$. If $x \in T_i$ and $y \in T_j$ with $i \neq j$, then $xy = yx$ acts as $x \otimes y$ on $V_i \otimes V_j$, and the form is -2 times the product of $\epsilon_i(\cdot, x\,\cdot)$, $\epsilon_j(\cdot, y\,\cdot)$ and ϵ_k; the first two are symmetric because x and y lie in $\mathfrak{sp}(\epsilon) = \mathfrak{sl}_2$, and $x \mapsto \epsilon(\cdot, x\,\cdot)$ is an isomorphism $\mathfrak{sl}_2 \to S^2V^\vee$, so μ maps $T_i \otimes T_j$ isomorphically onto U_{ij}. If $i = j$, then $xy + yx = \operatorname{tr}(xy)$ on V_i, because x and y are trace-free 2×2 matrices, and $\mu(x \otimes y) = -\operatorname{tr}(xy)\,\theta$.

For (ii), m_e is self-adjoint for q_λ, and the adjoint of a map out of $(T \otimes T, q_\lambda \otimes q_\lambda)$ is $(m_\lambda \otimes m_\lambda)^{-1}$ times its adjoint for $q_1 \otimes q_1$, so $\mu_e\mu_e^\dagger = \mu\,(m_g \otimes m_g)\,\mu^{\dagger_1}$ with $\mu^{\dagger_1}$ the adjoint for $q_1 \otimes q_1$. This map is G-equivariant, so it preserves $\mathbb{Q}\theta$ and each U_{ij}, and $\mu^{\dagger_1}$ carries U_{ij} into $(T_i \otimes T_j) \oplus (T_j \otimes T_i)$, on which $m_g \otimes m_g$ is multiplication by $\sigma_i(g)\sigma_j(g) = \mathrm{N}(g)/\sigma_k(g)$. Hence $\mu_e\mu_e^\dagger$ is $\mathrm{N}(g)\sigma_k(g)^{-1}$ times $\mu\mu^{\dagger_1}$ on U_{ij}. The summands of $H^2(X, \bar{\mathbb{Q}})$ are pairwise non-isomorphic and self-dual, so B is nondegenerate on U_{ij}, and by Schur's lemma the pull-back of B along the isomorphism $\mu_0 = \mu|_{T_i \otimes T_j}$ is $c_k\, q_1 \otimes q_1$ with $c_k \neq 0$; then $\mu_0\mu_0^{\dagger_1} = c_k$ on U_{ij}. As μ is symmetric in its two arguments and the exchange of the factors is an

isometry of $q_1 \otimes q_1$, $\mu^{\dagger_1} u = \mu_0^{\dagger_1} u + s\,\mu_0^{\dagger_1} u$ with s that exchange, and $\mu\mu^{\dagger_1} = 2c_k$ on U_{ij}. On the split model $V = \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2$, $\psi = \epsilon \otimes \epsilon \otimes \epsilon$, the permutations of the three factors preserve ψ, hence θ, B and μ, and also $q_1 = \sum_i \operatorname{tr}(x_i y_i)$, while they permute the U_{ij}; so $c_1 = c_2 = c_3$, and $\mu\mu^{\dagger_1}$ is one and the same nonzero number ν on the three summands U_{ij}, rational because $\mu\mu^{\dagger_1}$ is defined over $\mathbb{Q}$. By Proposition 3.2(ii) an endomorphism of U acting on U_{ij} by $\nu \mathrm{N}(g)/\sigma_k(g)$ is the element $\nu \mathrm{N}(g) g^{-1}$ of F. ∎

Figure 16 draws Proposition 3.2 and Lemma 3.3 over the three factors of the Hodge group.

By the Kuga–Satake class of a projective K3 surface S we mean the Hodge class on $S \times \mathrm{KS}(S) \times \mathrm{KS}(S)$ that induces the embedding of [I, Proposition 11.7] for $T(S)_{\mathbb{Q}}$; its algebraicity is the Kuga–Satake Hodge conjecture for S ([I, Remark 11.8]).

Theorem 3.4 (One K3 surface for the two classes)**.** *In Setup 3.1, let $\lambda \in F$ be totally positive.*

(i) *There is a projective K3 surface S_λ of Picard number thirteen and an isomorphism of Hodge structures $T(S_\lambda)_{\mathbb{Q}} \cong T(-1)$ carrying the intersection form to q_λ. Its Kuga–Satake variety is isogenous to X^{32}.*

(ii) *If the Kuga–Satake class of S_λ is algebraic, then the real multiplication by F on $T(S_\lambda)$ is induced by algebraic cycles on $S_\lambda \times S_\lambda$, and every Hodge class in $H^4(X \times X, \mathbb{Q})$ is algebraic; in particular the two classes of Proposition 2.74(ii) are algebraic.*

Proof. (i) The quadratic space (T, q_λ) has dimension nine and signature $(2, 7)$ by Proposition 3.2(iii), and H^2 of a K3 surface with its intersection form has dimension 22 and signature $(3, 19)$. Over $\mathbb{Q}_p$ a nondegenerate quadratic space embeds isometrically in every nondegenerate space of dimension at least three larger, and over $\mathbb{R}$ the signatures allow it, so by the Hasse–Minkowski theorem for representations of forms by forms [OMe00, §§63, 66] there is an isometric embedding of (T, q_λ) into $\Lambda_{\mathbb{Q}}$, where Λ is the K3 lattice. The line $\ell = T(-1)^{2,0} \subset T_{\mathbb{C}} \subset \Lambda_{\mathbb{C}}$ satisfies $(\ell, \ell) = 0$ and $(\sigma, \bar{\sigma}) > 0$ for $0 \neq \sigma \in \ell$, so by the surjectivity of the period map there is a K3 surface S_λ with a marking $H^2(S_\lambda, \mathbb{Z}) \cong \Lambda$ that carries $H^{2,0}(S_\lambda)$ to ℓ [Huy16, Chs. 6–7]. A rational class is of type $(1, 1)$ exactly when it is orthogonal to ℓ, and its component in T is then a rational class of type $(1, 1)$ in $T(-1)$, that is a G-invariant vector of $T = \operatorname{Lie} G$; there is none, since G is semisimple. So $\mathrm{NS}(S_\lambda)_{\mathbb{Q}} = T^\perp$, of dimension 13 and signature $(1, 12)$, and $T(S_\lambda)_{\mathbb{Q}} = T$ with the form q_λ. A class of positive square in $\mathrm{NS}(S_\lambda)$ makes S_λ projective [Huy16, Ch. 8].

For the Kuga–Satake variety, note first that over $\bar{\mathbb{Q}}$ the space T is the orthogonal sum of the three ternary spaces T_i, so $C(T)$ is the graded tensor product of the $C(T_i)$. Scaling a ternary form does not change its even Clifford algebra, so D^1 is the spin group of $(D^0, \lambda \operatorname{trd})$ for every λ, and $\operatorname{Res}_{F/\mathbb{Q}} D^1$ maps to $\mathrm{Spin}(T, q_\lambda)$ by multiplication in the Clifford algebra, with the same kernel as its map to $\mathrm{GL}(V)$, so that G is a subgroup of $\mathrm{Spin}(T, q_\lambda)$; as a representation of the i-th factor, $C(T_i)$ is four copies of V_i. Hence $C^+(T)$ is $V^{\oplus 32}$ as a representation of G over $\bar{\mathbb{Q}}$, and over $\mathbb{Q}$ as well, because V is absolutely irreducible. The Hodge structures on $C^+(T) = H^1(\mathrm{KS}(S_\lambda), \mathbb{Q})$ and on $V^{\oplus 32} = H^1(X^{32}, \mathbb{Q})$ are both given by lifts of weight one of the same homomorphism $\mathbb{S} \to \mathrm{MT}(T)_{\mathbb{R}}$ through $G \cdot \mathbb{G}_m$, and two such lifts coincide.[17] So $H^1(\mathrm{KS}(S_\lambda), \mathbb{Q}) \cong H^1(X^{32}, \mathbb{Q})$ as Hodge structures, and $\mathrm{KS}(S_\lambda)$ is isogenous to X^{32}. For the form used by Galluzzi this is her theorem [Gal00]; van Geemen proves the corresponding splitting for every K3 surface with real multiplication [vG08].

[17] They differ by a homomorphism from the anisotropic torus $\mathbb{S}/\mathbb{G}_m$ to the centre of $G \cdot \mathbb{G}_m$, whose identity component is $\mathbb{G}_m$, and such a homomorphism is trivial.

(ii) Write $S = S_\lambda$ and $A = \mathrm{KS}(S)$, and let Z_S be an algebraic cycle on $S \times A \times A$ inducing the Kuga–Satake embedding $T(S) \to \mathrm{End}\, H^1(A)$. Composing Z_S with the graph of an isogeny $A \to X^{32}$ and with the graphs of the inclusions and projections of the factors of X^{32} gives algebraic cycles on $S \times X \times X$ inducing the 32^2 block components of the embedding, which are morphisms of Hodge structures $T(S) \to \mathrm{End}(V)$, up to a Tate twist; both lie in the tensor category generated by $H^1(X)$, so the blocks are equivariant for $\mathrm{MT}(X) = G \cdot \mathbb{G}_m$. Over $\bar{\mathbb{Q}}$, $\mathrm{End}(V) = S^2V \oplus \bigwedge^2 V$, and $T_i \cong S^2V_i$ occurs once in S^2V, as $S^2V_i \otimes \bigwedge^2 V_j \otimes \bigwedge^2 V_k$, and not at all in $\bigwedge^2 V$. So each block takes values in T, and through the identification of (i) it is a G-endomorphism of T, that is $m_{e_{ab}}$ for some $e_{ab} \in F$ by Proposition 3.2(iii). Over $\bar{\mathbb{Q}}$ the embedding is therefore $v \mapsto v \otimes N_i$ on T_i, acting on $H^1(A) = V \otimes \mathbb{Q}^{32}$, with $N_i = (\sigma_i(e_{ab}))_{a,b}$. The embedding Ψ of [I, Proposition 11.7] satisfies the Clifford relation $\Psi(v)\Psi(w) + \Psi(w)\Psi(v) = 2q_\lambda(v_0, v_0)\, q_\lambda(v, w)$, because $\Psi(v)\Psi(w)x = vwx\, v_0^2$. Take nonzero $v \in T_i$ and $w \in T_j$ with $i \neq j$. As endomorphisms of V they commute and $vw \neq 0$, so the left side is $vw \otimes (N_iN_j + N_jN_i)$, while the right side is zero because T_i and T_j are orthogonal; hence N_i and N_j anticommute. For $v \in T_i$ with $q_\lambda(v, v) \neq 0$ the relation reads $v^2 \otimes N_i^2 = q_\lambda(v_0, v_0)\, q_\lambda(v, v)$, and v^2 is a nonzero scalar on V_i, so N_i^2 is a nonzero scalar. Three pairwise anticommuting invertible matrices are linearly independent: multiplying a relation $\sum_i c_iN_i = 0$ by N_j on either side and adding gives $2c_jN_j^2 = 0$. If the e_{ab} spanned a subspace of F of dimension at most two, the three N_i would lie in the span of at most two rational matrices, which is impossible; so the e_{ab} span F, and two of them, e and e', are not rational multiples of each other. Then $(e/e')^2$ is irrational, since a cubic field contains no irrational element with rational square, so e^2/λ and e'^2/λ are not both rational. Fix a block $\kappa = m_e$ with $e^2/\lambda \notin \mathbb{Q}$, induced by a cycle Γ.

Consider first the real multiplication. The trace form $(a, b) \mapsto \mathrm{tr}(ab)$ on $\mathrm{End}(V)$ is induced by an algebraic class: for correspondences a, b on X the Lefschetz trace formula gives $\mathrm{tr}(a_*b_* \mid H^1) = -\deg((a \circ b \circ \pi_1) \cdot \Delta_X)$, where π_1 is the Künneth projector onto H^1, algebraic on an abelian variety [Kle68, §2]. So the adjoint $\kappa^\dagger$ of κ, for this form and the intersection form on S, is induced by a cycle built from the transpose of Γ, π_1 and Δ_X, and $\kappa^\dagger\kappa$ is an algebraic endomorphism of $T(S)$. An element x of T_i acts on the eight-dimensional space V through a two-dimensional factor, with multiplicity four, so $\mathrm{tr}_V(xy) = 4\,\mathrm{Tr}_{F/\mathbb{Q}}\,\mathrm{trd}(xy)$, and

$$q_\lambda(\kappa^\dagger\kappa x, y) = \mathrm{tr}_V(ex \cdot ey) = 4\,\mathrm{Tr}_{F/\mathbb{Q}}(e^2\,\mathrm{trd}(xy)) = q_\lambda(4e^2\lambda^{-1}x, y),$$

that is $\kappa^\dagger\kappa = m_{4e^2/\lambda}$. By the choice of e this element is irrational, so it generates the cubic field F and every m_a is a polynomial in $\kappa^\dagger\kappa$. Composing with the projector of $H^2(S)$ onto $T(S)$, which is algebraic because it is the diagonal minus $[\mathrm{pt} \times S]$, $[S \times \mathrm{pt}]$ and $\sum_r d_r \times d_r^\vee$ over dual bases of $\mathrm{NS}(S)_\mathbb{Q}$, gives algebraic cycles on $S \times S$ inducing the real multiplication.

Consider next the classes on $X \times X$. The map $\mu_\kappa = \mu \circ (\kappa \otimes \kappa)$ from $T(S) \otimes T(S)$ to $H^2(X, \mathbb{Q})$ is induced by an algebraic cycle on $S \times S \times X$, obtained from $\Gamma \times \Gamma$, the divisor $m^*\theta - p_1^*\theta - p_2^*\theta$ on $X \times X$, whose class is ψ in $H^1 \otimes H^1$, and the diagonal of X by pull-back, push-forward and intersection, because μ uses nothing but ψ and the composition of endomorphisms of V. The form $q_\lambda \otimes q_\lambda$ is the intersection form of $S \times S$ on $T(S) \otimes T(S)$ and B is the intersection form of X twisted by θ^2, so the adjoint $\mu_\kappa^\dagger$ is induced by cup product with θ^2 followed by the transposed cycle, and $E = \mu_\kappa\mu_\kappa^\dagger$ is an algebraic endomorphism of $H^2(X, \mathbb{Q})$. Since $\kappa = m_e$, Lemma 3.3(ii) says that E preserves $\mathbb{Q}\theta$ and acts on U as $\nu\mathrm{N}(g)g^{-1}$ with $g = e^2/\lambda$, and this element of F is not rational, since g is not. The projection π_θ onto $\mathbb{Q}\theta$ is algebraic, being given by the class $\theta \times \theta$ suitably normalised, so the algebraic Hodge endomorphisms of $H^2(X, \mathbb{Q})$ contain 1, π_θ and $E(1 - \pi_\theta)$, which

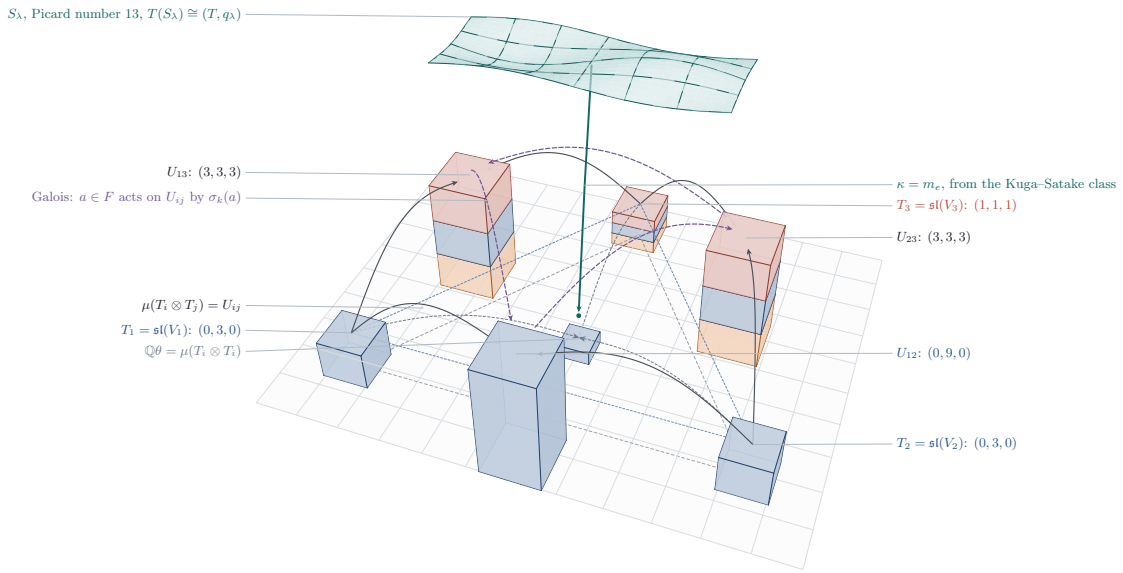


FIGURE 16. Proposition 3.2 and Lemma 3.3, drawn over the triangle of the three factors of the Hodge group; every height is a dimension. At the corners stand the summands $T_i = \mathfrak{sl}(V_i)$ of $T = \operatorname{Lie} G$, banded by Hodge type after the Tate twist $T(-1)$: blue for $(1,1)$, clay for $(2,0)$ and amber for $(0,2)$, so that only T_3 carries the K3 type. On the edges stand the three summands U_{ij} of the primitive part of $H^2(X)$, of dimension nine, U_{12} of pure type $(1,1)$ and the other two in three bands of three; in the middle is the line of θ. The solid arcs are the product map μ, which carries $T_i \otimes T_j$ onto U_{ij}, and the dashed ones carry $T_i \otimes T_i$ to $\mathbb{Q}\theta$. The violet arcs are the Galois group permuting the U_{ij} as it permutes the embeddings of F, which is why the two exceptional classes may be taken to be the trace-zero part of a cubic field. Above, the K3 surface S_λ of Theorem 3.4, whose Kuga–Satake class, if algebraic, comes down to T as the map $\kappa = m_e$.

generate $\mathbb{Q} \times F$, that is all of them by Proposition 3.2(ii). By the same part every Hodge class in $H^2(X) \otimes H^2(X)$ is then algebraic. The Hodge classes of $H^4(X \times X, \mathbb{Q})$ in the other Künneth components number one in each of $H^a \otimes H^{4-a}$, $a \neq 2$, by the count in the proof of Proposition 2.74, and are products of divisor classes. ■

Remark 3.5 (The Kuga–Satake class in the literature). Theorem 3.4 takes the algebraicity of the Kuga–Satake class of S_λ as its input. The cases of the Kuga–Satake Hodge conjecture proved in the literature are these. They include the projective K3 surfaces with a Shioda–Inose structure, which relates the surface to an abelian surface; by Morrison these are the surfaces of Picard number 19 or 20, those of Picard number 18 with $T(S) \cong U \oplus T'$ and those of Picard number 17 with $T(S) \cong U^2 \oplus T'$ [Mor84, Theorem 6.3 and Corollary 6.4], and for these the proof uses in addition the Hodge conjecture for the powers of an abelian surface, whose Hodge classes are generated by divisor classes (see [MZ99]), and the fact that the Kuga–Satake variety of an abelian surface is isogenous to a power of it [Mor85]. With Buskin's theorem that rational Hodge isometries between K3 surfaces are algebraic [Bus19], this reaches the surfaces of Picard number at least 19, those of Picard number

18 with $T(S)_{\mathbb{Q}}$ isotropic, and those of Picard number 17 with $T(S)_{\mathbb{Q}}$ of Witt index two (Remark 3.17). It does not reach every surface of Picard number seventeen, as Morrison notes [Mor84, Remark 7.2]: for $T(S) = U \oplus \langle 6\rangle \oplus \langle -2\rangle \oplus \langle -2\rangle$, the transcendental lattice of the very general K3 surface of a family of dimension three [Mor84, Corollary 2.10], $T(S)_{\mathbb{Q}}$ has Witt index one, since $\langle 6, -2, -2\rangle$ is anisotropic over $\mathbb{Q}_3$: a primitive solution of $3x^2 = y^2 + z^2$ in $\mathbb{Z}_3$ would have $3 \mid y$ and $3 \mid z$, as -1 is not a square modulo 3, and then $3 \mid x$. The double planes branched along six lines, of Picard number at least sixteen, are settled by Paranjape's cycle; see [vG00]. Floccari proves the conjecture for the K3 surfaces attached to projective sixfolds of generalised Kummer type [Flo24], and then, for the K3 surfaces attached to all projective varieties of generalised Kummer type and for every projective K3 surface S with $T(S)_{\mathbb{Q}}$ embedding isometrically in $(U^3 \oplus \langle -m\rangle)_{\mathbb{Q}}$ for some $m > 0$, both the conjecture and the Hodge conjecture for all powers [Flo25, Theorems 1.2 and 5.11]; their transcendental lattices embed, after scaling in the first case, in a rational quadratic space of rank seven and signature $(3, 4)$, so their transcendental rank is at most six. Ingalls, Logan and Patashnick prove the conjecture, with the Hodge conjecture for the square, for a general K3 surface in $\mathbb{P}^4$ with fifteen ordinary double points and for the K3 surfaces joined to it by chains of rational maps of finite degree [ILP22, Theorems 1.5, 4.8 and 6.4]. Bolognesi and Laterveer prove it for the triple covers of a quadric branched along a curve of genus four, which form two families of dimension nine; these carry a non-symplectic automorphism of order three, so their fields of Hodge endomorphisms are CM fields [BL24]. The surfaces S_λ, by contrast, have transcendental rank nine and real multiplication by a cubic field, and none of these cases contains them: a Shioda–Inose structure needs $T(S) \cong T(A)$ for an abelian surface A, hence a transcendental rank of at most five; the surfaces of Floccari, of Paranjape and of Ingalls, Logan and Patashnick have transcendental rank at most six; and the field of Hodge endomorphisms of S_λ is totally real, not CM.

Schlickewei made the real multiplication algebraic on those double planes that have it, necessarily by a field of degree at most two, from Paranjape's cycle and van Geemen's description of their Kuga–Satake varieties [Sch10, vG08], and Varesco showed that when the Kuga–Satake Hodge conjecture holds for two hyperkähler varieties, every Hodge similarity between their transcendental lattices is algebraic up to the Lefschetz isomorphism [Var23], which reaches a quadratic field through its square root. More generally, if the Hodge endomorphisms of $T(S)$ form a totally real field E, then E acts by self-adjoint maps [Zar83], so $g \in E$ is a similarity, with multiplier μ, exactly when $g^2 = \mu \in \mathbb{Q}$; the Hodge similarities of $T(S)$ to itself, and with them all composites of Hodge similarities between transcendental lattices of K3 surfaces that start and end at $T(S)$, therefore span exactly the maximal multiquadratic subfield E_{mq} of E, the compositum of its quadratic subfields. For a cubic field $E_{\mathrm{mq}} = \mathbb{Q}$: a cubic field contains no irrational element with rational square, so no irrational element of it acts as a similarity. In the proof of Theorem 3.4 the element $\kappa^\dagger\kappa = m_{4e^2/\lambda}$ plays the role that the similarities play for quadratic fields, and the Clifford relations are what make it irrational. Buskin's theorem [Bus19] does not give the real multiplication either. A Hodge isometry of $T(S_\lambda)$ is a Hodge endomorphism, hence some m_a by Proposition 3.2(iii), and it is an isometry only when $a^2 = 1$. For $b \in F^\times$ Buskin's theorem does give algebraic cycles between S_λ and $S_{\lambda b^{-2}}$ that realise m_b, but every composite of them that returns to S_λ is again a Hodge isometry of $T(S_\lambda)$, so it multiplies by an element of square one.

The surfaces S_λ include those of Zhu, who computes the integral structure that the fourfold induces on T and studies the K3 surfaces of Picard number thirteen that carry it

[Zhu18]; the form of that structure is the corestriction of the reduced norm of D^0, which up to sign is $\frac{1}{2}q_1$. Theorem 3.4 applies to them as it stands, because the irrationality of e^2/λ comes from the anticommuting blocks and not from λ. The explicit families closest to these are those of van Geemen and Schütt, with real multiplication by the cyclic cubic fields $\mathbb{Q}(\zeta_7+\zeta_7^{-1})$ and $\mathbb{Q}(\zeta_9+\zeta_9^{-1})$ and Picard numbers four and ten, on which the real multiplication is induced by explicit algebraic cycles [vGS25, Theorem 1.2 and Section 4.8]; their Picard numbers are not thirteen, and the Kuga–Satake class, which Theorem 3.4 needs and algebraic real multiplication does not supply, is not treated there.

By Theorem 3.4, then, the first case of (F2) is implied by one instance of the Kuga–Satake Hodge conjecture, for one K3 surface of Picard number thirteen; by Proposition 3.12 below that instance is equivalent to the first case of (F2) at X together with a link, and it is not a reduction of it. The sixfold link of [BB26b] replaces the Kuga–Satake variety of dimension 2^7 by an abelian sixfold and shows what the replacement does and does not give.

Theorem 3.4 trades the two exceptional classes for one Kuga–Satake class. The next results make this exchange precise. If the exceptional classes are algebraic, so is every Hodge class on every power of X (Theorem 3.7); if the Kuga–Satake class is algebraic, so is every Hodge class on every product $S_\lambda^a \times X^b$ (Corollary 3.11). The Kuga–Satake class is algebraic exactly when the exceptional classes are and, in addition, some algebraic cycle links S_λ to a power of X (Proposition 3.12), and no such link can pass through an abelian variety of dimension at most five (Proposition 3.13). The theorem and the corollary are implications, and Remark 3.17 collects the forms they give to the first case of (F2).

Proposition 3.6 (Where T occurs)**.** *In Setup 3.1, write* $\mathrm{Hom}_{\mathrm{Hdg}}$ *for morphisms of rational Hodge structures.*

(i) *T is isomorphic to no subquotient of $H^k(X,\mathbb{Q})(j)$, for any k and j.*

(ii) *The F-vector space $\mathrm{Hom}_{\mathrm{Hdg}}(T, H^2(X\times X,\mathbb{Q})(1))$ is one-dimensional, and the copy of T in $H^2(X\times X,\mathbb{Q})(1)$ lies in $H^1(X)\otimes H^1(X)$. For n odd and for $n\in\{0,16\}$ the Hodge structure T is isomorphic to no subquotient of any $H^n(X\times X,\mathbb{Q})(j)$.*

(iii) *$\mathrm{Hom}_{\mathrm{Hdg}}(T,\mathrm{End}\,V)$ is a one-dimensional F-vector space, spanned by the inclusion. For $\lambda\in F$ totally positive, after an isogeny $\mathrm{KS}(S_\lambda)\to X^{32}$ the Kuga–Satake embedding of $T(S_\lambda)$ is a 32×32 matrix of maps $m_{e_{ab}}$ with $e_{ab}\in F$, and $\mathrm{KS}(S_\lambda)$ is isogenous to B^4, where B is isogenous to X^8 and $\mathrm{End}^0(B)=M_8(\mathbb{Q})$.*

Proof. After the twists all the Hodge structures involved have weight zero and lie in the tensor category generated by V, whose Mumford–Tate group is $G\cdot\mathbb{G}_m$ with $\mathbb{G}_m$ acting trivially in weight zero; so their morphisms and subquotients are those of representations of G. Over $\bar{\mathbb{Q}}$, $T=T_1\oplus T_2\oplus T_3$ with $T_i\cong S^2V_i$, and the Galois group permutes the three summands transitively. Hence T is irreducible, a representation of G defined over $\mathbb{Q}$ contains T_1, T_2 and T_3 with one and the same multiplicity m, and its space of G-maps from T, of $\mathbb{Q}$-dimension $3m$, is a vector space over $\mathrm{End}_G(T)=F$ (Proposition 3.2(iii)) of dimension m.

For (i), write $V=V_1\otimes W$ with $W=V_2\otimes V_3$, on which $\mathrm{SL}(V_2)\times\mathrm{SL}(V_3)$ acts through $\mathrm{SO}(W,q)$, q the product of the two symplectic forms. By the skew Cauchy formula $\bigwedge^k V=\bigoplus_\lambda S_\lambda V_1\otimes S_{\lambda'}W$, over the partitions λ of k with at most two rows and at most four columns, and $S_\lambda V_1\cong S^{\lambda_1-\lambda_2}V_1$ as a representation of $\mathrm{SL}(V_1)$. So T_1 occurs in $\bigwedge^k V$ only through $\lambda=(2,0)$, $(3,1)$ or $(4,2)$, and only if $S_{\lambda'}W$ has a nonzero invariant under $\mathrm{SL}(V_2)\times\mathrm{SL}(V_3)$. As representations of $\mathrm{SL}(W)$, $S_{(1,1)}W=\bigwedge^2 W$, $S_{(2,2,1,1)}W\cong\bigwedge^2 W$, and $S_{(2,1,1)}W\cong\mathfrak{sl}(W)$, the kernel of the contraction $\bigwedge^3 W\otimes W\to\bigwedge^4 W$ with $\bigwedge^3 W\cong W^*$.

Restricted to $\mathrm{SO}(W,q)$, $\bigwedge^2 W = \mathfrak{so}(W) = \mathfrak{sl}(V_2)\oplus\mathfrak{sl}(V_3)$ and $\mathfrak{sl}(W) = \mathfrak{so}(W)\oplus S^2V_2\otimes S^2V_3$, the second summand being the trace-free part of S^2W; none of these has a nonzero invariant. So T_1 occurs in no $\bigwedge^k V$, and by the same argument with the factors exchanged neither do T_2 and T_3. As G is reductive, the subquotients of $\bigwedge^k V$ are its summands; that is (i).

For (ii), $H^n(X\times X) = \bigoplus_{a+b=n}\bigwedge^a V\otimes\bigwedge^b V$. The element of G that is -1 on the first factor $\mathrm{SL}(V_1)$ and 1 on the other two acts on V by -1, hence on $H^n(X\times X)$ by $(-1)^n$, and trivially on T; and H^0, H^{16} are trivial representations. In $H^2(X\times X) = \bigwedge^2 V\oplus(V\otimes V)\oplus\bigwedge^2 V$ the summands $\bigwedge^2 V$ contain no T_i by (i), and $V\otimes V = S^2V\oplus\bigwedge^2 V$ with

$$S^2(V_1\otimes V_2\otimes V_3) = S^2V_1\otimes S^2V_2\otimes S^2V_3\oplus\bigoplus_{i=1}^{3} S^2V_i\otimes\bigwedge^2 V_j\otimes\bigwedge^2 V_k,$$

$\{i,j,k\} = \{1,2,3\}$, where $\bigwedge^2 V_j$ is trivial. So each T_i occurs exactly once in $H^2(X\times X)$, inside $V\otimes V$, and $m=1$. For (iii), $\mathrm{End}\,V = S^2V\oplus\bigwedge^2 V$ contains each T_i once, and the proof of Theorem 3.4(ii) shows that the blocks of the Kuga–Satake embedding are maps $m_{e_{ab}}$. Finally $\mathrm{KS}(S_\lambda)$ is isogenous to $X^{32} = (X^8)^4$ by Theorem 3.4(i), and $\mathrm{End}^0(X^8) = M_8(\mathrm{End}^0 X) = M_8(\mathbb{Q})$. This is the splitting $\mathrm{KS}\sim B^{2^{d-1}}$, $d=[F:\mathbb{Q}]$, which van Geemen proves for K3 surfaces with real multiplication [vG08]; there $H^1(B,\mathbb{Q})$ is the corestriction to $\mathbb{Q}$ of the even Clifford algebra of T over F, here D, and over $\bar{\mathbb{Q}}$ one has $\mathrm{Cor}_{F/\mathbb{Q}}(D) = \bigotimes_k D\otimes_{F,\sigma_k}\bar{\mathbb{Q}} = \bigotimes_k\mathrm{End}(V_k) = \mathrm{End}(V)$, which is $V^{\oplus 8}$ under left multiplication by G. ∎

Theorem 3.7 (Every power of a Mumford fourfold)**.** *In Setup 3.1, suppose that the two exceptional classes of Proposition 2.74(ii) are algebraic on $X\times X$. Then every Hodge class on X^n is algebraic, for every $n\geq 1$.*

Proof. Over $\bar{\mathbb{Q}}$ keep $V = V_1\otimes V_2\otimes V_3$ and $\psi = \epsilon_1\otimes\epsilon_2\otimes\epsilon_3$, as in the proof of Lemma 3.3. For $m\geq 1$ regard $V^{\otimes m} = H^1(X)^{\otimes m}$ as the Künneth component of $H^m(X^m,\mathbb{Q})$ of multidegree $(1,\dots,1)$, whose projector is algebraic [Kle68, §2]. Call an endomorphism of $V^{\otimes m}$ algebraic if it is the action on $V^{\otimes m}$ of an algebraic cycle on $X^m\times X^m$ that preserves $V^{\otimes m}$. Composing with the Künneth projector, the algebraic endomorphisms form a $\mathbb{Q}$-algebra, and they carry algebraic classes in $V^{\otimes m}$ to algebraic classes.

Step 1: reduction to multilinear invariants. Let $n\geq 1$ and $p\geq 0$. The map $V^{\otimes 2p}\otimes(\mathbb{Q}^n)^{\otimes 2p}\to\bigwedge^{2p}(V\otimes\mathbb{Q}^n) = H^{2p}(X^n,\mathbb{Q})$ sending $(v_1\otimes\cdots\otimes v_{2p})\otimes(a_1\otimes\cdots\otimes a_{2p})$ to $(v_1\otimes a_1)\wedge\cdots\wedge(v_{2p}\otimes a_{2p})$ is surjective and G-equivariant for the trivial action on $\mathbb{Q}^n$, so it is surjective on G-invariants, G being reductive. For $a_1,\dots,a_{2p}\in\mathbb{Z}^n$ the image of $t\otimes(a_1\otimes\cdots\otimes a_{2p})$ is f^*t, where $f\colon X^n\to X^{2p}$ is the homomorphism with rows $a_1,\dots,a_{2p}$, and such tensors span $(\mathbb{Q}^n)^{\otimes 2p}$. So $\mathrm{Hdg}^p(X^n)$ is spanned by the classes f^*t with f a homomorphism and $t\in(V^{\otimes 2p})^G$, and it suffices to show that every element of $(V^{\otimes 2p})^G$ is algebraic.

Step 2: degree four. The space $(V^{\otimes 4})^G$ has dimension eight, the product of the dimensions of the three spaces of SL_2-invariants in $V_k^{\otimes 4}$. Let $\mathcal{P}_0\subset(V^{\otimes 4})^G$ be the span of the multilinear components of the classes $f^*\xi$, for $\xi\in\mathrm{Hdg}^2(X\times X)$ and $f\colon X^4\to X^2$ a homomorphism, and of the products of two divisor classes of X^4. By Proposition 2.74(ii) $\mathrm{Hdg}^2(X\times X)$ is spanned by the six products of divisor classes and the two exceptional classes, so under the hypothesis every element of $\mathcal{P}_0$ is algebraic. For $t\in V^{\otimes 4}$, read as a correspondence from the first two factors of X^4 to the last two, let $\hat{t}$ be the endomorphism $v\mapsto t_*\big((\theta^3\times\theta^3)\cup v\big)$ of $V\otimes V$. The pairing $(v,w)\mapsto\int_X v\cup w\cup\theta^3$ on V is nondegenerate by hard Lefschetz and G-invariant, so $t\mapsto\hat{t}$ is an isomorphism $(V^{\otimes 4})^G\to\mathrm{End}_G(V\otimes V)$, and $\hat{t}$ is algebraic

when t is. We show that the $\hat t$ and the composites $\hat u\hat t$, for $t,u\in\mathcal{P}_0$, span $\mathrm{End}_G(V\otimes V)$. Over $\bar{\mathbb{Q}}$, $\mathrm{End}_G(V\otimes V)=\bigotimes_k \mathrm{End}_{\mathrm{SL}_2}(V_k\otimes V_k)$ is the algebra of functions on the eight sign patterns $\varepsilon\in\{\pm1\}^3$, where ε_k is the eigenvalue of the exchange s_k of the two copies of V_k; the patterns with $\varepsilon_1\varepsilon_2\varepsilon_3=-1$, the odd ones, make up $\bigwedge^2 V$.

(a) Let $f=(x_1+x_2,x_3+x_4)\colon X^4\to X^2$. The multilinear component of $f^*\xi$ depends only on the Künneth component of ξ in $H^2(X)\otimes H^2(X)$, and it is that component read as a tensor $t\in\bigwedge^2V\otimes\bigwedge^2V$, so $\hat t$ vanishes on S^2V and takes values in $\bigwedge^2V$. These components fill $(H^2\otimes H^2)^G=\mathrm{End}_G(H^2)$, of dimension 4 by Proposition 3.2(ii); so the $\hat t$ span the functions supported on the four odd patterns.

(b) The map $g=(x_1+x_3,x_2+x_4)$ is f composed with the exchange of the factors 2 and 3 of X^4, which permutes the factors of $V^{\otimes4}$ up to sign. On $V_k^{\otimes4}$ the SL_2-invariants are spanned by $\epsilon_{13}\epsilon_{24}$, $\epsilon_{14}\epsilon_{23}$ and $\epsilon_{12}\epsilon_{34}=\epsilon_{13}\epsilon_{24}-\epsilon_{14}\epsilon_{23}$, where ϵ_{ab} is ϵ_k on the factors a and b; up to a common sign their hats are 1, s_k and $1-s_k$. The exchange of the factors 2 and 3 swaps the first and the third and changes the sign of the second, so on functions of ε_k it carries $\frac12(1-s_k)$ to $\frac12$ and $\frac12(1+s_k)$ to $\frac12-s_k$. Applied to the indicator of the odd pattern $(1,1,-1)$ it gives $\frac12(\frac12-s_1)(\frac12-s_2)$. The identity is $\hat t$ for the multilinear component of a product of two divisor classes, the pull-backs of the class ψ on $X\times X$ along the projections of X^4 to the factors $(1,3)$ and $(2,4)$, so $h=(1-2s_1)(1-2s_2)-1$ lies in the span of the $\hat t$, $t\in\mathcal{P}_0\otimes\bar{\mathbb{Q}}$. As $(\varepsilon_1,\varepsilon_2)$ runs through $(1,1)$, $(1,-1)$, $(-1,1)$, $(-1,-1)$, h takes the values $0,-4,-4,8$, so $h(h+4)/96$ is the indicator of $\varepsilon_1=\varepsilon_2=-1$; subtracting the indicator of $(-1,-1,-1)$, from (a), leaves that of $(-1,-1,1)$. The same argument with the factors V_k permuted gives the other two even patterns with two minus signs, and the indicator of $(1,1,1)$ is 1 minus the other seven.

So the $\hat t$ and $\hat u\hat t$ span $\mathrm{End}_G(V\otimes V)\otimes\bar{\mathbb{Q}}$. Since composition is bilinear and $\mathcal{P}_0\otimes\bar{\mathbb{Q}}$ is spanned by the same pull-backs and products of a basis of Hodge classes, the $\mathbb{Q}$-algebra of algebraic G-endomorphisms of $V\otimes V$ has dimension 8: every G-endomorphism of $V\otimes V$ is algebraic.

Step 3: the partial swaps. Over $\bar{\mathbb{Q}}$, $\mathrm{End}_G(V\otimes V)=\bigotimes_k\mathrm{End}_{\mathrm{SL}_2}(V_k\otimes V_k)$, and $\mathrm{End}_{\mathrm{SL}_2}(V_k\otimes V_k)$ is spanned by 1 and the exchange s_k of the two copies of V_k. In a basis of V_k the matrix units E_{ab} and E_{ba} are dual for the trace form of $\mathfrak{gl}(V_k)$, and $\sum_{a,b}E_{ab}\otimes E_{ba}=s_k$; removing the centre, spanned by the identity 1 with $\mathrm{tr}(1\cdot1)=2$, gives $s_k=\mathrm{Cas}_k+\frac12$, with $\mathrm{Cas}_k=\sum_c x_c\otimes x^c$ for dual bases (x_c), (x^c) of $\mathfrak{sl}(V_k)$ under the trace form. So the s_k come from the three summands T_k of T. The Galois group permutes the s_k as it permutes the embeddings σ_k, so for $a\in F$ the endomorphism $r_a=\sum_k\sigma_k(a)s_k$ is rational, hence algebraic by Step 2. For $m\ge2$ and $1\le i<m$ let $r_a^{(i)}$ and $s_k^{(i)}$ act as r_a and s_k on the factors i and $i+1$ of $V^{\otimes m}$ and as the identity on the others; $r_a^{(i)}$ is algebraic, being induced by the product of a cycle inducing r_a with the diagonals of the other factors. By Schur–Weyl duality $\mathrm{End}_{\mathrm{SL}_2}(V_k^{\otimes m})$ is the image of $\bar{\mathbb{Q}}[S_m]$, generated by the adjacent transpositions, and the commutant of SL_2^3 in $\bigotimes_k V_k^{\otimes m}$ is the tensor product of the three commutants; so $\mathrm{End}_G(V^{\otimes m})\otimes\bar{\mathbb{Q}}$ is generated by the $s_k^{(i)}$. For a basis a_1,a_2,a_3 of F the matrix $(\sigma_k(a_j))$ is invertible, the square of its determinant being the discriminant of F, so each $s_k^{(i)}$ is a $\bar{\mathbb{Q}}$-linear combination of the $r_{a_j}^{(i)}$. The $\mathbb{Q}$-algebra generated by the $r_a^{(i)}$ therefore lies in $\mathrm{End}_G(V^{\otimes m})$ and spans it over $\bar{\mathbb{Q}}$, so it is $\mathrm{End}_G(V^{\otimes m})$: every G-endomorphism of $V^{\otimes m}$ is algebraic. Its dimension is the cube of $\dim\mathrm{End}_{\mathrm{SL}_2}(V_k^{\otimes m})$, the sum of the squares of the multiplicities in $V_k^{\otimes m}$: $8=2^3$ for $m=2$ and $125=5^3$ for $m=3$, where $V_k^{\otimes3}=S^3V_k\oplus V_k^{\oplus2}$.

Step 4: all multilinear invariants. By the first fundamental theorem for SL_2 the invariants in $V_k^{\otimes 2m}$ are spanned by the images of $\epsilon_k^{\otimes m}$ under the permutations of the $2m$ factors. So $(V^{\otimes 2m})^G \otimes \bar{\mathbb{Q}}$ is spanned by the images of $\psi^{\otimes m} = \bigotimes_k \epsilon_k^{\otimes m}$ under the operators that permute the factors of the three $V_k^{\otimes 2m}$ separately, which lie in $\mathrm{End}_G(V^{\otimes 2m}) \otimes \bar{\mathbb{Q}}$, and the map $\mathrm{End}_G(V^{\otimes 2m}) \to (V^{\otimes 2m})^G$, $e \mapsto e(\psi^{\otimes m})$, is surjective. The class ψ in $V \otimes V$ is that of the divisor $m^*\theta - p_1^*\theta - p_2^*\theta$ on $X \times X$ (proof of Theorem 3.4(ii)), up to a nonzero factor, and $\psi^{\otimes m}$ is, up to sign, the product of its pull-backs along the projections of X^{2m} to the pairs of factors $(2j-1, 2j)$; so it is algebraic, and by Step 3 so is every element of $(V^{\otimes 2m})^G$. Step 1 concludes. ∎

Remark 3.8 (Why a composite is needed). Step 2 cannot be carried out with pull-backs and products alone. The skew Cauchy decomposition $\bigwedge^4(V \otimes \mathbb{Q}^4) = \bigoplus_\lambda S_\lambda V \otimes S_{\lambda'}\mathbb{Q}^4$, over the partitions λ of 4 with conjugates λ', is functorial in $\mathbb{Q}^4$, and its summand $S^4V \otimes \bigwedge^4 \mathbb{Q}^4$ contains exactly one line of Hodge classes of X^4: S_4 acts on the two-dimensional space of SL_2-invariants in $V_k^{\otimes 4}$ through its two-dimensional irreducible representation σ, and the trivial representation occurs once in $\sigma^{\otimes 3}$. That line is spanned by a class $\mathrm{Det} \in (V^{\otimes 4})^G$ on which the permutations of the factors of X^4 act by the sign, and whose restriction to the diagonal, on the split model $V = \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2$, is an invariant quartic form of SL_2^3, hence a nonzero multiple of Cayley's hyperdeterminant. Pull-backs along homomorphisms $X^4 \to X^2$ have no component in that summand, because $\bigwedge^4 \mathbb{Q}^2 = 0$, and products of divisor classes have none, because they are invariant under $\mathrm{Sp}(V, \psi)$ and S^4V has no $\mathrm{Sp}(V, \psi)$-invariant. In the notation of Step 2 the hats of the elements of $\mathcal{P}_0$ therefore satisfy one linear condition, which is $v_{(1,-1,-1)} + v_{(-1,1,-1)} + v_{(-1,-1,1)} = 3v_{(1,1,1)}$ for the values v on the even patterns, since the functions found in (a) and (b), those supported on the odd patterns, the constant, h and its two images under the permutations of the factors V_k, satisfy it and span a space of dimension 7. So $\mathcal{P}_0$ has dimension 7 and is the space of Hodge classes of X^4 in the multilinear parts of the other summands, and Det is reached only through a composite of correspondences, which involves a push-forward. The composite h^2 of Step 2 is one: the indicator of $(-1, -1, 1)$ violates the condition. This is where Theorem 3.7 differs from Proposition 2.24, in which the first fundamental theorem for $\mathrm{SL}(W)$ lets pull-backs and products suffice. We have not found Theorem 3.7 in the literature; it is an implication between two cases of the Hodge conjecture.

At the CM points of a Mumford family the conclusion holds without any hypothesis.

Proposition 3.9 (Powers at a CM point)**.** *Let $W \to C$ be a family as in Corollary 2.22 and $c \in C$ a CM point. Then the Hodge conjecture holds for X_c^n for every $n \geq 1$.*

Proof. The proof of Theorem 2.29 shows that for every $n \geq 1$ the ring $\mathrm{Hdg}^*(X_c^n)$ is generated by pull-backs of Hodge classes of the fourfold X_c along homomorphisms $X_c^n \to X_c$, and these are algebraic [Mar25a]. ∎

The statement is also contained in the theorem of [Li26] on the powers of CM fourfolds.

Corollary 3.10 (Powers and fibre powers of a Mumford family)**.** *Let $W \to C$ be as in Corollary 2.22 and, for $n \geq 1$, let $W^{(n)} = W \times_C \cdots \times_C W$, with n factors. The conditions of Corollary 2.30 are equivalent to each of the following.*

(e) *The Hodge conjecture holds for X_t^n for every $t \in C$ and every $n \geq 1$.*

(f) *$B(W^{(n)})$ holds for $L_{W^{(n)}}$ for every $n \geq 1$.*

Proof. (b) implies (e): by Corollary 2.30, (b) gives (c), so the exceptional classes of $X_t \times X_t$ are algebraic for every t. At a point t that is not a CM point the Hodge group of X_t

is G, as shown in the proof of Corollary 2.30, and Theorem 3.7 applies; at a CM point Proposition 3.9 does. (e) implies (f): $W^{(n)}$ is an abelian scheme over C with projective total space, and, as in the proof of Corollary 2.25, its monodromy invariant classes are Hodge classes on every fibre; by (e) they are algebraic on every fibre, hence restrictions of algebraic classes on $W^{(n)}$ by Lemma 2.27, and Theorem 2.21 gives $B(W^{(n)})$. (f) implies (a): take $n = 2$. ∎

Corollary 3.11 (Products with the K3 surface)**.** *In Setup 3.1, let $\lambda \in F$ be totally positive and $S = S_\lambda$ as in Theorem 3.4. If the Kuga–Satake class of S is algebraic, then the Hodge conjecture holds for $S^a \times X^b$ for all $a, b \geq 0$.*

Proof. By Theorem 3.4(ii) the exceptional classes are algebraic, so by Theorem 3.7 the Hodge conjecture holds for every power of X. In the proof of Theorem 3.4(ii) a block $\kappa = m_e$, $e \neq 0$, of the Kuga–Satake embedding is induced by a cycle on $S \times X \times X$; it is a map from $T(S)$ to $\operatorname{End} V$, which up to a twist is a Künneth component of $H^*(X \times X, \mathbb{Q})$. Its adjoint $\kappa^\dagger$ is algebraic, $\kappa^\dagger\kappa = m_{4e^2/\lambda}$, and every m_a, $a \in F$, is induced by a cycle on $S \times S$. So $L = m_{\lambda/(4e^2)}\kappa^\dagger$ is algebraic and $L\kappa$ is the identity of $T(S)$. The decomposition $H^*(S, \mathbb{Q}) = \mathbb{Q} \oplus \mathrm{NS}(S)_\mathbb{Q} \oplus T(S) \oplus H^4(S, \mathbb{Q})$ has algebraic projectors, as in that proof, so $H^*(S^a \times X^b, \mathbb{Q})$ is the sum of the summands $N \otimes T(S)^{\otimes r} \otimes H^*(X^b, \mathbb{Q})$, N a tensor product of $a - r$ of the three other pieces, each cut out by an algebraic projector. Each such N is spanned by products of algebraic classes and is of Tate type, so a Hodge class in the summand is a sum of products $p^*\eta \cup q^*\xi$ with η algebraic on S^{a-r} and ξ a Hodge class in $T(S)^{\otimes r} \otimes H^*(X^b, \mathbb{Q}) \subset H^*(S^r \times X^b, \mathbb{Q})$. For such ξ the class $(\kappa^{\otimes r} \otimes 1)\xi$ is a Hodge class on X^{2r+b}, hence algebraic, and $\xi = (L^{\otimes r} \otimes 1)(\kappa^{\otimes r} \otimes 1)\xi$ is algebraic. ∎

Corollary 3.11 strengthens Theorem 3.4(ii), which gives $X \times X$ only. Varesco proves an implication of the same kind, from the algebraicity of the Kuga–Satake correspondence to the Hodge conjecture for all powers, for families of K3 surfaces of generic Picard number sixteen [Var22]. The converse question, what the Kuga–Satake class asks for beyond the target, has a precise answer.

Proposition 3.12 (What the Kuga–Satake route asks)**.** *In the situation of Corollary 3.11, the Kuga–Satake class of S is algebraic if and only if*

(i) *the two exceptional classes of $X \times X$ are algebraic, and*
(ii) *(a link) for some $n \geq 1$ there is an algebraic cycle Z on $S \times X^n$ whose action Z_* is nonzero on $T(S)$.*

Proof. If the Kuga–Satake class is algebraic, (i) is Theorem 3.4(ii), and a nonzero block of the embedding, induced by a cycle on $S \times X \times X$, is a link with $n = 2$. Conversely assume (i) and (ii). By Theorem 3.7 the Hodge conjecture holds for every power of X. Since $T(S) \cong T(-1)$ is irreducible (proof of Proposition 3.6), Z_* maps $T(S)$ isomorphically onto a sub-Hodge structure of some $H^{2k}(X^n, \mathbb{Q})(k-1)$, which is a direct summand, polarisable Hodge structures being semisimple. Projecting onto it and composing with an isomorphism onto the copy of $T(-1)$ in $H^2(X \times X, \mathbb{Q})$ (Proposition 3.6(ii)) gives a morphism of Hodge structures $H^{2k}(X^n, \mathbb{Q})(k-1) \to H^2(X \times X, \mathbb{Q})$, that is a Hodge class on X^{n+2}, which is algebraic. Its composite with Z_* is an algebraic nonzero morphism $T(S) \to T(-1) \subset H^2(X \times X, \mathbb{Q})$, which is some m_e, $e \neq 0$, by Proposition 3.2(iii). The endomorphisms of $H^2(X \times X, \mathbb{Q})$ that act by m_a on this copy of $T(-1)$ and by zero on a Hodge complement are Hodge classes on X^4, hence algebraic, so every $m_a \colon T(S) \to H^1(X) \otimes H^1(X)$, $a \in F$, is algebraic, and so is its composite with cup product with θ^3 on one factor, which carries $H^1(X) \otimes H^1(X)$ onto the Künneth component of $H^8(X \times X, \mathbb{Q})$ that represents $\operatorname{End} V$.

By Proposition 3.6(iii), after an isogeny $\mathrm{KS}(S) \to X^{32}$ the Kuga–Satake embedding is a matrix of such maps, and the Kuga–Satake class is recovered from its blocks through the graphs of the isogeny and of the inclusions and projections of the factors of X^{32}; so it is algebraic. ∎

So the Kuga–Satake route is at least as strong as the first case of (F2) and is not a reduction of it: at the point X it asks for (i), which is the first case of (F2) there, and in addition for a link. The Hodge conjecture implies both (i) and (ii). The route helps only if the Kuga–Satake class of S_λ can be made algebraic by arguments proper to K3 surfaces. A link has to be a correspondence of a new kind: T occurs in no $H^k(X)$ (Proposition 3.6(i)), and the next proposition shows that it occurs in the cohomology of no abelian variety of dimension below six.

Proposition 3.13 (No small abelian carrier of T)**.** *In Setup 3.1, let Y be an abelian variety such that $T(-1)$ is isomorphic to a subquotient of $H^k(Y,\mathbb{Q})(j)$ for some k and j. Then $\dim Y \geq 6$. Consequently, if $\dim Y \leq 5$ and $S = S_\lambda$ is as in Theorem 3.4, every Hodge class on $S \times Y$, algebraic or not, acts by zero on $T(S)$, and no link as in Proposition 3.12(ii) is a composite $Z_2 \circ Z_1$ of algebraic cycles on $S \times Y$ and on $Y \times X^n$.*

Proof. Write $W = T(-1)$. As W is a subquotient of a tensor construction on $H^1(Y)$, $\mathrm{MT}(Y)$ maps onto $\mathrm{MT}(W)$, whose derived group is the adjoint group G^{ad} of G, acting on W through the adjoint representation. So the derived group of $\mathrm{Hg}(Y)$, an almost direct product of $\mathbb{Q}$-simple groups, maps onto G^{ad}. The image of each factor is a normal subgroup of the $\mathbb{Q}$-simple group G^{ad}, the images commute, and G^{ad} is not commutative, so exactly one factor M maps onto G^{ad}, with finite kernel. Over $\bar{\mathbb{Q}}$, M is isogenous to SL_2^3, with factors indexed by the embeddings $\sigma_1, \sigma_2, \sigma_3$ and permuted transitively by the Galois group, and the irreducible constituents of $H^1(Y,\bar{\mathbb{Q}})$ under M are products $R_1 \otimes R_2 \otimes R_3$ of irreducible representations of the three factors.

Let μ_Y be the Hodge cocharacter of Y, with the two weights 0 and 1 on $H^1(Y)$, and write $\mu_Y = \mu'\mu''$ with fractional cocharacters μ' of M and μ'' commuting with M. The image of μ'' in $\mathrm{MT}(W)$ commutes with G^{ad}, so it is central, and the image of μ' in G^{ad} is the component there of the Hodge cocharacter of W, which by Proposition 3.2(iii) is trivial on the factors at σ_1 and σ_2 and has weights $-1, 0, 1$ on T_3. So μ' is trivial on the first two factors of M and has weights $\pm\frac{1}{2}$ on the standard representation of the third. On $R_1 \otimes R_2 \otimes R_3$, tensored with an eigenvector of μ'' in its multiplicity space, the weights of μ_Y therefore fill an interval of length $\dim R_3 - 1$, so $\dim R_3 \leq 2$. The constituents form a Galois-stable multiset and the Galois group carries the third factor to each of the others, so every constituent is $V_I = \bigotimes_{i\in I} V_i$ for a subset $I \subseteq \{1,2,3\}$, with a multiplicity m_I that depends only on $|I|$, and $m_I \neq 0$ for some nonempty I, since M acts nontrivially.

If $m_I \neq 0$ for some I with $|I| = 2$, then $\dim H^1(Y) \geq 3 \cdot 4 = 12$. If $m_{\{1\}} \neq 0$, let $U = V_1 \otimes N$, $\dim N = m_{\{1\}}$, be the V_1-isotypic summand of $H^1(Y,\mathbb{C})$. The embedding σ_1 is real, so complex conjugation preserves the first factor of M and U is defined over $\mathbb{R}$. The connected group $\mathrm{MT}(Y)$ acts on M preserving each factor, and SL_2 has a single irreducible representation of dimension two, so U is stable under $\mathrm{MT}(Y)(\mathbb{R})$ and is a real sub-Hodge structure of $H^1(Y,\mathbb{R})$, and $U^{1,0}$ and $U^{0,1}$ are complex conjugate and both nonzero. Since μ' acts trivially on U, μ_Y acts on $V_1 \otimes N$ through N, so N carries both weights, $m_{\{1\}} \geq 2$, and $\dim H^1(Y) \geq 3 \cdot 2 \cdot 2 = 12$. Otherwise $H^1(Y,\bar{\mathbb{Q}})$ is $V^{\oplus m}$, $m \geq 1$, plus a trivial representation of M. If $m = 1$, every $H^k(Y)$ is, as a representation of M, a sum of copies of the $\bigwedge^a V$, none of which contains an $S^2 V_i$ (Proposition 3.6(i)), while W contains all three; so $m \geq 2$ and $\dim H^1(Y) \geq 16$. In every case $\dim Y = \frac{1}{2}\dim H^1(Y) \geq 6$.

For the rest, a Hodge class on $S \times Y$ acting nontrivially on the irreducible $T(S) \cong W$ carries it isomorphically onto a sub-Hodge structure of some $H^k(Y,\mathbb{Q})(j)$, and if $Z_2 \circ Z_1$ acts nontrivially on $T(S)$, so does Z_1. ∎

Among the powers of X the smallest carrier of T is $X \times X$, of dimension eight, where $T(-1)$ sits in H^2 (Proposition 3.6(ii)). Whatever is known about abelian varieties of dimension at most five, in particular Markman's theorem in dimension four [Mar25a], no link passes through one.

Remark 3.14 (The bound six as a statement about Hodge structures). The bound of Proposition 3.13 is, as far as we know, the best that Hodge theory gives. Let $K \subset D$ be a CM quadratic extension of F, χ_i the character of the i-th factor of the torus $\mathrm{Res}_{F/\mathbb{Q}} K^1$ over $\bar{\mathbb{Q}}$, and $H = \bigoplus_i (V_i\chi_i \oplus V_i\chi_i^{-1})$, of dimension twelve, with types $(1,0)$ on $V_i\chi_i$ and $(0,1)$ on $V_i\chi_i^{-1}$ for $i = 1, 2$, and with the Hodge cocharacter of V_3 on the two summands with $i = 3$: the signature $(2,0)$, $(2,0)$, $(1,1)$. The characters $\chi_i^{\pm 1}\chi_j^{\pm 1}$ with $i \neq j$ and $\chi_i^{\pm 2}$ are nontrivial, so the part of $\bigwedge^2 H$ on which the torus acts trivially is $\bigoplus_i V_i\chi_i \otimes V_i\chi_i^{-1} = \bigoplus_i (S^2 V_i \oplus \bigwedge^2 V_i)$, that is, $T_1 \oplus T_2 \oplus T_3$ plus three trivial summands. For $i = 1, 2$ the summand T_i is of type $(1,1)$, and $S^2 V_3$ has one dimension in each of the types $(2,0)$, $(1,1)$, $(0,2)$, so this copy of T has Hodge numbers $(1,7,1)$, those of $T(S_\lambda)$. We recall without proof that these are the Hodge structures on $H^1(Z,\mathbb{Q}) \cong D$ of the abelian sixfolds Z with multiplication by K over the Shimura curves of PEL type attached to K and D, which go back to Shimura, and that at a suitable member the copy of T is isomorphic to $T(S_\lambda)$, the Hodge structure on it depending only on the image of the Hodge cocharacter in G^{ad}. Nothing in the paper uses this, and it would not give a link: that would still need an algebraic cycle on $S_\lambda \times Z$ acting nontrivially on $T(S_\lambda)$.

Remark 3.15 (The K3 surfaces at the CM points). Apply the construction of Theorem 3.4(i) at a CM point c of a family as in Corollary 2.22, with the Hodge structure that X_c induces on T, and call the surface S_c. By the proof of Theorem 2.29 the Hodge group of X_c is $\mathrm{Res}_{F/\mathbb{Q}} L^1$ for a quadratic extension L of F contained in D; L is a CM field, since the torus is compact at σ_1 and σ_2, where D is definite, and at σ_3, where it contains the nontrivial image of the circle of the Hodge structure. The torus acts on $T = D^0$ by conjugation, with invariants $D^0 \cap L$, the three-dimensional space L^0 of elements of L of reduced trace zero. So $\mathrm{NS}(S_c)_{\mathbb{Q}} = T^{\perp} \oplus L^0$ has rank sixteen. Write $D = L \oplus Lj$ with $jx = \bar{x}j$ for $x \in L$. Then Lj is the q_λ-orthogonal complement of L^0 in D^0, because $\mathrm{trd}(yj) = 0$ for $y \in L$, so $T(S_c)_{\mathbb{Q}} = Lj$ has rank six; $t \in L^1$ acts on it by multiplication by t^2, so left multiplication by L consists of Hodge endomorphisms and $T(S_c)$ has complex multiplication by the sextic CM field L. At the CM points the Kuga–Satake question therefore lies among K3 surfaces of Picard number sixteen, among which the known cases recalled in Remark 3.5 are particular families. Membership of some S_c in one of them would not help: at c the exceptional classes are already algebraic (Theorem 2.29), and what is missing is the propagation along C.

The last reduction replaces the eightfold $X \times X$ by a fourfold.

Proposition 3.16 (A surface in X). *In Setup 3.1, let $i\colon \Sigma \to X$ be a smooth surface that is the intersection of two ample divisors, and L the cup product with θ. Then $i_* i^* = cL^2$ on $H^2(X,\mathbb{Q})$ for a rational number $c > 0$. For $a \in F$ let ρ_a be the Hodge endomorphism of $H^2(X,\mathbb{Q})$ that is zero on $\mathbb{Q}\theta$ and acts on U as a (Proposition 3.2(ii)), and put $\rho'_a = c^{-1} i^* \rho_a L^{-2} i_*$, a Hodge endomorphism of $H^2(\Sigma,\mathbb{Q})$, where L^{-2} inverts*

$L^2\colon H^2(X) \to H^6(X)$. Then ρ_a is induced by an algebraic cycle on $X \times X$ if and only if ρ'_a is induced by an algebraic cycle on $\Sigma \times \Sigma$. So the two exceptional classes of $X \times X$ are algebraic if and only if the Hodge classes in $H^2(\Sigma) \otimes H^2(\Sigma) \subset H^4(\Sigma \times \Sigma, \mathbb{Q})$ that correspond to the ρ'_a, $a \in F$, are algebraic on the fourfold $\Sigma \times \Sigma$.

Proof. By Proposition 2.74(i) the two divisors have classes $d_1\theta$ and $d_2\theta$ with $d_1, d_2 > 0$, so $i_*i^*\alpha = \alpha \cup [\Sigma] = d_1d_2\theta^2\alpha$ and $c = d_1d_2$. By hard Lefschetz $L^2\colon H^2(X) \to H^6(X)$ is an isomorphism, so i^* is injective on $H^2(X)$, and L^{-2} is induced by an algebraic cycle, X being an abelian variety [Kle68, §2]. For $\alpha \in H^2(X)$, $\rho'_a(i^*\alpha) = c^{-1}i^*\rho_aL^{-2}(cL^2\alpha) = i^*\rho_a\alpha$. If $\beta \in H^2(\Sigma)$ is orthogonal to $i^*H^2(X)$, then $\int_X i_*\beta \cup \gamma = \int_\Sigma \beta \cup i^*\gamma = 0$ for every $\gamma \in H^2(X)$, so $i_*\beta = 0$ and $\rho'_a\beta = 0$. Hence ρ'_a is $i^*\rho_a(i^*)^{-1}$ on $i^*H^2(X)$ and zero on its orthogonal complement, and $\rho_a = c^{-1}L^{-2}i_*\rho'_ai^*$. Both formulas are composites of correspondences with the graph of i, its transpose and L^{-2}, so ρ_a is algebraic if and only if ρ'_a is. The exceptional classes are algebraic if and only if every ρ_a is, by Proposition 3.2(ii), since 1 and the projection onto $\mathbb{Q}\theta$ are algebraic. The last statement uses Poincaré duality on Σ and the algebraicity of the Künneth projectors of a surface [Mur90]. ∎

This is the usual reduction to a complete intersection surface, and it does not make the problem easier. By adjunction K_Σ is the restriction of the sum of the two ample divisors, so Σ is a surface of general type, and the ρ'_a are Hodge classes of degree four on the self-product of a surface of general type.

Remark 3.17 (The forms of the first case of (F2)). Theorem 3.7, Corollary 3.11 and one direction of Proposition 3.12 are implications between cases of the Hodge conjecture; Proposition 3.9 follows from Markman's theorem and is also contained in [Li26]; the rest of the results from Proposition 3.6 to Proposition 3.16 are statements about Hodge structures and correspondences. Together they give the first case of (F2) many equivalent forms. By Corollary 2.30, Corollary 3.10 and Proposition 3.16 it is equivalent to each of the following: the algebraicity of the exceptional classes of $X_t \times X_t$ for uncountably many $t \in C$; the Hodge conjecture for every power of every fibre X_t; $B(W \times_C W)$; B for every fibre power of W; the Hodge conjecture for $W \times_C W$, for the curves of Corollary 2.30(d); and, for uncountably many t, the algebraicity of the classes ρ'_a on $\Sigma_t \times \Sigma_t$ for one complete intersection surface $\Sigma_t \subset X_t$. For a family defined over a number field it is also equivalent to the algebraicity, at one point $t \in C(\mathbb{C}) \setminus C(\bar{\mathbb{Q}})$, of one cycle on one power of X_t whose class is not fixed by the cyclic permutation of the three tensor factors of V (Proposition 3.47). It is implied by the Kuga–Satake Hodge conjecture for the surfaces S_λ attached to uncountably many fibres (Corollary 3.45(ii)), which by Proposition 3.12 is equivalent to it together with a link; the cases of the Kuga–Satake Hodge conjecture recalled in Remark 3.5 are of other shapes, and none has transcendental rank nine with real multiplication by a cubic field.

Two limits govern the constructions at hand. First, the Kuga–Satake class has a nonzero component in $T(S) \otimes H^*(A)$, A a power of X, whereas a product $p^*\alpha \cup q^*\beta$ of pull-backs of algebraic classes of S and of A has none, the algebraic classes of S lying in $H^0 \oplus \mathrm{NS}(S)_\mathbb{Q} \oplus H^4$; so no product of pull-backs gives it, whatever is known on A, and a new correspondence between S and an abelian variety is needed. Second, $\mathrm{Sp}(V, \psi)$ acts on $H^*(X^m, \mathbb{Q}) = \bigwedge^*(V \otimes \mathbb{Q}^m)$ through V, compatibly with products, with pull-back and push-forward along homomorphisms and with integration, hence with the composition of correspondences. The divisor classes of X^m, the Hodge classes of X and the Weil classes of every abelian variety isogenous to a power of X are invariant under it: for a CM field $E \subset M_m(\mathbb{Q}) = \mathrm{End}^0(X^m)$, each component over $\mathbb{C}$ of the top exterior power over E of $V \otimes \mathbb{Q}^m$ is the top exterior power of $V \otimes U$ for a complex vector space U on which V plays

no part, that is $(\det V)^{\otimes \dim U} \otimes (\det U)^{\otimes 8}$, and $\mathrm{SL}(V)$ acts trivially on it. So every class obtained from these by the operations above is $\mathrm{Sp}(V,\psi)$-invariant, while the exceptional classes are not, the $\mathrm{Sp}(V,\psi)$-invariants of $H^4(X\times X,\mathbb{Q})$ being the six products of divisor classes (proof of Proposition 2.74). Proposition 3.48 extends this to every abelian variety, with the Hodge classes of the abelian varieties without isogeny factor X and the Hodge classes of the powers of X fixed by the cyclic permutation of the three tensor factors among the generators, and with a subgroup $G\cdot A_3$ of $\mathrm{Sp}(V,\psi)$ in place of $\mathrm{Sp}(V,\psi)$. What escapes it are classes on products $X^m\times Z$ that are not invariant under $G\cdot A_3$ acting through V, for instance, with $T(-1)$ in $H^2(Z)$ as in Remark 3.14, the Hodge classes of $X\times X\times Z$ that pair the two copies of $T(-1)$.

The statement needed can be taken in either of two forms. One is (V) along C for the exceptional classes, from a CM point, where they are algebraic (Theorem 2.29), and Section 3.2 studies it. The other is (V) for the Kuga–Satake class along the seven-dimensional family of K3 surfaces whose Néron–Severi group contains $\mathrm{NS}(S_\lambda)$, on which it stays a Hodge class, from a member at which it is algebraic; it implies the Kuga–Satake Hodge conjecture for S_λ and with it everything above.

The Kuga–Satake class is algebraic at some members of this family. Call the family $\mathcal{K}$, keep the embedding $T\subset\Lambda_\mathbb{Q}$ of the proof of Theorem 3.4(i), and for $S\in\mathcal{K}$ let κ_S be the Kuga–Satake class of T with the Hodge structure that S induces on it, a Hodge class on $S\times A_S\times A_S$ with $H^1(A_S,\mathbb{Q})=C^+(T)$; at S_λ it is the class of Theorem 3.4. If $P=T(S)_\mathbb{Q}$ is smaller than T, then $N=P^\perp\cap T$ lies in $\mathrm{NS}(S)_\mathbb{Q}$. The complex structure on $C^+(T)$ is left multiplication by an element of $C^+(P\otimes\mathbb{R})$, which commutes with right multiplications; so $C^+(T)=C^+(P)\otimes C^+(N)\oplus C^-(P)\otimes C^-(N)$ is, as a Hodge structure, a sum of copies of $C^+(P)$, right multiplication by an anisotropic vector of P carrying $C^-(P)$ onto $C^+(P)$, and the map Ψ of [I, Proposition 11.7], with $v_0\in P$, carries N to Hodge endomorphisms of A_S, left multiplication by N commuting with that of $C^+(P\otimes\mathbb{R})$. So κ_S is algebraic if and only if the Kuga–Satake class of P is. By the argument recalled in Remark 3.5, κ_S is therefore algebraic at the members of Picard number at least 19, which are dense in $\mathcal{K}$, at those of Picard number 18 with $T(S)_\mathbb{Q}$ isotropic and at those of Picard number 17 with $T(S)_\mathbb{Q}$ of Witt index two. In the last two cases $T(S)_\mathbb{Q}\cong L_\mathbb{Q}$ for an even lattice $L=U\oplus\langle 2a\rangle\oplus\langle -2b\rangle$ or $U^2\oplus\langle -2b\rangle$, which embeds primitively in $U^3\subset\Lambda$. The surjectivity of the period map gives a K3 surface S' with $T(S')=L$, which has a Shioda–Inose structure [Mor84, Corollary 6.4], and a Hodge isometry $T(S)_\mathbb{Q}\cong T(S')_\mathbb{Q}$; by Witt's theorem it extends to a rational Hodge isometry $H^2(S,\mathbb{Q})\cong H^2(S',\mathbb{Q})$, which is algebraic by [Bus19]. The case of Picard number 17 fills subfamilies of dimension three. The space T has Witt index two: its signature $(2,7)$ allows no totally isotropic subspace of dimension three, and an anisotropic kernel of dimension at least seven would be indefinite, hence isotropic by the Hasse–Minkowski theorem [OMe00]. So $T=H\oplus H\oplus K$ with H a hyperbolic plane and K negative definite, and each of the countably many rational subspaces $W\subset T$ of signature $(2,3)$ and Witt index two, for instance $H\oplus H\oplus\mathbb{Q}k$ with $k\in K$, defines a subfamily $\mathcal{K}_W=\{S : T(S)_\mathbb{Q}\subseteq W\}$ of dimension three. At a very general point of a component of $\mathcal{K}_W$ one has $T(S)_\mathbb{Q}=W$, so κ is algebraic there. After a finite cover with level structure the component carries a family of K3 surfaces and of Kuga–Satake abelian varieties, on which κ is a flat section of Hodge classes, and, as in the proof of Lemma 2.27, the points at which it is algebraic form a countable union of images of components of relative Hilbert schemes, closed by properness; by the Baire category theorem one of them is the whole component, so κ is algebraic at every point of $\mathcal{K}_W$. The Mumford curve meets none of these loci: its surfaces have Picard number thirteen, and

sixteen at the CM points (Theorem 3.4(i), Remark 3.15). What is missing is (V) for κ along $\mathcal{K}$ from them, or a bound on the Hilbert data of cycles representing κ over a Zariski dense union of the $\mathcal{K}_W$, which would give κ on the whole of $\mathcal{K}$ by the same argument; neither is known.

3.2. **Rigidity and a numerical criterion for the Mumford classes.** The first case of (F2) is a propagation problem along one compact Shimura curve: the exceptional classes are algebraic at every CM point (Theorem 2.29), and they are algebraic on every member exactly when $B(W \times_C W)$ holds (Corollary 2.30). We ask here whether the propagation can be obtained without the Lefschetz standard conjecture, by one of the mechanisms that have proved other cases of the Hodge conjecture: degeneration, deformation into a larger family on which the class stays of Hodge type, specialisation to special points, or semiregularity. The first two cannot work; the third works exactly when the representing cycles have bounded degree; and the fourth reduces to one dimension count at one CM point. We do not prove that the classes are algebraic at a very general point.

Setup 3.18 (The classes ω_a)**.** Keep Setup 3.1 and let $W \to C$ be a family as in Corollary 2.22, with fibres X_t. Over $\bar{\mathbb{Q}}$ let $\pi_0, \pi_{12}, \pi_{13}, \pi_{23} \in H^2(X) \otimes H^2(X) \subset H^4(X \times X)$ be the classes that correspond, under the isomorphism $H^2 \otimes H^2 \cong \operatorname{End} H^2$ given by the form $\bigwedge^2 \psi$ on $H^2(X) = \bigwedge^2 V^\vee$, to the projections onto $\mathbb{Q}\theta$, U_{12}, U_{13} and U_{23} of Proposition 3.2(i), and for $a = (a_0, a_1, a_2, a_3)$ put

$$\omega_a = a_0\pi_0 + a_1\pi_{12} + a_2\pi_{13} + a_3\pi_{23}, \qquad L_0(a) = a_0 + a_1 + a_2 + a_3.$$

The four tensors are invariant under G and under the exchange τ of the two factors of $X \times X$, because $\bigwedge^2 \psi$ is symmetric; so each ω_a is a flat section over C of Hodge type at every point. The argument of Proposition 3.2(ii) shows that the rational ones are those with $a_0 \in \mathbb{Q}$ and $(a_1, a_2, a_3) = (\sigma_3(\alpha), \sigma_2(\alpha), \sigma_1(\alpha))$ for some $\alpha \in F$; we call such a class *exceptional* when $\alpha \notin \mathbb{Q}$. The products of divisor classes in $H^2 \otimes H^2$ are the rational ω_a with $\alpha \in \mathbb{Q}$, and π_0 is a multiple of $\mathrm{pr}_1^*\theta \cup \mathrm{pr}_2^*\theta$; so adding a rational multiple of π_0 to an exceptional class changes L_0 and not the question of algebraicity. For $c \in C$ write $Y = X_c \times X_c$, let $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\bigwedge^2 T_Y)$, of dimension $28 + 64 + 28 = 120$, act on $H^*(Y, \mathbb{C})$ by contraction as in [II, Setup 4.54], and let $\kappa \in H^1(T_Y)$ be the Kodaira–Spencer class at c of the family $W \times_C W \to C$. Put

$$e(\omega) = \dim HT^2(Y) - \dim \operatorname{Ann}_{HT^2(Y)}(\omega) = 120 - \dim \operatorname{Ann}_{HT^2(Y)}(\omega).$$

Lemma 3.19 (Linear relations among conjugates)**.** *Let F be a cubic number field with embeddings $\sigma_1, \sigma_2, \sigma_3$, let $\alpha \in F \setminus \mathbb{Q}$, and let $c_0, c_1, c_2, c_3, a_0 \in \mathbb{Q}$. The three conjugates $\sigma_i(\alpha)$ are pairwise distinct and nonzero, and if $c_0 a_0 + \sum_i c_i \sigma_i(\alpha) = 0$ then $c_1 = c_2 = c_3$.*

Proof. The minimal polynomial of α has degree three, since $[F : \mathbb{Q}]$ is prime, and is separable, so the conjugates are distinct; they are nonzero because $\alpha \neq 0$. Let Γ be the Galois group of a normal closure; it permutes the σ_i transitively by $\gamma \circ \sigma_i = \sigma_{\gamma(i)}$. Put $\beta = \alpha - \operatorname{Tr}_{F/\mathbb{Q}}(\alpha)/3$, so that $\beta \neq 0$ has trace zero, and $r = -c_0 a_0 - (c_1 + c_2 + c_3)\operatorname{Tr}_{F/\mathbb{Q}}(\alpha)/3 \in \mathbb{Q}$; the hypothesis reads $\sum_i c_i \sigma_i(\beta) = r$. Applying every $\gamma \in \Gamma$ and averaging, each σ_j occurs $|\Gamma|/3$ times in the place of each σ_i, so $r = \frac{1}{3}(\sum_i c_i)\sum_j \sigma_j(\beta) = 0$. The relations $c \in \mathbb{Q}^3$ with $\sum_i c_i \sigma_i(\beta) = 0$ form a Γ-stable subspace N containing $(1, 1, 1)$. The plane $\{\sum c_i = 0\}$ is an irreducible representation of Γ over $\mathbb{Q}$: being transitive on three letters, Γ is the alternating or the full symmetric group, so it contains the 3-cycles, which have no rational eigenvector in that plane. If N were larger than the line of $(1, 1, 1)$, it would therefore contain the plane and be all of $\mathbb{Q}^3$, and $\sigma_1(\beta) = 0$. Hence $c \in \mathbb{Q}(1, 1, 1)$. ∎

Lemma 3.20 (One exceptional class gives the others)**.** *Let $t \in C$. If one rational exceptional class ω is algebraic on $X_t \times X_t$, then every Hodge class in $H^2(X_t) \otimes H^2(X_t)$ is algebraic; in particular so are the two exceptional classes of Proposition 2.74.*

Proof. At a CM point every Hodge class of $X_t \times X_t$ is algebraic (Theorem 2.29). Let t be any other point, so that the Hodge group of X_t is G (proof of Corollary 2.30), and write $X = X_t$. For $x \in H^2(X) \otimes H^2(X)$ let E_x be the endomorphism $y \mapsto x_*(\theta^2 \cup y)$ of $H^2(X, \mathbb{Q})$, where x_* is the action of x as a correspondence. By Poincaré duality and the hard Lefschetz isomorphism $\theta^2 \colon H^2 \to H^6$, the map $x \mapsto E_x$ is a G-equivariant isomorphism onto $\operatorname{End} H^2(X, \mathbb{Q})$, so it carries the Hodge classes onto the Hodge endomorphisms, which are $\mathbb{Q} \times F$ by Proposition 3.2(ii). It carries the products of divisor classes, which are invariant under $\operatorname{Sp}(V, \psi)$, onto the $\operatorname{Sp}(V, \psi)$-equivariant endomorphisms, which are $\mathbb{Q} \times \mathbb{Q}$; hence it carries ω to an element (r, β) with $\beta \notin \mathbb{Q}$. The algebraic Hodge classes form a subspace A containing the products of divisor classes, and its image under $x \mapsto E_x$ is closed under composition: if x_1, x_2 are algebraic, then the composite correspondence $x_1 \circ \Delta_*(\theta^2) \circ x_2$ is algebraic, and so is its Künneth component in $H^2 \otimes H^2$, since the Künneth projectors of $X \times X$ are polynomials in the graphs of $[2] \times 1$ and $1 \times [2]$, which act on $H^i(X) \otimes H^j(X)$ by 2^i and 2^j (as in Step 1 of the proof of Theorem 2.21); that component x has $E_x = E_{x_1} E_{x_2}$. So the image of A is a subalgebra of $\mathbb{Q} \times F$ containing $\mathbb{Q} \times \mathbb{Q}$ and (r, β), hence $\mathbb{Q} \times \mathbb{Q}[\beta] = \mathbb{Q} \times F$, because β generates the cubic field F. Thus A is the space of all Hodge classes in $H^2(X) \otimes H^2(X)$. ∎

Theorem 3.21 (Rigidity of the exceptional classes)**.** *Let $c \in C$, $Y = X_c \times X_c$, and let $\omega = \omega_a$ be a rational exceptional class.*

(i) $\operatorname{Ann}_{H^1(T_Y)}(\omega) = \mathbb{C}\kappa$ *and* $\operatorname{Ann}_{H^{0,2}(Y)}(\omega) = 0$.

(ii) $\operatorname{Ann}_{H^0(\bigwedge^2 T_Y)}(\omega) = 0$ *if and only if* $L_0(a) = a_0 + \operatorname{Tr}_{F/\mathbb{Q}}(\alpha) \neq 0$. *In particular* $\operatorname{Ann}_{HT^2(Y)}(\omega) = \mathbb{C}\kappa$ *and* $e(\omega) = 119$ *whenever* $L_0(a) \neq 0$, *which can always be arranged by adding a rational multiple of* π_0.

(iii) *Let $\mathfrak{S}$ be the Siegel space of weight one Hodge structures on $H^1(Y, \mathbb{Q})$ polarised by $\psi \oplus \psi$, of dimension* 36. *The germ at Y of the locus in $\mathfrak{S}$ where ω stays of Hodge type is the germ of the curve $\{X_t \times X_t\}$ through Y.*

Proof. Notation. Over $\mathbb{C}$ write $V = V_1 \otimes V_2 \otimes V_3$ and $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$, as in the proof of Lemma 3.3, and identify V with $H^1(X_c, \mathbb{C})$ and with its dual through ψ. As at every point of C, the Hodge cocharacter lies in the third factor: $V_3^{1,0} = \mathbb{C}e$, $V_3^{0,1} = \mathbb{C}f$, $V^{1,0} = V_1 \otimes V_2 \otimes e$ and $V^{0,1} = V_1 \otimes V_2 \otimes f$. Let $N \in \mathfrak{sl}(V_3)$ be the nilpotent element with $Ne = f$, $Nf = 0$. Then $H^1(Y, \mathbb{C}) = V_{(1)} \oplus V_{(2)}$, two copies of V, with the form $\psi \oplus \psi$, and $\omega \in \bigwedge^2 V_{(1)} \otimes \bigwedge^2 V_{(2)}$.

Step 1: operators. For $J \subseteq \{1, 2, 3\}$ let $E_J \subset \operatorname{End} V$ be the tensor product of $\mathfrak{sl}(V_i)$ for $i \in J$ and of $\mathbb{C} \cdot 1$ for $i \notin J$. Then $\operatorname{End} V = \bigoplus_J E_J$, and the E_J are irreducible and pairwise non-isomorphic under conjugation by G. The ψ-adjoint of an element of $\mathfrak{sl}(V_i)$ is its negative, so E_J consists of ψ-skew operators when $|J|$ is odd and of ψ-symmetric ones when $|J|$ is even. The map $v \wedge w \mapsto \psi(v, \cdot)w - \psi(w, \cdot)v$ identifies $\bigwedge^2 V$ with the ψ-symmetric operators $E_\emptyset \oplus E_{12} \oplus E_{13} \oplus E_{23}$, which are $\mathbb{C}\theta$, U_{12}, U_{13}, U_{23}; a direct expansion shows that $\bigwedge^2 \psi$ becomes the form $\langle M, M' \rangle = \frac{1}{2} \operatorname{tr}(MM')$, and that the derivation action of $X \in \operatorname{End} V$ becomes $M \mapsto XM + MX^\psi$, that is $D_X M = [X, M]$ if X is ψ-skew and $D_X M = XM + MX$ if X is ψ-symmetric. In particular $D_X^\dagger = \mp D_X$ in the two cases, $\dagger$ denoting the adjoint for $\langle\,,\rangle$. By the Cayley–Hamilton theorem $XY = \frac{1}{2}[X, Y] + \frac{1}{2} \operatorname{tr}(XY)$ for $X, Y \in \mathfrak{sl}(V_i)$. Finally, by Setup 3.18, $\omega = \sum_j M_j \otimes P_a M^j$ for a basis (M_j) of $\bigwedge^2 V$

with dual basis (M^j), where P_a acts on $E_\emptyset, E_{12}, E_{13}, E_{23}$ by a_0, a_1, a_2, a_3. As P_a commutes with G, it preserves the Hodge types.

The space $\bigwedge^2 V^{0,1}$ consists of the ψ-symmetric operators that kill $V^{0,1}$ and take values in it, namely $\mathfrak{sl}(V_1)\otimes 1\otimes N \oplus 1\otimes\mathfrak{sl}(V_2)\otimes N \subset E_{13}\oplus E_{23}$, on which P_a is a_2 and a_3. The map $v\otimes w\mapsto \psi(v,\cdot)w$ identifies $V^{0,1}\otimes V^{0,1}$ with $\operatorname{Hom}(V^{1,0},V^{0,1}) = \operatorname{End} V_1 \otimes \operatorname{End} V_2 \otimes \mathbb{C}N = \bigoplus_{J\ni 3} E_J^-$, where $E_J^- = E_J \cap \operatorname{Hom}(V^{1,0},V^{0,1})$ has dimension $1,3,3,9$ for $J = \{3\},\{1,3\},\{2,3\},\{1,2,3\}$.

Step 2: the pieces. The space $HT^2(Y)$ acts as follows: $H^{0,2} = \bigwedge^2 H^{0,1}$ by the wedge product, $H^1(T_Y) = \operatorname{Hom}(H^{1,0},H^{0,1})$ by derivations, and $H^0(\bigwedge^2 T_Y) = \bigwedge^2 (H^{1,0})^\vee$, which $\psi\oplus\psi$ identifies with $\bigwedge^2 H^{0,1}$, by contraction. Write $\eta\in\bigwedge^2 H^{0,1}$ as $\eta_{11}+\eta_{12}+\eta_{22}$ with $\eta_{11}\in\bigwedge^2 V^{0,1}_{(1)}$, $\eta_{12}\in V^{0,1}_{(1)}\otimes V^{0,1}_{(2)}$, $\eta_{22}\in\bigwedge^2 V^{0,1}_{(2)}$, and $\xi\in H^1(T_Y)$ as a matrix (ξ_{st}) with $\xi_{st}\in\operatorname{Hom}(V^{1,0}_{(t)},V^{0,1}_{(s)})$. The products $\eta_{11}\wedge\omega$, $\eta_{12}\wedge\omega$, $\eta_{22}\wedge\omega$ lie in different summands $\bigwedge^p V_{(1)}\otimes\bigwedge^q V_{(2)}$, and so do the three contractions, and $\xi_{11}\omega+\xi_{22}\omega$, $\xi_{12}\omega$, $\xi_{21}\omega$. So each annihilator is computed piece by piece, and τ exchanges the pieces 11 and 22, 12 and 21. For a rational exceptional class, Lemma 3.19 gives: a_1,a_2,a_3 are pairwise distinct and nonzero, and no linear form in $a_0,\dots,a_3$ with rational coefficients that are not all equal on a_1,a_2,a_3 vanishes. We use this for $a_i\pm a_j$ $(i\neq j)$, a_2+3a_3, $3a_2+a_3$ and the three forms in Step 4.

Step 3: $H^{0,2}$. Let (x_j) be a basis of $\bigwedge^2 V^{1,0}$ and (x^j) the dual basis of $\bigwedge^2 V^{0,1}$. The component of ω in $\bigwedge^2 V^{1,0}_{(1)}\otimes\bigwedge^2 V^{0,1}_{(2)}$ is $\sum_j x_j\otimes Kx^j$, where $K = P_a$ on $\bigwedge^2 V^{0,1}$ is invertible because $a_2a_3\neq 0$. For $\eta_{11}\neq 0$ the part of $\eta_{11}\wedge\omega$ in $H^{2,2}(X_c)\otimes H^{0,2}(X_c)$ is $\sum_j(\eta_{11}\wedge x_j)\otimes Kx^j$, which is nonzero because $\bigwedge^2 V^{0,1}\otimes\bigwedge^2 V^{1,0}\to\bigwedge^4 V$ is injective; by τ the same holds for η_{22}. For $\eta_{12} = \sum_r v_r\otimes w_r$ the part of $\eta_{12}\wedge\omega$ in $H^{2,1}\otimes H^{0,3}$ is $\sum_{r,j}(v_r\wedge x_j)\otimes(w_r\wedge Kx^j)$. If it vanishes, then, as $V^{0,1}\otimes\bigwedge^2 V^{1,0}\to\bigwedge^3 V$ is injective and the Kx^j span $\bigwedge^2 V^{0,1}$, $\sum_r\phi(v_r)\,w_r\wedge y = 0$ for every linear form ϕ on $V^{0,1}$ and every $y\in\bigwedge^2 V^{0,1}$; as $\dim V^{0,1}=4$ this forces $\sum_r\phi(v_r)w_r = 0$ for every ϕ, so $\eta_{12}=0$. Hence $\operatorname{Ann}_{H^{0,2}(Y)}(\omega) = 0$.

Step 4: $H^0(\bigwedge^2 T_Y)$. Up to signs and one nonzero factor fixed by the normalisation of the contraction, $\iota_{\eta_{11}}\omega = \sum_j\langle\eta_{11},M_j\rangle P_aM^j = P_a\eta_{11}$, which is nonzero for $\eta_{11}\neq 0$ because $a_2a_3\neq 0$; likewise for η_{22}. For $\eta_{12} = v\otimes w$ the contraction is $\sum_j M_jv\otimes(P_aM^j)w$, and through $v\otimes w\mapsto\psi(v,\cdot)w$ it becomes the operator $Q_a(R) = \sum_j(P_aM^j)\,R\,M_j$ with $R = \psi(v,\cdot)w$, because the M_j are ψ-symmetric. For a basis (X_k) of $\mathfrak{sl}(V_i)$ and the dual basis (X^k) for the trace form, $\sum_k X^kRX_k$ is $\frac{3}{2}R$ for $R=1$ and $-\frac{1}{2}R$ for $R\in\mathfrak{sl}(V_i)$ (take $X_k = h,e,f$ and $X^k = h/2,f,e$). The dual basis of E_{ij} for $\langle\,,\rangle$ is the tensor product of these dual bases, and that of $E_\emptyset$ is $\frac{1}{4}$. Hence Q_a acts on E_J by the scalar

$$q_J(a) = \tfrac{1}{4}a_0 + \sum_{\{i,j\}} a_{\{i,j\}}\,\lambda_i\lambda_j, \qquad \lambda_i = -\tfrac{1}{2}\ (i\in J),\quad \lambda_i = \tfrac{3}{2}\ (i\notin J),$$

where $a_{\{1,2\}} = a_1$, $a_{\{1,3\}} = a_2$, $a_{\{2,3\}} = a_3$; on the four pieces E_J^- this is

$$4q_3 = a_0+9a_1-3a_2-3a_3, \qquad 4q_{13} = a_0-3a_1+a_2-3a_3,$$
$$4q_{23} = a_0-3a_1-3a_2+a_3, \qquad 4q_{123} = L_0(a).$$

The first three do not vanish. So the annihilator in $H^0(\bigwedge^2 T_Y)$ is zero when $L_0(a)\neq 0$, and is the piece E_{123}^- of η_{12}, of dimension nine, when $L_0(a)=0$.

Step 5: $H^1(T_Y)$, the diagonal pieces. Put $\Phi(x\otimes y)(z) = \langle z,x\rangle y$, so that $\Phi(\omega) = P_a$, $\Phi((D\otimes 1)\omega) = P_aD^\dagger$ and $\Phi((1\otimes D)\omega) = DP_a$. Thus $\xi_{11}\omega+\xi_{22}\omega = 0$ if and only if $P_aD^\dagger_{\xi_{11}} + D_{\xi_{22}}P_a = 0$. This condition is G-equivariant in $(\xi_{11},\xi_{22})\in\operatorname{End}V\oplus\operatorname{End}V$, so

its solutions are the pairs $(b_{11}X, b_{22}X)$ with $X \in E_J$ and (b_{11}, b_{22}) in a subspace $M_J \subseteq \mathbb{C}^2$; we need $J \ni 3$. For $X = \bigotimes X_i \in E_J$ and $M = \bigotimes M_i \in E_I$, the product rule above gives $XM \pm MX$ as $2^{1-|I\cap J|}$ times the sum, over $T \subseteq I \cap J$ with $|T| \equiv |J|$ modulo 2, of the tensor with factor $[X_i, M_i]$ for $i \in T$, $\operatorname{tr}(X_i M_i)$ for $i \in (I \cap J) \setminus T$, and the remaining factor of X or M elsewhere; this term lies in $E_{(I \triangle J)\cup T}$ and is nonzero for suitable X and M. On the block from E_I to $E_{I'}$ the condition reads $b_{22}a_I = b_{11}a_{I'}$ for $|J|$ odd and $b_{22}a_I = -b_{11}a_{I'}$ for $|J|$ even, with $a_\emptyset = a_0$, $a_{12} = a_1$, $a_{13} = a_2$, $a_{23} = a_3$.

- $J = \{3\}$: the blocks from E_{13} to E_{13} and from E_{23} to E_{23} give $(b_{22} - b_{11})a_2 = 0$, so $M_J = \mathbb{C}(1, 1)$.
- $J = \{1, 2, 3\}$: there is a block between any two distinct E_{12}, E_{13}, E_{23}, so $b_{22}a_1 = b_{11}a_2$ and $b_{22}a_2 = b_{11}a_1$, whence $(b_{11} + b_{22})(a_1 - a_2) = 0$ and $b_{11} = -b_{22}$; then $b_{22}(a_1 + a_2) = b_{22}(a_1 + a_3) = 0$, and $b_{22} \neq 0$ would give $a_2 = -a_1 = a_3$. So $M_J = 0$.
- $J = \{1, 3\}$: the blocks between E_{12} and E_{23} give $b_{22}a_1 + b_{11}a_3 = b_{22}a_3 + b_{11}a_1 = 0$, of determinant $a_1^2 - a_3^2 \neq 0$, so $M_J = 0$; and $J = \{2, 3\}$ likewise, with $a_1^2 - a_2^2 \neq 0$.

So on the diagonal pieces the annihilator is the line of (N, N), with N acting on both copies of V.

Step 6: $H^1(T_Y)$, the mixed pieces. The map ξ_{12} kills $V^{0,1}_{(2)}$, so the part of $\xi_{12}\omega$ in $H^{0,3}(X_c) \otimes H^{1,0}(X_c)$ comes from the component $\sum_j Kx^j \otimes x_j$ of ω in $\bigwedge^2 V^{0,1}_{(1)} \otimes \bigwedge^2 V^{1,0}_{(2)}$ only, and is $\sum_j Kx^j \wedge \xi_{12}(x'_j) \otimes x''_j$, where $x \mapsto x' \otimes x''$ is the coproduct $\bigwedge^2 \to V \otimes V$. Identify $V^{1,0}$ and $V^{0,1}$ with $W = V_1 \otimes V_2$ through e and f, with the symmetric form $q = \epsilon_1 \otimes \epsilon_2$ and a volume form; then $\bigwedge^2 W$ is the space of q-skew operators $\mathfrak{sl}(V_1) \otimes 1 \oplus 1 \otimes \mathfrak{sl}(V_2)$, the Hodge star is $+1$ on one summand and -1 on the other, and K is a_2 on the first and a_3 on the second. Pair the part above with $b \in W$ in the first factor, through the volume form, and with $\beta \in W$ in the second, through q. Writing $\operatorname{vol}(y \wedge y') = q(*y, y')$ on $\bigwedge^2 W$ turns the pairing into the trace of $x \mapsto K * \big(\xi_{12}(\iota_\beta x) \wedge b\big)$ on $\bigwedge^2 W$, and summing over orthogonal bases of the two summands gives, up to a nonzero factor and a sign, $q\big(c_a(\xi_{12})\beta, b\big)$ with

$$c_a(\xi) = a_2 \sum_k (X^k \otimes 1)\, \xi\, (X_k \otimes 1) - a_3 \sum_k (1 \otimes X^k)\, \xi\, (1 \otimes X_k)$$

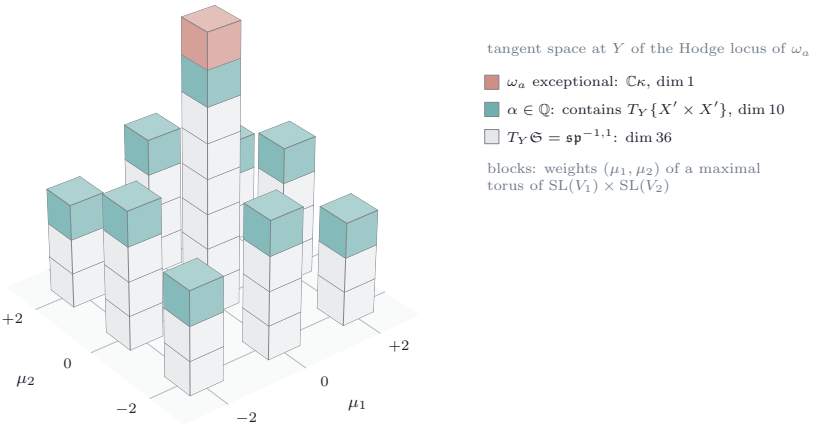
for $\xi \in \operatorname{End} W = \operatorname{End} V_1 \otimes \operatorname{End} V_2$. By the scalars of Step 4, c_a acts on $1 \otimes 1$, $\mathfrak{sl}(V_1) \otimes 1$, $1 \otimes \mathfrak{sl}(V_2)$, $\mathfrak{sl}(V_1) \otimes \mathfrak{sl}(V_2)$, that is on E_3^-, E_{13}^-, E_{23}^-, E_{123}^-, by

$$\tfrac{3}{2}(a_2 - a_3), \qquad -\tfrac{1}{2}(a_2 + 3a_3), \qquad \tfrac{1}{2}(3a_2 + a_3), \qquad -\tfrac{1}{2}(a_2 - a_3),$$

none of which vanishes. So $\xi_{12}\omega \neq 0$ for $\xi_{12} \neq 0$, and by τ the same holds for ξ_{21}.

Step 7: κ. The family of Hodge structures along C is the orbit of the Hodge structure at c under the real points of the third factor of G, so the Kodaira–Spencer class κ is the component of type $(-1, 1)$ of an element of $\mathfrak{sl}(V_3)$ acting on both copies of V, a nonzero multiple of (N, N); it is nonzero because the Hodge structures along C vary. By Steps 5 and 6, $\operatorname{Ann}_{H^1(T_Y)}(\omega) = \mathbb{C}\kappa$. With Steps 3 and 4 this proves (i) and (ii).

For (iii), a point of $\mathfrak{S}$ is a Hodge filtration F^1 on $H^1(Y, \mathbb{C})$, and ω is of Hodge type at it when $\omega \in F^2H^4$, a holomorphic condition. For weight one every tangent direction of $\mathfrak{S}$ at Y is an element ξ of $\mathfrak{sp}(\psi \oplus \psi)^{-1,1} \subset H^1(T_Y)$, and it moves F^2H^4 by the derivation ξ of $\bigwedge^* H^1$, which maps $H^{2,2}$ into $H^{1,3}$; so the Zariski tangent space of the Hodge locus is $\{\xi : \xi \lrcorner \omega = 0\}$, which is $\mathbb{C}\kappa$ by (i). This is the usual description of the tangent space of the Hodge locus of a class of type (p, p) as the kernel of the contraction into $H^{p-1,p+1}$ [Voi03, Chapter 5], in its simplest case: a variation of weight one is subject to no horizontality condition, so every direction of $\mathfrak{S}$ enters. The Hodge locus is an analytic germ whose



	E_3^-, dim 1	E_{13}^-, dim 3	E_{23}^-, dim 3	E_{123}^-, dim 9
$H^1(T_Y)$ diagonal	$b_{11} = b_{22}$: κ	$a_1^2 - a_3^2$	$a_1^2 - a_2^2$	a_1, a_2, a_3 distinct
$H^1(T_Y)$ mixed	$\frac{3}{2}(a_2 - a_3)$	$-\frac{1}{2}(a_2 + 3a_3)$	$\frac{1}{2}(3a_2 + a_3)$	$-\frac{1}{2}(a_2 - a_3)$
$H^0(\bigwedge^2 T_Y)$ mixed	$\frac{1}{4}(a_0 + 9a_1 - 3a_2 - 3a_3)$	$\frac{1}{4}(a_0 - 3a_1 + a_2 - 3a_3)$	$\frac{1}{4}(a_0 - 3a_1 - 3a_2 + a_3)$	$\frac{1}{4}L_0(a)$

$H^{0,2}$ and the unmixed pieces of $\bigwedge^2 T_Y$: the block $K = (a_2, a_3)$ of P_a. At a rational exceptional class no entry vanishes, by the lemma on conjugates, except $b_{11} - b_{22}$, which leaves κ, and L_0, which can be made nonzero by adding a multiple of π_0.

FIGURE 17. Theorem 3.21. Top: the tangent space $\mathfrak{sp}^{-1,1}$ of the Siegel space $\mathfrak{S}$ at $Y = X_c \times X_c$, of dimension 36, split into nine blocks by the weights of a maximal torus of $\mathrm{SL}(V_1) \times \mathrm{SL}(V_2)$, one cube for each dimension. The tangent space of the Hodge locus of an exceptional class is the line $\mathbb{C}\kappa$ of the Mumford curve, in the block of weight $(0, 0)$; for $\alpha \in \mathbb{Q}$ it contains the tangent space of the diagonal Siegel space, of dimension 10 (Remark 3.22). Bottom: the scalars of Steps 4 to 6 of the proof on the pieces E_J^- of $HT^2(Y)$; the pieces of $H^{0,2}$ and the unmixed pieces of $H^0(\bigwedge^2 T_Y)$ are controlled by the block K of P_a (Steps 3 and 4).

Zariski tangent space is a line, so it lies in a smooth curve germ; it contains the smooth curve germ $\{X_t \times X_t\}$, whose tangent line is $\mathbb{C}\kappa$; so the two germs are equal. ∎

Remark 3.22 (Comparison with the divisor classes). For $\alpha \in \mathbb{Q}$ the class ω_a is a combination of products of divisor classes and is invariant under the whole of $\mathrm{Sp}(V, \psi)$ acting diagonally on $V \oplus V$, so its Hodge locus in $\mathfrak{S}$ contains the diagonal Siegel space $\{X' \times X'\}$, of dimension 10; in the proof above this shows in the scalars $a_2 - a_3$ of Step 6, which vanish when $a_1 = a_2 = a_3$. The products of divisor classes therefore move with $X' \times X'$ for every X', whereas the exceptional classes move with $X_t \times X_t$ for $t \in C$ only and cease to be of Hodge type under every deformation of Y off the curve, however small.

Corollary 3.23 (No degeneration and no larger family)**.** *Let ω be a rational exceptional class on $X_c \times X_c$. Every connected complex analytic family of polarised abelian eightfolds through $X_c \times X_c$ on which ω stays of Hodge type lies, near each point of the curve, inside the curve $\{X_t \times X_t : t \in C\}$, which is compact. In particular no such family degenerates, and on no member of it other than the CM points of the curve is ω a combination of*

products of divisor classes or of Weil classes of CM fields acting on the member or on its quotients.

Proof. By Theorem 3.21(iii) the Hodge locus of ω coincides with the curve in a neighbourhood of every point of the curve, so a connected family through Y on which ω stays of Hodge type maps into the curve near each of its points over the curve; by connectedness and the identity theorem its image lies in the curve. The curve is compact because D is a division algebra, so that the arithmetic group of the Shimura curve is cocompact [Mum69]. Indeed, by Setup 3.1 the algebra D is definite at σ_1 and σ_2, so $D \otimes_{F,\sigma_1} \mathbb{R}$ is Hamilton's quaternions and D is not $M_2(F)$: it is a division algebra, split at the one real place σ_3. The norm one group of an order in D, embedded in $\mathrm{SL}_2(\mathbb{R})$ through σ_3, is a Fuchsian group of finite covolume with no parabolic element, since a parabolic $\gamma \neq \pm 1$ would make $\gamma \mp 1$ a nonzero nilpotent element of D; and a Fuchsian group of finite covolume without parabolic elements has compact quotient.

For the last assertion: a member of the curve other than a CM point has Hodge group G (proof of Corollary 2.30), and on it ω lies outside the span of the products of divisor classes and of the Weil classes of quotients, by Proposition 2.74(ii) and (iii). At a CM point the class is already algebraic (Theorem 2.29), and the CM points, being countably many, do not carry algebraicity to the other members (Lemma 2.27 needs uncountably many points). ∎

The same annihilator controls deformations that are not algebraic at all, and so decides whether the twistor lines along which Markman moves sheaves in his proof for Weil classes can be used here.

Proposition 3.24 (The Hodge locus among all complex tori, and twistor lines)**.** *Let ω be a rational exceptional class, $Y = X_c \times X_c$ with $c \in C$, and $\Lambda = H_1(Y, \mathbb{Z})$. Let $\mathcal{T}$ be the space of complex structures on the real torus $\Lambda_{\mathbb{R}}/\Lambda$, a complex manifold of dimension 64 with tangent space $H^1(T_Y)$ at Y, and $\mathcal{T}_\omega \subset \mathcal{T}$ the closed analytic subset on which ω is of type $(2,2)$.*

(i) *The connected component of $\mathcal{T}_\omega$ that contains Y is the universal cover $\mathfrak{H}$ of the Mumford curve, the set of squares of members of the family, biholomorphic to the upper half plane. Every connected complex analytic family of complex tori, algebraic or not, through a member of the Mumford family on which ω stays of Hodge type lies in the family.*

(ii) *No twistor line through a member of the Mumford family, and no compact connected complex curve in $\mathcal{T}$ through it, keeps ω of Hodge type.*

(iii) *Nevertheless $\mathcal{T}_\omega$ contains twistor lines. The algebra $D \otimes_{F,\sigma_1} \mathbb{R}$ is Hamilton's quaternions and acts on $V_{\mathbb{R}}$ through the first factor of $G(\mathbb{R})$; for a unit quaternion j with $j^2 = -1$, the complex structure $j \oplus j$ on $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$ makes ω of type $(2,2)$, and these structures form a twistor line. On each factor $\psi(x, jy)$ has signature $(4,4)$, and a member whose Mumford–Tate group is G is not an abelian variety. These twistor lines lie in components of $\mathcal{T}_\omega$ other than $\mathfrak{H}$.*

Proof. (i) The complex structures on $\Lambda_{\mathbb{R}}$ for which all the tensors invariant under G are of Hodge type are those J with $J^2 = -1$ in the real points of $G \cdot \mathbb{G}_m$, since a reductive group containing the homotheties is the stabiliser of its invariant tensors. Let $\mathfrak{H}$ be the connected component of this closed set that contains Y; it is closed in $\mathcal{T}$ and consists of the $J = 1 \otimes 1 \otimes j_3$ with $j_3 \in \mathrm{SL}_2(\mathbb{R})$, $j_3^2 = -1$, in the half plane that contains the structure of Y, that is, of the squares of the members of the Mumford family. At each of its points

the first order condition for ω to stay of type $(2,2)$ along $\xi \in H^1(T_Y)$ is $\xi \lrcorner \omega = 0$, as in the proof of Theorem 3.21(iii), where no polarisation was used, and by Theorem 3.21(i) the solutions form the line $\mathbb{C}\kappa$ tangent to $\mathfrak{H}$. So near each of its points $\mathcal{T}_\omega$ is the smooth curve germ of $\mathfrak{H}$, and $\mathfrak{H}$ is open in $\mathcal{T}_\omega$. Being also closed and connected, it is a connected component. A connected analytic family on which ω stays of Hodge type maps into $\mathcal{T}_\omega$, hence into $\mathfrak{H}$ once one of its points does.

(ii) A twistor line is a nonconstant holomorphic image of $\mathbb{P}^1$ in $\mathcal{T}$. If ω stayed of Hodge type along one through a point of $\mathfrak{H}$, or along any compact connected curve through it, the curve would lie in $\mathfrak{H}$ by (i), and a holomorphic map from a compact connected curve to the upper half plane is constant.

(iii) At σ_1 the quaternion algebra D is definite (Setup 3.1), so the first factor of $G(\mathbb{R})$ is the group of unit quaternions, and $V_\mathbb{R}$ restricted to it is a module over $D \otimes_{F,\sigma_1} \mathbb{R} \cong \mathbb{H}$. The unit quaternions j with $j^2 = -1$ form a two-sphere, and the complex structures given by left multiplication by them on a quaternionic vector space form a twistor line. Each j lies in $G(\mathbb{R})$, and so does the circle $\cos\vartheta + j\sin\vartheta$, which fixes ω; so ω is of type $(2,2)$. Inside $V_1 \otimes V_2 \otimes V_3$ the space $V_\mathbb{R}$ is the fixed space of a real structure that is quaternionic on the first two factors and real on the third, so $V_\mathbb{R} = U \otimes V_{3,\mathbb{R}}$ with U the real form of $V_1 \otimes V_2$. On U the form $g = \epsilon_1 \otimes \epsilon_2$ is symmetric and invariant under the compact group of pairs of unit quaternions, which acts irreducibly on $U \otimes \mathbb{C} = V_1 \otimes V_2$; so g is definite. The complex structures of the Mumford family are $1 \otimes h$, with h a complex structure on $V_{3,\mathbb{R}}$ for which $\epsilon_3(\cdot, h\cdot)$ is definite, and $\psi(x,(1\otimes h)y) = g \otimes \epsilon_3(\cdot, h\cdot)$ is definite. For $J = j \otimes 1 \otimes 1$ the form $g(\cdot,(j\otimes 1)\cdot)$ is alternating, $j \otimes 1$ being an isometry of g with square -1, and $\psi(x,Jy) = g(\cdot,(j\otimes 1)\cdot) \otimes \epsilon_3$ is the product of nondegenerate alternating forms on spaces of dimensions 4 and 2; as the product of two alternating forms on $\mathbb{R}^2$ is the determinant on $\mathbb{R}^2 \otimes \mathbb{R}^2 = M_2(\mathbb{R})$, of signature $(2,2)$, it has signature $(4,4)$. At a member whose Mumford–Tate group is G, the Hodge classes of degree two on $V_\mathbb{R} \oplus V_\mathbb{R}$ are the forms $\psi \otimes s$ with s a symmetric 2×2 matrix, by Proposition 2.74(ii), and the signature of $\psi(\cdot, J\cdot) \otimes s$ is $(4(p+q), 4(p+q))$ when s has signature (p,q), so none is definite, none is a polarisation, and the member is not an abelian variety. The members of the Mumford family are not on these lines, since $\psi(x,Jy)$ is definite at the former and of signature $(4,4)$ at the latter, so by (i) the lines lie in other components. ∎

Markman's proof for Weil classes on abelian fourfolds moves a sheaf along twistor lines on which its Chern character stays of Hodge type. By Proposition 3.24 no such line passes through the square of a member of the Mumford family: twistor lines on which ω is of Hodge type exist, but they carry non-algebraic tori whose Hodge cocharacter lies in a definite factor of D, and they lie in components of the Hodge locus that the Mumford family never meets. The Kuga–Satake class of Theorem 3.4 behaves differently. The Kuga–Satake construction applies to every Hodge structure of K3 type on (T, q_λ), so that class stays of Hodge type over the whole period domain of (T, q_λ), of dimension seven, and not only along the curve; among the classes attached to the target, it is the only one for which deformation methods of this kind are not excluded.

The same space T also makes the square a holomorphic symplectic variety, and this determines what the mechanism of van Geemen and Rapagnetta would have to supply here. They prove the Hodge conjecture for the general abelian fourfold of Weil type with trivial discriminant by pulling back the second Chern class of a hyperkähler sixfold along a map from the fourfold, which gives an algebraic class of codimension two that is not a product of divisor classes [vGR26]. For a projective hyperkähler manifold M write σ_M for

its symplectic form, q_M for its Beauville–Bogomolov form and $T(M)$ for the orthogonal complement of $\mathrm{NS}(M)_{\mathbb{Q}}$ in $H^2(M,\mathbb{Q})$ for q_M.

Proposition 3.25 (The square as a holomorphic symplectic variety)**.** *Let $c \in C$ be a point at which the Mumford–Tate group of $X = X_c$ is $G \cdot \mathbb{G}_m$, let $Y = X \times X$, and let $\Psi \in \bigwedge^2 V$ be the G-invariant bivector dual to ψ. For $x \in T$ put $\iota(x) = (x \otimes 1)\Psi$ in $V \otimes V = H^1(X,\mathbb{Q}) \otimes H^1(X,\mathbb{Q})$, a Künneth component of $H^2(Y,\mathbb{Q})$, and let $\iota_2 \colon S^2T \to H^4(Y,\mathbb{Q})$ be the map $xy \mapsto \iota(x) \cup \iota(y)$. Let $C_i \in S^2T_i$ be the Casimir element of the trace form of $\mathfrak{sl}(V_i)$, and for $\nu \in F$ let $c_\nu = \sum_i \sigma_i(\nu) C_i$, a rational element of S^2T; for $\nu \neq 0$ it is the element dual to the form $q_{1/\nu}$.*

(i) *ι is an embedding of Hodge structures $T(-1) \to H^2(Y,\mathbb{Q})$, its image is the only irreducible sub-Hodge structure of $H^2(Y,\mathbb{Q})$ with $h^{2,0} = 1$, and the holomorphic 2-form σ_0 spanning its part of type $(2,0)$ is symplectic.*

(ii) *$\iota_2(c_\nu) = \rho\,\omega_a$ with $a_0 = 3\,\mathrm{Tr}_{F/\mathbb{Q}}(\nu)$ and $\alpha = 4\nu - \mathrm{Tr}_{F/\mathbb{Q}}(\nu)$, in the notation of Setup 3.18, where ρ is a nonzero rational number independent of ν. So $\iota_2(c_\nu)$ is exceptional if and only if $\nu \notin \mathbb{Q}$, and every rational exceptional class is some $\iota_2(c_\nu)$ plus a rational multiple of π_0.*

(iii) *Let M be a projective hyperkähler manifold and $f \colon Y \dashrightarrow M$ a rational map with $f^*\sigma_M \neq 0$. Then f^* maps $T(M)$ isomorphically onto $\iota(T)$ and is compatible with products of classes in $T(M)$; f is generically finite onto a subvariety of dimension eight on which σ_M is generically nondegenerate; $b_2(M) \geq 10$; and $\dim M \geq 10$.*

(iv) *Suppose in (iii) that $M = S^{[n]}$ is the Hilbert scheme of n points on a projective K3 surface S, and let q_λ be the form on T that corresponds to the intersection form of S on $T(S) = T(M)$ under $\iota^{-1}f^*$ (Proposition 3.2(iii)). If $\lambda \notin \mathbb{Q}$, or if the real multiplication of $T(S)$ is induced by algebraic cycles on $S \times S$, then the Hodge conjecture holds for Y.*

Proof. (i) The map ι is G-equivariant because Ψ is invariant, and injective. Its image lies in S^2V: for x in the symplectic Lie algebra of ψ one has $(x \otimes 1)\Psi = -(1 \otimes x)\Psi$, and the exchange of the two factors carries $(x \otimes 1)\Psi$ to $-(1 \otimes x)\Psi = (x \otimes 1)\Psi$, because Ψ is alternating. An element g of $G \cdot \mathbb{G}_m$ multiplies Ψ by the multiplier $\chi(g)$ of ψ, so $(g \otimes g)\iota(x) = \chi(g)\,\iota(gxg^{-1})$: the map ι is equivariant for the adjoint action on T twisted by χ, which is the action defining $T(-1)$. Both Hodge structures come from the Hodge cocharacter through $G \cdot \mathbb{G}_m$, so ι is a morphism of Hodge structures from $T(-1)$. The sub-Hodge structures of $H^2(Y,\mathbb{Q})$ are its G-stable subspaces, since $\mathbb{G}_m$ acts by scalars. Over $\bar{\mathbb{Q}}$, $H^2(Y) = \bigwedge^2(V \otimes \bar{\mathbb{Q}}^2) = S^2V \oplus (\bigwedge^2 V)^{\oplus 3}$, and expanding $V \otimes V = \bigotimes_k (S^2V_k \oplus \bigwedge^2 V_k)$ gives $\bigwedge^2 V$ as the sum of the pieces of highest weights $(0,0,0)$, $(2,2,0)$, $(2,0,2)$, $(0,2,2)$ (an odd number of factors $\bigwedge^2 V_k$) and S^2V as the sum of those of highest weights $(2,2,2)$, $(2,0,0)$, $(0,2,0)$, $(0,0,2)$. So the isotypic pieces of $H^2(Y)$ have these eight highest weights and dimensions $3, 27, 27, 27, 27, 3, 3, 3$. The Hodge cocharacter acts through V_3, on which it has one weight of each type, so S^2V_3 has one dimension of type $(2,0)$ and $\bigwedge^2 V_3$ none; the parts of type $(2,0)$ of the eight pieces have dimensions $0, 0, 9, 9, 9, 0, 0, 1$. The last three pieces occur once, in S^2V, and are $\iota(T_1)$, $\iota(T_2)$, $\iota(T_3)$. So an irreducible summand of highest weight $(2,0,2)$ or $(0,2,2)$ contributes 3 to $h^{2,0}$, one of weight $(2,2,2)$ contributes 9, one of weight $(0,0,2)$ contributes 1, and the others contribute nothing. A rational sub-Hodge structure W is stable under the Galois group, which permutes the highest weights as it permutes the factors, transitively. So if $W \otimes \bar{\mathbb{Q}}$ has a summand of weight $(2,2,0)$, $(2,0,2)$ or $(0,2,2)$, it has summands of all three and $h^{2,0}(W) \geq 6$; if it has one of weight $(2,2,2)$, then $h^{2,0}(W) \geq 9$. If $h^{2,0}(W) = 1$, therefore, $W \otimes \bar{\mathbb{Q}}$ contains the piece $\iota(T_3)$

of weight $(0,0,2)$, which has multiplicity one, and with it $\iota(T_1)$ and $\iota(T_2)$; so $W \supseteq \iota(T)$, and $W = \iota(T)$ if W is irreducible. The same argument shows that $\iota(T)$ is irreducible: a nonzero sub-Hodge structure of it contains one of the $\iota(T_i)$ over $\bar{\mathbb{Q}}$, hence all three. The part of type $(2,0)$ of $\iota(T)$ is spanned by $\sigma_0 = \iota(E_3)$, where $E_3 \in \mathfrak{sl}(V_3) \otimes \mathbb{C}$ maps $V_3^{0,1}$ onto $V_3^{1,0}$. Write $\Psi = \Psi_1 \otimes \Psi_2 \otimes \Psi_3$ with $\Psi_k \in \bigwedge^2 V_k$ dual to ϵ_k, and $\Psi_3 = e_+ \otimes e_- - e_- \otimes e_+$ with $e_\pm$ spanning $V_3^{1,0}$ and $V_3^{0,1}$. Then $(E_3 \otimes 1)\Psi_3$ is a nonzero multiple of $e_+ \otimes e_+$, so σ_0 is a nonzero multiple of $\Psi_1 \otimes \Psi_2 \otimes e_+ \otimes e_+$: it lies in $H^{1,0}(X) \otimes H^{1,0}(X)$ and pairs $H^{1,0}$ of the first factor with $H^{1,0}$ of the second through the nondegenerate form dual to $\epsilon_1 \otimes \epsilon_2$ on $V_1 \otimes V_2$. So σ_0 has rank eight: it is a holomorphic symplectic form on Y.

(ii) For $x, y \in T$ the class $\iota(x) \cup \iota(y)$ lies in $H^2(X) \otimes H^2(X)$ and comes from the tensor $(x \otimes 1)\Psi \otimes (y \otimes 1)\Psi$ in $V^{\otimes 4}$ by antisymmetrising its two slots in the first factor of Y and its two slots in the second. Read it, as in Setup 3.18, as a map from the first-factor slots to the second-factor slots. Contraction with ϵ_k turns $\Psi_k \otimes \Psi_k$ into γ times the identity of $V_k \otimes V_k$ and, for $x, y \in T_k$, the tensor $(x \otimes 1)\Psi_k \otimes (y \otimes 1)\Psi_k$ into $\gamma\,(x \otimes y)$, with one and the same constant $\gamma \neq 0$, since x and y are antisymmetric for ϵ_k and the two signs cancel. So $\iota_2(C_1)$ corresponds, up to a nonzero constant γ' independent of the factor, to the restriction to $\bigwedge^2 V$ of $\mathrm{Cas}_1 \otimes 1 \otimes 1$, where $\mathrm{Cas}_1 = \sum_c x_c \otimes x^c = s_1 - \frac{1}{2}$ is the operator of Step 3 of the proof of Theorem 3.7, s_1 the exchange of the two copies of V_1. On $\bigwedge^2 V$ the exchange s_1 is -1 on $\mathbb{Q}\theta$ and U_{23} and $+1$ on U_{12} and U_{13}, so $\iota_2(C_1) = \rho\,(3\pi_0 - \pi_{12} - \pi_{13} + 3\pi_{23})$ with $\rho = -2\gamma'$, and permuting the factors gives the two cyclic permutations of this identity. Summing with the coefficients $\sigma_i(\nu)$ gives $\iota_2(c_\nu) = \rho\big(3\,\mathrm{Tr}(\nu)\pi_0 + \sum \sigma_k(4\nu - \mathrm{Tr}(\nu))\pi_{ij}\big)$, the sum over $\{i,j,k\} = \{1,2,3\}$ with $i < j$, which is $\rho\,\omega_a$ with the stated a_0 and α. For rational $\nu \neq 0$ both $\iota_2(c_\nu)$ and ω_a are rational and nonzero, so $\rho \in \mathbb{Q}$. Since $\mathrm{Tr}(\nu) \in \mathbb{Q}$, the element α is rational exactly when ν is; and $\mathrm{Tr}(\alpha) = \mathrm{Tr}(\nu)$, so $\nu = (\alpha + \mathrm{Tr}(\alpha))/4$ recovers ν from any $\alpha \in F$. Two rational classes with the same α differ by a multiple of π_0, and $\iota_2(c_{\nu/\rho}) = \omega_a$ because c_ν is linear in ν; so every rational exceptional class is some $\iota_2(c_\nu)$ plus a rational multiple of π_0. The Casimir element C_i is dual to the trace form on T_i, so c_ν is dual to $\sum_i \sigma_i(\nu)^{-1}\,\mathrm{tr}(x_i y_i) = q_{1/\nu}(x,y)$.

(iii) Resolve the indeterminacy of f by a sequence of blow-ups along smooth centres, $\pi\colon \tilde{Y} \to Y$ and $\tilde{f}\colon \tilde{Y} \to M$, and put $f^* = \pi_* \tilde{f}^*$, a morphism of Hodge structures. The kernel of π_* on $H^2(\tilde{Y}, \mathbb{Q})$ is spanned by the classes of the exceptional divisors, so $\alpha \mapsto \tilde{f}^*\alpha - \pi^* f^* \alpha$ is a morphism of Hodge structures from $H^2(M, \mathbb{Q})$ to a space of Hodge classes. The Hodge structure $T(M)$ is irreducible: a sub-Hodge structure $W \subseteq T(M)$ with $W^{2,0} = 0$ consists of rational classes of type $(1,1)$, which lie in $\mathrm{NS}(M)_{\mathbb{Q}} \cap T(M) = 0$; if $W^{2,0} \neq 0$, then $\sigma_M \in W_{\mathbb{C}}$, the orthogonal complement of W in $H^2(M, \mathbb{Q})$ is a sub-Hodge structure orthogonal to σ_M and $\bar{\sigma}_M$, hence of type $(1,1)$ and contained in $\mathrm{NS}(M)_{\mathbb{Q}}$, and $W \supseteq T(M)$. Since $T(M)$ is not spanned by Hodge classes, that morphism vanishes on it, and $\tilde{f}^*\alpha = \pi^* f^* \alpha$ for $\alpha \in T(M)$. Because $\tilde{f}^*$ is a ring homomorphism and $\pi_*\pi^*$ is the identity, $f^*(\alpha\beta) = f^*\alpha \cup f^*\beta$ for $\alpha, \beta \in T(M)$. The kernel of f^* on $T(M)$ is a sub-Hodge structure not containing σ_M, hence zero, and the image is irreducible with $h^{2,0} = 1$, hence $\iota(T)$ by (i). So $f^*\sigma_M$ is a nonzero multiple of σ_0 and has rank eight at every point of Y where f is defined; the rank of a pulled-back form is at most the dimension of the image, so f is generically finite onto its image Z, of dimension eight, and σ_M is generically nondegenerate on Z. As M is projective, $b_2(M) = \dim T(M) + \dim \mathrm{NS}(M)_{\mathbb{Q}} \geq 9 + 1$; this excludes the generalised Kummer varieties, whose second Betti number is seven [Bea83]. Finally $\dim M \geq \dim Z = 8$; suppose that $\dim M = 8$. Then f is dominant, and for $\alpha \in T(M)_{\mathbb{C}}$ the relation of Fujiki and Beauville

[Bea83, Fuj87] gives

$$\int_Y (f^*\alpha)^8 = \int_{\tilde{Y}} \tilde{f}^*(\alpha^8) = \deg(f)\int_M \alpha^8 = \deg(f)\, c_M\, q_M(\alpha)^4, \qquad c_M > 0.$$

Write $f^*\alpha = \iota(x)$ with $x = x_1 + x_2 + x_3 \in T_{\mathbb{C}}$ and $N_i = -\det x_i$. The form $x \mapsto q_M((f^*)^{-1}\iota(x))$ is invariant under G, so it is $q_\lambda(x,x) = \sum_i \sigma_i(\lambda)\,\mathrm{tr}(x_i^2) = 2\sum_i \sigma_i(\lambda) N_i$ for some $\lambda \in F^\times$ by Proposition 3.2(iii). On the other side, in a basis (e_i) of V one has $\iota(x) = \sum_{i,j} M_{ij}\, e_i^{(1)} \wedge e_j^{(2)}$, with M the product of the matrix of x and the constant invertible matrix of Ψ, and the 2-forms $e_i^{(1)} \wedge e_j^{(2)}$ commute, so $\iota(x)^8 = 8!\det(M)\prod_i e_i^{(1)} \wedge e_i^{(2)}$ is $8!\det(x)$ times a fixed volume form, and $\det(x_1 \otimes 1 \otimes 1 + 1 \otimes x_2 \otimes 1 + 1 \otimes 1 \otimes x_3) = \Delta(N_1, N_2, N_3)^2$ with $\Delta = N_1^2 + N_2^2 + N_3^2 - 2N_1N_2 - 2N_1N_3 - 2N_2N_3$; this is also the product of the eight numbers $\pm r_1 \pm r_2 \pm r_3$, where $\pm r_i$ are the eigenvalues of x_i and $r_i^2 = N_i$. The N_i are algebraically independent functions on $T_{\mathbb{C}}$, so the two expressions for $\int_Y (f^*\alpha)^8$ would make Δ^2 a nonzero constant multiple of the fourth power of the linear form $\sum_i \sigma_i(\lambda) N_i$, and then, by unique factorisation, Δ a multiple of its square. But Δ is a nondegenerate ternary quadratic form, hence irreducible. So $\dim M \neq 8$, and since the dimension of a hyperkähler manifold is even, $\dim M \geq 10$.

(iv) By [Bea83] the map $H^2(S,\mathbb{Q}) \to H^2(S^{[n]},\mathbb{Q})$ induced by the universal subscheme $Z_n \subset S \times S^{[n]}$ is injective, carries the intersection form to the Beauville–Bogomolov form, and has $H^2(S^{[n]},\mathbb{Q}) = H^2(S,\mathbb{Q}) \oplus \mathbb{Q}\delta$ with δ algebraic, so $T(M) = T(S)$. Let Δ_T be the Künneth component in $T(S) \otimes T(S)$ of the class of the diagonal of S; it is algebraic, being the class of the diagonal minus $[\mathrm{pt} \times S]$, $[S \times \mathrm{pt}]$ and $\sum_r d_r \times d_r^\vee$ over dual bases of $\mathrm{NS}(S)_{\mathbb{Q}}$. Its image under the correspondence $Z_n \times Z_n$, pulled back along the diagonal of M, is the class $\sum_a e_a \cup e_a^\vee$ over dual bases of $T(S)$ for the intersection form, which is therefore algebraic on M, and by (iii) its image under f^* is ι_2 of the element of S^2T dual to q_λ, that is $\iota_2(c_{1/\lambda})$, an algebraic class on Y. If $\lambda \notin \mathbb{Q}$ it is exceptional by (ii). If instead the real multiplication is algebraic, apply it to the first factor of Δ_T: for $b \in F$ the class $\sum_a (m_b e_a) \cup e_a^\vee$ is algebraic, and its image is $\iota_2(c_{b/\lambda})$, which is exceptional for $b \notin \mathbb{Q}\lambda$. In both cases Lemma 3.20 makes every Hodge class in $H^2(X) \otimes H^2(X)$ algebraic; the Hodge classes in the other Künneth components of $H^4(Y,\mathbb{Q})$ are products of divisor classes, as at the end of the proof of Theorem 3.4; and the Hodge classes of Y form a ring generated in degrees two and four (proof of Corollary 2.30). So every Hodge class of Y is algebraic. ∎

Remark 3.26 (The mechanism of van Geemen and Rapagnetta). By Proposition 3.25(ii) the exceptional classes are, up to rational multiples of π_0, the classes $\iota_2(c_\nu)$ with $\nu \notin \mathbb{Q}$, the images of the elements dual to the forms q_λ, $\lambda = 1/\nu$, on the part $\iota(T)$ of $H^2(Y)$ generated by the symplectic class; the form with λ rational gives a product of divisor classes. They are the analogues for (Y,σ_0) of the dual Beauville–Bogomolov class, whose pull-back is what [vGR26] uses. By part (iii), a transfer of that argument needs a projective hyperkähler manifold of dimension at least ten with transcendental part $T(-1)$, such as $S_\mu^{[n]}$ with $n \geq 5$ for a surface S_μ of Theorem 3.4, and a rational map from Y onto a symplectic subvariety of dimension eight. For $M = S_\mu^{[n]}$ the form of (iv) is q_{μ/e^2} for the $e \in F^\times$ with $\iota^{-1}f^* = m_e$ on $T(S_\mu) = T$, so $\lambda \notin \mathbb{Q}$ for every such map as soon as $\mu \notin \mathbb{Q}^\times F^{\times 2}$. No such map is known. It would give an algebraic cycle on $S_\mu \times X \times X$ inducing an isomorphism of $T(S_\mu)$ onto $\iota(T) \subset H^1(X) \otimes H^1(X)$, a correspondence of the kind that the Kuga–Satake class supplies in the proof of Theorem 3.4; so this route is the Kuga–Satake route in geometric form rather than an independent one. By (iii), maps to K3 surfaces and to hyperkähler eightfolds cannot serve.

Theorem 3.27 (A numerical criterion for the Mumford classes)**.** *Let $c \in C$, $Y = X_c \times X_c$, $\omega = \omega_a$ a rational exceptional class with $L_0(a) \neq 0$, and E a perfect complex on Y with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and no other component.*

(i) $\dim \mathrm{Ext}^2(E,E) \geq 119$.

(ii) *If $\dim \mathrm{Ext}^2(E,E) = 119$, then $\mathrm{ev}_E \colon HT^2(Y) \to \mathrm{Ext}^2(E,E)$ is surjective with kernel $\mathbb{C}\kappa$, and σ_E is injective. Conversely, if σ_E is injective, then ev_E has rank 119 and kernel $\mathbb{C}\kappa$, and E extends to first order along the curve.*

(iii) *If some such E at one point c has $\dim \mathrm{Ext}^2(E,E) = 119$, then ω_t is algebraic for every $t \in C$, the Hodge conjecture holds for $X_t \times X_t$ for every t, and $B(W \times_C W)$ holds.*

(iv) *At every CM point c some perfect complex E on Y has $\mathrm{ch}(E) = N\omega$ for some $N \neq 0$ and no other component.*

Proof. (i) and (ii). By the square $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,)\lrcorner\, \mathrm{ch}(E)$ of [II, Setup 4.54], $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$ by Theorem 3.21(ii), so ev_E has rank at least 119, which is (i). If $\dim \mathrm{Ext}^2(E,E) = 119$, then ev_E is onto, its kernel has dimension one and lies in $\mathbb{C}\kappa$, so equals it; and $\sigma_E(\mathrm{ev}_E(x)) = N\, x\lrcorner\, \omega$ vanishes only for $x \in \mathbb{C}\kappa = \ker \mathrm{ev}_E$, so σ_E is injective on $\mathrm{im}\, \mathrm{ev}_E = \mathrm{Ext}^2(E,E)$. Conversely, if σ_E is injective, then $\sigma_E(\mathrm{ev}_E(\kappa)) = N\kappa\lrcorner\omega = 0$ forces $\mathrm{ev}_E(\kappa) = 0$, so the kernel is $\mathbb{C}\kappa$ and the rank is 119; and ev_E restricted to $H^1(T_Y)$ is the obstruction map [Tod09], which therefore vanishes at κ, so that E extends to first order along the curve.

(iii) By (ii) σ_E is injective. The Chern character of E is $N\omega$, a flat section of Hodge type over the whole of C, so [II, Theorem 2.2] applies to every infinitesimal deformation of Y inside $W \times_C W \to C$. The proof of [II, Theorem 2.3] then applies word for word, with the stack of perfect complexes on the fibres of $W \times_C W \to C$ [Lie06] in place of that of $\mathcal{A} \to \mathcal{S}$, or the analytic Kuranishi germ if $\mathrm{Ext}^{<0}(E,E) \neq 0$: it is smooth over C at $[E]$, its image contains a nonempty open subset $U \subseteq C$, and for $t \in U$ a perfect complex E_t on $X_t \times X_t$ in the same connected component has $\mathrm{ch}(E_t) = N\omega_t$, so ω_t is algebraic, and by Lemma 3.20 so are both exceptional classes. Since U is uncountable, Corollary 2.30 gives the rest.

(iv) By Theorem 2.29 there is $\xi \in \mathrm{CH}^2(Y)_{\mathbb{Q}}$ with class ω, and the argument of [II, Proposition 2.1] gives E: the Chern character is an isomorphism $K_0(Y)_{\mathbb{Q}} \to \mathrm{CH}^*(Y)_{\mathbb{Q}}$ [Ful98, Ex. 15.2.16(b) and Cor. 18.3.2], so $\xi = \mathrm{ch}(u)$ for a unique u, and Nu is the class of a perfect complex for a suitable N. ∎

Part (iii) is the Mumford analogue of [II, Theorem 2.11(iii)]: one object, at one point, and one number, which is 119 here and $2n(2n-1)$ for a Weil family. For a Weil family that number is attained by no object ([II, Theorem 3.5]), because an object attaining it would have to deform to non-algebraic tori; the classes here are rigid, so that argument does not apply ([II, Remark 3.6]). The objects of (iv) exist, and the criterion asks that one of them, or another representative of the same class in K_0, attain the bound of (i). The rigidity of Theorem 3.21 is no obstacle: semiregularity moves an object along the Hodge locus of its Chern character, and the curve is exactly that locus, so a semiregular object would propagate along the curve and nowhere else, which is all that is needed.

Corollary 3.28 (Bounded data at CM points)**.** *Fix a relatively ample line bundle on $W \times_C W \to C$, and let ω be a rational exceptional class. The following are equivalent.*

(a) *ω_t is algebraic for every $t \in C$; equivalently, by Lemma 3.20 and Corollary 2.30, the Hodge conjecture holds for $X_t \times X_t$ for every $t \in C$.*

(b) *There are a finite set Φ of pairs $(\underline{P},\underline{q})$, each a tuple of Hilbert polynomials with a tuple of rational numbers of the same length, and infinitely many points $c\in C$ at each of which $\omega_c=\sum_i q_i[Z_i]$ for subschemes $Z_i\subset X_c\times X_c$ with Hilbert polynomials P_i, for some $(\underline{P},\underline{q})\in\Phi$.*

In particular the classes are algebraic on every member as soon as they are represented at infinitely many CM points by cycles of bounded degree.

Proof. For a pair $(\underline{P},\underline{q})$ let $\Sigma_{\underline{P},\underline{q}}\subseteq C$ be the set of points at which ω is represented with these data. The proof of [I, Theorem 16.4] applies verbatim to $(W\times_C W, C, \omega)$, as in Lemma 2.27, and shows that each $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed. If (b) holds, then, Φ being finite, one pair of Φ serves at infinitely many points, so its Σ is an infinite closed subset of the irreducible curve C, hence C, and (a) holds. If (a) holds, C is the union of the countably many $\Sigma_{\underline{P},\underline{q}}$, and C is uncountable, so one of them is infinite, hence equal to C; then (b) holds with Φ that one pair and every point of C. ∎

The proof of Theorem 2.29 represents the classes at every CM point, through Markman's theorem for abelian fourfolds [Mar25a] and the ring generated by divisor classes and pull-backs of Hodge classes of X_c along homomorphisms $X_c\times X_c\to X_c$, but with no control of degree; isogenies between CM members, which move CM points along their Hecke orbits, do not bound the degree either. By Corollary 3.28 such a bound, at infinitely many CM points, is equivalent to the target; without it the CM points, being countably many, say nothing about a very general member (Lemma 2.27 needs uncountably many points).

The next proposition says which endomorphisms those pull-backs have to use. At a CM point the fourfold is of Weil type, its two Hodge classes of degree four that are not products of divisor classes form its Weil line, and the exceptional classes of the square are out of reach of everything built from that line with the Weil field alone: they need the real multiplication by the totally real quartic algebra of Rosati-fixed endomorphisms, and all of it.

Proposition 3.29 (The Weil structure at a CM point, and what the cycles must use)**.** *Let $c\in C$ be a CM point, $X=X_c$, $Y=X\times X$, and let e_w, $w\in\{\pm1\}^3$, be a weight basis of $V=H^1(X,\mathbb{C})$ for the Hodge torus T of X, as in the proof of Theorem 2.29, normalised so that $\psi(e_w,e_{-w})=\pm1$, with $e_w^{(1)}$, $e_w^{(2)}$ its two copies in $H^1(Y,\mathbb{C})$. Let T_+ be the set of the four weights with an even number of entries -1, let $T_-=-T_+$ be the set of the other four, let $t_\pm=\prod_{w\in T_\pm}e_w\in H^4(X,\mathbb{C})$, the factors in a fixed order, and let $\dagger$ be the Rosati involution of θ on $\mathrm{End}^0(X)$. For $\varphi\in\mathrm{End}^0(X)$ write φ_w for its eigenvalue on e_w.*

(i) *T_+ and T_- are the only sets of four weights with sum zero that contain no pair of opposite weights, and $\mathrm{Hdg}^2(X)\otimes\mathbb{C}$ is the direct sum of the span of the products of divisor classes and of $\mathbb{C}t_+\oplus\mathbb{C}t_-$. The elements of $\mathrm{End}^0(X)$ with φ_w constant on T_+ and constant on T_- form an imaginary quadratic field k_c; (X,k_c,θ) is of $(k_c,-1,2)$-Weil type, and its Weil line is $W(X)=(\mathbb{C}t_+\oplus\mathbb{C}t_-)\cap H^4(X,\mathbb{Q})$. The fixed algebra K_c^+ of $\dagger$ consists of the elements with $\varphi_w=\varphi_{-w}$ for every w; it is a product of totally real fields of total degree four, and $\mathrm{End}^0(X)=K_c^+\otimes_{\mathbb{Q}}k_c$.*

(ii) *For $\varphi,\psi\in\mathrm{End}^0(X)$ let $f^*_{\varphi,\psi}$ be the ring homomorphism $H^*(X,\mathbb{Q})\to H^*(Y,\mathbb{Q})$ that sends $e\in H^1(X,\mathbb{Q})$ to $\mathrm{pr}_1^*\varphi^*e+\mathrm{pr}_2^*\psi^*e$; for $\varphi,\psi\in\mathrm{End}(X)$ it is the pull-back along the homomorphism $\varphi\circ\mathrm{pr}_1+\psi\circ\mathrm{pr}_2\colon Y\to X$, and every homomorphism $Y\to X$ is of this form. For a $\mathbb{Q}$-subalgebra $R\subseteq\mathrm{End}^0(X)$ let $\Sigma_R\subseteq\mathrm{Hdg}^2(Y)$ be the $\mathbb{Q}$-span of the products of two divisor classes of Y and of the classes $f^*_{\varphi,\psi}W(X)$ with $\varphi,\psi\in R$.*

Then $\Sigma_{k_c} = \Sigma_{\mathbb{Q}}$, the span obtained from the maps $(x, y) \mapsto ax + by$ alone, has dimension 110 and meets the span of $\pi_0, \pi_{12}, \pi_{13}, \pi_{23}$ in the plane of the classes with $a_1 = a_2 = a_3$; it contains no exceptional class.

(iii) *For a $\mathbb{Q}$-subalgebra $R \subseteq \mathrm{End}^0(X)$ the following are equivalent:* (a) *Σ_R contains an exceptional class;* (b) *$\Sigma_R = \mathrm{Hdg}^2(Y)$, which has dimension 132;* (c) *R separates the weights of T_+, that is, for any two distinct $w, w' \in T_+$ some $\varphi \in R$ has $\varphi_w \neq \varphi_{w'}$; equivalently, R and k_c generate $\mathrm{End}^0(X)$. In particular $\Sigma_{K_c^+} = \mathrm{Hdg}^2(Y)$, while Σ_R contains no exceptional class if R is a proper subalgebra of K_c^+ or a subalgebra of k_c.*

Proof. (i) The four weights $d_1 = (1,1,1)$, $d_2 = (1,-1,-1)$, $d_3 = (-1,1,-1)$, $d_4 = (-1,-1,1)$ of T_+ have sum zero, and any three of them are linearly independent, so every linear relation among them is a multiple of $d_1 + d_2 + d_3 + d_4 = 0$. A set of four weights containing no opposite pair contains exactly one of d_i, $-d_i$ for each i, say $\varepsilon_i d_i$, and has sum zero only if $\sum_i \varepsilon_i d_i = 0$, that is, if all ε_i are equal: the set is T_+ or T_-. The Hodge classes of X of degree four, tensored with $\mathbb{C}$, are spanned by the monomials $\prod_{w\in S} e_w$ over the sets S of four weights with sum zero (proof of Theorem 2.29); the divisor classes are spanned by the four monomials $e_w e_{-w}$ of weight zero and degree two, their products are the six monomials of the sets that are unions of two opposite pairs, and the two remaining sets are T_+ and T_-.

The torus T is defined over $\mathbb{Q}$, so the Galois group of $\bar{\mathbb{Q}}$ acts linearly on its character lattice and permutes the set of the eight weights of V. A linear map that permutes the weights carries opposite pairs to opposite pairs and sets with sum zero to sets with sum zero, hence carries $\{T_+, T_-\}$ to itself. Over $\bar{\mathbb{Q}}$ the algebra $\mathrm{End}^0(X) \otimes \bar{\mathbb{Q}} = \mathrm{End}_T(V_{\bar{\mathbb{Q}}})$ consists of the diagonal matrices $\sum_w x_w p_w$ in the weight basis, p_w the projection to the weight space of w, and $\gamma \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ acts by $\gamma(\sum_w x_w p_w) = \sum_w \gamma(x_w) p_{\gamma w}$, because γ carries the weight space of w to that of γw. As ψ is T-invariant, $\psi(e_w, e_{w'}) = 0$ unless $w' = -w$, so the adjoint of $\sum_w x_w p_w$ for ψ, which is the Rosati involution, is $\sum_w x_{-w} p_w$. The subalgebra A_W of the diagonal matrices that are constant on T_+ and on T_-, and the subalgebra A^+ of those with $x_w = x_{-w}$ for all w, are therefore stable under the Galois group and under †; so $k_c = A_W \cap \mathrm{End}^0(X)$ and $K_c^+ = A^+ \cap \mathrm{End}^0(X)$ are $\mathbb{Q}$-forms of A_W and A^+, of dimensions 2 and 4, and K_c^+ is the fixed algebra of †. On A_W the involution † exchanges the two scalars. The Rosati involution is positive: $\mathrm{Tr}(xx^\dagger) > 0$ for $x \neq 0$, with Tr the trace on V [BL04, Section 5.1]. If k_c were not a field it would contain the idempotent $p_+ = \sum_{w\in T_+} p_w$, and $\mathrm{Tr}(p_+ p_+^\dagger) = \mathrm{Tr}(p_+ p_-) = 0$; so k_c is a quadratic field on which † is the nontrivial automorphism. If k_c were real, $x = \sqrt{d} \in k_c$ would satisfy $x^\dagger = -x$ and $\mathrm{Tr}(xx^\dagger) = -\mathrm{Tr}(x^2) = -8d < 0$. So k_c is imaginary quadratic and † is complex conjugation on it, which is condition (iii) of [I, Definition 3.1]. The eigenspaces of $k_c \otimes \mathbb{C}$ on V are $V_\pm = \bigoplus_{w\in T_\pm} \mathbb{C} e_w$. The Hodge decomposition of V is the eigenspace decomposition of the Hodge cocharacter, which takes values in the diagonal torus $T_{\mathbb{C}} \cdot \mathbb{G}_m$, so every e_w is of pure type $(1,0)$ or $(0,1)$, and t_+ is of type $(p, 4-p)$ with $p = \dim(V_+ \cap H^{1,0}(X))$. Since t_+ is a Hodge class, $p = 2$: each eigenvalue of $\sqrt{-d}$ has multiplicity two on $H^{1,0}(X)$, which is condition (ii) of [I, Definition 3.1], and condition (i) is $\dim X = 4$. The Weil line of [I, Definition 3.6] is $(\bigwedge^4 V_+ \oplus \bigwedge^4 V_-) \cap H^4(X, \mathbb{Q}) = (\mathbb{C}t_+ \oplus \mathbb{C}t_-) \cap H^4(X, \mathbb{Q})$; it has dimension two by [I, Proposition 3.7(i)], so it is a rational complement of the products of divisor classes in $\mathrm{Hdg}^2(X)$. The algebra K_c^+ is commutative and semisimple, being a subalgebra of $\mathrm{End}^0(X)$, whose complexification is commutative, and its trace form $\mathrm{Tr}(xy) = \mathrm{Tr}(xy^\dagger)$ is positive definite; so every factor field E of K_c^+ has $\mathrm{Tr}_{E/\mathbb{Q}}(x^2) > 0$ for $x \neq 0$ and is totally real. Finally the multiplication map $K_c^+ \otimes_{\mathbb{Q}} k_c \to \mathrm{End}^0(X)$ is an isomorphism,

because over $\bar{\mathbb{Q}}$ it is the isomorphism $\bar{\mathbb{Q}}^{\{\text{opposite pairs}\}} \otimes \bar{\mathbb{Q}}^{\{T_+,T_-\}} \to \bar{\mathbb{Q}}^{\{\text{weights}\}}$ induced by the bijection that sends a weight to the pair and the tetrahedron containing it, a bijection because the two weights of a pair lie in different tetrahedra, T_- being $-T_+$.

(ii) Extend the notation: for diagonal $\varphi, \psi \in \mathrm{End}^0(X) \otimes \mathbb{C}$ with entries φ_w, ψ_w let $f^*_{\varphi,\psi}$ be the ring homomorphism $H^*(X,\mathbb{C}) \to H^*(Y,\mathbb{C})$ with $e_w \mapsto \varphi_w e_w^{(1)} + \psi_w e_w^{(2)}$; for $\varphi, \psi \in \mathrm{End}^0(X)$ this is the map of the statement, since $\varphi^* e_w = \varphi_w e_w$, and every homomorphism $Y \to X$ is $\varphi \circ \mathrm{pr}_1 + \psi \circ \mathrm{pr}_2$ for its restrictions φ, ψ to the two factors. Then

$$f^*_{\varphi,\psi} t_\pm = \prod_{w \in T_\pm} (\varphi_w e_w^{(1)} + \psi_w e_w^{(2)}) = \sum_{S \subseteq T_\pm} \pm \Big(\prod_{w \in S} \varphi_w \Big) \Big(\prod_{w \in T_\pm \setminus S} \psi_w \Big)\, e_S^{(1)}\, e_{T_\pm \setminus S}^{(2)}, \tag{3}$$

where $e_S^{(1)} e_{T\setminus S}^{(2)}$ is the product of the $e_w^{(1)}$, $w \in S$, and the $e_w^{(2)}$, $w \in T \setminus S$, taken in the order of T, and the sign is that of the shuffle. We call these 32 products the tetrahedron monomials. The Hodge classes of Y of degree four, tensored with $\mathbb{C}$, are spanned by the monomials of weight zero in the sixteen classes $e_w^{(i)}$ (proof of Theorem 2.29). Such a monomial whose four weights contain an opposite pair is the product of two monomials of weight zero and degree two, that is, of two divisor classes: there are four of the form $e_w^{(1)} e_w^{(2)} e_{-w}^{(1)} e_{-w}^{(2)}$ and 96 with two distinct opposite pairs and a choice of copy for each of the four weights. A weight repeated twice, $2w + w' + w'' = 0$, forces $w' = w'' = -w$, so the remaining monomials have four distinct weights with no opposite pair, and by (i) they are the tetrahedron monomials. Hence $\dim \mathrm{Hdg}^2(Y) = 132$, the products of divisor classes span the 100 monomials of the first two kinds, and the tetrahedron monomials form a basis of the quotient Ξ of $\mathrm{Hdg}^2(Y) \otimes \mathbb{C}$ by the span of the products of divisor classes. Two distinct weights of a tetrahedron T differ in exactly two coordinates, and for $S = \{a,b\} \subseteq T$ the complement $T \setminus S$ has the same set of differing coordinates, because $a + b$ and the sum of the complement are opposite. Writing $D(a,b) \subseteq \{1,2,3\}$ for the set of coordinates in which two weights differ, the six tetrahedron monomials of T with two factors from each copy fall into three pairs $\{e_S^{(1)} e_{T\setminus S}^{(2)}, e_{T\setminus S}^{(1)} e_S^{(2)}\}$, indexed by the two-element set $D(S) = D(T \setminus S)$.

For $\varphi, \psi \in k_c \otimes \mathbb{C} = A_W$, with values $\varphi_\pm$, $\psi_\pm$ on $T_\pm$, (3) reads $f^*_{\varphi,\psi} t_\pm = \sum_{k=0}^4 \varphi_\pm^k \psi_\pm^{4-k} s_{\pm,k}$, where $s_{\pm,k}$ is the sum, with the signs of (3), of the tetrahedron monomials of $T_\pm$ with k factors from the first copy. Conversely the classes $(a\,\mathrm{pr}_1 + b\,\mathrm{pr}_2)^* t_\pm$ for five values of a/b span the $s_{\pm,k}$, a Vandermonde matrix being invertible. Since $W(X) \otimes \mathbb{C} = \mathbb{C}t_+ \oplus \mathbb{C}t_-$, the complexifications of Σ_{k_c} and of $\Sigma_{\mathbb{Q}}$ are both spanned by the products of divisor classes and the ten classes $s_{\pm,k}$, which are supported on disjoint nonempty sets of tetrahedron monomials; so $\Sigma_{k_c} = \Sigma_{\mathbb{Q}}$ has dimension $100 + 10 = 110$.

For the four π we use the description of Setup 3.18. Fix a total order on the weights, and for an endomorphism A of $H^2(X,\mathbb{C}) = \bigwedge^2 V$ let

$$c(A) = \sum_{a<b} A(e_a e_b) \otimes (e_a e_b)^\vee \in H^2(X) \otimes H^2(X)$$

be its class, where $(e_a e_b)^\vee$ is the dual basis for the form $\bigwedge^2 \psi$. As $\bigwedge^2 \psi(e_a e_b, e_c e_d) = \psi(e_a, e_c)\psi(e_b, e_d) - \psi(e_a, e_d)\psi(e_b, e_c)$ vanishes unless $\{c,d\} = \{-a,-b\}$, we have $(e_a e_b)^\vee = \chi(a)\chi(b)\, e_{-a} e_{-b}$ with $\chi(w) = \psi(e_w, e_{-w}) = \pm 1$, the two factors in the order matching (a,b). Let s_t, $t = 1,2,3$, be the involution of $V \otimes V = \bigotimes_{u=1}^3 (V_u \otimes V_u)$ that exchanges the two factors V_t. The three s_t commute, and $s_1 s_2 s_3$ is the exchange of the two slots of $V \otimes V$, which is -1 on $\bigwedge^2 V$; so each s_t preserves $\bigwedge^2 V$, on which $s_1 s_2 s_3 = -1$, and $\bigwedge^2 V$ is the sum of the four joint eigenspaces with eigenvalues $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ of product -1: these are

$\mathbb{C}\theta = \bigwedge^2 V_1 \otimes \bigwedge^2 V_2 \otimes \bigwedge^2 V_3$ for $(-1,-1,-1)$ and $U_{ij} = S^2V_i \otimes S^2V_j \otimes \bigwedge^2 V_k$ for $\varepsilon_i = \varepsilon_j = 1$, $\varepsilon_k = -1$. The projection onto the eigenspace of $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ is $\prod_t (1 + \varepsilon_t s_t)/2$, which by the relation $s_1 s_2 s_3 = -1$ equals $(1 + \varepsilon_1 s_1 + \varepsilon_2 s_2 + \varepsilon_3 s_3)/4$. So

$$\pi_0 = c\Big(\frac{1 - s_1 - s_2 - s_3}{4}\Big), \qquad \pi_{12} = c\Big(\frac{1 + s_1 + s_2 - s_3}{4}\Big),$$

and π_{13}, π_{23} likewise, with the plus signs at s_1, s_3 and at s_2, s_3. Let a^t be the weight a with its t-th coordinate changed; then $s_t(e_a e_b) = e_a e_b$ if $t \notin D(a,b)$ and $s_t(e_a e_b) = e_{a^t} e_{b^t}$ if $t \in D(a,b)$. Under the Künneth map to $H^4(Y, \mathbb{C})$ the term of the pair $\{a, b\}$ in $c(s_t)$ is, up to sign, the monomial $e^{(1)}_{a'} e^{(1)}_{b'} e^{(2)}_{-a} e^{(2)}_{-b}$ with $\{a', b'\} = \{a, b\}$ if $t \notin D(a,b)$ and $\{a', b'\} = \{a^t, b^t\}$ otherwise. If $t \notin D(a,b)$, or $D(a,b) = \{t\}$, in which case $a^t = b$, or $D(a,b) = \{1,2,3\}$, in which case $b = -a$ and the monomial is $e^{(1)}_{a^t} e^{(1)}_{-a^t} e^{(2)}_{-a} e^{(2)}_{a}$, the monomial contains an opposite pair and is a product of two divisor classes. In the remaining case $D(a,b) = \{t, t'\}$ with $t' \neq t$; then $b = a^{tt'}$, and the four weights a^t, $b^t = a^{t'}$, $-a$, $-b = a^{t''}$, where $\{t, t', t''\} = \{1,2,3\}$, are pairwise non-opposite, since any two of them differ in exactly two coordinates, and they have sum zero, because $a^t + b^t$ and $a + b$ agree outside the t-th coordinate, where both vanish. So

$$m_{ab} = \chi(a)\chi(b)\, e^{(1)}_{a^t} e^{(1)}_{b^t} e^{(2)}_{-a} e^{(2)}_{-b}, \qquad t = \min D(a,b),$$

is a tetrahedron monomial with $D(a^t, b^t) = D(-a, -b) = D(a,b)$, and since $s_{t'}(e_a e_b) = e_{a^{t'}} e_{b^{t'}} = e_{b^t} e_{a^t} = -e_{a^t} e_{b^t}$, the term of the pair in $c(s_{t'})$ is $-m_{ab}$, while its term in $c(s_{t''})$ is a product of divisor classes. For each two-element set D put $u_D = \sum_{D(a,b)=D} m_{ab}$, a sum over the four pairs with difference set D: two of these pairs lie in each tetrahedron and a^t lies in the other one, so u_D has two monomials in each tetrahedron, all with coefficients ± 1, and the three u_D are supported on disjoint sets of monomials; the two monomials of u_D in T are those of the pair of T indexed by D. In Ξ this gives $c(1) \equiv 0$, $c(s_1) \equiv u_{12} + u_{13}$, $c(s_2) \equiv -u_{12} + u_{23}$, $c(s_3) \equiv -u_{13} - u_{23}$, hence

$$\pi_0 \equiv 0, \quad \pi_{12} \equiv \tfrac{1}{2}(u_{13} + u_{23}), \quad \pi_{13} \equiv \tfrac{1}{2}(u_{12} - u_{23}), \quad \pi_{23} \equiv -\tfrac{1}{2}(u_{12} + u_{13}), \tag{4}$$

and $\omega_a \equiv \frac{1}{2}((a_2 - a_3)u_{12} + (a_1 - a_3)u_{13} + (a_1 - a_2)u_{23})$ in Ξ: on the monomials of the pair of $T_\pm$ indexed by $\{1,2\}$, $\{1,3\}$, $\{2,3\}$ the class ω_a has coefficients of absolute values $|a_2 - a_3|/2$, $|a_1 - a_3|/2$, $|a_1 - a_2|/2$.

Now suppose that $\omega_a \in \Sigma_{\mathbb{Q}} \otimes \mathbb{C}$. Its image in Ξ is a combination of the $s_{\pm,k}$, and since every monomial of ω_a has two factors from each copy, only $s_{+,2}$ and $s_{-,2}$ occur: $\omega_a \equiv x\, s_{+,2} + y\, s_{-,2}$. On the six monomials of T_+ the right side has coefficients of absolute value $|x|$. So the three differences $a_2 - a_3$, $a_1 - a_3$, $a_1 - a_2$ have a common absolute value r, and as $a_1 - a_3 = (a_1 - a_2) + (a_2 - a_3)$, $r \in \{0, 2r\}$; hence $r = 0$ and $a_1 = a_2 = a_3$. Conversely the classes with $a_1 = a_2 = a_3$ are the combinations of π_0 and $\pi_{12} + \pi_{13} + \pi_{23}$, which are invariant under $\mathrm{Sp}(V, \psi)$ and so are combinations of the products of divisor classes $\psi_{11}\psi_{22}$ and ψ_{12}^2 (proof of Proposition 3.2(ii)); they lie in $\Sigma_{\mathbb{Q}}$. An exceptional class has for (a_1, a_2, a_3) the three distinct conjugates of an element of $F \setminus \mathbb{Q}$ (Lemma 3.19), so it does not lie in $\Sigma_{\mathbb{Q}}$.

(iii) We first show that (c) is equivalent to R and k_c generating $\mathrm{End}^0(X)$, and that R separates the weights of T_+ if and only if it separates those of T_-. For $\varphi \in \mathrm{End}^0(X)$ the eigenvalue on e_{-w} is $\varphi_{-w} = (\varphi^\dagger)_w$; since $\mathrm{End}^0(X)$ is a product of CM fields on which † is complex conjugation, and every embedding of a CM field into $\mathbb{C}$ commutes with complex conjugation, $\varphi_{-w} = \overline{\varphi_w}$. Hence $\varphi_w = \varphi_{w'}$ if and only if $\varphi_{-w} = \varphi_{-w'}$, and R separates T_+ if and only if it separates T_-. The elements of k_c are constant on each tetrahedron and

separate the two, and a subalgebra of $\bar{\mathbb{Q}}^{\{\text{weights}\}}$ containing the constants is the algebra of functions constant on the blocks of the partition it induces; so R separates T_+ exactly when $R \cdot k_c \otimes \bar{\mathbb{Q}}$ separates all eight weights, that is, when $R \cdot k_c = \mathrm{End}^0(X)$.

(c) implies (b). If R separates T_+, then $R \otimes \mathbb{C}$ maps onto $\mathbb{C}^{T_+}$ under $\varphi \mapsto (\varphi_w)_{w \in T_+}$, so the entries φ_w, ψ_w, $w \in T_+$, are eight independent coordinates on the image of $(R \otimes \mathbb{C})^2$. The sixteen coefficients in (3) are distinct monomials in these coordinates, hence linearly independent functions of $(\varphi, \psi) \in (R \otimes \mathbb{C})^2$; so no nonzero linear form on the span of the sixteen tetrahedron monomials of T_+ vanishes on all $f^*_{\varphi,\psi} t_+$, and these classes span that space. The same holds for T_-, and with the 100 products of divisor classes this is all of $\mathrm{Hdg}^2(Y) \otimes \mathbb{C}$. The map $(\varphi, \psi) \mapsto f^*_{\varphi,\psi} t_\pm$ is polynomial and $R \times R$ is Zariski dense in $(R \otimes \mathbb{C})^2$, so a linear form vanishing on its values at rational pairs vanishes on all values, and finitely many pairs $\varphi, \psi \in R$ give the same span. The $\mathbb{Q}$-span of the rational classes $f^*_{\varphi,\psi} W(X)$, $\varphi, \psi \in R$, and of the products of divisor classes is then a $\mathbb{Q}$-subspace of $\mathrm{Hdg}^2(Y)$ whose complexification is the whole space, hence is $\mathrm{Hdg}^2(Y)$.

(b) implies (a), as $\mathrm{Hdg}^2(Y)$ contains the exceptional classes.

(a) implies (c). Suppose that two distinct weights $u, u' \in T_+$ have $\varphi_u = \varphi_{u'}$ for every $\varphi \in R$, and let v, v' be the other two weights of T_+. In (3) the sets $S = \{u, v\}$ and $S' = \{u', v\}$ then have the same coefficient for every $\varphi, \psi \in R$, up to the signs of the shuffles, which do not depend on φ, ψ. Hence every class in $\Sigma_R \otimes \mathbb{C}$ has coefficients of equal absolute value on the tetrahedron monomials $e^{(1)}_S e^{(2)}_{T_+ \setminus S}$ and $e^{(1)}_{S'} e^{(2)}_{T_+ \setminus S'}$, the products of divisor classes involving no tetrahedron monomial. The pairs $\{S, T_+ \setminus S\}$ and $\{S', T_+ \setminus S'\}$ are different partitions of T_+, so $D(S) \neq D(S')$, and for an exceptional class ω_a in Σ_R two of the three numbers $|a_2 - a_3|$, $|a_1 - a_3|$, $|a_1 - a_2|$ would be equal. Each of the six resulting equations, $a_2 - a_3 = \pm(a_1 - a_3)$ and the four others, is a linear relation $c_1 a_1 + c_2 a_2 + c_3 a_3 = 0$ with rational coefficients not all equal, which Lemma 3.19 forbids for the conjugates of an element of $F \setminus \mathbb{Q}$. So (a) fails, and this proves the equivalence.

For the last assertions: $K_c^+ \otimes \bar{\mathbb{Q}}$ consists of all functions on the opposite pairs, and T_+ meets every pair once, so K_c^+ separates T_+; a proper subalgebra R of K_c^+ has $R \otimes \bar{\mathbb{Q}}$ a proper subalgebra of the functions on the four pairs, which identifies two pairs and hence two weights of T_+; and a subalgebra of k_c is constant on T_+. ∎

Figure 18 draws the weights of part (i), and Figure 19 the Hodge classes of parts (ii) and (iii).

The CM points fall into two kinds, according to whether the fourfold is simple; at the points of the second kind the real multiplication comes from the fixed cubic field F.

Lemma 3.30 (Split CM points)**.** *Let $c \in C$ be a CM point, and let $L \subset D$ be the quadratic extension of F whose norm one torus $\mathrm{Res}_{F/\mathbb{Q}} \ker(\mathrm{N}_{L/F})$ is the Hodge group of X_c (proof of Theorem 2.29). Then L is a CM field, and the following are equivalent:* (a) *X_c is not simple;* (b) *K_c^+ is not a field;* (c) *$L = Fk$ for an imaginary quadratic field $k \subseteq L$. In that case $k_c \cong k$, $K_c^+ \cong \mathbb{Q} \times F$, and X_c is isogenous to $E \times B$, where E is an elliptic curve with complex multiplication by k and B is an abelian threefold of CM type on which the factor F of K_c^+ acts, a real multiplication by F; the Weil line $W(X_c)$ lies in $H^1(E) \otimes H^3(B)$. At every other CM point K_c^+ is a totally real quartic field. Points of the second kind exist for infinitely many k: for every imaginary quadratic field k such that $k \otimes_{\mathbb{Q}} F_v$ is a field at each finite place v of F at which D is ramified.*

Proof. Write T for the Hodge group. By Proposition 3.29(i), $\mathrm{End}^0(X_c) = K_c^+ \otimes k_c$ is a product of CM fields on which † is complex conjugation, and T lies in the centraliser of

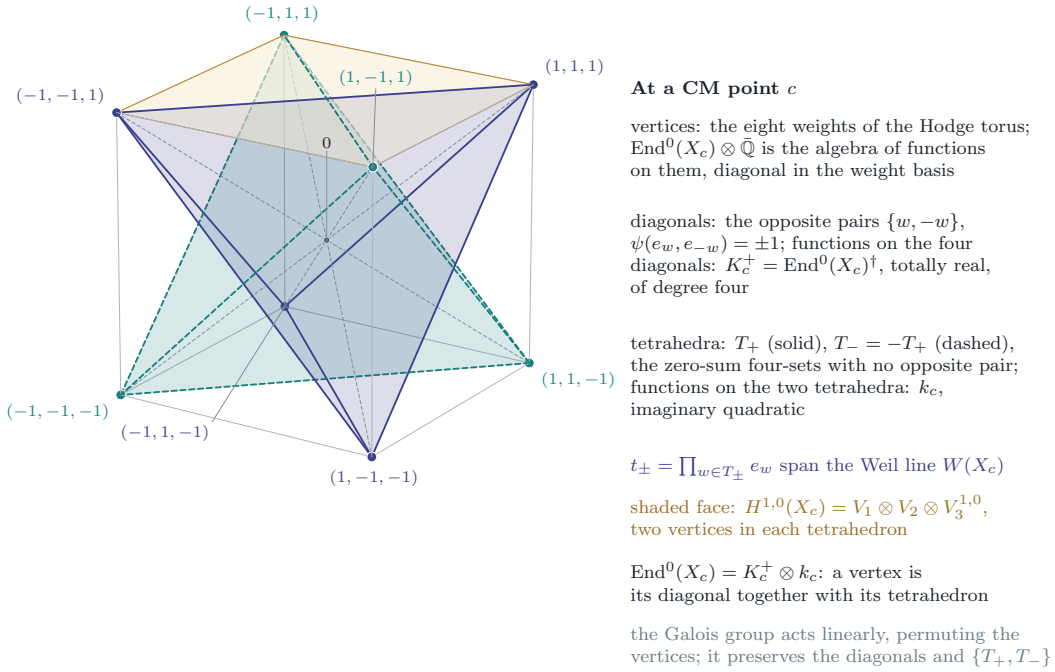
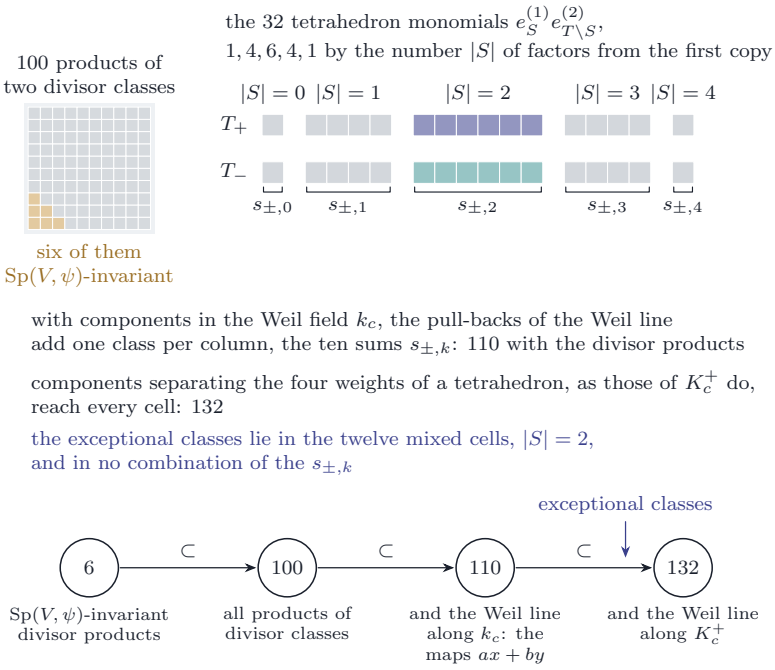


FIGURE 18. Proposition 3.29(i). The eight weights $(\pm1, \pm1, \pm1)$ of the Hodge torus at a CM point are the vertices of a cube. The four diagonals are the pairs of opposite weights, paired by the polarisation; the two inscribed tetrahedra T_+ (an even number of entries -1) and $T_- = -T_+$ are the only zero-sum four-sets without an opposite pair, and $t_\pm = \prod_{w\in T_\pm} e_w$ span the Weil line. The Galois group acts by linear maps permuting the vertices, so it preserves the diagonals and the pair of tetrahedra: the functions constant on the tetrahedra form the imaginary quadratic Weil field k_c, those constant on the diagonals the totally real quartic algebra K_c^+ of Rosati-fixed endomorphisms, and $\mathrm{End}^0(X_c) = K_c^+ \otimes k_c$ because a vertex is determined by its diagonal and its tetrahedron.

$\mathrm{End}^0(X_c)$, which over $\bar{\mathbb{Q}}$ is the diagonal torus, and in $\mathrm{Sp}(V, \psi)$; so $T(\mathbb{R})$ lies in the compact group of the $x \in (\mathrm{End}^0(X_c) \otimes \mathbb{R})^\times$ with $xx^\dagger = 1$. If $L \otimes_{F,\sigma} \mathbb{R}$ were $\mathbb{R} \times \mathbb{R}$ at a real place σ, the factor of $T(\mathbb{R})$ at σ would be the norm one group of $\mathbb{R} \times \mathbb{R}$ over $\mathbb{R}$, which is not compact. Hence $L \otimes_{F,\sigma} \mathbb{R} = \mathbb{C}$ at each real place, and L is a CM field.

Over $\bar{\mathbb{Q}}$, $T = \prod_\sigma T_\sigma$ with $T_\sigma = \ker(\mathrm{N}_{L/F}) \otimes_{F,\sigma} \bar{\mathbb{Q}}$ the diagonal torus of $\mathrm{SL}_2(\bar{\mathbb{Q}}) = (D \otimes_{F,\sigma} \bar{\mathbb{Q}})^1$ for the decomposition $L \otimes_{F,\sigma} \bar{\mathbb{Q}} = \bar{\mathbb{Q}} \times \bar{\mathbb{Q}}$ by the two embeddings of L over σ, one of them chosen and called τ_σ, the other $\bar\tau_\sigma$. The factor V_σ of $V = \bigotimes_\sigma V_\sigma$ (Setup 3.1) has the weights τ_σ and $\bar\tau_\sigma = \tau_\sigma^{-1}$ on T_σ, with eigenvectors $v_{\sigma,\pm}$, and $e_w = \bigotimes_\sigma v_{\sigma,w_\sigma}$. An element γ of the Galois group carries V_σ to $V_{\gamma\sigma}$ and $v_{\sigma,+}$ to $v_{\gamma\sigma,\varepsilon_\gamma(\sigma)}$, where $\varepsilon_\gamma(\sigma) = 1$ if $\gamma\tau_\sigma = \tau_{\gamma\sigma}$ and -1 if $\gamma\tau_\sigma = \bar\tau_{\gamma\sigma}$; so it acts on the weights by $(\gamma w)_{\gamma\sigma} = \varepsilon_\gamma(\sigma) w_\sigma$.

If $L = Fk$, then $\mathrm{Hom}(L, \bar{\mathbb{Q}}) = \mathrm{Hom}(F, \bar{\mathbb{Q}}) \times \mathrm{Hom}(k, \bar{\mathbb{Q}})$ with the Galois group acting on both factors, and with $\tau_\sigma = \sigma \otimes \iota_0$ for a fixed embedding ι_0 of k the sign $\varepsilon_\gamma(\sigma) = \varepsilon_\gamma$ does not depend on σ: it is 1 if γ fixes k and -1 otherwise. Then γ permutes the coordinates of a weight and changes all its signs when it moves k. It preserves each tetrahedron if it



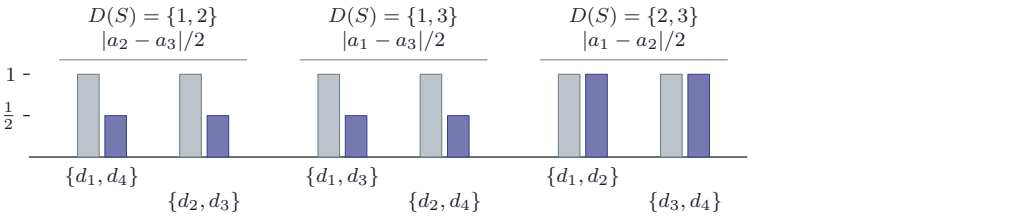


FIGURE 19. Proposition 3.29(ii) and (iii). Top: the 132 Hodge classes of degree four on the square at a CM point, one cell per monomial of weight zero. The 100 products of divisor classes are the cells with an opposite pair; the 32 tetrahedron monomials $e_S^{(1)} e_{T\setminus S}^{(2)}$ are arranged by tetrahedron and by the number $|S|$ of factors from the first copy. Pull-backs of the Weil line along homomorphisms with components in k_c add only the ten symmetric sums $s_{\pm,k}$, one per column, giving 110; components separating the four weights of a tetrahedron, as those of K_c^+ do, separate the monomials and give all 132. Bottom: on the six mixed monomials of one tetrahedron, grouped in the three pairs by the coordinates in which the two weights of S differ, the symmetric sum $s_{+,2}$ has coefficients ± 1 throughout, while an exceptional class ω_a has coefficients of absolute value $|a_2 - a_3|/2$, $|a_1 - a_3|/2$, $|a_1 - a_2|/2$, drawn for $\pi_{12} - \pi_{13}$; two equal absolute values would be a linear relation among the conjugates of an irrational element of F.

fixes k and exchanges the two otherwise, so $k_c \otimes \bar{\mathbb{Q}} = \bar{\mathbb{Q}}^{\{T_+,T_-\}}$ is the algebra of the Galois set $\mathrm{Hom}(k,\bar{\mathbb{Q}})$, and $k_c \cong k$. It carries the pair $\{\pm(1,1,1)\}$ to itself, so the projection p_E onto $\mathbb{C}e_{(1,1,1)} \oplus \mathbb{C}e_{(-1,-1,-1)}$ lies in $\mathrm{End}^0(X_c)$, and by Poincaré reducibility X_c is isogenous to $E \times B$ with $H^1(E,\mathbb{C})$ the image of p_E, of dimension two, so E is an elliptic curve, on whose H^1 the field k_c acts by two conjugate scalars: E has complex multiplication by k. The threefold B has $\mathrm{End}^0(B) \otimes \bar{\mathbb{Q}}$ containing the diagonal algebra of the six remaining weights, of dimension $6 = 2\dim B$, so B is of CM type. Each of the six remaining weights has exactly one coordinate different from the other two, and γ carries the opposite pair of the weight with distinguished coordinate t to the pair with distinguished coordinate γt; so the Galois set of the four opposite pairs is the fixed point $\{\pm(1,1,1)\}$ together with a copy of $\mathrm{Hom}(F,\bar{\mathbb{Q}})$, and $K_c^+ \cong \mathbb{Q} \times F$, the factor $\mathbb{Q}$ acting on $H^1(E)$ and the factor F on $H^1(B)$. Finally T_+ consists of $(1,1,1)$ and three weights of B, and T_- of $(-1,-1,-1)$ and three weights of B, so $t_\pm \in H^1(E) \otimes H^3(B)$ and $W(X_c)$ lies there. This proves that (c) implies (a) and (b), with the asserted structure.

X_c is simple exactly when $\mathrm{End}^0(X_c) = K_c^+ \otimes k_c$ is a field, and since every factor field E' of K_c^+ is totally real, $E' \otimes k_c$ is a field; so (a) and (b) are equivalent. Suppose (b). Then the Galois group does not act transitively on the four opposite pairs, whose algebra is K_c^+. It acts transitively on the three coordinates, as on the embeddings of F, so some γ permutes them cyclically. For each coordinate t the four pairs fall into two sets of two, according to the value of the coordinate t on the representatives in T_+, and this assigns the three coordinates bijectively to the three partitions of the four pairs into two sets of two, compatibly with the Galois action, which preserves or exchanges T_+ and T_-. So the permutation of the pairs induced by γ has order divisible by three, hence equal to three: it is a cycle of length three and fixes exactly one pair. The orbits of the Galois group on the pairs are therefore either one orbit or two, of sizes one and three, and by hypothesis some pair $\{w,-w\}$ is fixed: $\gamma w = \delta_\gamma w$ with $\delta_\gamma = \pm 1$ for every γ. Replace the choice of τ_σ by $\tau'_\sigma = \tau_\sigma$ if $w_\sigma = 1$ and $\tau'_\sigma = \bar{\tau}_\sigma$ if $w_\sigma = -1$; for the new choice w becomes $(1,1,1)$, and the new signs are $\varepsilon'_\gamma(\sigma) = \varepsilon_\gamma(\sigma) w_\sigma w_{\gamma\sigma} = \delta_\gamma$ for all σ. So γ carries the set $\Phi = \{\tau'_\sigma\}$ of three embeddings of L to Φ if $\delta_\gamma = 1$ and to $\bar{\Phi}$ if $\delta_\gamma = -1$. The map $\gamma \mapsto \delta_\gamma$ is a character of the Galois group; it is nontrivial, for otherwise Φ would be a Galois stable set of three of the six embeddings and L would not be a field; and complex conjugation carries every embedding τ of the CM field L to $\bar{\tau}$, hence Φ to $\bar{\Phi}$, so the quadratic field k cut out by the character is imaginary. Let ι_0, $\bar{\iota}_0$ be the embeddings of k, with ι_0 fixed by the kernel of the character. Then $\tau \mapsto (\tau|_F, \iota_0$ if $\tau \in \Phi$ and $\bar{\iota}_0$ otherwise) is a Galois equivariant bijection $\mathrm{Hom}(L,\bar{\mathbb{Q}}) \to \mathrm{Hom}(F,\bar{\mathbb{Q}}) \times \mathrm{Hom}(k,\bar{\mathbb{Q}})$, so $L \cong F \otimes_{\mathbb{Q}} k$, that is, $L = Fk$ for a copy of k in L, which is (c). At a CM point where (c) fails, K_c^+ is therefore a field, quartic and totally real by Proposition 3.29(i).

Existence. Let k be imaginary quadratic. By the theorem of Albert, Brauer, Hasse and Noether [PR94, §1.5], the CM field Fk splits D, and so embeds in D over F, exactly when $Fk \otimes_F F_v$ is a field at every place v of F at which D is ramified, a quadratic extension of a local field splitting the local quaternion division algebra. At the two real places where D is definite this holds because k is imaginary, and at the finite ramified places it is the hypothesis. Given the finitely many residue characteristics p of these places, choose for each a quadratic extension k_p of $\mathbb{Q}_p$ not contained in F_v for any v above p: this is possible since the completions F_v above p have degrees over $\mathbb{Q}_p$ adding up to three, so at most one of them contains a quadratic extension of $\mathbb{Q}_p$, and then exactly one, while $\mathbb{Q}_p$ has at least three. Infinitely many imaginary quadratic fields k have the completions k_p at these p, and for them $F_v \otimes_{\mathbb{Q}_p} k_p$ is a field. For such k let $T_k = \mathrm{Res}_{F/\mathbb{Q}} \ker(\mathrm{N}_{Fk/F}) \subset G$, a maximal

torus with $T_k(\mathbb{R})$ compact. A compact subgroup of $G(\mathbb{R})$ fixes a point x of the symmetric space of $G(\mathbb{R})$, and the stabiliser of x is the centraliser of $h_x(\mathbb{S})$; so $h_x(\mathbb{S})$ centralises $T_k(\mathbb{R})$, which is Zariski dense in T_k, and lies in $T_k(\mathbb{R})$. The Hodge group of the fibre X_c over the image c of x is the smallest $\mathbb{Q}$-subgroup of G containing $h_x(U(1))$, so it lies in T_k; it has rank three (proof of Theorem 2.29), so it equals T_k, and c is a CM point with $L = Fk$. ■

Remark 3.31 (What the degree bound has to control).

(1) Markman's theorem [Mar25a], which the proof of Theorem 2.29 invokes, produces the cycles on X_c through its Weil structure for k_c. Proposition 3.29 says that this structure is blind to the exceptional classes of the square: everything obtained from the Weil line and from divisors by pull-back along homomorphisms with components in k_c, in particular along the maps $(x, y) \mapsto ax + by$, and by products, spans 110 of the 132 Hodge classes and misses every exceptional class; and a representation of an exceptional class by pull-backs of the Weil line and products of divisors must use homomorphisms whose components, with k_c, generate the whole endomorphism algebra. At a CM point the exceptional classes are thus a statement about the real multiplication K_c^+, as they are at a general point about the real multiplication by F (Proposition 3.2).

(2) For $\varphi, \psi \in \mathrm{End}(X_c)$ and a cycle Z of codimension two on X_c, the degree $\int_Y f_{\varphi,\psi}^*[Z] \cup \theta_Y^6$ is, by (3) applied to the monomials of $[Z]$, a polynomial of degree four in the eigenvalues φ_w, ψ_w, which are algebraic integers. For cycles of this shape the bound of Corollary 3.28 therefore asks, at infinitely many CM points, for endomorphisms with bounded eigenvalues whose components generate $\mathrm{End}^0(X_c)$ over k_c. Only finitely many algebraic integers of degree at most eight have bounded conjugates, so the same finitely many minimal polynomials would have to occur at infinitely many points. At the split points of Lemma 3.30 the algebra $K_c^+ = \mathbb{Q} \times F$ does not depend on the point and the real multiplication is by F on the threefold factor; the representation of the proof of Theorem 2.29 then uses, besides elements of F, the isogeny $X_c \to E \times B$, a cycle representing the Weil line of $E \times B$ in $H^1(E) \otimes H^3(B)$, and divisor classes, and these three are what varies with the point.

(3) At a split point the Weil line is a Hodge class in $H^1(E) \otimes H^3(B)$, a correspondence between an elliptic curve with complex multiplication by k and a CM threefold with real multiplication by F, and its algebraicity is again Markman's theorem for the fourfold $E \times B$. A construction of that correspondence with a degree bounded independently of k, together with a bound on the isogeny and on the divisor classes, would give the target through Corollary 3.28. We do not have one.

Remark 3.32 (Routes to the Mumford target). We list the routes to the Mumford target that avoid the Lefschetz standard conjecture, with what is known about each; Figure 20 summarises them.

(1) *Degeneration.* Closed off: the base is compact and the Hodge locus is the base (Corollary 3.23), so no family on which ω stays of Hodge type approaches the boundary, and there is no limit mixed Hodge structure to specialise to.

(2) *Deformation into a larger Hodge locus*, on which ω would become a product of divisor classes or a Weil class, hence algebraic by a known theorem, and could then be carried back. Closed off by Theorem 3.21(iii), already to first order.

(3) *Hochschild deformations*: twisting $D^b(Y)$ by a B-field or a Poisson bivector, as in the gerby and noncommutative deformations used for Weil classes. Closed off: an

object with $\mathrm{ch}(E) = N\omega$ extends along no direction of $HH^2(Y)$ except that of the curve (Corollary 3.41(i)).

(4) *Constructions from line bundles*: cones, tensor products, duals, pull-backs and push-forwards along homomorphisms, and Fourier–Mukai functors whose kernels are built in the same way. Closed off at every point of the curve: everything so built has a Chern character invariant under the Lefschetz group, and the exceptional classes are not (Theorem 3.36).

(5) *Twistor and hyperholomorphic deformation*, the mechanism of Markman's proof for Weil classes: move a sheaf along twistor lines on which its Chern character stays of Hodge type. Closed off on the square: no twistor line through a member of the Mumford family keeps ω of Hodge type, and the twistor lines that do keep it of Hodge type carry non-algebraic tori and lie in other components of the Hodge locus (Proposition 3.24).

(6) *Specialisation.* The class is algebraic at every CM point (Theorem 2.29), on every power of X_c [Li26, Theorem 1.3]; this suffices exactly when the degrees are bounded (Corollary 3.28), and no bound is known. The cycles that represent the class at a CM point must use the real multiplication by the Rosati-fixed algebra K_c^+, all of it; the Weil structure of the fourfold, with which Markman's theorem produces its cycles, reaches 110 of the 132 Hodge classes and none of the exceptional ones (Proposition 3.29, Remark 3.31).

(7) *Reduction modulo p.* At every closed point of every reduction of the curve the class is a $\mathbb{Q}_\ell$-combination of algebraic cycles (Proposition 3.42), by Li's theorem [Li26, Theorem 1.1]; this suffices, with no lifting step, exactly when the Hilbert data of the cycles are bounded on a Zariski dense set of closed points of the arithmetic model (Theorem 3.43), and no bound is known.

(8) *Semiregularity.* Reduced to one dimension count at one CM point (Theorem 3.27); an object attaining it has large odd self-extensions (Theorem 3.34), is not built from line bundles (Proposition 3.35) and splits only in a constrained way (Corollary 3.41); no such object is known.

(9) *Kuga–Satake.* Reduced to the Kuga–Satake class of one K3 surface of Picard number thirteen (Theorem 3.4), which the known cases of the Kuga–Satake Hodge conjecture do not reach (Remark 3.5). Its geometric form is the mechanism of van Geemen and Rapagnetta [vGR26], pull-back of the dual Beauville–Bogomolov class of a hyperkähler manifold: up to multiples of π_0, the exceptional classes are exactly such twisted dual forms of the part of H^2 of the square generated by its symplectic class, and a rational map onto a symplectic subvariety of some $S_\mu^{[n]}$, $n \geq 5$, would suffice; K3 surfaces and hyperkähler eightfolds cannot serve (Proposition 3.25, Remark 3.26).

(10) *Motivated classes.* Every Hodge class on an abelian variety is motivated [And96b, Théorème 0.6.2], so ω_t is motivated for every t; a motivated class is algebraic once the Lefschetz involutions of the auxiliary varieties are (Proposition 2.16). Along the family the auxiliary variety that suffices is $W \times_C W$ (Theorem 2.28), and its Lefschetz standard conjecture is equivalent to the target (Corollary 2.30): the route moves the problem and does not shorten it.

Five of the ten routes are closed off by theorems. Of the other five, two work at special points only, the CM points and the closed points of the reductions modulo p, and each suffices exactly when a degree bound holds; two are reduced to a single object, one Kuga–Satake class or one perfect complex with $\dim \mathrm{Ext}^2 = 119$ at one CM point; and one is

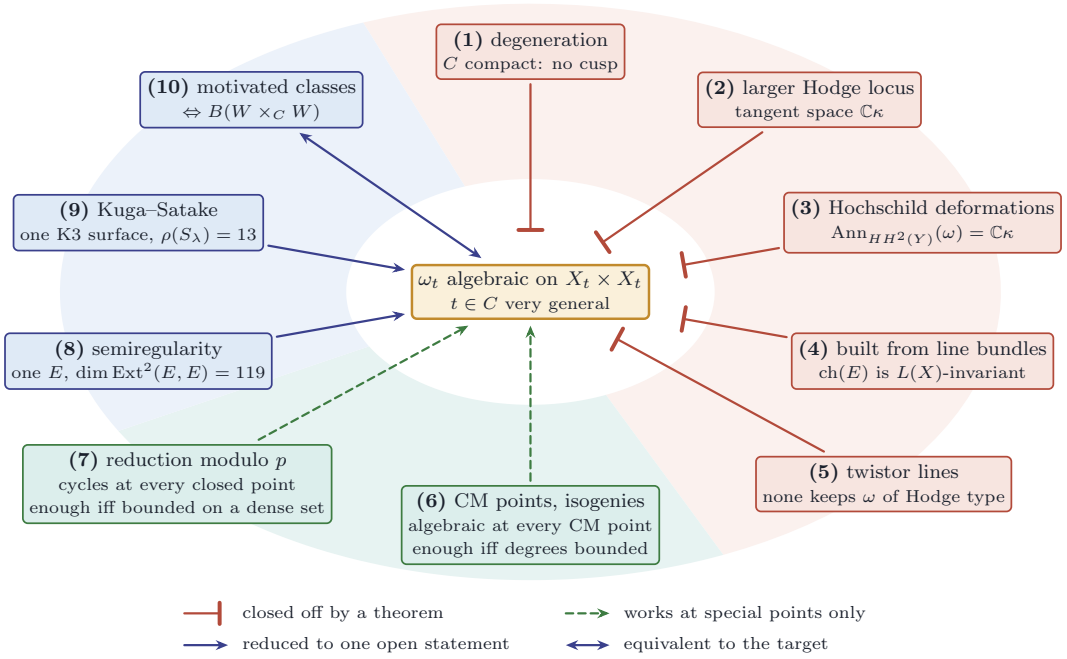


FIGURE 20. Remark 3.32: the ten routes to the Mumford target that avoid the Lefschetz standard conjecture. Clay, with a crossed dashed spoke: a route closed off by a theorem of Section 3.2 or Section 3.3. Grass: a route that works at special points only, at the CM points or at the closed points of the reductions modulo p, and suffices exactly when a degree bound holds (Corollary 3.28, Theorem 3.43). Indigo: a route reduced to one statement, of the same kind as the target or equivalent to it.

equivalent to the target. They are not five separate obligations: three of them are the target itself in other forms and the other two are stronger inputs that imply it, so any one of them would suffice (Corollary 3.45, Remark 3.46).

3.3. **Objects, reduction modulo p and the motivic group.** Theorem 3.27 reduces the semiregularity route to one object at one CM point. We first derive what any object with the right Chern character must satisfy, whether or not it is semiregular: no sequence of standard operations produces it from line bundles; it extends along no first order deformation of $D^b(Y)$ other than the direction of the curve; if it is semiregular, it splits only into summands with no Ext^2 between them; and its self-extensions are large in every degree, with an odd part that the Euler characteristic forces beyond what Hochschild cohomology supplies. We then turn to reduction modulo p. By Li's proof of the Tate conjecture for powers of abelian fourfolds over finite fields, the classes are represented by cycles at every closed point of every reduction of the curve, and the target is equivalent to a bound on those cycles on a Zariski dense set of closed points. Finally, the motivic group of X is one of four groups, and the kinds of algebraic classes known on products of X with other abelian varieties generate no exceptional class. None of this constructs such an object or proves the target.

Lemma 3.33 (The contraction ranks are symmetric)**.** *Let A be an abelian variety of dimension g and $c \in H^*(A,\mathbb{C})$ a class each of whose components has Hodge type (p,p) for some p. For $k \geq 0$ put $r_k(c) = \dim(HT^k(A) \lrcorner c)$, with the action of* [II, Setup 2.7]*. Then $r_k(c) = r_{g-k}(c)$ for $0 \leq k \leq g$, and $r_k(c) = 0$ for $k > g$.*

Proof. Write $H^{1,0} = H^{1,0}(A)$ and $A' = H^{0,1}(A)$, both of dimension g, so that $H^*(A,\mathbb{C}) = \bigwedge^* H^{1,0} \otimes \bigwedge^* A'$ and $HT^*(A) = \bigwedge^*(A' \oplus T)$ with $T = (H^{1,0})^\vee$. Fix a generator of $\bigwedge^g A'$ and let $\star\colon \bigwedge^q A' \to \bigwedge^{g-q} A'^\vee$ send η to $\zeta \mapsto \eta \wedge \zeta$. Under $\Phi = \mathrm{id} \otimes \star$, wedging with $z \in A'$ becomes, up to a sign fixed on each bigraded piece, contraction with z on the second factor, while T still contracts the first. So Φ identifies the action of $HT^*(A)$ with the action of $\bigwedge^* U^\vee$ by interior products on $\bigwedge^* U$, where $U = H^{1,0} \oplus A'^\vee$ has dimension $2g$, up to signs that do not change ranks. A class of type (p,p) goes to $\bigwedge^p H^{1,0} \otimes \bigwedge^{g-p} A'^\vee \subset \bigwedge^g U$, so $\eta = \Phi(c)$ lies in the middle degree whatever the p of its components. Hence $r_k(c)$ is the rank of $\bigwedge^k U^\vee \to \bigwedge^{g-k} U$, $x \mapsto \iota_x \eta$, which is zero for $k > g$. Its transpose for the natural pairings is $\bigwedge^{g-k} U^\vee \to \bigwedge^k U$, $y \mapsto \pm \iota_y \eta$, because $\langle \iota_x \eta, y\rangle = \langle \eta, x \wedge y\rangle = \pm\langle \iota_y \eta, x\rangle$. A map and its transpose have the same rank. ∎

Theorem 3.34 (The self-extensions of an object with an exceptional Chern character)**.** *Let $c \in C$ and $Y = X_c \times X_c$, and write $e_k = \dim \mathrm{Ext}^k(E,E)$ for a perfect complex E on Y.*

(i) *For every E and every k, $e_k \geq r_k(\mathrm{ch}\, E)$ and $e_k = e_{8-k}$.*

(ii) *For a rational exceptional class $\omega = \omega_a$ with $L_0(a) \neq 0$: $r_0 = r_8 = 1$, $r_1 = r_7 = 16$ and $r_2 = r_6 = 119$.*

(iii) *Let ω be a rational exceptional class and $\mathrm{ch}(E) = N\omega$ with $N \neq 0$. Then*

$$e_1 + e_3 = e_0 + e_2 + \tfrac{1}{2} e_4 + \sum_{j \geq 1} (-1)^j e_{-j}.$$

If $\mathrm{Ext}^{<0}(E,E) = 0$, for instance if E is a sheaf, then $\sum_{k \text{ odd}} e_k = \sum_{k \text{ even}} e_k$, and $e_1 + e_3 \geq 120 + \frac{1}{2} e_4$ when $L_0(a) \neq 0$.

Proof. (i) The identity $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$ of [II, Setup 2.7] holds in every degree, so the kernel of ev_E on $HT^k(Y)$ lies in the annihilator of $\mathrm{ch}(E)$, and the image of $HT^k(Y)$ in $\mathrm{Ext}^k(E,E)$ has dimension at least $r_k(\mathrm{ch}\, E)$. Serre duality with the trivial canonical bundle gives $e_k = e_{8-k}$.

(ii) $r_0 = 1$ because $\omega \neq 0$, and $r_2 = 119$ is Theorem 3.21(ii). If $w \in HT^1(Y)$ annihilates ω, so does $w' \wedge w$ for every $w' \in HT^1(Y)$, since the annihilator is a left ideal; for $w \neq 0$ these products span a space of dimension 15, which cannot lie in $\mathrm{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$. So $r_1 = 16$, and Lemma 3.33 gives r_8, r_7, r_6.

(iii) Since $\mathrm{ch}(E)$ has only a component in H^4, $\mathrm{ch}(E^\vee)\,\mathrm{ch}(E)$ lies in $H^8(Y)$ and $\chi(E,E) = \int_Y \mathrm{ch}(E^\vee)\,\mathrm{ch}(E) = 0$, the Todd class being trivial. By (i) and Serre duality, which also pairs Ext^{-j} with Ext^{8+j},

$$0 = \chi(E,E) = 2(e_0 - e_1 + e_2 - e_3) + e_4 + 2\sum_{j \geq 1} (-1)^j e_{-j},$$

which is the displayed identity. With no negative extensions the odd and even parts are equal, and $e_0 \geq 1$, $e_2 \geq 119$ by (i) and (ii) when $L_0(a) \neq 0$. ∎

Proposition 3.35 (Divisor classes do not reach the exceptional classes)**.** *Let $t \in C$ and let ω be a rational exceptional class. Then ω_t is not in the $\mathbb{Q}$-span of products of divisor classes of $X_t \times X_t$: that span meets the rational Hodge classes of $H^2 \otimes H^2$ exactly in the classes with $\alpha \in \mathbb{Q}$. Consequently no perfect complex whose Chern character lies in the*

subring generated by divisor classes, in particular no bounded complex of direct sums of line bundles, has $\mathrm{ch}(E) = N\omega$ *with* $N \neq 0$*. At a CM point* c *the fourfold* X_c *has eight Hodge classes in* H^4*, of which the products of divisor classes span six.*

Proof. At a point with Hodge group G, the divisor classes of $X_t \times X_t$ are the three classes of Proposition 2.74(ii), all invariant under $\mathrm{Sp}(V,\psi)$, and so are their products; the $\mathrm{Sp}(V,\psi)$-invariant part of $H^2 \otimes H^2$ is spanned by π_0 and $\pi_{12}+\pi_{13}+\pi_{23}$, the classes with $a_1 = a_2 = a_3$. At a CM point the Hodge group is a maximal torus of G (proof of Theorem 2.29), so the Hodge classes are the monomials of weight zero. The proof of Proposition 3.29(ii) counts them: 16 divisor classes $e_w^{(i)} e_{-w}^{(j)}$, 132 Hodge classes in degree four, of which the products of divisor classes span 100, and modulo that span $\omega_a \equiv \frac{1}{2}((a_2-a_3)u_{12} + (a_1-a_3)u_{13} + (a_1-a_2)u_{23})$ with u_{12}, u_{13}, u_{23} linearly independent. So the span meets that of the four π in exactly the classes with $a_1 = a_2 = a_3$. For a rational class these coordinates are the conjugates of α, which by Lemma 3.19 are equal only if $\alpha \in \mathbb{Q}$. The Chern character of a complex in the subring generated by divisor classes has its component in H^4 in the span of products of two divisor classes, which excludes $N\omega$. For X_c itself the monomials of weight zero in H^4 are the six products $e_w e_{-w} e_{w'} e_{-w'}$ over two opposite pairs, which are products of divisor classes, and the two tetrahedron monomials. ∎

At a CM point the Hodge classes of X_c in H^4 that are not products of divisor classes are among the classes that the proof of Theorem 2.29 pulls back along homomorphisms $X_c \times X_c \to X_c$; they are algebraic by Markman's theorem [Mar25a]. Li proves, independently and more generally, that every Hodge class on every power of every complex CM abelian fourfold is algebraic [Li26, Theorem 1.3], which gives Theorem 2.29 for every power X_c^n and not only for the square.

By Proposition 3.35, an object with $\mathrm{ch}(E) = N\omega$ cannot have its Chern character in the subring generated by divisor classes, so it cannot be assembled from line bundles alone. The next theorem shows that no sequence of the standard operations of the derived category, applied to line bundles on powers of X_t, gets any further.

Theorem 3.36 (Objects generated by line bundles)**.** *Let* $t \in C$*,* $X = X_t$*, and let* $L(X) \subseteq \mathrm{Sp}(V,\psi)$ *be the Lefschetz group of* X*, the centraliser of* $\mathrm{End}^0(X)$ *in* $\mathrm{Sp}(V,\psi)$*. Let* $\mathcal{A}$ *be the class of abelian varieties isogenous to a power of* X*; through an isogeny* $X^n \to B$ *the group* $L(X)$ *acts diagonally on* $H^*(B,\mathbb{Q})$*, and the action does not depend on the isogeny. Let* $\mathcal{L}$ *be the smallest class of objects of the categories* $D^b(B)$*,* $B \in \mathcal{A}$*, that contains the line bundles and is closed under shifts, cones, derived tensor products, derived duals, translations, the functors* Lf^* *and* Rf_* *for homomorphisms* f *between members of* $\mathcal{A}$*, and the Fourier–Mukai functors* $\Phi_K \colon D^b(B) \to D^b(B')$ *with kernel* $K \in \mathcal{L}$ *on* $B \times B'$*.*

(i) *Every object of* $\mathcal{L}$ *has an* $L(X)$*-invariant Chern character.*

(ii) $L(X) = \mathrm{Sp}(V,\psi)$ *if* t *is not a CM point, and* $L(X)$ *is a maximal torus of* $\mathrm{Sp}(V,\psi)$ *if it is. In both cases the* $L(X)$*-invariant classes on* $X \times X$ *form the subring generated by divisor classes, of dimensions* $1,3,6,10,15,10,6,3,1$*, respectively* $1,16,100,304,454,304,100,16,1$*, in the degrees* $0,2,\ldots,16$*.*

(iii) *No rational exceptional class* ω_t *is* $L(X)$*-invariant. Hence for no object* $E \in \mathcal{L}$ *on* $X \times X$ *and no* $N \neq 0$ *is* $\mathrm{ch}(E) - N\omega_t$ *invariant under* $L(X)$*; in particular* $\mathrm{ch}(E) \neq N\omega_t$*.*

(iv) *The* $\mathrm{Sp}(V,\psi)$*-module generated by* $\pi_{12}-\pi_{13}$ *is irreducible, of highest weight* $2\varpi_2$ *and dimension* 308*, and it meets the span of the four* π *in the plane of the classes with* $a_0=0$ *and* $a_1+a_2+a_3=0$.

Proof. (i) Two isogenies $X^n\to B$ differ by an element of $\mathrm{End}^0(X^n)=M_n(\mathrm{End}^0(X))$, which commutes with $L(X)$ acting diagonally, so the action is well defined. For the same reason a homomorphism f between members of $\mathcal{A}$ is $L(X)$-equivariant on H^1, so f^* is equivariant on H^*, and so is f_*, which is f^* conjugated by Poincaré duality, since $L(X)\subseteq\mathrm{SL}(V)$ fixes the orientation classes. Translations act trivially on cohomology. The Chern character is additive on distinguished triangles and multiplicative on derived tensor products, $\mathrm{ch}(E^\vee)=\sum_i(-1)^i\,\mathrm{ch}_i(E)$, $\mathrm{ch}(Lf^*E)=f^*\,\mathrm{ch}(E)$, and $\mathrm{ch}(Rf_*E)=f_*\,\mathrm{ch}(E)$ by the Grothendieck–Riemann–Roch theorem [Ful98, Theorem 15.2], the Todd class of an abelian variety being 1; and $\Phi_K=Rp_{2*}(p_1^*(\,\cdot\,)\otimes^L K)$. So the objects with $L(X)$-invariant Chern character form a class with all the closure properties, and it remains to see that it contains the line bundles, that is, that $c_1(\mathcal{O}(D))$ is $L(X)$-invariant. Pulled back along an isogeny, which is injective on rational cohomology, c_1 becomes a Hodge class of degree two on some X^n, an element of the invariants of the Hodge group in $\bigwedge^2(V^\vee\otimes\mathbb{Q}^n)=\bigwedge^2V^\vee\otimes S^2\mathbb{Q}^n\oplus S^2V^\vee\otimes\bigwedge^2\mathbb{Q}^n$. If t is not a CM point, the Hodge group is G, whose invariants in $\bigwedge^2V$ are the multiples of ψ and in S^2V are zero, since every summand of $S^2(V_1\otimes V_2\otimes V_3)$ contains a factor S^2V_i; so the Hodge classes of degree two are $\mathrm{Sp}(V,\psi)$-invariant. If t is a CM point, a monomial of degree two in a weight basis has weight zero for the Hodge torus exactly when its two weights are opposite, and then it has weight zero for $L(X)$, by the description in (ii).

(ii) If t is not a CM point, $\mathrm{End}^0(X)=\mathbb{Q}$ and $L(X)=\mathrm{Sp}(V,\psi)$. At a CM point $\mathrm{End}^0(X)\otimes\mathbb{C}$ is the algebra of diagonal matrices in a weight basis (proof of Theorem 2.29), and ψ pairs each weight with its opposite only, so $L(X)_{\mathbb{C}}$ is the torus of diagonal matrices $\mathrm{diag}(\lambda_w)$ with $\lambda_w\lambda_{-w}=1$, of rank four. For t not CM, the first fundamental theorem for $\mathrm{Sp}(V,\psi)$ [GW09, Chapter 5] shows that its invariants in $\bigwedge^*(V\otimes\mathbb{Q}^2)$ are spanned by the products of the three quadratic invariants, which are the divisor classes. Their dimensions follow from the skew Cauchy formula $\bigwedge^{2m}(V\otimes\mathbb{Q}^2)=\bigoplus_\lambda S_\lambda V\otimes S_{\lambda'}\mathbb{Q}^2$, over the partitions $\lambda=(2^r,1^s)$ of $2m$ with at most two columns, and the rule that $S_\lambda V$ contains the trivial representation of $\mathrm{Sp}(V,\psi)$, then once, exactly when both columns of λ have even length at most 8 [GW09]: the invariants of degree $2m$ have dimension $\sum(s+1)$, over the even $r,s\geq0$ with $2r+s=2m$ and $r+s\leq8$, which is $1,3,6,10,15,10,6,3,1$ for $m=0,\dots,8$. At a CM point the invariants are spanned by the monomials of weight zero for $L(X)$; a set of weights $\pm e_j$ with sum zero is a union of opposite pairs, so each such monomial is, up to sign, a product of monomials of degree two and weight zero, which are divisor classes. For each opposite pair $\{w,-w\}$ the four vectors $e^{(1)}_{\pm w}$, $e^{(2)}_{\pm w}$ give one monomial of weight zero in degree 0, four in degree 2 and one in degree 4, so the numbers of these monomials are the coefficients of $(1+4y+y^2)^4$.

(iii) If t is not a CM point, the $\mathrm{Sp}(V,\psi)$-invariant combinations of the π are those with $a_1=a_2=a_3$ (Proposition 3.35), and the coefficients of ω_t are the distinct conjugates of α (Lemma 3.19). At a CM point use the map $x\mapsto E_x$ of the proof of Lemma 3.20, which is $\mathrm{Sp}(V,\psi)$-equivariant: it carries ω_t to an element (r,β) of $\mathbb{Q}\times F$ with $\beta\notin\mathbb{Q}$, acting on U_{ij} by the conjugate $\sigma_k(\beta)$ (Proposition 3.2(ii)), and ω_t is $L(X)$-invariant only if E_{ω_t} preserves the weight spaces of $L(X)$ in $\bigwedge^2V$. Take weight vectors $v=x\otimes y\otimes z$ and $w=x\otimes y'\otimes z'$ of weights $(1,1,1)$ and $(1,-1,-1)$ for the Hodge torus. They are not opposite, so $v\wedge w$ spans a weight space of $L(X)$, every weight of $L(X)$ on V having multiplicity one. But $v\wedge w=(x\cdot x)\otimes(y\cdot y'\otimes z\wedge z'+y\wedge y'\otimes z\cdot z')$ has nonzero components in U_{12} and in U_{13},

on which E_{ω_t} acts by $\sigma_3(\beta) \neq \sigma_2(\beta)$. So the line is not preserved. The rest of (iii) follows from (i).

(iv) Write $H^2(X) = \mathbb{Q}\theta \oplus P$, with P the primitive part, the irreducible $\mathrm{Sp}(V,\psi)$-module of highest weight ϖ_2 and dimension 27. Through the form $\bigwedge^2\psi$ the four π are self-adjoint endomorphisms of $H^2(X)$, and $\pi_{12}-\pi_{13}$ kills θ and is a trace-free self-adjoint endomorphism of P, an element of S^2P orthogonal to the invariant. Restricting $S^2(\bigwedge^2 V) = S_{(2,2)}V \oplus \bigwedge^4 V$ from $\mathrm{GL}(V)$ to $\mathrm{Sp}(V,\psi)$ gives, with dimensions $336 = 308+27+1$ and $70 = 42+27+1$, the decomposition $S^2P = V(2\varpi_2) \oplus V(\varpi_4) \oplus V(\varpi_2) \oplus \mathbb{Q}$. The element $\pi_{12}-\pi_{13}$ is G-invariant. The summand $V(\varpi_2) \cong P$ has no G-invariant, since $(\bigwedge^2 V)^G = \mathbb{Q}\theta$, and neither has $V(\varpi_4)$, which lies in $\bigwedge^4 V$, whose G-invariants are the line of θ^2 (Proposition 2.74(i)), invariant under all of $\mathrm{Sp}(V,\psi)$. So $\pi_{12}-\pi_{13}$ is a nonzero vector of the irreducible module $V(2\varpi_2)$, which it generates; its dimension is 308 by Weyl's dimension formula. The same argument puts every trace-free combination of $\pi_{12},\pi_{13},\pi_{23}$ in $V(2\varpi_2)$, while π_0 and $\pi_{12}+\pi_{13}+\pi_{23}$ are $\mathrm{Sp}(V,\psi)$-invariant and independent modulo it; so adding the four π raises the dimension by two. ∎

Figure 21 compares the Hodge classes of $X_t \times X_t$ with the subring generated by divisor classes, degree by degree.

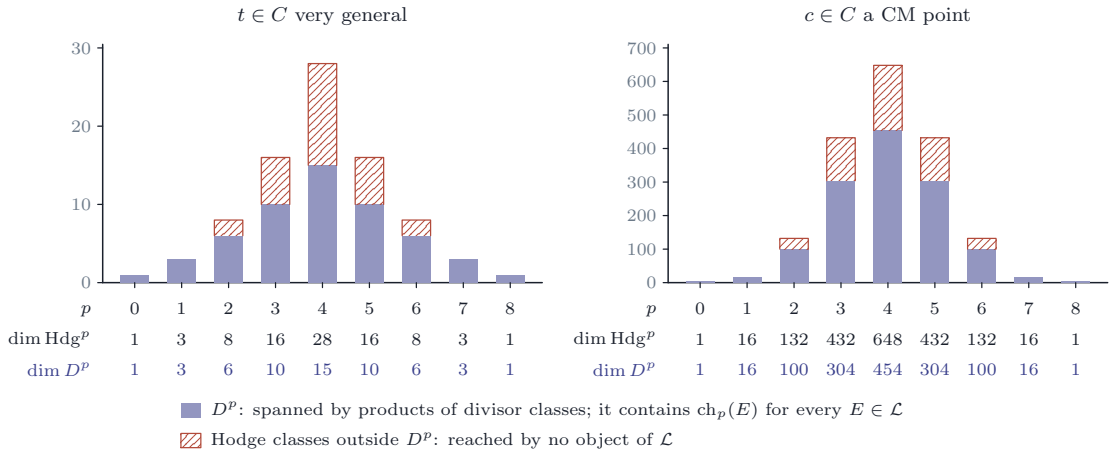


FIGURE 21. Theorem 3.36(ii). The Hodge classes of $X_t \times X_t$ degree by degree, against the subring D^p generated by divisor classes, which contains the Chern character of every object of $\mathcal{L}$: at a very general point of the Mumford curve (left) and at a CM point (right). The hatched classes, the exceptional ones among them, are reached by no object of $\mathcal{L}$.

Remark 3.37 (Where a construction has to leave $\mathcal{L}$). Theorem 3.36 extends Proposition 3.35 from sums of line bundles to everything the standard operations make out of them, at every point of the curve, including the CM points, where the exceptional classes are algebraic. There the cycles that represent them are not Chern characters of objects of $\mathcal{L}$, and neither are the two Hodge classes of X_c in degree four that are not products of divisor classes. At a point that is not CM, every Chern character of an object of $\mathcal{L}$ lies in the isotypic component of the trivial representation of $\mathrm{Sp}(V,\psi)$, while by Theorem 3.36(iv) an object with $\mathrm{ch}(E) = N\omega$ must have a nonzero component in the one of highest weight $2\varpi_2$. Every construction of such an object therefore contains a step that does not commute

with the Lefschetz group on cohomology: passing to a direct summand, a subobject or a Jordan–Hölder factor, choosing a point of a moduli space, or deforming from a member with more endomorphisms. The last is what semiregularity does along the curve (Theorem 3.27).

Lemma 3.38 (The summands of a semiregular object)**.** *Let Z be a smooth projective variety and E_1, E_2 perfect complexes on Z. On $\operatorname{Ext}^2(E_1 \oplus E_2, E_1 \oplus E_2) = \bigoplus_{i,j} \operatorname{Ext}^2(E_j, E_i)$ the semiregularity map is $\sigma_{E_1 \oplus E_2}(\zeta) = \sigma_{E_1}(\zeta_{11}) + \sigma_{E_2}(\zeta_{22})$. So $\sigma_{E_1 \oplus E_2}$ is injective if and only if σ_{E_1} and σ_{E_2} are injective, their images meet only in zero, and $\operatorname{Ext}^2(E_1, E_2) = \operatorname{Ext}^2(E_2, E_1) = 0$.*

Proof. The Atiyah class is natural, so $\operatorname{At}(E_1 \oplus E_2)$ is the diagonal sum of $\operatorname{At}(E_1)$ and $\operatorname{At}(E_2)$. In $\sigma(\zeta) = \sum_q \frac{1}{q!} \operatorname{tr}(\zeta \circ \operatorname{At}^q)$ of [II, Theorem 2.2] the composite $\zeta \circ \operatorname{At}^q$ has diagonal blocks $\zeta_{ii} \circ \operatorname{At}(E_i)^q$, and the trace of a block matrix of morphisms is the sum of the traces of its diagonal blocks. In particular the off-diagonal blocks ζ_{12} and ζ_{21} lie in the kernel, so injectivity forces them to vanish; on the diagonal blocks the map is $(\zeta_{11}, \zeta_{22}) \mapsto \sigma_{E_1}(\zeta_{11}) + \sigma_{E_2}(\zeta_{22})$, which is injective exactly when both summands are injective and their images meet only in zero. ∎

The condition on the images cannot be dropped, and the objects that show it are the simplest ones: the structure sheaves of abelian subvarieties of half dimension, each of which is semiregular on its own.

Lemma 3.39 (Abelian subvarieties of half dimension are semiregular)**.** *Let A be an abelian variety of dimension $2n$ and $B \subseteq A$ an abelian subvariety of dimension n. Then $\operatorname{ev}_{\mathcal{O}_B} \colon HT^2(A) \to \operatorname{Ext}^2(\mathcal{O}_B, \mathcal{O}_B)$ is surjective, its kernel is $\operatorname{Ann}_{HT^2}([B])$, of dimension $6n^2 - n$, and $\sigma_{\mathcal{O}_B}$ is injective.*

Proof. The normal bundle of B is trivial, so $\operatorname{ch}(\mathcal{O}_B) = [B]$ by the Grothendieck–Riemann–Roch theorem for the embedding, and $\operatorname{Ext}^*(\mathcal{O}_B, \mathcal{O}_B) \cong \bigwedge^* \mathbb{C}^{2n}$ (proof of [II, Corollary 3.17]), so $\dim \operatorname{Ext}^2(\mathcal{O}_B, \mathcal{O}_B) = \binom{2n}{2} = 2n^2 - n$, while $\dim HT^2(A) = 2\binom{2n}{2} + 4n^2 = 8n^2 - 2n$. Choose a complementary abelian subvariety B'; the isogeny $B \times B' \to A$ identifies $H^1(A, \mathbb{C})$ with $H^1(B) \oplus H^1(B')$ compatibly with the Hodge decompositions, identifies $HT^2(A)$ with $HT^2(B \times B')$ compatibly with contraction, and carries $[B]$ to a nonzero multiple of $1 \otimes [\mathrm{pt}_{B'}]$, where $[\mathrm{pt}_{B'}]$ spans $\bigwedge^{2n} H^1(B')$. Write $T = T_B \oplus T_{B'}$ for the tangent space and $H^{0,1}(A) = H^{0,1}(B) \oplus H^{0,1}(B')$. A class $\zeta \in H^{0,2}(A) = \bigwedge^2 H^{0,1}(A)$ has $\zeta \cup [B] = \zeta_{BB} \otimes [\mathrm{pt}_{B'}]$, with ζ_{BB} its component in $\bigwedge^2 H^{0,1}(B)$, because every factor from $H^{0,1}(B')$ is killed by the top class of B'; so the annihilator in $H^{0,2}$ has dimension $\binom{2n}{2} - \binom{n}{2}$. For $a \otimes v \in H^{0,1}(A) \otimes T = H^1(T_A)$ the contraction is $a \wedge (v \lrcorner [B]) = a_B \otimes (v_{B'} \lrcorner [\mathrm{pt}_{B'}])$, since v_B kills $[\mathrm{pt}_{B'}]$ and $a_{B'} \wedge (v_{B'} \lrcorner [\mathrm{pt}_{B'}])$ would have type $(n-1, n+1)$ on the n-fold B'; the map $v \mapsto v_{B'} \lrcorner [\mathrm{pt}_{B'}]$ has rank n, as does $a \mapsto a_B$, so the contraction has rank n^2 on the space $H^{0,1}(A) \otimes T$ of dimension $4n^2$ and the annihilator there has dimension $3n^2$. For $\vartheta \in \bigwedge^2 T$ only the component in $\bigwedge^2 T_{B'}$ contracts $[\mathrm{pt}_{B'}]$ nontrivially, and injectively; so the annihilator in $\bigwedge^2 T$ has dimension $\binom{2n}{2} - \binom{n}{2}$. Altogether $\dim \operatorname{Ann}_{HT^2}([B]) = 2\binom{2n}{2} - 2\binom{n}{2} + 3n^2 = 6n^2 - n$. By the square $\sigma_E \circ \operatorname{ev}_E = (\,\cdot\,) \lrcorner \operatorname{ch}(E)$ of [II, Setup 4.54] the kernel of $\operatorname{ev}_{\mathcal{O}_B}$ lies in $\operatorname{Ann}_{HT^2}([B])$, so $\operatorname{ev}_{\mathcal{O}_B}$ has rank at least $(8n^2 - 2n) - (6n^2 - n) = 2n^2 - n = \dim \operatorname{Ext}^2(\mathcal{O}_B, \mathcal{O}_B)$: it is surjective and its kernel is the annihilator. If $\sigma_{\mathcal{O}_B}(y) = 0$, write $y = \operatorname{ev}(x)$; then $x \lrcorner [B] = 0$, so x lies in the annihilator, which is the kernel, and $y = 0$. ∎

Remark 3.40 (Sums of abelian subvarieties). Let $B_1, \dots, B_r \subseteq A$ be abelian subvarieties of dimension $n \geq 3$, translated so that any two of them meet in finitely many points or not at all, which generic translates achieve, and let $E = \bigoplus_j \mathcal{O}_{B_j}[k_j]$. The sheaves $\mathcal{E}xt^q(\mathcal{O}_{B_i}, \mathcal{O}_{B_j})$

for $i \neq j$ are supported on $B_i \cap B_j$ and vanish for $q \neq n$, by the Koszul resolution of $\mathcal{O}_{B_i}$ restricted to B_j, so $\operatorname{Ext}^2(\mathcal{O}_{B_i}[k_i], \mathcal{O}_{B_j}[k_j]) = H^{2+k_j-k_i-n}(B_i \cap B_j, \mathcal{E}xt^n)$ vanishes when all k_j are equal and $n \geq 3$, and when $|k_i - k_j| \leq 1$ and $n \geq 4$; and each $\sigma_{\mathcal{O}_{B_j}}$ is injective by Lemma 3.39. In these cases every hypothesis of Lemma 3.38, applied to one summand and the sum of the others, holds except the one on the images. Now let $v \in HT^2(A)$ satisfy $v \lrcorner \operatorname{ch}(E) = \sum_j (-1)^{k_j} v \lrcorner [B_j] = 0$ with $v \lrcorner [B_1] \neq 0$. Then $\zeta = \bigoplus_j \operatorname{ev}_{\mathcal{O}_{B_j}}(v)$ is nonzero, because v is not in the annihilator of $[B_1]$, and $\sigma_E(\zeta) = \sum_j (-1)^{k_j} v \lrcorner [B_j] = 0$: the images of the $\sigma_{\mathcal{O}_{B_j}}$ meet, and E is not semiregular. This is exactly the situation at a member of a Weil family on which a Weil class is a combination of classes of abelian subvarieties of dimension n, as on a product of elliptic curves: a tangent direction v of the family keeps $\operatorname{ch}(E)$ of Hodge type but moves any $[B_j]$ outside $\mathbb{Q}\theta^n \oplus W(A)$, and [II, Corollary 3.17] excludes such sums for this reason. The sum is obstructed although its summands are not.

Corollary 3.41 (Hochschild deformations and direct summands)**.** *Let $c \in C$, $Y = X_c \times X_c$, $\omega = \omega_a$ a rational exceptional class with $L_0(a) \neq 0$, and E a perfect complex on Y with $\operatorname{ch}(E) = N\omega$, $N \neq 0$.*

(i) *E extends to the first order deformation of $D^b(Y)$ given by $x \in HH^2(Y) = HT^2(Y)$ only if $x \in \mathbb{C}\kappa$. No B-field, no Poisson bivector and no commutative deformation off the curve carries it.*

(ii) *If σ_E is injective and $E = E_1 \oplus E_2$ with both summands nonzero, then $\operatorname{Ext}^2(E_1, E_2) = \operatorname{Ext}^2(E_2, E_1) = 0$, each σ_{E_i} is injective, each E_i extends to first order along the curve, and $\kappa \lrcorner \operatorname{ch}(E_i) = 0$. If moreover $\dim \operatorname{Ext}^2(E, E) = 119$, then $\dim \operatorname{Ext}^2(E_i, E_i) = 120 - \dim \operatorname{Ann}_{HT^2}(\operatorname{ch} E_i)$, and*

$$\operatorname{Ann}_{HT^2}(\operatorname{ch} E_1) + \operatorname{Ann}_{HT^2}(\operatorname{ch} E_2) = HT^2(Y), \qquad \operatorname{Ann}_{HT^2}(\operatorname{ch} E_1) \cap \operatorname{Ann}_{HT^2}(\operatorname{ch} E_2) = \mathbb{C}\kappa.$$

Proof. (i) The kernel of ev_E on $HH^2(Y)$ is the space of deformations of $D^b(Y)$ along which E deforms to first order [Tod09], [Mar25a, Section 8.3]. If $\operatorname{ev}_E(x) = 0$ then $N\, x \lrcorner \omega = \sigma_E(\operatorname{ev}_E(x)) = 0$, so $x \in \mathbb{C}\kappa$ by Theorem 3.21(ii). The three kinds of deformation named are the three summands of $HT^2(Y)$.

(ii) The first two assertions are Lemma 3.38. By Theorem 3.27(ii), $\operatorname{ev}_E(\kappa) = 0$, and ev_E is the diagonal sum of ev_{E_1} and ev_{E_2}, so each vanishes at κ; then $\kappa \lrcorner \operatorname{ch}(E_i) = \sigma_{E_i}(\operatorname{ev}_{E_i}(\kappa)) = 0$. Put $e(x) = 120 - \dim \operatorname{Ann}_{HT^2}(x)$, the rank of contraction into x. It is subadditive in x, and by Theorem 3.34(i) $\dim \operatorname{Ext}^2(E_i, E_i) \geq e(\operatorname{ch} E_i)$. The cross terms vanish, so

$$119 = \dim \operatorname{Ext}^2(E, E) = \sum_i \dim \operatorname{Ext}^2(E_i, E_i) \geq e(\operatorname{ch} E_1) + e(\operatorname{ch} E_2) \geq e(N\omega) = 119,$$

and every inequality is an equality. Hence $\dim \operatorname{Ann}(\operatorname{ch} E_1) + \dim \operatorname{Ann}(\operatorname{ch} E_2) = 121$. The intersection of the two annihilators lies in $\operatorname{Ann}(N\omega) = \mathbb{C}\kappa$ and contains κ, so it is $\mathbb{C}\kappa$, and the sum has dimension 120. ∎

Part (ii) plays for the Mumford classes the role of Theorem 2.32: a semiregular object attaining the bound either does not split, or splits into two pieces whose annihilators fill $HT^2(Y)$ and meet only in the direction of the curve. The two annihilators are the analogue of the two branches P and Q of [II, Remark 2.10].

Proposition 3.42 (Tate classes at every closed point of every reduction)**.** *There are a finitely generated subring $R \subset \mathbb{C}$, a model $W_R \to C_R$ of $W \to C$ over R, with C_R smooth and projective over R with geometrically connected fibres and W_R an abelian scheme, projective over C_R, and a finite surjective morphism $C'_R \to C_R$ from a second curve of the*

same kind, with the following properties. Put $Y' = (W_R \times_{C_R} W_R) \times_{C_R} C'_R$*, with projection* $\pi' \colon Y' \to C'_R$*, and let* ω *be a rational exceptional class. For every prime* ℓ *there is a global section* ω_ℓ *of the lisse sheaf* $R^4\pi'_*\mathbb{Q}_\ell(2)$ *on* $C'_R[1/\ell]$ *whose restriction to* $C'_R \otimes_R \mathbb{C}$ *is the pull-back of* $\omega \otimes 1$*; write* $\omega_\ell(\bar{x})$ *for its value at a geometric point over a point* x*.*

(i) *At a closed point* x *of* $C'_R[1/\ell]$ *the residue field* $\kappa(x)$ *is finite,* Y'_x *is the square of an abelian fourfold over it, and* $\omega_\ell(\bar{x})$ *is fixed by* $\mathrm{Gal}(\bar{\kappa}/\kappa(x))$*.*

(ii) *At such a point* $\omega_\ell(\bar{x})$ *is a* $\mathbb{Q}_\ell$*-linear combination of classes of algebraic cycles of codimension two on* Y'_x*, defined over* $\kappa(x)$*.*

(iii) *Let* s *be a closed point of* $\operatorname{Spec} R[1/\ell]$ *and* η_s *the generic point of the reduction* C'_s*. Then* $\omega_\ell(\bar{\eta}_s)$ *is fixed by the Galois group of the function field* $\kappa(\eta_s)$*, and it is not a* $\mathbb{Q}_\ell$*-linear combination of products of divisor classes of* $Y'_{\bar{\eta}_s}$*.*

Every closed point x *of a reduction* C_s *of* C *is the image of a closed point* x' *of* C'_s*, and* $Y'_{x'} = Y_x \otimes_{\kappa(x)} \kappa(x')$*; so (i) and (ii) hold at every closed point of every reduction of* C*, after a finite extension of its residue field.*

Proof. The models. The family $W \to C$ is defined over a subfield $k_1 \subset \mathbb{C}$ finitely generated over $\mathbb{Q}$, say as $W_1 \to C_1$. Let K_1 be the function field of C_1 and fix an embedding over k_1 of an algebraic closure $\bar{K}_1$ into $\mathbb{C}$; it sends the geometric generic point of C_1 to a point $t \in C$, and $Y_t = X_t \times X_t$ is the base change of $Y_{\bar{K}_1}$. The class ω_t is a Hodge class on an abelian variety, hence absolutely Hodge, so it comes from an absolutely Hodge class of $Y_{\bar{K}_1}$, and $\mathrm{Gal}(\bar{K}_1/K_1)$ acts on these through a finite quotient [Del82, Theorem 2.11 and Proposition 2.9]. Let $K' \subset \bar{K}_1$ be a finite extension of K_1 on which the action is trivial, k' the algebraic closure of k_1 in K', and C' the smooth projective curve over k' with function field K'; it is geometrically connected over k' and maps finitely onto $C_1 \otimes_{k_1} k'$. Spreading out $W_1 \to C_1$, C' and this map over a finitely generated subring of k', and inverting one element of it, gives a normal R and models with the stated properties.

The section. By smooth and proper base change $R^4\pi'_*\mathbb{Q}_\ell(2)$ is lisse on $C'_R[1/\ell]$. Its stalk at the geometric generic point $\operatorname{Spec} \bar{K}_1$ is $H^4(Y_{\bar{K}_1}, \mathbb{Q}_\ell(2)) = H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell$, and $\omega_t \otimes 1$ is the ℓ-adic component of an absolutely Hodge class, so it is fixed by $\mathrm{Gal}(\bar{K}_1/K')$. The scheme $C'_R[1/\ell]$ is normal and integral with function field K', so $\mathrm{Gal}(\bar{K}_1/K')$ maps onto its fundamental group [Stacks, Tag 0BQI], and the vector extends to a global section ω_ℓ. On the connected curve $C'_R \otimes_R \mathbb{C}$ it agrees with the pull-back of $\omega \otimes 1$, the two being global sections of the same local system that agree at a point over t.

(i) and (ii). A scheme of finite type over $\mathbb{Z}$ has finite residue fields at its closed points. A global section of a lisse sheaf is fixed by the fundamental group of every point, which is (i); and (ii) is Li's theorem [Li26, Theorem 1.1], with $n = r = 2$, for the abelian fourfold that is the fibre of W_R at the image of x, base changed to $\kappa(x)$.

(iii) The first assertion holds for the same reason as (i). For the second, let $\bar{\eta}$ be a geometric generic point of C'_R and $\bar{s}$ a geometric point over s. The morphism $C'_R \to \operatorname{Spec} R$ is flat and proper with geometrically connected and geometrically reduced fibres, so the specialisation map $\pi_1(C'_{\bar{\eta}}) \to \pi_1(C'_{\bar{s}})$ is surjective [Stacks, Tag 0C0P], and it is compatible with the actions of the two groups on the first cohomology of the fibres of W_R, identified by cospecialisation. The image of $\pi_1(C'_{\bar{\eta}})$ contains that of $\pi_1(C'_{\mathbb{C}})$, the closure of the image of the topological fundamental group of a finite cover of C, whose Zariski closure contains $G_{\mathbb{Q}_\ell}$ because the connected monodromy group of $W \to C$ is G (proof of Corollary 2.22). So the Zariski closure of the monodromy of W_R over $C'_{\bar{s}}$ contains $G_{\mathbb{Q}_\ell}$. A divisor class of $Y'_{\bar{\eta}_s}$ is defined over a finite extension of $\kappa(\eta_s)$, so it is fixed by an open subgroup of the Galois group, whose image in $\pi_1(C'_s)$ is open [Stacks, Tag 0BQI] and meets $\pi_1(C'_{\bar{s}})$

in an open subgroup; the Zariski closure of the image of an open subgroup has the same identity component, so the class is $G_{\mathbb{Q}_\ell}$-invariant. The G-invariants of $\bigwedge^2(V \oplus V)$ are the three divisor classes of Proposition 2.74(ii), which are $\mathrm{Sp}(V,\psi)$-invariant, so every product of divisor classes of $Y'_{\bar\eta_s}$ is $\mathrm{Sp}(V,\psi)$-invariant. The class $\omega_\ell(\bar\eta_s)$ corresponds to $\omega \otimes 1$, whose coefficients a_1, a_2, a_3 are the distinct conjugates of α (Lemma 3.19), and the $\mathrm{Sp}(V,\psi)$-invariant classes among the ω_a are those with $a_1 = a_2 = a_3$ (Proposition 3.35). The last sentence of the statement holds because $C'_R \to C_R$ is finite and surjective. ∎

At the generic point of each reduction the class is therefore a Tate class over a global function field that is not a combination of products of divisor classes, exactly as over $\mathbb{C}$, while at every closed point it is a combination of classes of cycles. The next theorem shows that a bound on those cycles is all that is missing, with no passage through p-adic Hodge theory.

Theorem 3.43 (The Mumford target is a degree bound in positive characteristic)**.** *Keep Proposition 3.42, fix a prime ℓ and a relatively ample line bundle on $Y' \to C'_R$, and for a finite set Φ of tuples of Hilbert polynomials say that ω is* Φ-represented *at a point x of $C'_R[1/\ell]$ if $\omega_\ell(\bar x) = \sum_i q_i \,\mathrm{cl}(Z_i)$ for some $q_i \in \mathbb{Q}_\ell$ and closed subschemes $Z_i \subset Y'_{\bar x}$ of pure codimension two whose Hilbert polynomials form a tuple in Φ. The following are equivalent.*

(a) *ω_t is algebraic for every $t \in C$.*

(b) *For some finite Φ, the closed points of $C'_R[1/\ell]$ at which ω is Φ-represented are Zariski dense in C'_R.*

(c) *For some finite Φ and some Zariski dense set S of closed points of $\operatorname{Spec} R[1/\ell]$, the class ω is Φ-represented at infinitely many closed points of C'_s for every $s \in S$.*

For a single closed point s of $\operatorname{Spec} R[1/\ell]$, the class ω is Φ-represented at infinitely many closed points of C'_s, for some finite Φ, if and only if $\omega_\ell(\bar\eta_s)$ is a $\mathbb{Q}_\ell$-linear combination of classes of algebraic cycles on $Y'_{\bar\eta_s}$.

Proof. One standard property of the ℓ-adic cycle class is used beyond those over a field: if $\mathcal{O}$ is a discrete valuation ring in which ℓ is invertible, $\mathcal{Y} \to \operatorname{Spec}\mathcal{O}$ is smooth and projective, and $\mathcal{Z} \subset \mathcal{Y}$ is a closed subscheme flat over $\mathcal{O}$, then the specialisation isomorphism of ℓ-adic cohomology carries the cycle class of the geometric generic fibre of $\mathcal{Z}$ to that of its geometric special fibre [SGA $4\frac{1}{2}$, Cycle]. The specialisation isomorphism also carries the values of a global section of a lisse sheaf to each other, by its construction.

Step 1: the represented loci are closed. For a tuple $\underline{P} = (P_1, \dots, P_m)$ let $\mathcal{H}_{\underline{P}}$ be the fibre product over $C'_R[1/\ell]$ of the relative Hilbert schemes $\mathrm{Hilb}^{P_i}(Y'/C'_R[1/\ell])$. It is projective over $C'_R[1/\ell]$ [Gro61], so it has finitely many connected components. On a component T, with universal subschemes $\mathcal{Z}_i$, put $A(h) = \{q \in \mathbb{Q}_\ell^m : \sum_i q_i \,\mathrm{cl}(\mathcal{Z}_{i,\bar h}) = \omega_\ell(\bar h)\}$ for $h \in T$. If h' is a specialisation of h, there is a discrete valuation ring mapping to T with generic point to h and closed point to h' [Stacks, Tag 054F], and the property above, applied to the pull-backs of Y' and of the $\mathcal{Z}_i$, gives $A(h) = A(h')$. Any two points of the connected noetherian scheme T are joined by a chain of specialisations and generisations, through the generic points of its irreducible components, so A is constant on T, say A_T. Let $\Sigma_{\underline{P}}$ be the union of the images in $C'_R[1/\ell]$ of the components T with $A_T \neq \emptyset$. It is closed, as a finite union of images of proper morphisms, and it is the set of points at which ω is $\{\underline{P}\}$-represented: at a point x of the image of such a T, a closed point of the fibre of T over x has residue field finite over $\kappa(x)$ and gives subschemes of $Y'_{\bar x}$. Put $\Sigma_\Phi = \bigcup_{\underline{P}\in\Phi} \Sigma_{\underline{P}}$.

Step 2: (b) implies (a). Σ_Φ is closed and contains a dense set, so it is $C'_R[1/\ell]$ and contains the generic point ξ of C'_R, whose residue field is K'. So $\omega_t \otimes 1 = \omega_\ell(\bar\xi) = \sum_i q_i \,\mathrm{cl}(Z_i)$

with $q \in \mathbb{Q}_\ell^m$ and subschemes Z_i of $Y_{\bar{K}_1}$, and so of Y_t. The cycle class maps are compatible with the comparison isomorphism $H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell \cong H^4(Y_t, \mathbb{Q}_\ell)$ [Del82, Section 1 and Example 2.1(a)], so the linear system $\omega_t = \sum_i q_i [Z_i]$, which has rational coefficients, is solvable over $\mathbb{Q}_\ell$, hence over $\mathbb{Q}$, and ω_t is algebraic. The Z_i are defined over a finite extension of K', so they spread out to subschemes, flat over a nonempty open subset [Stacks, Tag 052A], of the pull-back of $W \times_C W$ to a finite cover of C, and by the local constancy of the cycle class used in the proof of [I, Theorem 16.4], ω is algebraic at every point of a nonempty Zariski open subset of C, which is uncountable. Corollary 2.30 gives (a).

Step 3: (c) implies (b). Let D be the set of closed points at which ω is Φ-represented and $\bar{D}$ its closure. For $s \in S$ the set $D \cap C'_s$ is infinite, so its closure is the irreducible curve C'_s, and $C'_s \subseteq \bar{D}$. The points u of $\operatorname{Spec} R[1/\ell]$ over which the fibre of $\bar{D}$ has dimension at least one form a closed set, by the upper semicontinuity of the dimension of the fibres of a proper morphism [Stacks, Tag 0D4I]; it contains the dense set S, so it contains the generic point. The generic fibre of $\bar{D}$ is then a closed subset of dimension one of the irreducible curve $C'_R \otimes_R k'$, so it contains ξ, and $\bar{D} = C'_R[1/\ell]$.

Step 4: (a) implies (c). By (a) and Step 1, applied over $\bar{K}_1$ with the Betti class in place of the ℓ-adic one, ω_t is the class of a rational combination of subschemes of $Y_{\bar{K}_1}$, defined over a finite extension L of K'. Enlarge R to a finitely generated subring of the algebraic closure of k' in L, let $C''_R \to C'_R$ be the normalisation in L, and spread the subschemes out over C''_R; by generic flatness [Stacks, Tag 052A] they are flat over a dense open subset U of $C''_R[1/\ell]$, with constant Hilbert polynomials. By Step 1 applied on U, ω is represented by their fibres at every point of U. The image of U in $C'_R[1/\ell]$ contains a dense open subset U', every closed point of U' is the image of a closed point of U, and U' meets C'_s in a nonempty open, hence infinite, subset for every s in a dense open subset of $\operatorname{Spec} R[1/\ell]$.

The single reduction. If ω is Φ-represented at infinitely many closed points of C'_s, then $\Sigma_\Phi \cap C'_s$ is closed and infinite, so it is C'_s and contains η_s. Conversely, cycles at $\bar{\eta}_s$ are defined over a finite extension of $\kappa(\eta_s)$, and spreading them out as in Step 4 over a finite cover of C'_s represents ω, with fixed Hilbert polynomials, at every closed point of a nonempty open subset of C'_s. ∎

Remark 3.44 (What is left in positive characteristic). By Proposition 3.42(ii) the class is represented at every closed point of $C'_R[1/\ell]$, so Theorem 3.43 turns the Mumford target into a statement of uniformity alone: the Hilbert data of some representation must stay in one finite set on a Zariski dense set of closed points, for instance at infinitely many closed points of infinitely many reductions. The passage to characteristic zero is then done by the specialisation of cycle classes along C'_R, and neither the p-adic variational Hodge conjecture nor a lifting of cycles is needed. Li's theorem gives no bound: where its proof carries cycles into a prescribed isogeny class, it does so by an isogeny whose degree it does not control [Li26, proof of Theorem 1.2, Step 3].

Nor do the natural representatives at the supersingular points supply the bound. At a closed point x whose fibre is supersingular, the Frobenius eigenvalues on $H^2(Y'_{\bar{x}}, \mathbb{Q}_\ell(1))$ are roots of unity, so by Tate's theorem [Tat66] this space is spanned by divisor classes, and $\omega_\ell(\bar{x}) \in \bigwedge^4 H^1$ is a combination of products of two of them. But for a closed point s of $\operatorname{Spec} R[1/\ell]$ and a finite set of Hilbert polynomials, only finitely many closed points of C'_s carry such a combination $\sum_i q_i \operatorname{cl}(D_i) \operatorname{cl}(D'_i)$ with effective divisors D_i, D'_i whose Hilbert polynomials lie in that set: otherwise Step 1 of the proof of Theorem 3.43, run on relative Hilbert schemes of divisors, would give one at $\bar{\eta}_s$, against Proposition 3.42(iii). This holds

whether or not the supersingular points are dense. They are dense in C'_R when R has dimension one. Indeed, for every prime $p \geq 5$ not dividing the discriminants of F and D, which is all but finitely many, the supersingular points of the reduction at a prime over p of Kisin's integral model of the Shimura curve, with hyperspecial level at p, are its whole Newton jumping locus and number $(p^r - 1)(g - 1)$ on each component of genus g, r being the local degree of F [SZ11, Theorem 1.2 and Corollary 5.17]. For all but finitely many s the curve C'_s maps finitely onto such a component, of genus at least two, so it has at least $p - 1$ supersingular points, while a closed subset of C'_R of dimension one meets all but finitely many of the C'_s in boundedly many points.

Within one reduction the bound is equivalent to the class being algebraic at the generic point of C'_s (Theorem 3.43), a Tate class over a global function field that is not a combination of products of divisor classes (Proposition 3.42(iii)). This is the codimension two case of the variational Tate conjecture along C'_s, which Morrow proves for divisors in its crystalline form [Mor19]. From one reduction alone, the way back to characteristic zero would be the p-adic variational Hodge conjecture for $W \times_C W$ over the Witt vectors of $\kappa(s)$, with rational rather than ℓ-adic coefficients and with the crystalline class of the cycle matching the de Rham class of ω. Bloch, Esnault and Kerz lift such a class in $K_0 \otimes \mathbb{Q}$ of the special fibre to the formal completion for $p > 15$, the dimension being nine, and the algebraisation of the formal class is the remaining step of their programme [BEK14, Theorem 1.3]. The route therefore runs through the uniform bound or through this conjecture.

Five of the routes of Remark 3.32 survive the rigidity theorems: *specialisation*, *reduction modulo p*, *semiregularity*, *Kuga–Satake* and *motivated classes*. The next corollary shows that they are not five separate obligations.

Corollary 3.45 (The five routes reduce to one statement)**.** *Let $W \to C$ be as in Corollary 2.22, let ω be a rational exceptional class, and let the* Mumford target *be the first case of* (F2)*: the Hodge conjecture for $X_t \times X_t$ for every $t \in C$.*

(i) *Each of the following is equivalent to the Mumford target: ω_t is algebraic for uncountably many $t \in C$; $B(W \times_C W)$ holds for $L_{W \times_C W}$; ω is represented with Hilbert data in one finite set at infinitely many points of C, for instance at infinitely many CM points; ω is represented with Hilbert data in one finite set on a Zariski dense set of closed points of the arithmetic model of Theorem 3.43.*

(ii) *Each of the following implies the Mumford target: a perfect complex E at one point of C with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, no other component, and $\dim \mathrm{Ext}^2(E, E) = 119$; the algebraicity, for uncountably many t, of the Kuga–Satake class of the surface S_λ of Theorem 3.4 attached to X_t; a rational map as in Proposition 3.25(iv), under its hypotheses, from $X_t \times X_t$ for uncountably many t.*

(iii) *There is a countable set $\Sigma_0 \subset C$ containing the CM points such that the Mumford target holds as soon as the Hodge conjecture holds for $X_c \times X_c$ at one point $c \notin \Sigma_0$.*

Proof. (i) The first two are Corollary 2.30, with Lemma 3.20 for one class in place of two, the third is Corollary 3.28 and the fourth is Theorem 3.43. (ii) The first is Theorem 3.27(iii). The second and the third give the Hodge conjecture for $X_t \times X_t$ at each of the uncountably many points, by Theorem 3.4(ii) and Proposition 3.25(iv), and then (i) applies. (iii) In the proof of Corollary 3.28 the set of points at which ω is algebraic is the union of the countably many closed sets $\Sigma_{\underline{P},\underline{q}}$, each finite or equal to C. Let Σ_0 be the union of the finite ones and of the CM points. If ω_c is algebraic at a point $c \notin \Sigma_0$, then c lies in some $\Sigma_{\underline{P},\underline{q}}$ that is not finite, which is C, and (i) applies. ∎

Remark 3.46 (Which routes are needed). Of the five routes, specialisation, reduction modulo p and motivated classes are the Mumford target itself in three other forms, by Corollary 3.45(i) and, for the motivated classes, Corollary 2.30; semiregularity and Kuga–Satake are stronger inputs that imply it. So any one route suffices. Each of the three forms is the whole of the problem, a bound on Hilbert data or a case of (L) that is equivalent to the target, and the two stronger inputs ask for an object and for a cycle.

The target can also be read at a single member, through the motivic group of X rather than through cycles on the family; it then becomes the question of which of four groups this is.

Proposition 3.47 (Four possible motivic groups)**.** *In Setup 3.1, let M be the group of the Tannakian category of homological motives generated by $h(X)$, with the Betti realisation as fibre functor, let $M^1 = M \cap \mathrm{Sp}(V,\psi)$, and let N be the normaliser of G in $\mathrm{Sp}(V,\psi)$. Over $\bar{\mathbb{Q}}$ identify V_1, V_2 and V_3 with one plane, carrying ϵ_1, ϵ_2, ϵ_3 to one form, and let $\zeta \in \mathrm{Sp}(V,\psi)(\bar{\mathbb{Q}})$ be the cyclic permutation $v_1 \otimes v_2 \otimes v_3 \mapsto v_3 \otimes v_1 \otimes v_2$ of the three factors. It normalises G, and other identifications change it by an element of $G(\bar{\mathbb{Q}})$; so it acts through V on the Hodge classes of every power of X, independently of the choice.*

(i) *N/G is a form of the symmetric group S_3, $M = \mathbb{G}_m \cdot M^1$, and M^1 is one of G, $G \cdot A_3$, N and $\mathrm{Sp}(V,\psi)$, where $G \cdot A_3$ is the inverse image in N of the subgroup of order three of N/G. The last three contain ζ.*

(ii) *The following are equivalent: $M^1 = G$; the Hodge conjecture holds for X^n for every $n \geq 1$; the two exceptional classes of $X \times X$ are algebraic; some algebraic class on some power of X is not fixed by ζ.*

(iii) *Let $\mathrm{Det} \in (V^{\otimes 4})^G \subset H^4(X^4,\mathbb{Q})$ be the Hodge class of Remark 3.8, and write r_1 also for the Hodge class $t \in (V^{\otimes 4})^G$ with $\hat{t} = r_1 = s_1 + s_2 + s_3$, in the notation of Steps 2 and 3 of the proof of Theorem 3.7. Both are fixed by N and not by $\mathrm{Sp}(V,\psi)$, and each of them is algebraic if and only if $M^1 \neq \mathrm{Sp}(V,\psi)$. So the algebraicity of either excludes the last case only; by this argument it does not give the exceptional classes, which are not algebraic when M^1 is $G \cdot A_3$ or N.*

(iv) *Let $W \to C$ be as in Corollary 2.22, and suppose that it is defined over a number field; this holds for the restriction of the universal family of a fine moduli space of polarised abelian fourfolds with level structure to the normalisation of one of the Shimura curves of [Mum69] in it, and for its pull-backs to finite étale covers. Then the set Σ_0 constructed in the proof of Corollary 3.45(iii) lies in $C(\bar{\mathbb{Q}})$, and the Mumford target holds as soon as, at one point $t \in C(\mathbb{C}) \setminus C(\bar{\mathbb{Q}})$, some algebraic cycle on some power of X_t has a class that is not fixed by ζ.*

Proof. The group. On an abelian variety homological and numerical equivalence coincide [Lie68], so the homological motives generated by $h(X)$ and the Lefschetz motive form a semisimple abelian category [Jan92]. The Künneth projectors are algebraic [Kle68, §2], so after the usual change of sign in the commutativity constraint this is a Tannakian category over $\mathbb{Q}$, with the Betti realisation as a fibre functor, faithful by the definition of homological equivalence. Its group M is reductive, the category being semisimple, and the M-invariants in the realisation of a motive P are the image of $\mathrm{Hom}(\mathbf{1}, P)$ [DM82, Theorem 2.11]; so for all m and p the M-invariants of $H^{2p}(X^m,\mathbb{Q})(p)$ are the classes of algebraic cycles. The group M acts faithfully on $V \oplus \mathbb{Q}(1)$ and fixes the polarisation class, so it lies in $\mathrm{GSp}(V,\psi)$ and acts on $\mathbb{Q}(1)$ through the multiplier. Through ψ and the Künneth projectors every tensor space of $V \oplus \mathbb{Q}(1)$ is a sum of direct summands of spaces $H^k(X^k,\mathbb{Q})(j)$ cut out by

algebraic projectors, so the M-invariants there are algebraic classes, hence Hodge classes, fixed by $\mathrm{MT}(X) = \mathbb{G}_m \cdot G$; as a reductive group is the subgroup fixing its invariants in the tensor spaces [Del82, Proposition 3.1], $\mathrm{MT}(X) \subseteq M$ and $G \subseteq M^1$. An element of M with multiplier ν is a square root of ν times an element of M^1, so $M = \mathbb{G}_m \cdot M^1$. The homotheties act trivially on the spaces $H^{2p}(X^m, \mathbb{Q})(p)$, so there the algebraic classes are the M^1-invariants, as the Hodge classes are the G-invariants.

The groups between G and $\mathrm{Sp}(V,\psi)$. Over $\bar{\mathbb{Q}}$ the Lie algebra $\mathfrak{sp}(V,\psi) \cong S^2V$ is the sum of $\mathrm{Lie}\, G = T_1 \oplus T_2 \oplus T_3$, where $T_i \cong S^2V_i \otimes \bigwedge^2 V_j \otimes \bigwedge^2 V_k$, and of $S^2V_1 \otimes S^2V_2 \otimes S^2V_3$, which is an irreducible representation of G of dimension 27; so a G-stable subspace of $\mathfrak{sp}(V,\psi)$ containing $\mathrm{Lie}\, G$ is $\mathrm{Lie}\, G$ or $\mathfrak{sp}(V,\psi)$. If H is a closed subgroup with $G \subseteq H \subseteq \mathrm{Sp}(V,\psi)$, its Lie algebra is such a subspace. If it is $\mathrm{Lie}\, G$, then $H^\circ = G$, which H normalises, so $H \subseteq N$; if it is $\mathfrak{sp}(V,\psi)$, then $H^\circ = \mathrm{Sp}(V,\psi)$, which is connected, so $H = \mathrm{Sp}(V,\psi)$.

The normaliser. Since V is absolutely irreducible, the centraliser of G in $\mathrm{Sp}(V,\psi)$ is $\{\pm 1\}$, which lies in G as the image of $(-1,1,1)$, and $G(\bar{\mathbb{Q}})$ maps onto the group of inner automorphisms of $G_{\bar{\mathbb{Q}}}$. So $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}})$ embeds in the outer automorphism group of $G_{\bar{\mathbb{Q}}}$, which permutes the three factors, SL_2 having no outer automorphism. The six permutations of the factors, after identifications as above, preserve $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$ and normalise G, so $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}}) \cong S_3$. The Galois group acts on it by conjugation through its action on the three factors, which is transitive, and the subgroups of S_3 stable under conjugation by a transitive subgroup are 1, A_3 and S_3. So the algebraic subgroups of N containing G are G, $G \cdot A_3$ and N, and with the previous paragraph this proves (i), ζ mapping to an element of order three of N/G.

(ii) If $M^1 = G$, then $M = \mathrm{MT}(X)$, and every Hodge class on every power of X is algebraic. The Hodge conjecture for $X \times X$ includes the exceptional classes. If these are algebraic, so is every rational exceptional class ω_a of Setup 3.18, and ω_a is not fixed by ζ: ζ permutes the three factors cyclically, so it fixes θ and carries $U_{ij} = S^2V_i \otimes S^2V_j \otimes \bigwedge^2 V_k$ onto $U_{\zeta(i)\zeta(j)}$; labelling the factors so that ζ sends the i-th to the $(i+1)$-th, it carries π_{12} to π_{23}, π_{23} to π_{13} and π_{13} to π_{12} and fixes π_0. So it fixes ω_a only if $a_1 = a_2 = a_3$, whereas these are the distinct conjugates of an irrational element of F (Lemma 3.19). Finally, if $M^1 \neq G$, then M^1 contains ζ by (i), and every algebraic class, being M^1-invariant, is fixed by ζ.

(iii) The class Det spans $(S^4V)^G \otimes \bigwedge^4 \mathbb{Q}^4$, where $(S^4V)^G$ is a line (Remark 3.8). As N normalises G, it preserves this line, and it fixes it, because $N(\bar{\mathbb{Q}})$ is generated by $G(\bar{\mathbb{Q}})$ and the permutations of the factors, which fix Cayley's hyperdeterminant, a symmetric function of the three factors, and the restriction of Det to the diagonal is a nonzero multiple of it (Remark 3.8). $\mathrm{Sp}(V,\psi)$ fixes no nonzero element of S^4V, which would be a nonzero quartic form constant on the single orbit $V \setminus \{0\}$. The permutations of the factors conjugate s_k into $s_{\sigma(k)}$, so they commute with r_1, as does G; and $t \mapsto \hat{t}$ is $\mathrm{Sp}(V,\psi)$-equivariant, since the pairing $\int_X v \cup w \cup \theta^3$ on V is G-invariant and alternating, hence a multiple of the form ψ. So the class r_1 is fixed by N. On $S^2V \subset V \otimes V$ the endomorphism r_1 acts by 3 on $S^2V_1 \otimes S^2V_2 \otimes S^2V_3$ and by -1 on the T_i, since s_k is 1 on S^2V_k and -1 on $\bigwedge^2 V_k$; as S^2V is irreducible under $\mathrm{Sp}(V,\psi)$, the class r_1 is not $\mathrm{Sp}(V,\psi)$-invariant. If $M^1 \neq \mathrm{Sp}(V,\psi)$, then $M^1 \subseteq N$ by (i), and the two classes are M^1-invariant, hence algebraic; if $M^1 = \mathrm{Sp}(V,\psi)$, they are not. When M^1 is $G \cdot A_3$ or N it contains ζ, which fixes no exceptional class.

(iv) A fine moduli space as in the statement is defined over a number field, and the curve contains infinitely many CM points, which are algebraic, an abelian variety with complex multiplication being defined over a number field; so the curve is the Zariski closure of a set of algebraic points and is defined over $\bar{\mathbb{Q}}$, and so are its normalisation, the restricted

family and their finite étale covers. Now let $W \to C$ be defined over $\bar{\mathbb{Q}}$. The moduli map of the family is a finite morphism defined over $\bar{\mathbb{Q}}$, so the CM points of C, whose images are algebraic, lie in $C(\bar{\mathbb{Q}})$. The other points of Σ_0 lie in the finite sets $\Sigma_{\underline{P},\underline{q}}$ of the proof of Corollary 3.28, images of unions of connected components of fibre products of relative Hilbert schemes of $W \times_C W \to C$. These are base changes of schemes over $\bar{\mathbb{Q}}$, and each connected component of the base change is the base change of a connected component over $\bar{\mathbb{Q}}$, which is algebraically closed; so a finite $\Sigma_{\underline{P},\underline{q}}$ consists of algebraic points. If $t \notin C(\bar{\mathbb{Q}})$, then $t \notin \Sigma_0$, the Hodge group of X_t is G, and (ii) for X_t gives the Hodge conjecture for $X_t \times X_t$; Corollary 3.45(iii) gives the target. ∎

The four cases differ already on X^4 and X^6 (Figure 22). Over $\bar{\mathbb{Q}}$, $(V^{\otimes 2m})^G$ is the tensor product of three copies of the space of SL_2-invariants of $(\bar{\mathbb{Q}}^2)^{\otimes 2m}$, of dimension $c = 2$ for $m = 2$ and $c = 5$ for $m = 3$, with a basis of non-crossing pairings that ζ and the transpositions of the factors permute. So the algebraic classes in the multilinear part $V^{\otimes 2m} \subset H^{2m}(X^{2m}, \mathbb{Q})$ have dimension c^3, $(c^3 + 2c)/3$, $(c^3 + 3c^2 + 2c)/6$ and $(2m-1)(2m-3)\cdots 1$ when M^1 is G, $G \cdot A_3$, N and $\mathrm{Sp}(V, \psi)$, the last by the first fundamental theorem for $\mathrm{Sp}(V, \psi)$: 8, 4, 4, 3 on X^4 and 125, 45, 35, 15 on X^6. The middle two count the orbits of the basis under $\langle \zeta \rangle$ and under S_3: ζ fixes the c triples of equal pairings and a transposition fixes c^2 of them. If M were known to be connected, M^1 would be connected too, since $-1 \in G$ lies in its identity component; it would then be G or $\mathrm{Sp}(V, \psi)$, and the algebraicity of Det alone would give the Hodge conjecture for every power of X. The argument does not decide whether M is connected. It is the standard use of the motivic group as the stabiliser of the algebraic classes, and we claim no priority for Proposition 3.47. It proves no new case of the Hodge conjecture: it says what one new cycle on one power of one member would have to do.

The same invariance explains why the classes known or expected to be algebraic on abelian varieties stop short of the exceptional classes, on all abelian varieties at once.

Proposition 3.48 (The system generated by the known classes)**.** *Keep Setup 3.1, and ζ and $G \cdot A_3$ as in Proposition 3.47. For an abelian variety A let $A_X \subseteq A$ be the largest abelian subvariety isogenous to a power of X, and A' the identity component of the intersection of the kernels of the homomorphisms $A \to X$. Then $A_X \times A' \to A$ is an isogeny,* $\mathrm{Hom}(X, A') = 0$*, and*

$$H^1(A, \mathbb{Q}) = (V \otimes \mathrm{Hom}^0(A_X, X)) \oplus H^1(A', \mathbb{Q}),$$

the first summand embedded by $v \otimes f \mapsto f^* v$. Let $\mathrm{GL}(V)$ act on $H^*(A, \mathbb{Q}) = \bigwedge^* H^1(A, \mathbb{Q})$ through the factor V. Let $\mathcal{S}$ be the smallest family of subspaces $\mathcal{S}(A) \subseteq H^*(A, \mathbb{Q})$, over all complex abelian varieties A, that is closed under linear combinations, cup products, exterior products, and pull-back and push-forward along homomorphisms, and contains:

(a) *the divisor classes;*
(b) *for every subfield $E \subseteq \mathrm{End}^0(A)$, the rational classes whose extension to $\bar{\mathbb{Q}}$ lies in $\bigoplus_\sigma \bigwedge^d H^1(A, \bar{\mathbb{Q}})_\sigma$, the sum over the embeddings $\sigma \colon E \to \bar{\mathbb{Q}}$ of the top exterior powers of the eigenspaces, $d = \dim_E H^1(A, \mathbb{Q})$; among them are the Weil classes;*
(c) *the Hodge classes of the abelian varieties without isogeny factor X;*
(d) *the Hodge classes of the powers of X that are fixed by ζ, among them* Det *and r_1.*

Then every class of $\mathcal{S}$ is invariant under $G \cdot A_3$ acting through V, and the classes obtained from (a), (b) *and* (c) *alone are invariant under $\mathrm{Sp}(V, \psi)$. The Hodge classes in $\mathcal{S}(X^m)$ are the Hodge classes of X^m fixed by ζ, and $\mathcal{S}$ contains no exceptional class of $X \times X$.*

$M = \mathbb{G}_m \cdot M^1$, $M^1 = M \cap \mathrm{Sp}(V,\psi)$	dim of the algebraic classes in $V^{\otimes 2m}$: X^4, $m=2$	X^6, $m=3$	formula, $c=2$ or 5	$\zeta \in M^1$	Det, r_1 algebraic	exceptional classes algebraic
$\mathrm{Sp}(V,\psi)$	3	15	$(2m-1)!!$	yes	no	no
dim: $9 < 36$						
N	4	35	$(c^3+3c^2+2c)/6$	yes	yes	no
index 2						
$G \cdot A_3$	4	45	$(c^3+2c)/3$	yes	yes	no
index 3						
$G = \mathrm{Hg}(X)$	8	125	c^3	no	yes	yes

$M^1 = G \Leftrightarrow$ the Hodge conjecture for every $X^n \Leftrightarrow$ the two exceptional classes of $X \times X$ are algebraic
$\Leftrightarrow$ some algebraic class on some power of X is not fixed by ζ

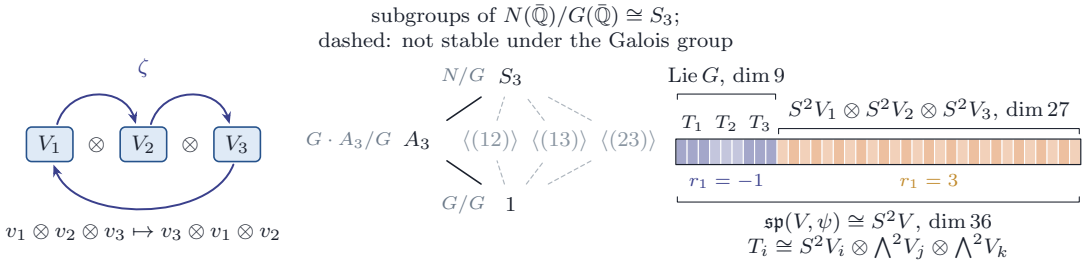


FIGURE 22. The four possible groups $M^1 = M \cap \mathrm{Sp}(V,\psi)$ of Proposition 3.47, where $M = \mathbb{G}_m \cdot M^1$, as the chain $G \subset G \cdot A_3 \subset N \subset \mathrm{Sp}(V,\psi)$: the first two inclusions have index 3 and 2, and N, whose identity component is G, has dimension 9 against 36. For each case the table gives the dimension of the algebraic classes in the multilinear part $V^{\otimes 2m}$ of $H^{2m}(X^{2m},\mathbb{Q})$ for $m=2$ and $m=3$, from the $c=2$, respectively $c=5$, non-crossing pairings; whether M^1 contains the cyclic permutation ζ of the three tensor factors; whether Det and r_1 are algebraic; and whether the two exceptional classes of $X \times X$ are. Only $M^1 = G$ makes the exceptional classes algebraic, and it is equivalent to the Hodge conjecture for every power of X (Proposition 3.47(ii)); the proposition does not decide which of the four cases occurs. Below, from left to right: ζ; the subgroups of $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}}) \cong S_3$, of which only 1, A_3 and S_3 are stable under the Galois group, which acts transitively on the three factors; and $\mathfrak{sp}(V,\psi) = \mathrm{Lie}\, G \oplus S^2V_1 \otimes S^2V_2 \otimes S^2V_3$, $36 = 9+27$, the second summand irreducible under G, with the eigenvalues -1 and 3 of r_1 on the two summands.

Proof. Up to isogeny $A = A_X \times A'$ by Poincaré reducibility, with A_X isogenous to some X^k and no isogeny factor X in A', so $\mathrm{Hom}(X, A') = \mathrm{Hom}(A', X) = 0$, X being simple. As $\mathrm{End}^0(X) = \mathbb{Q}$, $\mathrm{Hom}^0(A_X, X)$ has dimension k and the displayed map is an isomorphism. A homomorphism $h\colon A \to B$ is, up to isogeny, $h_X \times h'$ with $h_X\colon A_X \to B_X$ and $h'\colon A' \to B'$, and h^* acts on H^1 as $(1 \otimes (f \mapsto f \circ h_X)) \oplus h'^*$, which commutes with $\mathrm{GL}(V)$; so pull-backs are equivariant, and so are cup and exterior products, $\mathrm{GL}(V)$ acting on $H^*(A \times B) = H^*(A) \otimes H^*(B)$ diagonally. The orientation class of A spans $\bigwedge^{\mathrm{top}} H^1(A) = (\det V)^{\otimes k} \otimes (\det \mathrm{Hom}^0(A_X, X))^{\otimes 8} \otimes \bigwedge^{\mathrm{top}} H^1(A')$, on which $\mathrm{SL}(V)$ acts trivially, so Poincaré duality and push-forwards are $\mathrm{SL}(V)$-equivariant. Since $G \cdot A_3 \subseteq \mathrm{Sp}(V, \psi) \subseteq \mathrm{SL}(V)$, the classes invariant under either group form a family closed under all the operations, and it remains to look at the generators.

(a) A divisor class of A is the sum of its Künneth components in $\bigwedge^2 H^1(A_X)$, $H^1(A_X) \otimes H^1(A')$ and $\bigwedge^2 H^1(A')$, which are Hodge classes. The middle one corresponds to a homomorphism from A' to the dual of A_X, which is zero. The first is a Hodge class of A_X, so a G-invariant of $\bigwedge^2(V \otimes \mathbb{Q}^k) = S^2V \otimes \bigwedge^2 \mathbb{Q}^k \oplus \bigwedge^2 V \otimes S^2\mathbb{Q}^k$; as $(S^2V)^G = 0$ and $(\bigwedge^2 V)^G = \mathbb{Q}\psi$ (proof of Theorem 3.36(i)), it lies in $\psi \otimes S^2\mathbb{Q}^k$, which $\mathrm{Sp}(V, \psi)$ fixes. On the third $\mathrm{GL}(V)$ acts trivially. (b) The field E acts on $Q = \mathrm{Hom}^0(A_X, X)$ and on $H^1(A')$, and the σ-eigenspace of $H^1(A, \bar{\mathbb{Q}})$ is $V \otimes Q_\sigma \oplus H^1(A', \bar{\mathbb{Q}})_\sigma$. Its top exterior power is $(\det V)^{\otimes \dim Q_\sigma} \otimes (\det Q_\sigma)^{\otimes 8} \otimes \det H^1(A', \bar{\mathbb{Q}})_\sigma$, on which $\mathrm{SL}(V)$ acts trivially. (c) For such an abelian variety $A_X = 0$, and $\mathrm{GL}(V)$ acts trivially. (d) The Hodge classes of X^m are the invariants of G, which acts through V, and those fixed by ζ are invariant under the group $G \cdot A_3$ that G and ζ generate.

So every class of $\mathcal{S}$ is $G \cdot A_3$-invariant, and those obtained from (a), (b) and (c) are $\mathrm{Sp}(V, \psi)$-invariant. The Hodge classes of X^m in $\mathcal{S}$ contain those of (d) and are fixed by ζ, so they are exactly the Hodge classes fixed by ζ; and an exceptional class of $X \times X$ is not fixed by ζ (proof of Proposition 3.47(ii)). ∎

So no argument that uses only these inputs and these operations reaches the exceptional classes. The inputs include all Weil classes, whether or not they are known to be algebraic, and with them Markman's cycles [Mar25a]; the whole Hodge conjecture for the abelian varieties without isogeny factor X; and the algebraicity of Det and r_1. A proof needs a cycle outside this system: on a product of a power of X with an abelian variety, a class that is not invariant under $G \cdot A_3$ acting through V, or a cycle through a variety that is not an abelian variety, such as the Kuga–Satake class of S_λ would provide (Proposition 3.12). Proposition 3.48 extends the second limit of Remark 3.17; it is an obstruction to a method, not to the target, and we claim no priority for it.

The same invariance, used the other way round, shows that the statement (F2) of Corollary 2.71 is not smaller than the conjecture for abelian varieties. For an abelian variety Y let $R(Y) \subseteq H^*(Y, \mathbb{Q})$ be the subring generated by divisor classes and Weil classes of quotients, as in Remark 2.72.

Proposition 3.49 (Statement (F2) is the conjecture for abelian varieties)**.** *Statement* (F2) *of Corollary 2.71 implies the Hodge conjecture for every abelian variety, and is therefore equivalent to it.*

Proof. The conjecture for abelian varieties contains (F2). For the converse, recall from Remark 2.72 that (F2) gives every Hodge class of degree $2q$ on Y as soon as one Hodge class of degree $2q$ on Y lies outside $R(Y)$. Let X be as in Setup 3.1, and let ζ act on the cohomology of every abelian variety through the factor V, as in Proposition 3.48. Every class of $R(Y)$ lies in $\mathcal{S}(Y)$, so it is fixed by ζ, for every Y; and an exceptional class

$e_0 \in H^4(X \times X, \mathbb{Q})$ is not (Proposition 3.48). Let θ be an ample class on $X \times X$, which ζ fixes. If $\zeta e_0 = \lambda e_0$, then $\lambda \neq 1$ and $e = e_0 + \theta^2$ has $\zeta e = \lambda e_0 + \theta^2$, which is not a multiple of e; otherwise put $e = e_0$. In either case e is a Hodge class of degree 4 on $X \times X$ with $\zeta e \notin \bar{\mathbb{Q}}\, e$.

Let A be an abelian variety and $\beta \neq 0$ a Hodge class of degree $2p$ on A, and put $Y = A \times X \times X$. As ζ acts on $H^*(Y) = H^*(A) \otimes H^*(X \times X)$ diagonally, a class $\beta \otimes e$ in $R(Y)$ would satisfy $\zeta\beta \otimes \zeta e = \beta \otimes e$, which forces $\zeta e \in \bar{\mathbb{Q}}\, e$. So the Hodge class $\beta \otimes e$ lies outside $R(Y)$, and (F2) makes it algebraic. In the same way $e \cup \theta^4 \notin R(X \times X)$, since otherwise $(\zeta e - e) \cup \theta^4 = 0$ and the hard Lefschetz theorem would give $\zeta e = e$; so (F2) makes every Hodge class of degree 12 on $X \times X$ algebraic. By the Hodge–Riemann bilinear relations the intersection pairing between the Hodge classes of degrees 4 and 12 on $X \times X$ is nondegenerate, so there is a Hodge class e' of degree 12 with $\int_{X\times X} e \cup e' = 1$. Then

$$\beta = \mathrm{pr}_{A*}\big((\beta \otimes e) \cup \mathrm{pr}^*_{X\times X} e'\big),$$

with pr_A and $\mathrm{pr}_{X\times X}$ the projections of Y, and β is algebraic. ∎

Remark 3.50 (The rule set with Proposition 3.49). Theorem 2.70 is a statement about the rule set of Setup 2.69, and Proposition 3.49 is a rule that the set does not contain. Adjoined to it, the rule changes the fifth minimal set of Theorem 2.70(ii): the minimal sufficient sets become

$$\{\mathrm{F3}\}, \quad \{\mathrm{L}, \mathrm{M}\}, \quad \{\mathrm{L}, \mathrm{F3}'\}, \quad \{\mathrm{V}, \mathrm{F3}'\}, \quad \{\mathrm{F2}, \mathrm{F3}'\},$$

and $\mathrm{P2}_{\mathrm{split}}$ lies in none of them. Indeed the new rule has the premise (F2) and the conclusion **AB**, the conjecture for abelian varieties, so the sets C_1, C_2 and C_4 of the proof of Theorem 2.70 stay closed, and a sufficient set meets H_1, H_2 and H_4. The sets meeting these three that are minimal with this property are the five above, by the argument of that proof without H_3, and each is sufficient: $\{\mathrm{F2}, \mathrm{F3}'\}$ gives **AB** by the new rule and then **HC** by Proposition 2.35(i). So the results of this paper on Weil classes do not shorten a route to the conjecture. What they do is reduce one case of it, the algebraicity of the Weil classes of every CM field, to the single statement $\mathrm{P2}_{\mathrm{split}}$, by a route that does not pass through (F2). Like (F3), which is the whole conjecture, (F2) is the whole conjecture for abelian varieties, carried by the classes that are hardest to reach; the exceptional classes of $X \times X$ are its smallest instance, and the proof above shows that the classes $\beta \otimes e$ built from them already carry all of it.

4. The main theorem in every family

[I, Theorem 1.1] describes the algebraic locus of the Weil classes in one family: it contains an explicit CM point, it is dense, it is a countable union of closed algebraic strata, and it is the whole family as soon as it has an interior point. The first two properties use the Weil classes; the last two hold for every flat class on every smooth projective family (Lemma 4.2). This section carries the theorem to the families that carry (F2) and (F3′), and records what each construction of the paper contributes there.

For (F2) the families are abelian schemes with a member of CM type. Every Hodge class of every abelian variety lies in such a family (Proposition 4.4), and (F2) is equivalent to the statement that in every one of them the algebraic locus of a flat Hodge class that is algebraic at a CM member has an interior point (Theorem 4.5). The base points of [I, Theorem 17.5] are the CM seeds of the Weil families, and André's theorem makes the Weil classes of the split families seeds for every Hodge class of CM type. For (F3′) the cohomology of a variety splits into the part reached from abelian varieties by algebraic

correspondences and a complement, and (F3′) is the statement that the complement carries no Hodge class (Proposition 4.7). On the first varieties where (F3′) is open, the squares of K3 surfaces with real multiplication, the algebraic locus of the real multiplication contains every CM point, is dense and stable under a group of rational Hodge isometries, and is either everything or meagre (Theorem 4.9). In both cases what is left is the interior point, and Theorem 4.10 lists what the constructions of the paper give towards it.

4.1. **Algebraic loci.**

Definition 4.1 (Algebraic locus)**.** Let S be a smooth connected complex quasi-projective variety, $f\colon \mathcal{X} \to S$ a smooth projective morphism, and ξ a global section of $R^{2p}f_*\mathbb{Q}$ over S^{an}.[18] The *algebraic locus* of ξ is

$$\Sigma(\xi) = \{\, s \in S :\ \xi_s \text{ is the class of an algebraic cycle with rational coefficients on } \mathcal{X}_s \,\}.$$

Lemma 4.2 (The dichotomy in every family)**.** *In Definition 4.1 the following hold.*

(i) $\Sigma(\xi)$ *is the union of countably many Zariski closed subsets of* S*.*

(ii) *Either* $\Sigma(\xi) = S$*, or* $\Sigma(\xi)$ *is meagre in* S^{an}*.*

(iii) *The following are equivalent:* $\Sigma(\xi) = S$*;* $\Sigma(\xi)$ *has an interior point;* $\Sigma(\xi)$ *is not meagre;* ξ_s *is algebraic for every* s *outside a countable union of proper closed analytic subsets of* S*. If* $\Sigma(\xi)$ *is dense, they are also equivalent to* $\Sigma(\xi)$ *being closed.*

Proof. (i) The proof of [I, Theorem 16.4] applies with $(\mathcal{A}, \mathcal{S}, \omega)$ replaced by $(\mathcal{X}, S, \xi)$, as in Lemma 2.27: for a tuple $\underline{P}$ of Hilbert polynomials the fibre product of the relative Hilbert schemes $\mathrm{Hilb}^{P_i}(\mathcal{X}/S)$ is projective over S [Gro61], the class of a flat family of cycles is a flat section, two flat sections over a connected base that agree at one point agree everywhere, and the image of a proper morphism is closed. So the set of points at which $\xi_s = \sum_i q_i[Z_i]$ with Z_i of Hilbert polynomial P_i is Zariski closed, and $\Sigma(\xi)$ is the union of these strata over the countably many pairs $(\underline{P}, \underline{q})$.

(ii) and (iii). A Zariski closed subset of the irreducible variety S that has an interior point in S^{an} is all of S, and a proper one is nowhere dense. If no stratum is S, then $\Sigma(\xi)$ is a countable union of nowhere dense closed sets, hence meagre, and has no interior point, since S^{an} is a Baire space. If one stratum is S, then $\Sigma(\xi) = S$. This gives (ii) and the equivalence of the first three conditions of (iii). The complement of a countable union of proper closed analytic subsets is not meagre, by the Baire category theorem, so the fourth condition implies the third, and it is implied by the first. Finally, a dense set that is closed is S. ∎

So the dichotomy of [I, Theorem 1.1(iii)] and the equivalences of [I, Theorem 1.1(iv)] hold for every flat class on every smooth projective family; what [I, Theorem 1.1] adds for the Weil classes is a point of $\Sigma(\xi)$ and the density, and neither is automatic, since $\Sigma(\xi)$ may be empty.

[18] By Ehresmann's theorem f is a locally trivial fibration of differentiable manifolds, so $R^{2p}f_*\mathbb{Q}$ is a local system with fibre $H^{2p}(\mathcal{X}_s, \mathbb{Q})$ at s. A global section is a class ξ_s on every fibre that is locally constant under the identifications of nearby fibres, equivalently a class on one fibre fixed by the monodromy. The set of points s at which ξ_s is of type (p,p) is in general a proper subset of S, and it is algebraic by [I, Theorem B.6].

4.2. **(F2) is propagation from CM points.**

Lemma 4.3 (Points of CM type are dense)**.** *Let G be a connected reductive group over $\mathbb{Q}$, let $u_0\colon \mathrm{U}(1) \to G_{\mathbb{R}}$ be a homomorphism, and let $\mathcal{D}$ be the orbit of u_0 under conjugation by $G(\mathbb{R})^+$, with the topology of $G(\mathbb{R})^+/\mathrm{Stab}(u_0)$. Then the points $u \in \mathcal{D}$ for which $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$ for some maximal torus T of G defined over $\mathbb{Q}$ are dense in $\mathcal{D}$.*

Proof. Let $u_1 \in \mathcal{D}$ and let W be a neighbourhood of u_1. The centraliser of the torus $u_1(\mathrm{U}(1))$ in $G_{\mathbb{R}}$ is connected and reductive, and a maximal torus of it defined over $\mathbb{R}$ is a maximal torus T_1 of $G_{\mathbb{R}}$ containing $u_1(\mathrm{U}(1))$. Write $\mathfrak{g}$ for the Lie algebra of G and $\mathfrak{t}_1 \subseteq \mathfrak{g}_{\mathbb{R}}$ for that of T_1, and choose a regular element $Y_1 \in \mathfrak{t}_1$. The map $\varphi\colon G(\mathbb{R})^+ \times \mathfrak{t}_1 \to \mathfrak{g}_{\mathbb{R}}$, $(g, Y) \mapsto \mathrm{Ad}(g)Y$, has differential $(Z, Y') \mapsto [Z, Y_1] + Y'$ at $(1, Y_1)$, which is onto because $\mathfrak{g}_{\mathbb{R}} = \mathfrak{t}_1 \oplus [\mathfrak{g}_{\mathbb{R}}, Y_1]$ for a regular semisimple Y_1. So for neighbourhoods N_1 of 1 in $G(\mathbb{R})^+$ with $N_1 u_1 N_1^{-1} \subseteq W$, and N_2 of Y_1 in $\mathfrak{t}_1$ consisting of regular elements, the image $\varphi(N_1 \times N_2)$ contains a nonempty open subset of $\mathfrak{g}_{\mathbb{R}}$. The rational points $\mathfrak{g}(\mathbb{Q})$ are dense in $\mathfrak{g}_{\mathbb{R}}$, so that open subset contains some $Y \in \mathfrak{g}(\mathbb{Q})$, and $Y = \mathrm{Ad}(g)Y'$ with $g \in N_1$ and $Y' \in N_2$. As Y is rational and regular semisimple, the identity component T of its centraliser in G is a maximal torus defined over $\mathbb{Q}$, and $T_{\mathbb{R}} = gT_1g^{-1}$ because the centraliser of Y' has identity component T_1. Then $u = gu_1g^{-1}$ lies in W and $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$. ∎

For an abelian variety the lemma is used with G its Hodge group, the smallest algebraic $\mathbb{Q}$-subgroup of $\mathrm{GL}(H^1(A, \mathbb{Q}))$ whose real points contain the image of the homomorphism $\mathrm{U}(1) \to \mathrm{GL}(H^1(A, \mathbb{R}))$ that defines the Hodge structure. If $\mathrm{Hg}(A)$ lies in a $\mathbb{Q}$-torus T, then A is of CM type.[19] Indeed T lies in a maximal $\mathbb{Q}$-torus T' of $\mathrm{GL}(H^1(A, \mathbb{Q}))$, which is the group of units of a commutative semisimple $\mathbb{Q}$-subalgebra $L \subseteq \mathrm{End}(H^1(A, \mathbb{Q}))$ of dimension $2 \dim A$; the algebra L commutes with $T' \supseteq \mathrm{Hg}(A)$, so it lies in the commutant of $\mathrm{Hg}(A)$, which is $\mathrm{End}^0(A)$.

Proposition 4.4 (A family with a CM member through every Hodge class)**.** *Let A_0 be a complex abelian variety and γ a Hodge class of degree $2p$ on A_0. There are a smooth connected complex quasi-projective variety S, a polarised abelian scheme $\pi\colon \mathcal{A} \to S$, a global section ξ of $R^{2p}\pi_*\mathbb{Q}$ that is a Hodge class at every point, and points s_0, $c \in S$, such that $\mathcal{A}_{s_0} \cong A_0$ with $\xi_{s_0} = \gamma$ under this isomorphism and $\mathcal{A}_c$ is of CM type.*

Proof. Let $\ell \geq 1$ be an integer with $\ell\gamma$ integral. We construct a section with value $\ell\gamma$ at s_0 and divide it by ℓ at the end; until then γ stands for $\ell\gamma$. Fix a polarisation of A_0, put $V = H_1(A_0, \mathbb{Q})$, and fix a level $N \geq 3$. As in [I, Setup 16.1], the quotient $\mathcal{S}_g$ of the Siegel space $\mathcal{H}$ of complex structures on $V_{\mathbb{R}}$ compatible with the polarisation, by the principal congruence subgroup of level N of the stabiliser of $H_1(A_0, \mathbb{Z})$, is a smooth quasi-projective variety carrying a universal family $f\colon \mathcal{A}_g \to \mathcal{S}_g$, polarised by the fixed polarisation [BB66]; let $a_0 \in \mathcal{S}_g$ be the point of A_0 and $J_0 \in \mathcal{H}$ a lift, through which the fibre of $R^{2p}f_*\mathbb{Z}$ at every point of the image of $\mathcal{H}$ is identified with $H^{2p}(A_0, \mathbb{Z}) = \bigwedge^{2p} H_1(A_0, \mathbb{Z})^\vee$ by parallel transport in the simply connected space $\mathcal{H}$. Put $\mathbb{V} = R^{2p}f_*\mathbb{Z}$, a polarised variation of $\mathbb{Z}$-Hodge structure. By [I, Theorem B.6] the connected component $\mathcal{C}$ of its locus of Hodge classes that contains (a_0, γ) is a closed algebraic subvariety of the total space of $\mathbb{V}$; it has finitely many irreducible components.

[19] For an elliptic curve $G = \mathrm{SL}_2$ and $\mathcal{D}$ is the upper half-plane. The compact torus $u_\tau(\mathrm{U}(1))$ fixing τ lies in the real points of a maximal $\mathbb{Q}$-torus of SL_2 exactly when τ is imaginary quadratic, and the torus is then the group of elements of norm one of $\mathbb{Q}(\tau)$. So the points of Lemma 4.3 are the curves with complex multiplication, and the lemma is the familiar density of the imaginary quadratic points in the upper half-plane.

Let $G = \mathrm{Hg}(A_0)$, which lies in the symplectic group of the polarisation and fixes γ [Del82, §3], let $u_0\colon \mathrm{U}(1) \to G_{\mathbb{R}}$, $u_0(e^{i\theta}) = \cos\theta + J_0 \sin\theta$, be the homomorphism defining the Hodge structure, and let $\mathcal{D} = \{gJ_0g^{-1} : g \in G(\mathbb{R})^+\}$. For $g \in G(\mathbb{R})^+$ the complex structure $J = gJ_0g^{-1}$ is compatible with the polarisation and positive for it, since g preserves the polarisation, so $\mathcal{D} \subseteq \mathcal{H}$; and $\mathcal{D}$ is the orbit of Lemma 4.3 when J is identified with $u_J = gu_0g^{-1}$. For $J \in \mathcal{D}$ the group $u_J(\mathrm{U}(1))$ lies in $G(\mathbb{R})$, so $\mathrm{Hg}(A_J) \subseteq G$ fixes γ, which is therefore a Hodge class on A_J. The map sending $J \in \mathcal{D}$ to the image of J in $\mathcal{S}_g$ together with γ is thus a continuous map from the connected space $\mathcal{D}$ to the locus of Hodge classes of $\mathbb{V}$, so it takes values in $\mathcal{C}$. By Lemma 4.3 there are points $J_k \in \mathcal{D}$ converging to J_0 with $u_{J_k}(\mathrm{U}(1))$ in the real points of a $\mathbb{Q}$-torus of G; then $\mathrm{Hg}(A_{J_k})$ lies in that torus and A_{J_k} is of CM type. One of the finitely many irreducible components of $\mathcal{C}$, say $\mathcal{C}_0$, contains the images of infinitely many J_k, and, being closed, also their limit (a_0, γ).

Let $\rho\colon S \to \mathcal{C}_0$ be a resolution of singularities [Hir64]: S is smooth, connected and quasi-projective, since $\mathcal{C}_0$ is irreducible and quasi-projective, and ρ is projective and surjective. Let $\pi\colon \mathcal{A} \to S$ be the pull-back of $\mathcal{A}_g$ along $S \to \mathcal{C}_0 \to \mathcal{S}_g$, a polarised abelian scheme. A point of $\mathcal{C}_0$ is a pair (a, v) with $v \in \mathbb{V}_a$ of Hodge type, and $(a, v) \mapsto v$ is a flat section of the pull-back of $\mathbb{V}$ to $\mathcal{C}_0$, because the total space of a local system is a covering space of the base; its pull-back to S, divided by ℓ, is a flat section ξ of $R^{2p}\pi_*\mathbb{Q}$ of Hodge type at every point. Take s_0 over (a_0, γ) and c over the image of some J_k in $\mathcal{C}_0$. ■

Theorem 4.5 ((F2) is propagation from CM points)**.** *The following are equivalent.*

(i) (F2).

(ii) *The Hodge conjecture holds for every complex abelian variety.*

(iii) *For every polarised abelian scheme $\pi\colon \mathcal{A} \to S$ over a smooth connected complex quasi-projective variety and every global section ξ of $R^{2p}\pi_*\mathbb{Q}$, if ξ_s is algebraic at one point $s \in S$, then $\Sigma(\xi) = S$.*

(iv) *The same, for the points s at which $\mathcal{A}_s$ is of CM type: if ξ_c is algebraic at one point c with $\mathcal{A}_c$ of CM type, then $\Sigma(\xi) = S$.*

(v) *For every π and ξ as in* (iii) *with ξ_c algebraic at a point c with $\mathcal{A}_c$ of CM type, $\Sigma(\xi)$ has an interior point.*

Proof. (i) and (ii) are equivalent by Proposition 3.49. If (ii) holds and ξ_s is algebraic, then ξ_t is a Hodge class at every t by the first part of the proof of Proposition 2.17, which uses only the theorem of the fixed part and the semisimplicity of polarisable Hodge structures, and it is algebraic by (ii); so (ii) implies (iii). (iv) is the case of (iii) at points of CM type, and (iv) and (v) are equivalent by Lemma 4.2(iii).

Assume (iv). Let F be a CM field and $n \geq 2$, and let δ_0 be the trivial class, the discriminant of a hermitian space that is a sum of n hyperbolic planes, each of determinant -1. The family of (F, n, δ_0)-Weil type over its Shimura variety with a level structure, as in [I, Setup 17.1] and [I, Setup 16.1], is a polarised abelian scheme over a smooth connected quasi-projective base, on which the Weil classes are global sections ([I, Proposition 17.3(i)]). By [I, Theorem 1.1(i)] they are algebraic at a member isogenous to a product of abelian varieties with complex multiplication by F, which is of CM type, and by (iv) they are algebraic at every member. Let A be an abelian variety of split Weil type for F in the sense of [Mil20, §2.1]: F acts on it with $\dim_F H^1(A, \mathbb{Q}) = 2n$ and $\dim V_\sigma^{1,0} = n$ for every embedding σ, and for some polarisation the hermitian form of [I, Remark 17.2] has a totally isotropic subspace of dimension n, so that it is a sum of n hyperbolic planes and A is of (F, n, δ_0)-Weil type. By [I, Proposition 17.3(iv)] the Hodge structure $H^1(A, \mathbb{Q})$ is isomorphic, compatibly with F, to that of a member A_s of the family, so there is an isogeny

$A \to A_s$ commuting with F; its pull-back maps $W_F(A_s)$ onto $W_F(A)$ and algebraic classes to algebraic classes. So every abelian variety of split Weil type has algebraic Weil classes, for $n \geq 2$ by this argument and for $n = 1$ by the theorem of Lefschetz on $(1,1)$ classes. By André's theorem [Mil20, Theorem 1 and its proof], every Hodge class on an abelian variety B of CM type is a sum $\sum f_\Delta^*(t_\Delta)$ of pull-backs along homomorphisms $f_\Delta \colon B \to A_\Delta$ of Weil classes t_Δ on abelian varieties A_Δ of split Weil type for CM fields F_Δ; so the Hodge conjecture holds for every abelian variety of CM type. Now let A_0 be any abelian variety and γ a Hodge class on it, and take S, π, ξ, s_0, c from Proposition 4.4. Then ξ_c is algebraic, so by (iv) $\xi_{s_0} = \gamma$ is algebraic. This is (ii). ∎

Figure 23 draws the theorem in one family.

The implication from (iii) to (ii) is the argument of Deligne and André as Milne records it [Mil20, §5, Theorem 3 and Remark 2].[20] In that argument the families can be taken to be polarised abelian schemes and the starting points to be of CM type, which is what (iv) records, and the converse holds. The seeds are explicit: for the Weil classes they are the base points of [I, Theorem 17.5], and for every other Hodge class they are the CM members supplied by Proposition 4.4, at which the class is algebraic once the Weil classes of the split families are. For the Weil classes of the split family of a CM field, whose base point is of CM type, statement (v) is the condition of [I, Theorem 1.1(iv)]. So the condition of [I, Theorem 1.1(iv)], required in every polarised abelian scheme with a member of CM type at which the class is algebraic, is exactly (F2); it is not a reduction of (F2) to something smaller, because the interior point it asks for is the conclusion of (iii).

4.3. **(F3′) and the part reached from abelian varieties.**

Definition 4.6 (The abelian part)**.** For a smooth complex projective variety X and $p \geq 0$ let $H^{2p}_{\mathrm{ab}}(X) \subseteq H^{2p}(X,\mathbb{Q})$ be the sum of the images of the maps $\Gamma_* \colon H^k(B,\mathbb{Q}) \to H^{2p}(X,\mathbb{Q})$, $y \mapsto \mathrm{pr}_{X*}([\Gamma] \cup \mathrm{pr}_B^* y)$, over all complex abelian varieties B, the point included, all k, and all algebraic cycles Γ on $B \times X$ with rational coefficients and of the codimension that lands in degree $2p$.

Proposition 4.7 ((F3′) is a statement about the complement of the abelian part)**.** *Let X be a smooth complex projective variety and $p \geq 0$.*

(i) *$H^{2p}_{\mathrm{ab}}(X)$ is a sub-Hodge structure of $H^{2p}(X,\mathbb{Q})$, spanned by the images of finitely many of the maps Γ_*.*

(ii) $\mathrm{Ab}^p(X) = \mathrm{Hdg}^p(X) \cap H^{2p}_{\mathrm{ab}}(X)$.

(iii) *Let $T^{2p}_{\mathrm{ab}}(X)$ be a complement of $H^{2p}_{\mathrm{ab}}(X)$ in $H^{2p}(X,\mathbb{Q})$ as a Hodge structure. Then* (F3′) *holds for X in degree $2p$ if and only if $T^{2p}_{\mathrm{ab}}(X)$ contains no nonzero Hodge class, in particular whenever the Hodge group of $T^{2p}_{\mathrm{ab}}(X)$ fixes no nonzero vector.*

(iv) *X lies in the class $\mathcal{A}$ of Proposition 2.35(iii) if and only if $T^{2p}_{\mathrm{ab}}(X) = 0$ for every p and the odd cohomology is reached in the same way.*

Proof. (i) An algebraic cycle Γ of codimension e on $B \times X$, with $\dim B = b$, induces a morphism of Hodge structures $H^k(B,\mathbb{Q}) \to H^{k+2e-2b}(X,\mathbb{Q})(e-b)$, so its image is a

[20] Deligne's proof that Hodge classes on abelian varieties are absolutely Hodge uses families of the same kind [Del82]. By his Principle B [Del82, §2] the property of being absolutely Hodge passes from one member of a family to every member, so a family through a given Hodge class with a member of CM type reduces the question to that member. Algebraicity is not known to pass from one member to all, and (iii) is precisely that property.

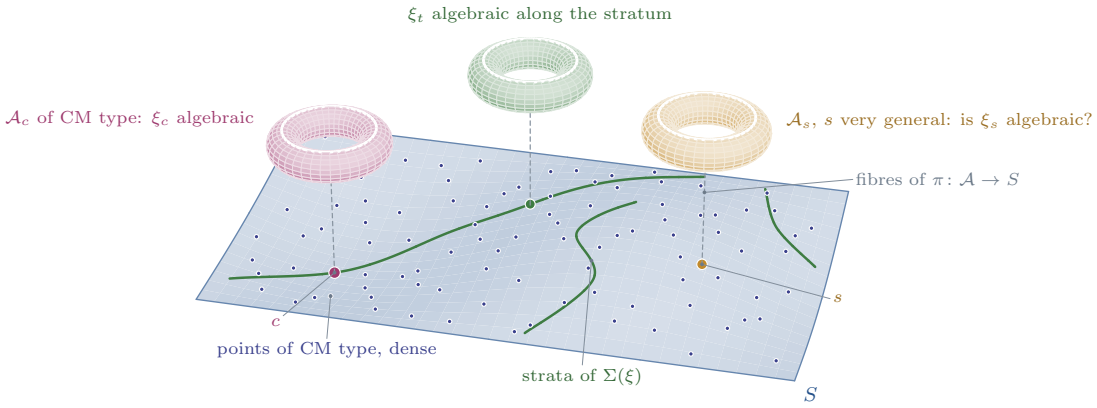


FIGURE 23. Theorem 4.5 in one family. The sheet is the base S of a polarised abelian scheme $\pi\colon \mathcal{A} \to S$ through a given Hodge class, as in Proposition 4.4; the small dots are its points of CM type, dense by Lemma 4.3, and the green curves are strata of the algebraic locus $\Sigma(\xi)$ of the flat class ξ (Lemma 4.2). Above three points the fibres are drawn as tori, and the loop on each stands for ξ, carried from fibre to fibre by parallel transport. At the CM member $\mathcal{A}_c$ the class is algebraic, and it stays algebraic along the stratum through c. At a very general member $\mathcal{A}_s$ it is algebraic exactly when $\Sigma(\xi)$ has an interior point, and (F2) is the statement that this happens in every such family.

sub-Hodge structure; a sum of sub-Hodge structures is one, and a finite-dimensional space is spanned by finitely many of them.

(ii) Every $\Gamma_*\beta$ with β a Hodge class lies in both sets, so $\mathrm{Ab}^p(X)$ is contained in the intersection. Conversely take (B_j, Γ_j, k_j), $1 \le j \le r$, as in (i), and let

$$\Phi = \sum_j \Gamma_{j*}\colon \bigoplus_j H^{k_j}(B_j, \mathbb{Q})(t_j) \longrightarrow H^{2p}_{\mathrm{ab}}(X),$$

with the Tate twists t_j that make it a morphism of pure Hodge structures of weight $2p$; it is surjective. Polarisable Hodge structures form a semisimple category, so Φ has a section σ that is a morphism of Hodge structures. For a Hodge class $\gamma \in H^{2p}_{\mathrm{ab}}(X)$ the class $\sigma(\gamma)$ is of type (p,p), so each component $\beta_j \in H^{k_j}(B_j, \mathbb{Q})$ is a Hodge class, k_j being even, and $\gamma = \sum_j \Gamma_{j*}\beta_j \in \mathrm{Ab}^p(X)$.

(iii) A complement exists by semisimplicity, and the Hodge classes of $H^{2p}(X, \mathbb{Q})$ are the sums of those of the two summands; by (ii) the Hodge classes of the first summand are $\mathrm{Ab}^p(X)$. So $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$ exactly when the second has none, and a Hodge class of $T^{2p}_{\mathrm{ab}}(X)$ is a vector fixed by its Hodge group [Del82, §3]. (iv) is the definition of $\mathcal{A}$ read degree by degree. ∎

So (F3′) asks nothing of the abelian part, where by (ii) and Proposition 3.49 the Hodge conjecture reduces to (F2), and it asks of the complement that it carry no Hodge class. The abelian part contains every algebraic class, taking B a point, so under the Hodge conjecture the complement carries none. To prove (F3′) for X in degree $2p$ without it, one has to show that every Hodge class of X is a sum of classes $\Gamma_*\beta$, that is, to construct algebraic cycles Γ on products $B \times X$ that reach it, which is the step Remark 2.36 isolates.

Proposition 2.35(iii), Proposition 2.42 and Theorem 2.52 are cases in which the complement is zero, and Proposition 2.39 and Proposition 2.40 cases in which the conjecture itself makes its Hodge classes algebraic.

4.4. **The first open case of (F3′) in families.** The frontier of (F3′) in Remark 2.68 begins with $S \times S$ and $S^{[2]}$ for a K3 surface S with real multiplication. These surfaces come in families over Hermitian symmetric domains, and the argument of [I, Theorem 1.1] applies to them with rational Hodge isometries in place of isogenies.

Setup 4.8 (A family with real multiplication)**.** Let E be a totally real field of degree $r \geq 2$ and T a finite-dimensional $\mathbb{Q}$-vector space with an action of E and a nondegenerate symmetric form q for which E acts by self-adjoint maps, so that $q = \mathrm{Tr}_{E/\mathbb{Q}}\, q_E$ for a unique E-bilinear form q_E. Put $m = \dim_E T \geq 3$, and suppose that q_E has signature $(2, m-2)$ at one real place σ_1 of E and is negative definite at the others, and that (T, q) embeds isometrically in $\Lambda_{\mathbb{Q}}$, where Λ is the K3 lattice; write $N = T^{\perp} \cap \Lambda$. Let $T_1 = T \otimes_{E,\sigma_1} \mathbb{R}$ and

$$\mathcal{D}_E = \{\, [\omega] \in \mathbb{P}(T_1 \otimes_{\mathbb{R}} \mathbb{C}) :\ q(\omega,\omega) = 0,\ q(\omega,\bar{\omega}) > 0 \,\}^{+},$$

one of its two components, a bounded symmetric domain of dimension $m-2$. For $\omega \in \mathcal{D}_E$ let S_ω be a K3 surface with a marking $H^2(S_\omega, \mathbb{Z}) \cong \Lambda$ carrying $H^{2,0}(S_\omega)$ to $\mathbb{C}\omega$, which exists by the surjectivity of the period map [Huy16, Chs. 6–7]; it is projective, since N has signature $(1, 21 - rm)$ and so contains a class of positive square, which is of type $(1,1)$ [Huy16, Ch. 8]. Each $e \in E$ acts on $T_1 \otimes \mathbb{C}$ by the scalar $\sigma_1(e)$ and is self-adjoint, so it preserves the Hodge decomposition of T, and $T \subseteq H^2(S_\omega, \mathbb{Q})$ is a sub-Hodge structure containing $T(S_\omega)_{\mathbb{Q}}$. Write $\tilde{e}_\omega \in H^4(S_\omega \times S_\omega, \mathbb{Q})$ for the class of the endomorphism $e \oplus 0$ of $H^2(S_\omega, \mathbb{Q}) = T \oplus N_{\mathbb{Q}}$, a Hodge class, and let

$$\Sigma_E = \{\, \omega \in \mathcal{D}_E :\ \tilde{e}_\omega \text{ is algebraic for every } e \in E \,\}$$

be the algebraic locus of the real multiplication; as $e \mapsto \tilde{e}_\omega$ is $\mathbb{Q}$-linear, it suffices to ask this for e in a $\mathbb{Q}$-basis of E. The set Σ_E does not depend on the choice of the surfaces S_ω. If (S, φ) and (S', φ') are two marked choices for ω, then $\psi = \varphi'^{-1}\varphi$ is a rational Hodge isometry $H^2(S, \mathbb{Q}) \to H^2(S', \mathbb{Q})$, induced by an algebraic class Z on $S \times S'$ by Buskin's theorem [Bus19], and Z^t induces the adjoint ψ^{-1}; the composite $Z \circ \tilde{e} \circ Z^t$ acts on $H^2(S', \mathbb{Q})$ as $\psi(e \oplus 0)\psi^{-1}$, which is $e \oplus 0$ for the marking φ', and by zero on H^0 and H^4, so it is the class $\tilde{e}'$ of S', and $\tilde{e}$ is algebraic if and only if $\tilde{e}'$ is. Let $G = \mathrm{Res}_{E/\mathbb{Q}} \mathrm{SO}(T, q_E)$, so that $G(\mathbb{R}) = \mathrm{SO}(T_1) \times \prod_{\sigma \neq \sigma_1} \mathrm{SO}(T_\sigma)$; write $G(\mathbb{R})^+$ for the identity component of this Lie group, which acts transitively on $\mathcal{D}_E$ through its first factor, and $\Gamma_{\mathbb{Q}} = G(\mathbb{Q}) \cap G(\mathbb{R})^+$.

The surfaces with real multiplication of [vG08, vGS25] lie in families of this kind,[21] and so do the surfaces S_λ of Theorem 3.4, for which E is cubic, $m = 3$ and $\mathcal{D}_E$ is a disc.

Theorem 4.9 (The algebraic locus of a real multiplication)**.** *Keep Setup 4.8.*

(i) *The points ω at which S_ω has complex multiplication, that is, at which the Hodge group of $T(S_\omega)_{\mathbb{Q}}$ is commutative, are dense in $\mathcal{D}_E$, and they lie in Σ_E.*

(ii) *Σ_E is stable under $\Gamma_{\mathbb{Q}}$, and $\Gamma_{\mathbb{Q}}$ is dense in $G(\mathbb{R})^+$.*

(iii) *Σ_E is dense in $\mathcal{D}_E$, and either $\Sigma_E = \mathcal{D}_E$ or Σ_E is meagre.*

[21] For a projective K3 surface S the transcendental part $T(S)_{\mathbb{Q}}$ is an irreducible Hodge structure whose algebra of Hodge endomorphisms is a field, totally real or CM [Zar83]; S has real multiplication when this field is totally real and not $\mathbb{Q}$. The space $T(S)_{\mathbb{Q}}$ is then a vector space over the field, so its degree divides $\dim T(S)_{\mathbb{Q}}$, and a very general member of a family of Setup 4.8 has $T(S_\omega)_{\mathbb{Q}} = T$ and real multiplication by exactly E, as the proof of Theorem 4.9 shows.

(iv) *The following are equivalent: Σ_E is closed; Σ_E has an interior point; Σ_E is not meagre; $\Sigma_E = \mathcal{D}_E$; the Hodge conjecture holds for $S_\omega \times S_\omega$ for every ω outside a countable union of proper closed analytic subsets of $\mathcal{D}_E$. When they hold, the Hodge conjecture and* (F3′) *hold for $S_\omega \times S_\omega$ and $S_\omega^{[2]}$ at every ω at which $\mathrm{End}_{\mathrm{Hdg}}(T(S_\omega)_{\mathbb{Q}})$ is E or a CM field.*

Proof. We first record two facts about the very general point. The Hodge structure of S_ω on T is given by $u_\omega\colon \mathrm{U}(1) \to \mathrm{SO}(T_{\mathbb{R}})$, acting by z^2 on $\mathbb{C}\omega$, by z^{-2} on $\mathbb{C}\bar\omega$ and trivially on their orthogonal complement; it commutes with E, lies in $G(\mathbb{R})^+$, and $u_{g\omega} = g u_\omega g^{-1}$ for $g \in G(\mathbb{R})^+$.

(a) *No nonzero $t \in T$ is orthogonal to every $\omega \in \mathcal{D}_E$.* The lines $\mathbb{C}\omega$, $\omega \in \mathcal{D}_E$, span $T_1 \otimes \mathbb{C}$, since $\mathcal{D}_E$ is a nonempty open subset of the nondegenerate quadric $q(\omega,\omega) = 0$ in $\mathbb{P}(T_1 \otimes \mathbb{C})$, so such a t would have zero image in T_1; but $T \to T_1$ is injective, since T is a free E-module and σ_1 is injective.

(b) *An endomorphism f of T that is an endomorphism of Hodge structures at every $\omega \in \mathcal{D}_E$ is multiplication by an element of E.* Such an f commutes with every u_ω, hence with the elements X_ω of the Lie algebra of $G(\mathbb{R})$ that span the Lie algebras of the groups $u_\omega(\mathrm{U}(1))$. They lie in $\mathfrak{so}(T_1) \oplus 0$, which acts on $T' = \bigoplus_{\sigma \neq \sigma_1} T_\sigma$ by zero, and their span is stable under $G(\mathbb{R})^+$, since $u_{g\omega} = g u_\omega g^{-1}$, so it is an ideal of $\mathfrak{so}(T_1)$. That ideal has nonzero projection to every simple ideal, as the eigenvalues $2i, -2i, 0, \dots, 0$ of X_ω on $T_1 \otimes \mathbb{C}$ show also for $m = 4$, where $\mathfrak{so}(T_1) \otimes \mathbb{C} \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ and $T_1 \otimes \mathbb{C}$ is the tensor product of the two standard representations; so it is all of $\mathfrak{so}(T_1)$, and f commutes with $\mathfrak{so}(T_1)$. For $m \geq 3$ the representation of $\mathfrak{so}(T_1)$ on $T_1 \otimes \mathbb{C}$ is irreducible and nontrivial, so f preserves $T_1 = \mathfrak{so}(T_1) \cdot T_{\mathbb{R}}$ and T' and acts on $T_1 \otimes \mathbb{C} = T \otimes_{E,\sigma_1} \mathbb{C}$ by a scalar λ. An automorphism τ of $\mathbb{C}$, acting on $T \otimes_{\mathbb{Q}} \mathbb{C}$ through the second factor, commutes with f and maps $T \otimes_{E,\sigma_1} \mathbb{C}$ onto $T \otimes_{E,\tau\sigma_1} \mathbb{C}$, so f acts on the latter by $\tau(\lambda)$; and every embedding of E is of the form $\tau\sigma_1$. So $f \otimes 1$ is multiplication by an element e of $E \otimes_{\mathbb{Q}} \mathbb{C}$ on $T \otimes_{\mathbb{Q}} \mathbb{C} = T \otimes_E (E \otimes_{\mathbb{Q}} \mathbb{C})$, and e is fixed by $\mathrm{Aut}(\mathbb{C})$ because f is defined over $\mathbb{Q}$; hence $e \in E$.

For $t \in T$ the set of ω with $t \perp \omega$ is a closed analytic subset of $\mathcal{D}_E$, and for $f \in \mathrm{End}_{\mathbb{Q}}(T)$ so is the set where f is an endomorphism of Hodge structures, namely where $f(\omega) \in \mathbb{C}\omega$ and $f^*(\omega) \in \mathbb{C}\omega$, f^* the adjoint. By (a) and (b) these sets are proper for $t \neq 0$ and $f \notin E$. There are countably many of them, and outside their union $\mathrm{NS}(S_\omega)_{\mathbb{Q}} \cap T = 0$, so $T(S_\omega)_{\mathbb{Q}} = T$ and $\mathrm{End}_{\mathrm{Hdg}}(T(S_\omega)_{\mathbb{Q}}) = E$.

(i) The Hodge group of T acts trivially on $T \cap \mathrm{NS}(S_\omega)_{\mathbb{Q}}$, whose orthogonal complement in T is $P = T(S_\omega)_{\mathbb{Q}}$, so it is commutative exactly when the Hodge group H of P is. The group G is connected and reductive, and $\omega \mapsto u_\omega$ identifies $\mathcal{D}_E$ with the orbit of Lemma 4.3, since $g \in G(\mathbb{R})^+$ fixes u_ω exactly when it preserves the line $\mathbb{C}\omega$, the eigenspace of $u_\omega(z)$ for z^2. By Lemma 4.3 the points at which $u_\omega(\mathrm{U}(1))$ lies in the real points of a maximal $\mathbb{Q}$-torus of G are dense; at them the Hodge group of T, the smallest algebraic $\mathbb{Q}$-subgroup whose real points contain $u_\omega(\mathrm{U}(1))$, lies in that torus, so these points have complex multiplication. Now let ω be any point at which H is commutative. The Hodge structure P is irreducible [Zar83], so P is an irreducible representation of the torus H, whose rational points are Zariski dense in it. The $\mathbb{Q}$-subalgebra L of $\mathrm{End}(P)$ that $H(\mathbb{Q})$ generates is therefore commutative, and P is a simple faithful L-module; so L is a field with $\dim_L P = 1$, and its commutant $E' = \mathrm{End}_{\mathrm{Hdg}}(P)$ is L. By [Zar83] E' is totally real or CM. If it were totally real, the embedding τ through which it acts on the line $H^{2,0}(S_\omega)$ would be real, and that line would lie in $P \otimes_{E',\tau} \mathbb{C}$, which is one-dimensional and stable under complex conjugation, so it would also contain the different line $H^{0,2}(S_\omega)$, which is impossible. So E' is a CM

field, the Hodge conjecture holds for $S_\omega \times S_\omega$ by Proposition 2.39(i), and the Hodge classes $\tilde{e}_\omega$ are algebraic.

(ii) Let $g \in \Gamma_\mathbb{Q}$ and $\omega \in \Sigma_E$. Then $g\omega \in \mathcal{D}_E$, and $\phi = \mathrm{id}_N \oplus g$ is a rational isometry of $\Lambda_\mathbb{Q} = N_\mathbb{Q} \oplus T$ carrying $\mathbb{C}\omega$ to $\mathbb{C}g\omega$, so through the markings it is a rational Hodge isometry $H^2(S_\omega, \mathbb{Q}) \to H^2(S_{g\omega}, \mathbb{Q})$, induced by an algebraic class $Z \in H^4(S_\omega \times S_{g\omega}, \mathbb{Q})$ [Bus19], whose transpose induces ϕ^{-1}. As g is E-linear, $\phi \circ (e \oplus 0) \circ \phi^{-1} = e \oplus 0$, so, as in Setup 4.8, $\tilde{e}_{g\omega} = Z \circ \tilde{e}_\omega \circ Z^t$ is algebraic.

For the density of $\Gamma_\mathbb{Q}$, let Y be an $E \otimes \mathbb{R}$-linear map of $T \otimes \mathbb{R}$ that is skew-adjoint for q and has $1 - Y$ invertible. Its Cayley transform $c(Y) = (1 + Y)(1 - Y)^{-1}$ is $E \otimes \mathbb{R}$-linear, has $c(Y)^* c(Y) = 1$ because $1 - Y$ is the adjoint of $1 + Y$, and has determinant one for the same reason, so $c(Y) \in G(\mathbb{R})$; and c is a homeomorphism onto the open neighbourhood $\{g \in G(\mathbb{R}) : 1 + g \text{ invertible}\}$ of 1, with inverse $g \mapsto (g - 1)(g + 1)^{-1}$. The E-linear skew-adjoint maps of T are dense among these Y, and c maps them into $G(\mathbb{Q})$. So $\Gamma_\mathbb{Q}$ is dense in a connected open neighbourhood of 1, its closure is an open subgroup of $G(\mathbb{R})^+$, and it is all of $G(\mathbb{R})^+$.

(iii) and (iv). Density follows from (i). If Σ_E contains a nonempty open set U_0, then $\Sigma_E = \mathcal{D}_E$: for $\omega \in \mathcal{D}_E$ choose $\omega' \in U_0$ and $g_1 \in G(\mathbb{R})^+$ with $g_1\omega' = \omega$; the set of $g \in G(\mathbb{R})^+$ with $g^{-1}\omega \in U_0$ is open and contains g_1, so by (ii) it contains some $g \in \Gamma_\mathbb{Q}$, and $\omega = g(g^{-1}\omega) \in \Sigma_E$ by (ii).

It remains to show that Σ_E has an interior point as soon as it is not meagre. Fix $h \in N$ with $h^2 > 0$ and $h \cdot \delta \neq 0$ for every $\delta \in N$ with $\delta^2 = -2$, which exists because the hyperplanes $\delta^\perp$ are locally finite in the positive cone of $N_\mathbb{R}$ [Huy16, Ch. 8]. Let $\mathcal{D}^\circ$ be the set of $\omega \in \mathcal{D}_E$ that are orthogonal to no $\delta \in \Lambda$ with $\delta^2 = -2$ and $h \cdot \delta = 0$. Writing such a δ as $n + t$ with $n \in N_\mathbb{Q}$ and $t \in T$, we have $t \neq 0$, because $N_\mathbb{Q} \cap \Lambda = N$, and $\delta \perp \omega$ exactly when $t \perp \omega$; so by (a) the complement of $\mathcal{D}^\circ$ lies in a countable union of proper closed analytic subsets and is meagre.

Let $\omega \in \mathcal{D}^\circ$. No class of $\mathrm{NS}(S_\omega)$ of square -2 is orthogonal to h, so after composing the marking with an isometry $\pm w$, w a product of reflections in such classes, which fixes ω and so changes only the choice of S_ω, the class h is that of an ample line bundle L on S_ω [Huy16, Ch. 8]. The deformations of the pair (S_ω, L) form a smooth family over a base that the local period map identifies with a neighbourhood of ω in the domain of periods orthogonal to h [Huy16, Ch. 6], and that domain contains $\mathcal{D}_E$, since $h \in N$. Restricting to $\mathcal{D}_E$, we obtain an open neighbourhood U of ω in $\mathcal{D}_E$ and a smooth family $\mathcal{X}_U \to U$ of K3 surfaces with markings, whose fibre over u is a choice of S_u, carrying a line bundle that restricts to L on S_ω. On S_ω the bundle $L^{\otimes 3}$ is very ample and has no higher cohomology [Huy16, Ch. 2]; by semicontinuity and base change, after shrinking U, its direct image is free and its sections embed $\mathcal{X}_U$ in $\mathbb{P}^M \times U$ over U. The classes $\tilde{e}$ are flat sections of the fourth cohomology of $\mathcal{X}_U \times_U \mathcal{X}_U$, which is projective over U. As in the proof of Lemma 4.2(i), with the Hilbert schemes of $\mathbb{P}^M \times \mathbb{P}^M$ [Gro61], which are projective, for each e in a $\mathbb{Q}$-basis of E the set of $u \in U$ at which $\tilde{e}_u = \sum_i q_i[Z_i]$ with rational q_i and Z_i of Hilbert polynomial P_i is the image, under a proper map, of a union of connected components of a fibre product of relative Hilbert spaces over U, and so it is closed in U. Taking the union over the countably many $(\underline{P}, \underline{q})$ and the intersection over the basis, $\Sigma_E \cap U$ is a countable union of subsets of U that are closed in U.

The sets U constructed for the points of $\mathcal{D}^\circ$ cover it, and countably many of them, U_i, suffice, since $\mathcal{D}_E$ has a countable basis. If Σ_E is not meagre, neither is $\Sigma_E \cap \mathcal{D}^\circ$, so some $\Sigma_E \cap U_i$ is not meagre, and by the Baire category theorem in U_i one of the countably many closed subsets of U_i that make it up has an interior point. This proves (iii), and in (iv) the

equivalence of the first four conditions, since a dense closed set is everything. The fifth condition implies the third, because at each point where the Hodge conjecture holds for $S_\omega \times S_\omega$ the Hodge classes $\tilde{e}_\omega$ are algebraic, and the complement of a countable union of proper closed analytic subsets is not meagre.

If $\Sigma_E = \mathcal{D}_E$ and $\mathrm{End}_{\mathrm{Hdg}}(T(S_\omega)_\mathbb{Q}) = E$, then E preserves $T(S_\omega)_\mathbb{Q}$, the smallest sub-Hodge structure of T whose complexification contains $H^{2,0}(S_\omega)$, since each $e \neq 0$ maps it onto a sub-Hodge structure of the same dimension that contains $H^{2,0}(S_\omega)$; and each $e \in E$ acts on it through the algebraic class $\pi_T \circ \tilde{e}_\omega \circ \pi_T$, where π_T is the algebraic projector onto $T(S_\omega)_\mathbb{Q}$ of the proof of Proposition 2.39; so Proposition 2.39(iii) gives the Hodge conjecture for $S_\omega \times S_\omega$ and $S_\omega^{[2]}$, and with it (F3′). If the field is CM, Proposition 2.39(i) gives the same. Outside the countable union found at the beginning of the proof the field is E, so the fourth condition implies the fifth. ∎

On the families of van Geemen and Schütt with real multiplication by $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ and $\mathbb{Q}(\zeta_9 + \zeta_9^{-1})$ the real multiplication is induced by explicit cycles [vGS25, Theorem 1.2 and Section 4.8], and on the double planes branched along six lines that have real multiplication, necessarily by a quadratic field here, by Schlickewei's cycles [Sch10, vG08]. Whenever the members on which such cycles are known contain a nonempty open subset of $\mathcal{D}_E$, (iv) gives $\Sigma_E = \mathcal{D}_E$. For the family of the surfaces S_λ, Theorem 4.9 gives the real multiplication at every CM point and on a dense set stable under $\Gamma_\mathbb{Q}$, and Theorem 3.4(ii) gives it at S_λ if the Kuga–Satake class of S_λ is algebraic; whether Σ_E is closed there is open. Since Σ_E is stable under the group $\Gamma_\mathbb{Q}$, every $\Gamma_\mathbb{Q}$-orbit lies either in Σ_E or in its complement: the Hodge isometries of (ii) move the real multiplication along an orbit and never bring it to an orbit where it is not yet algebraic, and by Remark 3.5 the composites of Hodge similarities that return to $T(S_\lambda)$ span only the maximal multiquadratic subfield of E, which is $\mathbb{Q}$ for a cubic field.

4.5. **What the constructions reach.**

Theorem 4.10 (Where (F2) and (F3′) stand)**.**

(i) What carries over. *In every polarised abelian scheme of Theorem 4.5 the algebraic locus of a flat class is a countable union of closed algebraic strata and is either everything or meagre (Lemma 4.2); in the Weil families it is moreover dense and contains an explicit CM point (*[I, Theorem 1.1]*); in every family of Setup 4.8 the algebraic locus of the real multiplication is dense, contains every CM point, is stable under $\Gamma_\mathbb{Q}$ and is either everything or meagre (Theorem 4.9); and every Hodge class of every abelian variety lies in a polarised abelian scheme with a CM member (Proposition 4.4).*

(ii) (F2). *It holds if and only if the algebraic locus of every flat Hodge class algebraic at a CM member of a polarised abelian scheme has an interior point (Theorem 4.5). In the Weil families the constructions of the paper that supply an interior point are: an object with injective semiregularity map, or satisfying the flatness condition, at one point (*[II, Theorem 2.3], [I, Theorem 17.7], [II, Proposition 3.15]*); a bound on the Hilbert data of representatives along the Hecke orbit of a base point (*[I, Theorem 16.9]*), which is equivalent to its conclusion (*[I, Theorem 16.17]*); and the Lefschetz standard conjecture for the total space over a curve through a base point, which implies it and, when the monodromy is Zariski dense, follows from it (Corollary 2.31). For the squares of Mumford fourfolds the last two take the forms of Theorem 3.43 and Corollary 2.30. Markman's sheaves are objects of the first kind, and with them he*

proves the algebraicity of the Weil classes on every abelian fourfold of Weil type and on the split sixfolds of every imaginary quadratic field [Mar25a, Mar25b]*; with* [MZ99] *this gives the Hodge conjecture for abelian varieties of dimension at most five. The open cases named in Section 2 include the non-split Weil sixfolds of imaginary quadratic fields, the Weil eightfolds of quartic CM fields, where an object of the first kind needs* $\dim \mathrm{Ext}^2(E,E) \geq 88$ *(*[II, Corollary 5.15]*), and the squares of Mumford fourfolds, where the classes are algebraic at every CM point (Theorem 2.29) and on every member once one Kuga–Satake class is algebraic (Theorem 3.4), or once one complex at one CM point has* Ext^2 *of dimension* 119 *(Theorem 3.27); on the square of a Mumford fourfold the two exceptional classes are algebraic exactly when one class on a product with an abelian sixfold is (the sixfold link of* [BB26b]*).*

(iii) (F3′). *It holds if and only if, for every variety and every degree, the complement of the abelian part carries no Hodge class (Proposition 4.7). The complement is zero on the class* $\mathcal{A}$*, which contains the Fermat and Delsarte hypersurfaces of every dimension and the cyclic covers branched along them (Proposition 2.42, Corollary 2.53), and it carries no Hodge class on the varieties of Proposition 2.43, Proposition 2.44 and Proposition 2.46, where* (F3′) *is proved, and of Proposition 2.39 and Proposition 2.40, where the conjecture itself is. On the first open varieties,* $S \times S$ *and* $S^{[2]}$ *for a K3 surface with real multiplication, the algebraic locus of the real multiplication is dense, contains every CM point and is either the whole family or meagre (Theorem 4.9); the real multiplication is algebraic on the members of the families of* [vGS25] *and on the quadratic double planes of* [Sch10]*, and whether the locus is closed is open for the family of* S_λ*, except for the two cyclic cubic fields of conductor* 7 *and* 9 *with the algebra split at one real place, where the dihedral correspondences of* [BB26b] *make the locus the whole family. Beyond the varieties of abelian type, a Hodge class off the abelian part is removed only by a new algebraic cycle on a product* $B \times X$ *(Remark 2.36).*

Neither (F2) nor (F3′) *is proved here, and by Theorem 5.1 together they are the Hodge conjecture.*

Proof. Each statement is the result cited with it. For (ii), Markman's theorems are recalled in the introduction and their reach in Section 2, where the non-split sixfolds and the fields of degree at least four are among the open entries, and [II, Theorem 5.14] and [II, Corollary 5.15] give the number 88. For (iii), the known cases of the real multiplication are those recalled in Remark 3.5. ∎

5. The closure theorem

We collect the results of the paper in one theorem. Its first clause is [I, Theorem 1.1]; the second and third give the base of that theorem in its sharpest form; the fourth and fifth say what closing the algebraic locus requires and which constructions cannot supply it; and the last places the Weil classes inside the Hodge conjecture and states what [I, Theorem 1.1] becomes when it is carried to every family. The theorems we quote from the literature are used as theorems: the semiregularity theorem of Buchweitz and Flenner [BF03], in the form for perfect complexes that Pridham proves [Pri24]; the theorem of Cattani, Deligne and Kaplan [CDK95]; the theorem of Baily and Borel [BB66]; the theorem of Pila, Shankar and Tsimerman on the André–Oort conjecture [PST21]; the description of the Hodge rings of abelian varieties of dimension at most five by Moonen and Zarhin [MZ99]; Markman's theorems on abelian varieties of Weil type [Mar25a, Mar25b]; the theorem of Bando and

Siu on stable reflexive sheaves [BS94], Kuranishi's theorem in the form of Goldman and Millson [Kur65, GM88], Grivaux's Chern classes of coherent sheaves [Gri10] and Voisin's theorem on Weil tori [Voi02b]; and the theorems of Deligne, André and Milne recorded in Section 2.6 [Del71, And96b, Mil20].

Theorem 5.1 (Closure)**.** *Let F be a CM field of degree $2m$, let $n \geq 2$, and let δ be a discriminant. Write $K = \mathbb{Q}(\sqrt{-d})$ when $m = 1$.*

(i) The algebraic locus. *The algebraic locus Σ_F of the Weil classes contains an explicit CM point at which they are polynomials with rational coefficients in divisor classes, it is dense, it is a countable union of closed algebraic strata, and it is either the whole domain or meagre. It is closed if and only if the Hodge conjecture holds for the very general member of the family, and then the Weil classes are algebraic on every member (*[I, Theorem 1.1]*).*

(ii) One class on one connected domain. *For an abelian $2n$-fold A of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ one has $\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = 1$ for $k \neq n$ and 3 for $k = n$, and the Hodge conjecture for A in every degree is the algebraicity of the single class ω_1 of* [I, (11)]*, written out with 2^{2n-1} integer coefficients. For the family of the triple (K, n, δ) the conjecture is equivalent to the algebraicity of the absolutely Hodge class ω_s at every point s of the connected period domain $\mathcal{D}_{n,n}$ of dimension n^2 (*[I, Theorem 4.14]*,* [I, Corollary 4.15]*,* [I, Theorem 15.14]*,* [I, Corollary 15.35]*).*

(iii) A base point in every family. *Every family of every CM field contains a member isogenous to a product of CM abelian m-folds at which the Weil classes are polynomials in divisor classes (*[I, Theorem 17.5]*). For an imaginary quadratic field every family also contains a member with quaternionic multiplication extending K, at which the Weil classes are the real and imaginary parts of the n-th power of one explicit $(1, 1)$ class, and every product of n abelian surfaces of Weil type is a member whose Weil classes are an explicit integral polynomial in $2n$ divisor classes (*[I, Theorem 15.25]*,* [I, Theorem 15.27]*,* [I, Theorem 15.10]*).*

(iv) Closing the locus. *If at every point of the orbit $G_F(\mathbb{Q}) \cdot s_0$ the Weil class is represented by a cycle whose Hilbert data comes from one finite set, then $\Sigma_F = \mathcal{D}_F$; this hypothesis is equivalent to its conclusion, and the transported cycle does not satisfy it whenever a component of the base cycle has finite stabiliser (*[I, Theorem 16.9]*,* [I, Theorem 16.17]*,* [I, Theorem 16.15]*). The bound is needed only along one sequence of CM points that leaves every proper special subvariety, and such a sequence exists inside the orbit (*[I, Theorem 17.14]*). The locus of members with smaller Hodge group, which contains every member at which this paper exhibits algebraic Weil classes, contains the base point, is dense, stable under $G_F(\mathbb{Q})$ and a countable union of closed algebraic strata, and is meagre, so closing the locus requires the algebraicity of the Weil classes at a member with full Hodge group (*[I, Proposition 17.12]*,* [I, Corollary 17.13]*). If at a base point some perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the divisor classes of the family has injective semiregularity map, or satisfies the flatness condition of* [II, Proposition 3.15]*, then $\Sigma_F = \mathcal{D}_F$ (*[II, Theorem 2.3]*,* [I, Theorem 17.7]*). When the Chern character is exactly a multiple of a Weil class, no complex at any member of any family, for any CM field, has injective semiregularity map (*[II, Theorem 3.5]*,* [II, Theorem 5.4]*). In the polarised form, for an imaginary quadratic field and $n \geq 3$, every object has $\dim \mathrm{Ext}^2(E, E) \geq (4 + \rho)n^2 - 2n$, with $\rho \leq 3$ the rank of the Hankel matrix of the coefficients, and an object attaining the bound meets the criterion (*[II, Theorem 3.9]*);*

at a quartic CM field and $n = 2$ the number takes fifteen values, an object meeting the criterion has $\dim \mathrm{Ext}^2(E, E) \geq 88$, and the characters of rank 88 whose Hodge locus leaves the polarised family are exactly those of θ^4 shape, which the Weil tori exclude ([II, Theorem 5.14], [II, Corollary 5.15], [II, Proposition 5.17], [II, Corollary 5.18]*). The statement is needed only for the split families of the CM fields of degree at least four, which carry the Weil classes of every CM field by descent and scalar extension (Proposition 2.13), and for one family it is equivalent to the Lefschetz standard conjecture for the total space of the family over a curve through a base point (Corollary 2.31).*

(v) Constructions that do not reach a Weil class. *Each of the following fails to produce a Weil class on a general member, in the sense and under the hypotheses of the theorem cited, which Section 2 delimits:*

- (a) *an object restricted from a projective space to the dual and transported by the Poincaré kernel (*[I, Theorem 7.3]*);*
- (b) *equivariant for the field with the norm character (*[I, Theorem 5.6]*);*
- (c) *an object whose Chern character lies on the Weil line and on which $H^2(\mathcal{O}_A)$ acts faithfully, whether through semiregularity or through the weakened criterion of* [Mar25b, Question 11.4] *(*[I, Theorem 9.7], [I, Corollary 9.18], [I, Theorem 9.19]*);*
- (d) *an object with a line bundle summand off the polarisation line, in particular a sum of line bundles (*[I, Theorem 9.20], [II, Corollary 4.21]*);*
- (e) *an object whose Chern character is effective at a base point (*[II, Theorem 4.5]*);*
- (f) *induction through hyperplane sections, in either direction (*[I, Theorem 10.3], [I, Theorem 10.8]*);*
- (g) *Kuga–Satake from a weight two sub-structure of $H^2(A, \mathbb{Q})$ (*[I, Proposition 11.11], [I, Corollary 11.12]*);*
- (h) *a divisorial argument at a general member of a family of nontrivial discriminant (*[I, Proposition 15.18]*);*
- (i) *a secant object whose support has two smooth branches of codimension two meeting transversally at a point, which includes the candidate of* [Mar25b, Section 12] *(*[II, Theorem 4.61], [II, Corollary 4.63], [II, Corollary 4.77]*);*
- (j) *a secant object of rank one whose support is a complete intersection of two divisors (*[II, Theorem 4.81]*), or a secant object quasi-isomorphic to a two-term complex of sums of powers of the polarisation or of simple semi-homogeneous bundles (*[II, Theorem 4.89], [II, Theorem 4.92]*); on the fourfold, a secant sheaf $\mathcal{I}_S(3\Theta)$ with smooth support unless $d \leq 9$, that is $d \in \{1, 3, 5, 7\}$ in* [II, Setup 4.51]*, and one whose ideal sheaf is presented by two bundles with classes in the subring generated by line bundles unless $d \equiv 15, 23 \pmod{24}$ (*[II, Corollary 4.68], [II, Corollary 4.84]*);*
- (k) *a representative of the Weil K-theory class that is a locally free sheaf on a smooth germ of codimension at least two of its support, or whose support has codimension at least n (*[II, Theorem 2.5], [II, Theorem 2.13]*);*
- (l) *at a quartic CM field, a complex with secant character that is locally free on a smooth curve at some point, in particular the explicit pair of* [Mar25c, Example 11.2.7] *(*[II, Theorem 5.28], [II, Corollary 5.30]*);*
- (m) *a convolution of fewer than $2n$ line bundles with nondegenerate differences (*[II, Theorem 4.26]*); on E_0^6, every convolution of the Weil pieces with three*

multiples of the polarisation, and on E_0^8 *every one whose pieces have their shifts within six consecutive values (*[II, Theorem 4.35]*,* [II, Corollary 4.48]*).*

(vi) The Weil classes inside the conjecture. *The Hodge conjecture is equivalent to the conjunction of* (F2)*, the algebraicity of the Hodge classes of abelian varieties outside the subring generated by divisor classes and Weil classes of quotients, and* (F3′)*, the algebraicity of every Hodge class modulo the images of Hodge classes of abelian varieties under algebraic correspondences; and* (F2) *alone is equivalent to the Hodge conjecture for abelian varieties (Proposition 3.49, Proposition 2.35). The statement* (F3′) *holds on every variety whose cohomology is reached from abelian varieties by correspondences, on every variety of dimension at most three, on every fourfold and fivefold whose zero-cycles are supported in dimension at most three or that is dominated by a variety of the first kind, on every smooth ample divisor of odd dimension in a variety of the first kind, and on every smooth Delsarte hypersurface of every dimension and every cyclic cover branched along one, and on every complete intersection of diagonal hypersurfaces of Vandermonde type, where the Hodge conjecture follows for two diagonal hypersurfaces of degree* 3*,* 4 *or* 6 *in* $\mathbb{P}^6$ *and for the very general member of degree* 2 *in every dimension, of degree* 3 *or* 4 *up to dimension seven and of degree* 6 *up to dimension five (Proposition 2.35, Proposition 2.42, Proposition 2.43, Proposition 2.44, Proposition 2.46, Proposition 2.47, Corollary 2.53, Theorem 2.55, Corollary 2.56, Theorem 2.66). The first instance of* (F2)*, the two exceptional classes on the square of a Mumford fourfold, is algebraic at every CM point of the family, is algebraic on every member once one Kuga–Satake class is algebraic, and is algebraic on every member exactly when the Lefschetz standard conjecture holds for one ninefold (Theorem 2.29, Theorem 3.4, Corollary 2.30). Over the rule set formed by the implications of the paper and of the literature, together with Proposition 3.49, the minimal sets of statements from which the conjecture follows are* $\{\mathrm{F3}\}$*,* $\{\mathrm{L},\mathrm{M}\}$*,* $\{\mathrm{L},\mathrm{F3}'\}$*,* $\{\mathrm{V},\mathrm{F3}'\}$ *and* $\{\mathrm{F2},\mathrm{F3}'\}$*, the first two being each equivalent to it (Theorem 2.70, Remark 3.50). Carried to every family,* [I, Theorem 1.1] *gives exactly these two statements back:* (F2) *is equivalent to the statement that in every polarised abelian scheme with a member of CM type the algebraic locus of a flat Hodge class that is algebraic at such a member has an interior point (Theorem 4.5);* (F3′) *is equivalent to the absence of Hodge classes off the part of the cohomology reached from abelian varieties by algebraic correspondences (Proposition 4.7); and on the squares of K3 surfaces with real multiplication, the first varieties on which* (F3′) *is open, the algebraic locus of the real multiplication is dense, contains every CM point and is either the whole family or meagre (Theorem 4.9), and it is the whole family for real multiplication by* $\sqrt{2}$*,* $\sqrt{3}$ *and* $\sqrt{6}$ *in families, for complex multiplication by* $\mathbb{Q}(\sqrt{-5})$ *and* $\mathbb{Q}(\sqrt{-7})$ *on the Fricke families, and for the cubic fields* $\mathbb{Q}(\zeta_7)^+$ *and* $\mathbb{Q}(\zeta_9)^+$ *on the surfaces of the Mumford fourfolds of the algebra split at one real place* [BB26b].[22]

Proof. Each clause is the theorem or theorems cited with it, and no clause is used in the proof of another. For (i), [I, Theorem 1.1] is proved at the end of [I, Section 17.3]. For (ii), [I, Theorem 4.14] counts the Hodge classes, [I, Corollary 4.15] reduces every degree to ω_1, [I,

[22]These are the statements (L), (M) and (V) of Definition 2.14: $B(X)$ of [I, Definition 10.6] for every smooth projective X; every Hodge class motivated in the sense of André [And96b, §2.1]; and, for every smooth projective morphism f to a smooth connected base, a flat section of $R^{2p}f_*\mathbb{Q}$ that is algebraic at one point is algebraic at every point. The statement (F3) is the Hodge conjecture for the varieties that are not abelian, which is equivalent to the whole conjecture (Proposition 2.33).

Theorem 4.9(i)] shows that ω_1 has integer coefficients and exactly 2^{2n-1} nonzero terms, [I, Theorem 15.14(a)] and (b) are the two implications for the family, and [I, Proposition B.8] gives the connectedness and the dimension. For (iii), [I, Theorem 17.5] gives the CM point, [I, Theorem 15.25(v)] the cycle at a quaternionic point, [I, Theorem 15.27] places such a point in every family by the discriminant computation of [I, Theorem 15.25(iv)] and [I, Lemma 15.26], and [I, Theorem 15.10] treats products.

For (iv), the bounded criterion is [I, Theorem 16.9] with [I, Theorem 16.17] and [I, Theorem 16.15], and its form for a CM field is [I, Theorem 17.7(iv)]; the escaping sequence is [I, Theorem 17.14], and the special locus is [I, Proposition 17.12] with [I, Corollary 17.13]. The semiregularity criterion is [II, Theorem 2.3] with [II, Remark 3.6], carried to every CM field by [I, Theorem 17.7(v)]; the flatness form is [II, Proposition 3.15]. The pure form is excluded by [II, Theorem 3.5], which rests on [II, Lemma 3.2], [II, Lemma 3.3] and [II, Proposition 3.4], and for every CM field by [II, Theorem 5.4]. The number of the polarised form is [II, Theorem 3.9(i)], (iv) and (v); the quartic rank and the Hodge loci are [II, Theorem 5.14], [II, Corollary 5.15], [II, Proposition 5.17] and [II, Corollary 5.18]. The reduction to the split families is Proposition 2.13, and the equivalence with the Lefschetz standard conjecture is Corollary 2.31. For (v), each item is the statement of the theorem cited, under its own hypotheses, which Section 2 delimits item by item.

For (vi), Proposition 2.35(i) gives the Hodge conjecture from (F3′) and the conjecture for abelian varieties, and Proposition 3.49 gives the latter from (F2); the converse implications hold because (F2) is a case of the conjecture and (F3′) is weaker than it. The cases of (F3′) are Proposition 2.35(iii) and (iv), Proposition 2.42, Proposition 2.43, Proposition 2.44, Proposition 2.46, Corollary 2.53, Theorem 2.55, Corollary 2.56 and Theorem 2.66. The statements on the Mumford classes are Theorem 2.29, Theorem 3.4(ii) and Corollary 2.30. The statements on the families are Theorem 4.5, Proposition 4.7 and Theorem 4.9; the families on which the real multiplication is algebraic are treated in [BB26b]. The minimal sets are Theorem 2.70(ii) with the rule of Proposition 3.49 adjoined, as computed in Remark 3.50, and Theorem 2.70(vi) shows that {F3} and {L, M} are each equivalent to the conjecture. ■

The theorem arranges the paper around [I, Theorem 1.1]. Clause (ii) is the problem in its smallest form, one explicitly written class on one connected domain, and clause (iii) shows that the base case is never the difficulty: every family has a base point, for every field, every dimension and every discriminant, with a cycle written down rather than an existence statement. Clause (iv) reduces the closedness of the locus to one object at one point and computes the number that object must attain, and clause (v) shows that the constructions that fail do so for identifiable structural reasons: a divisor-generated Chern character cannot leave the divisor subring, a norm eigenclass sits in the wrong summand of the character grading, a Chern character on the Weil line has an annihilator exactly as large as the family, a support with two transversal branches of codimension two through a point carries a local obstruction that the compensating scalar cannot meet, and a curve along which an object is locally free is tangent to at most one of the two eigenplanes of the real multiplication. Clause (vi) shows where the Weil classes stand in the whole conjecture: the conjecture for abelian varieties is carried by the classes beyond the Weil lines, and the rest is algebraicity modulo abelian varieties, which holds on every variety whose cohomology comes from abelian varieties, on the Delsarte hypersurfaces of every dimension and on the diagonal complete intersections of Vandermonde type (Figure 24). Section 4 shows that the structure [I, Theorem 1.1] finds in the Weil families, a CM seed, density and the

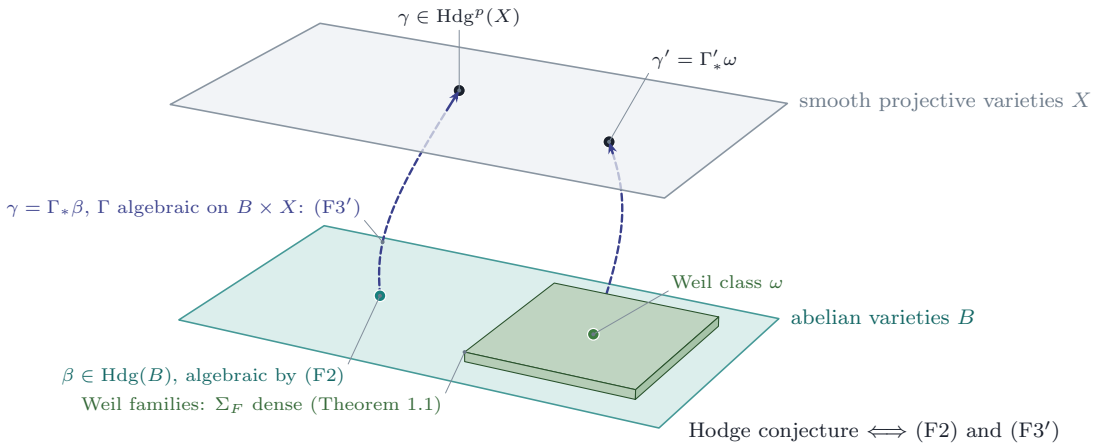


FIGURE 24. The two halves of the conjecture in Theorem 5.1(vi). A Hodge class γ on a smooth projective variety X is the image $\Gamma_*\beta$ of a Hodge class β on an abelian variety B under an algebraic correspondence Γ on $B \times X$, which is (F3′); the class β is algebraic, which is (F2), equivalently the conjecture for abelian varieties (Proposition 3.49). The raised plate on the lower sheet stands for the families of Weil type, on which [I, Theorem 1.1] makes the algebraic locus of the Weil class ω dense; the dashed part of each correspondence is the part behind the upper sheet.

dichotomy, is present in every family that carries (F2) or (F3′), and that in each of them the one missing property is the interior point that closes the locus (Theorem 4.10).

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Ethics approval and consent. This article reports no research involving human participants, human data or animals, so ethics approval and consent to participate or to publish were not required. The results are original and have not been published elsewhere except in the earlier paper named on the title page, which this paper and its companions replace, and the work of others is credited by citation throughout.

Data availability. The archive [BMB26] is deposited on Zenodo under the DOI 10.5281/zenodo.23225483. It holds programs that repeat much of the finite arithmetic of this paper and its companions [I, II] as an independent check, a Lean 4 certificate of part of that arithmetic, the Macaulay2 computations of local Ext algebras with their output, and the generators and TikZ sources of the figures; its programs perform 2218 checks, all of which pass, and the Lean kernel checks four hundred and nine theorems, each depending on no axiom or on propositional extensionality alone. Deep Bhattacharjee computed all the calculations, in C, Python, Julia, Macaulay2, Lean 4 and shell scripts, with the assistance of Claude Code, an AI model developed by Anthropic. The archive is immutable at that DOI; the command `python3 code/verify_all.py` in it runs every check and ends with their count, and `lean HodgeObstruction.lean` in its folder `lean` checks the certificate and prints the axioms of each theorem. The Lean theorems certify finite arithmetic only: rank certificates, enumerations, lattice computations and polynomial identities; no geometric argument of the three papers is formalised, and every proof in the text was written and checked by the authors. The authors are responsible for the mathematical statements and their proofs. No other data were generated or analysed.

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